

The background of the slide is a faded, light blue image. On the left side, there is a stack of several Bitcoin coins. The top coin is clearly visible, showing the Bitcoin logo (a stylized 'B' with two vertical bars) and the text 'BITCOIN' and 'DECENTRALIZED'. To the right of the coins, there is a faint, grid-like pattern resembling a digital ledger or blockchain data. The word 'Blockchain' is written in a large, bold, black serif font, centered horizontally. A thin horizontal line is positioned below the text.

Blockchain

Contents

Module A - Blockchain



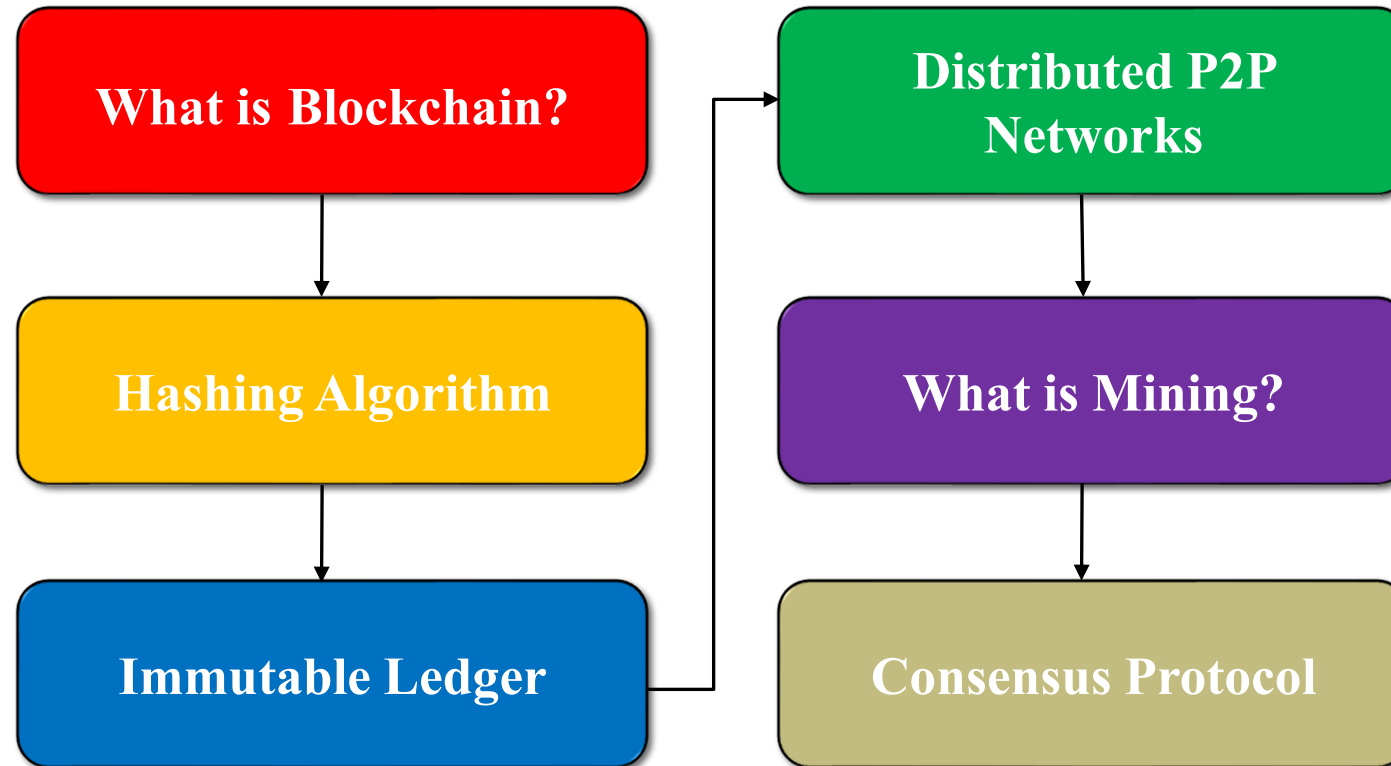
Module B - Cryptocurrency



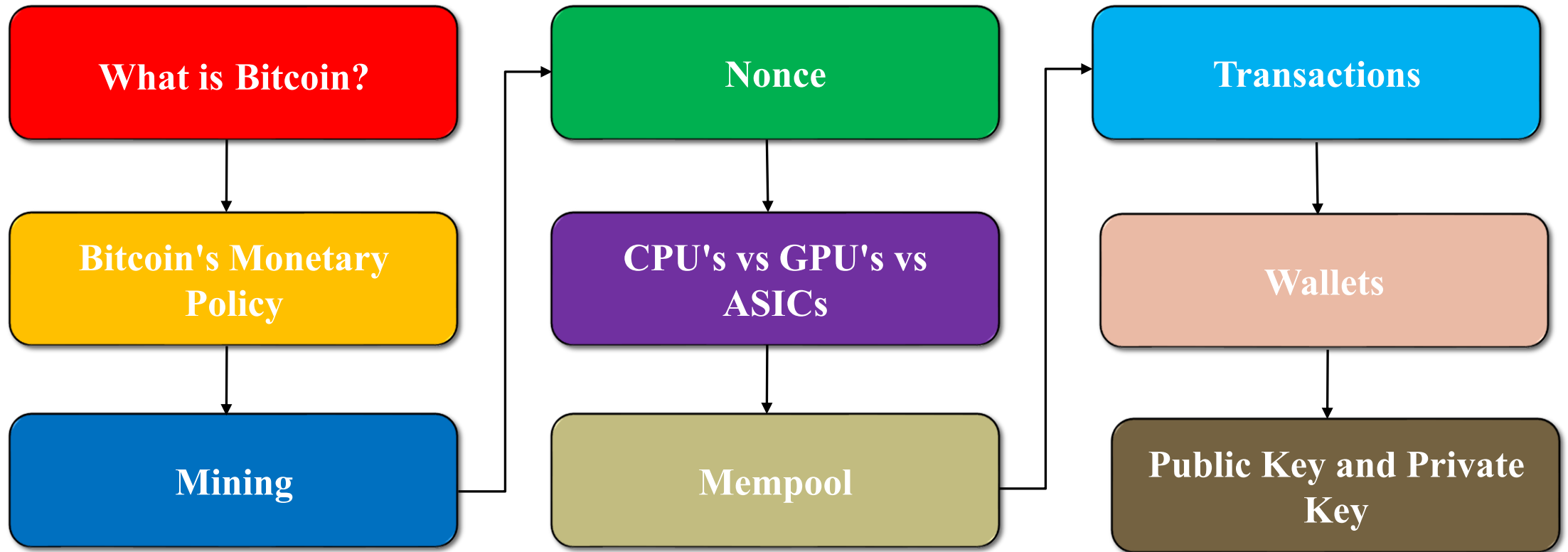
Module C – Smart Contract



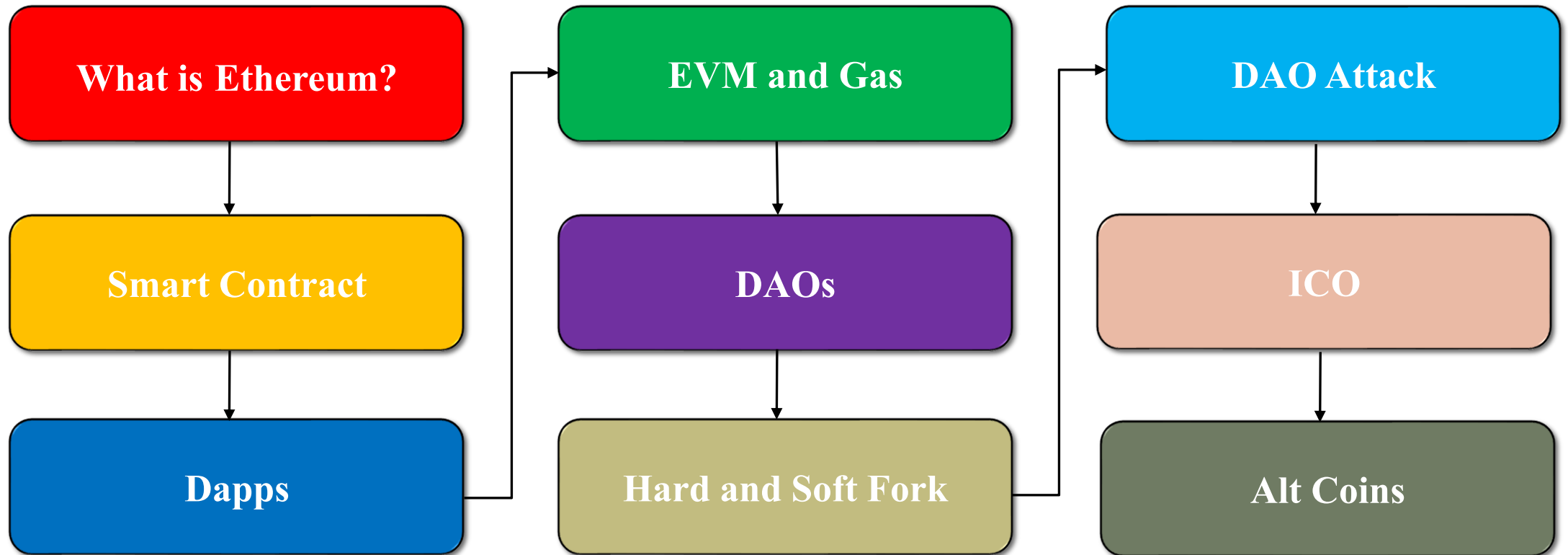
Contents – Module A

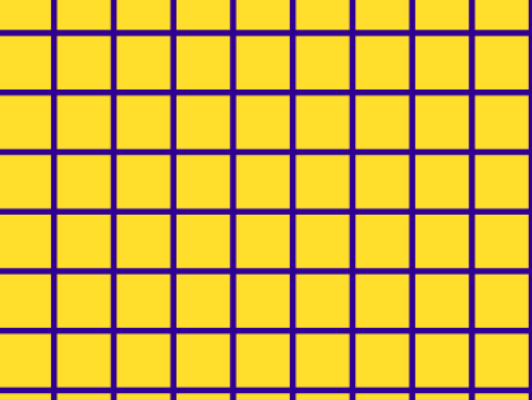


Contents – Module B



Contents – Module C





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Why study blockchain ?

Story of internet



Story of internet



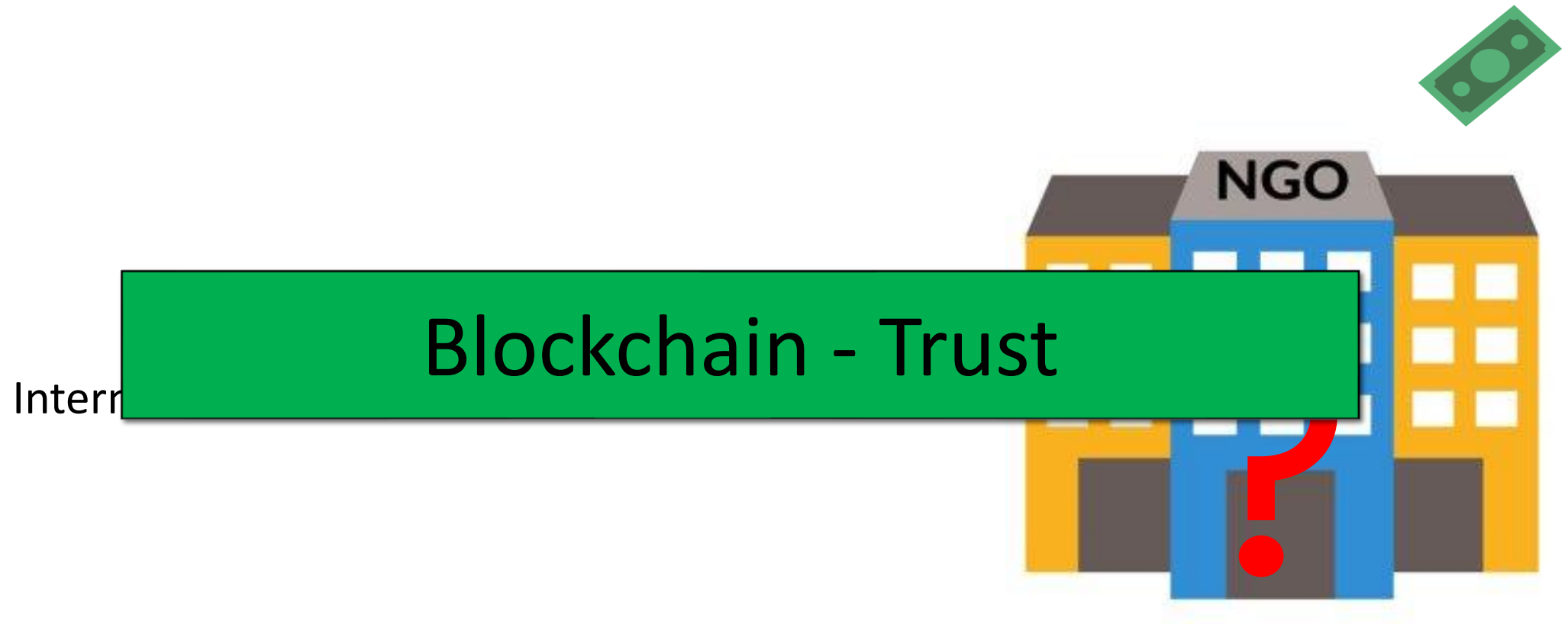
Trust

Story of internet

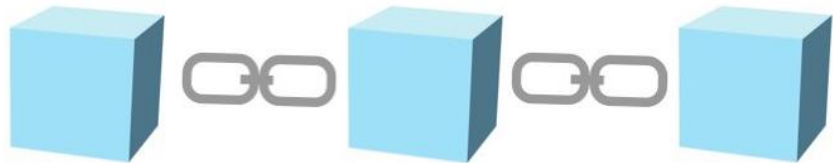
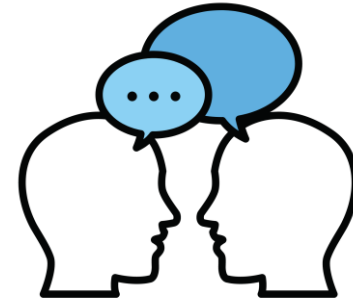
Internet Transaction



TRUST issues with internet



Disruptive Technology



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Why should I study Blockchain?

- Because Blockchain is a disruptive technology.

Q) What is a disruptive technology?



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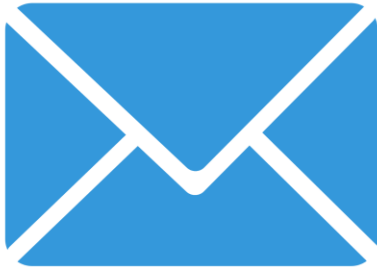


Why should I study Blockchain?



Why should I study Blockchain?

Internet:



Email



Social Media

Why should I study Blockchain?

Internet → Communication

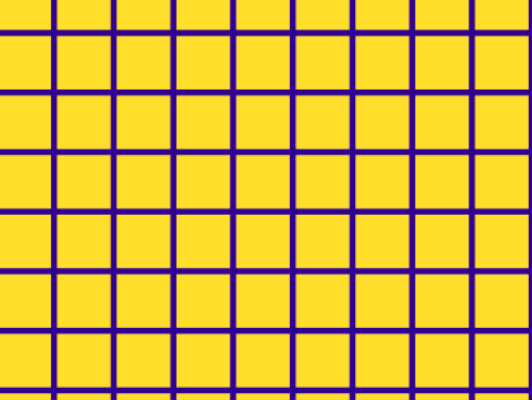
Blockchain → Trust

Why should I study Blockchain?



Why should I study Blockchain?





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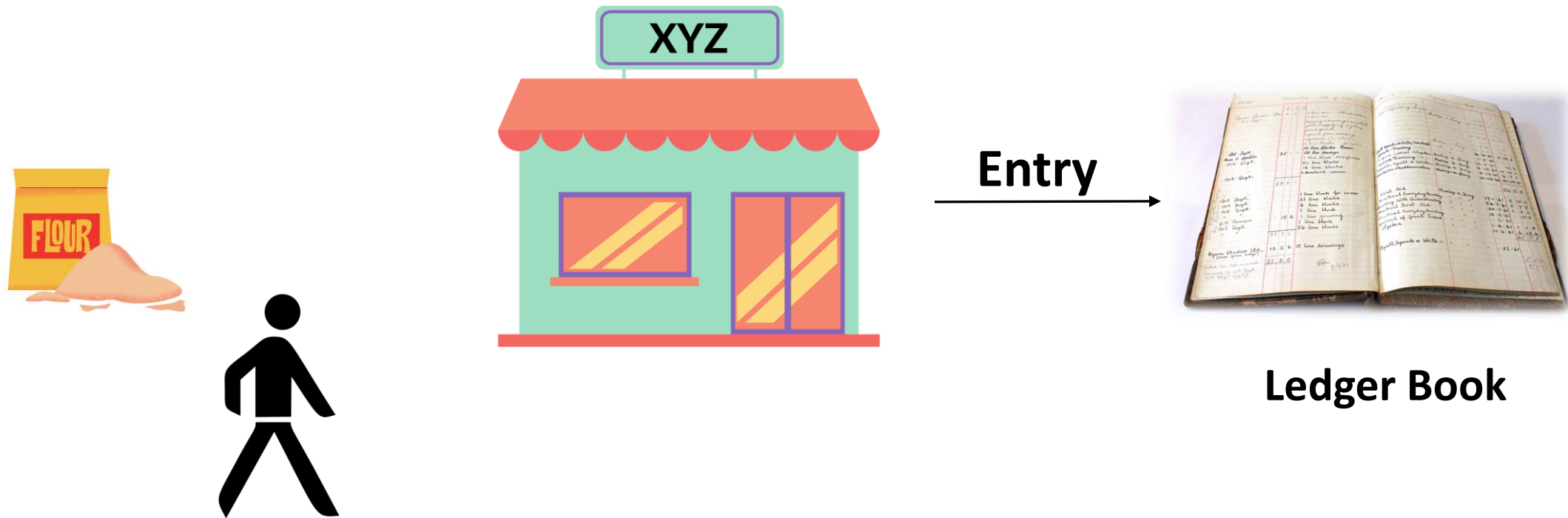


Stuart Haber

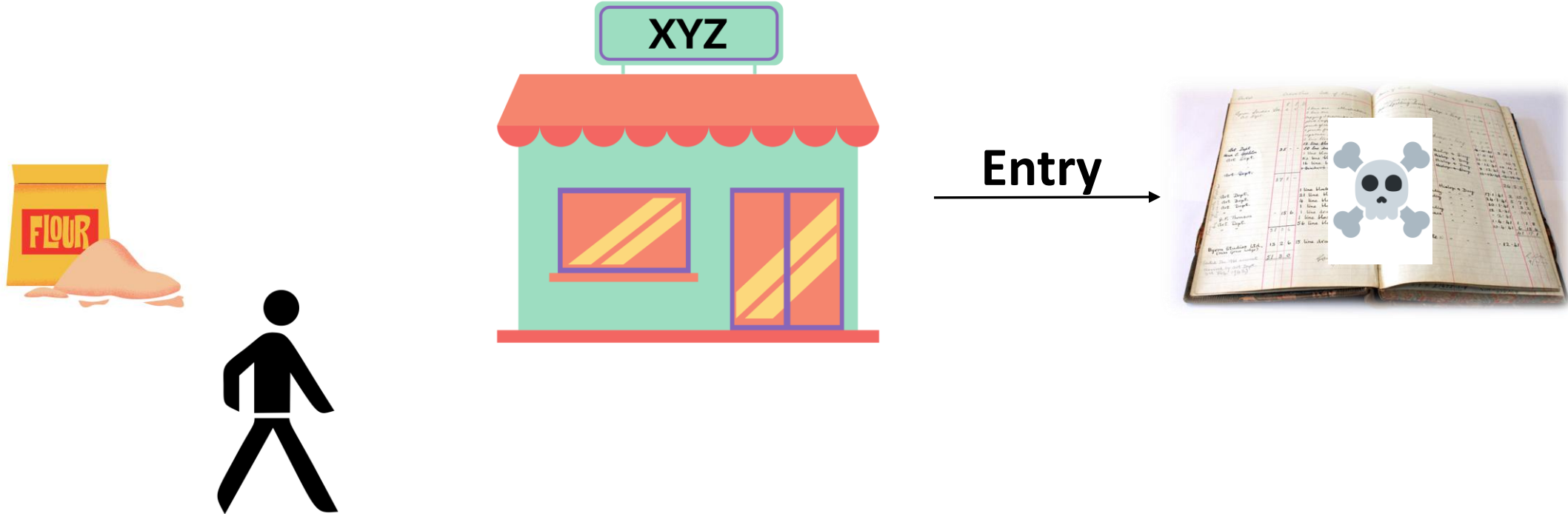


W. Scott Stornetta

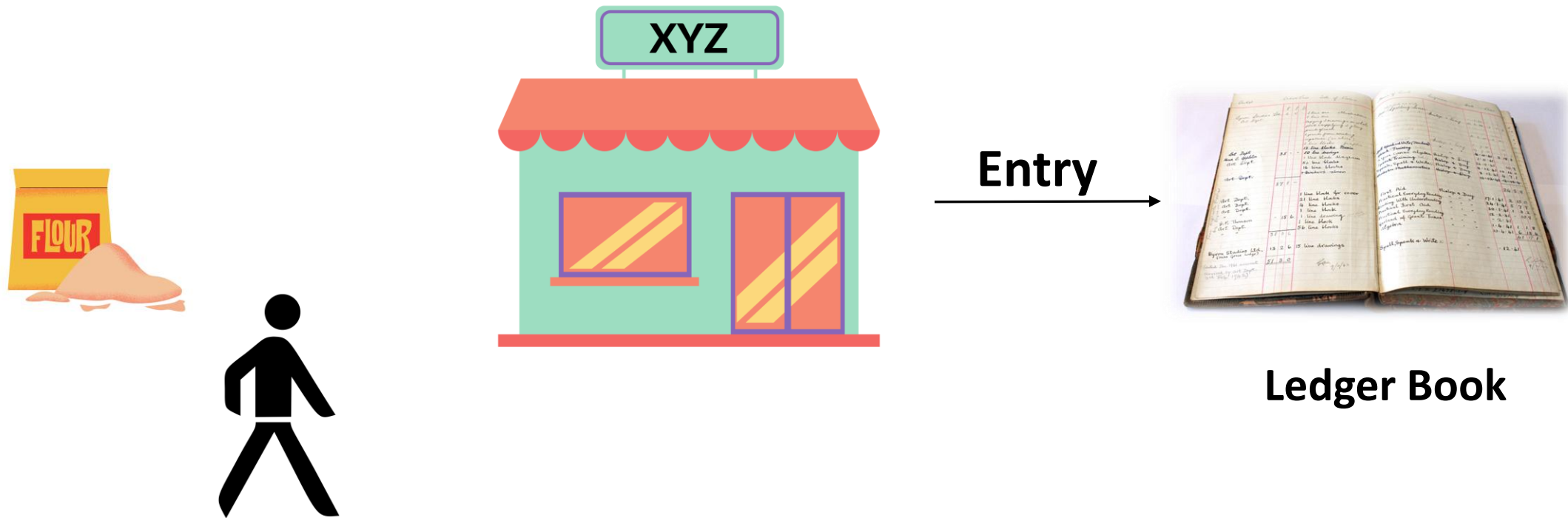
What is Blockchain?



What is Blockchain?



What is Blockchain?





What is Blockchain?

Inventors Of Blockchain

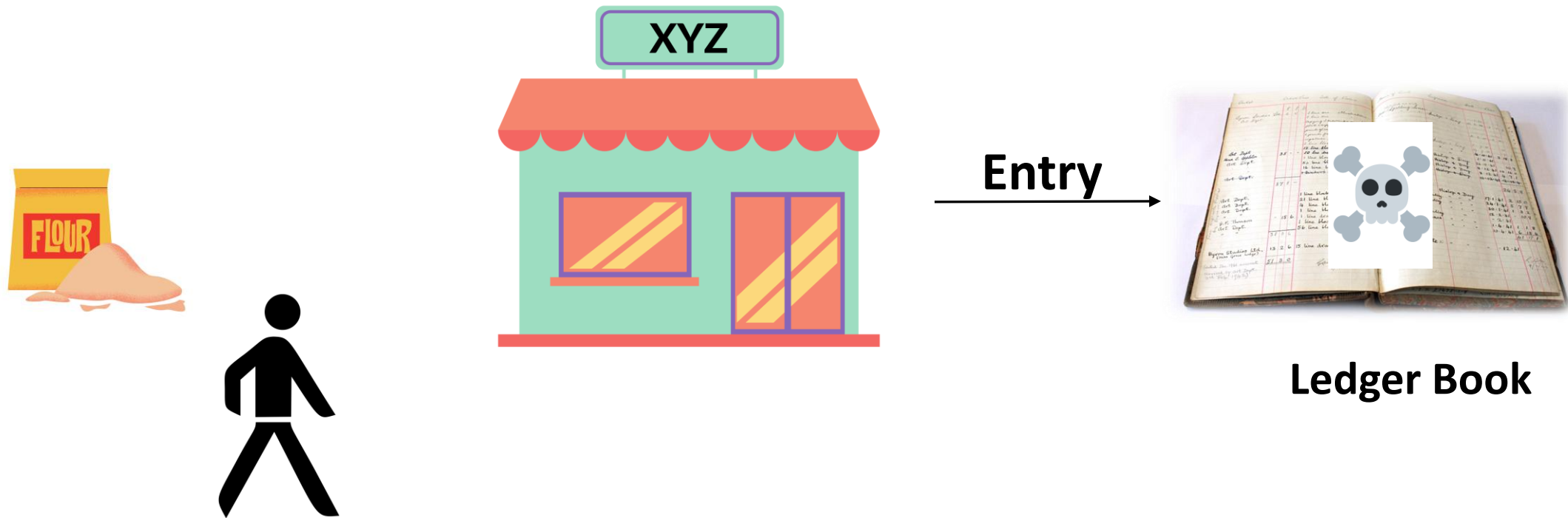


Stuart Haber



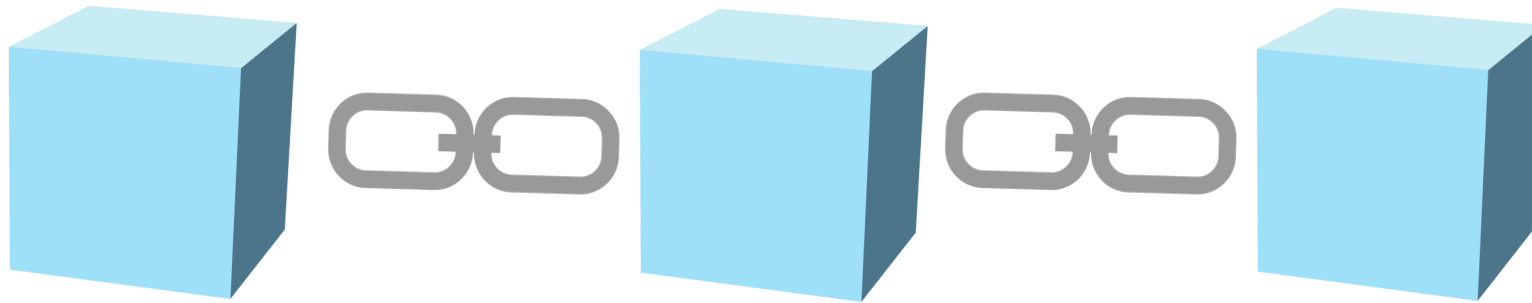
W. Scott Stornetta

What is Blockchain?

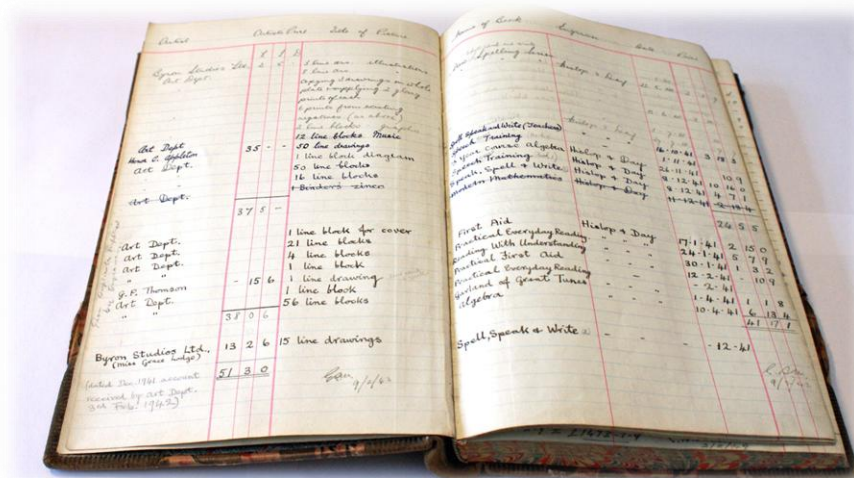


What is Blockchain?

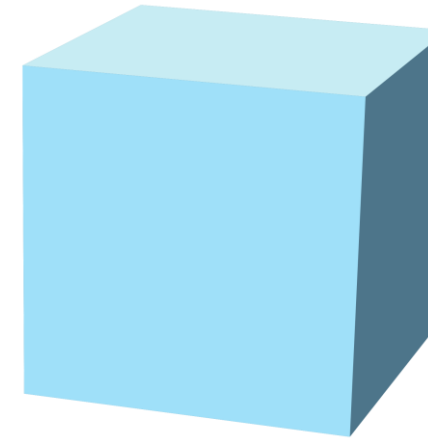
Blockchain is a **distributed immutable ledger** which is completely **transparent**.



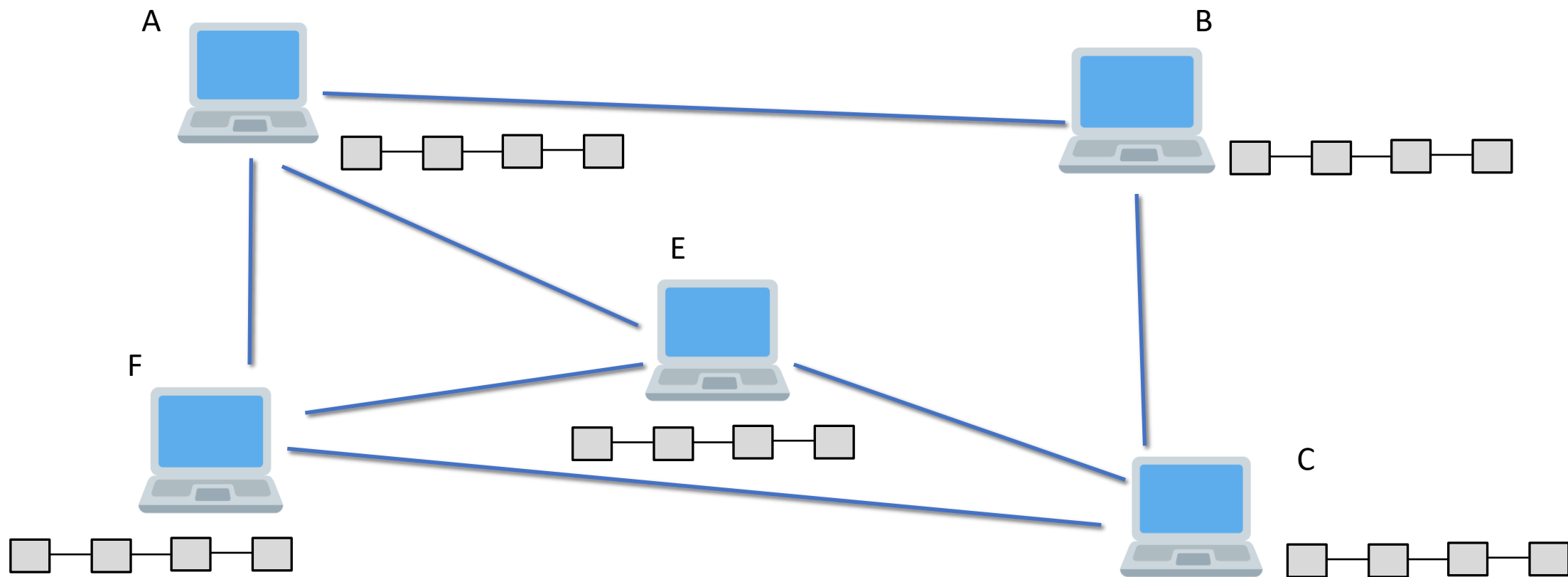
What is Blockchain?

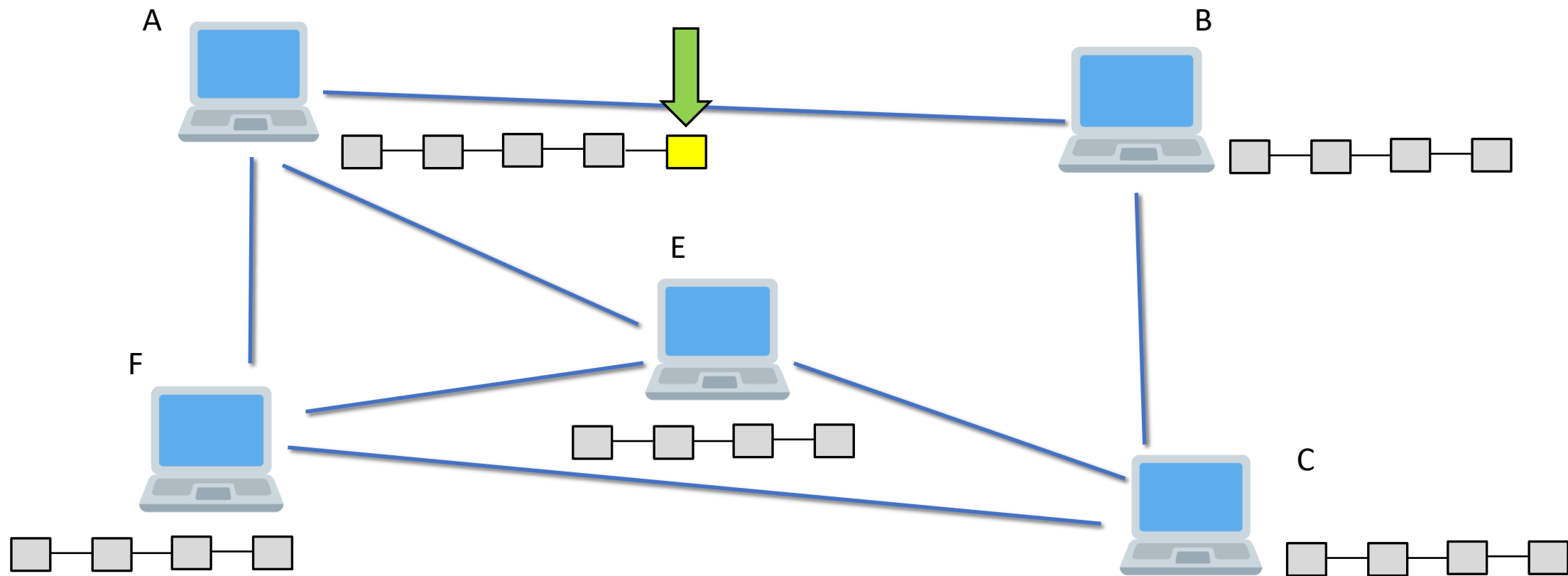


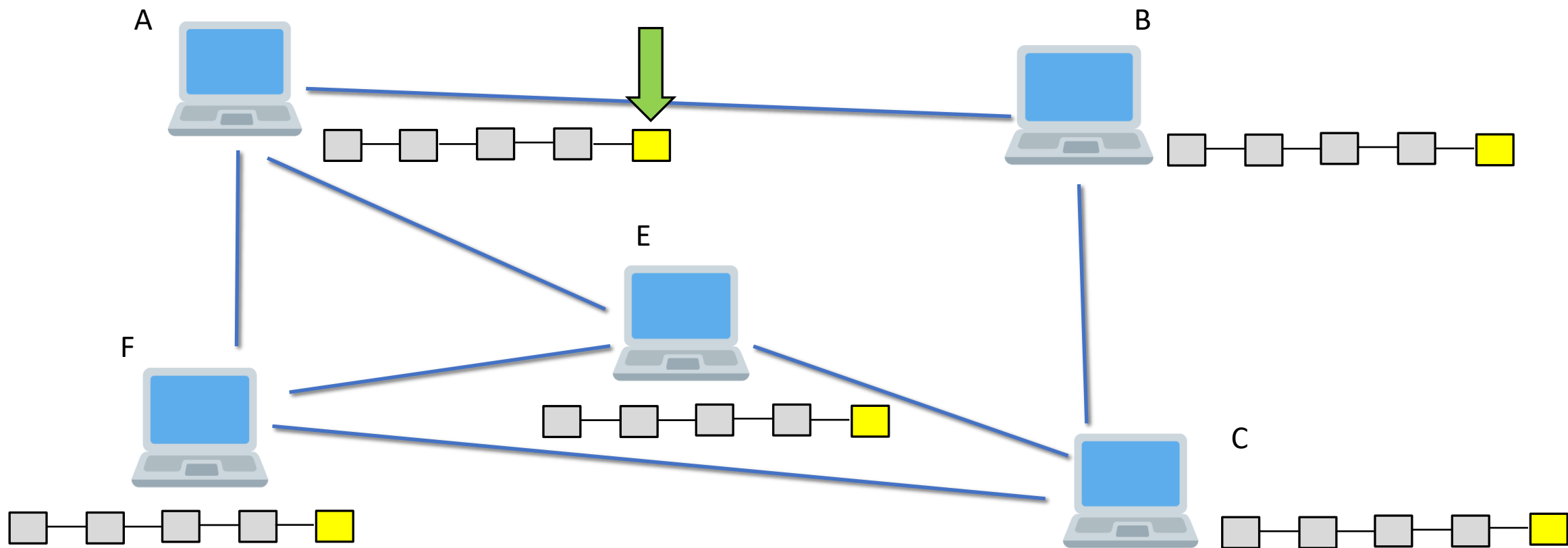
Ledger Book

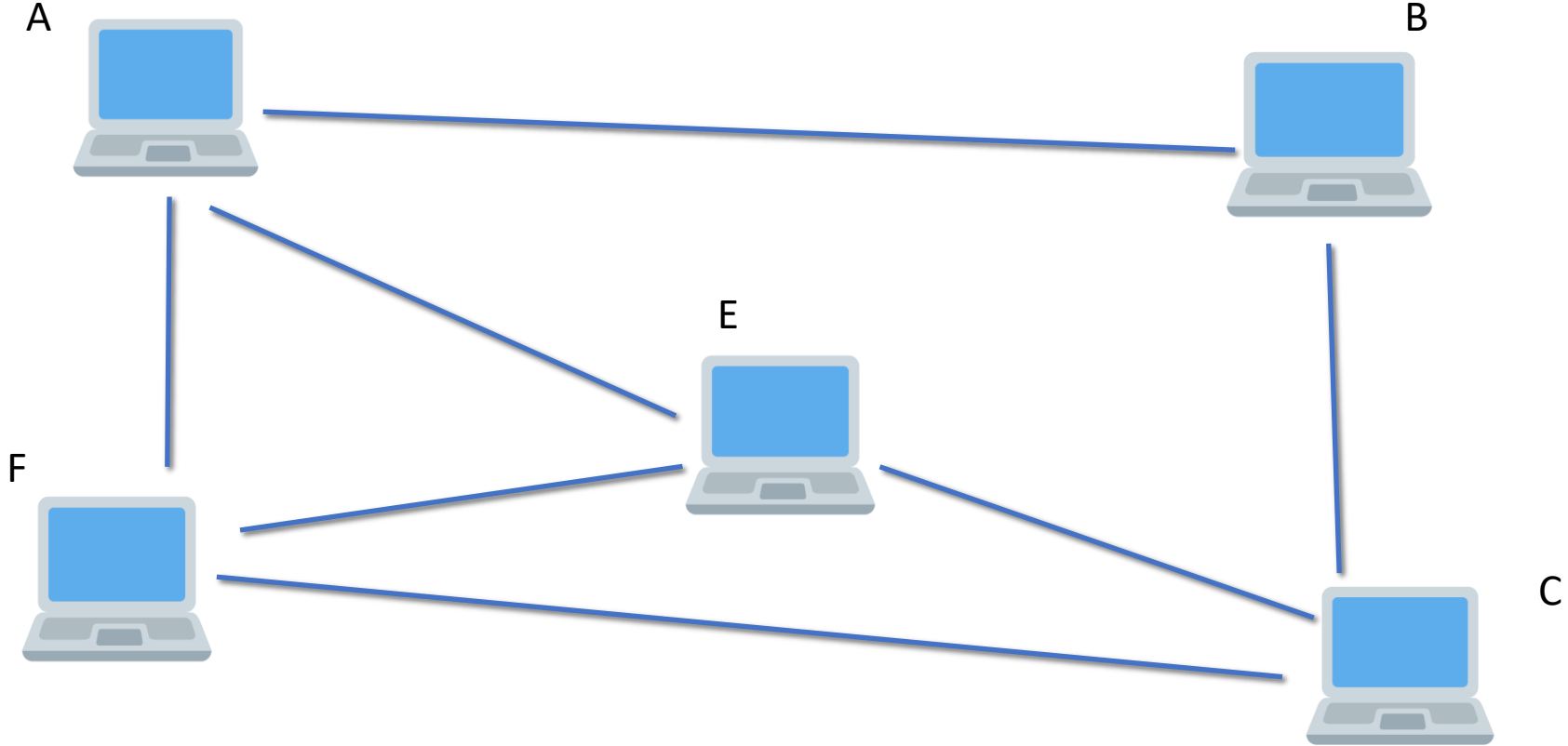


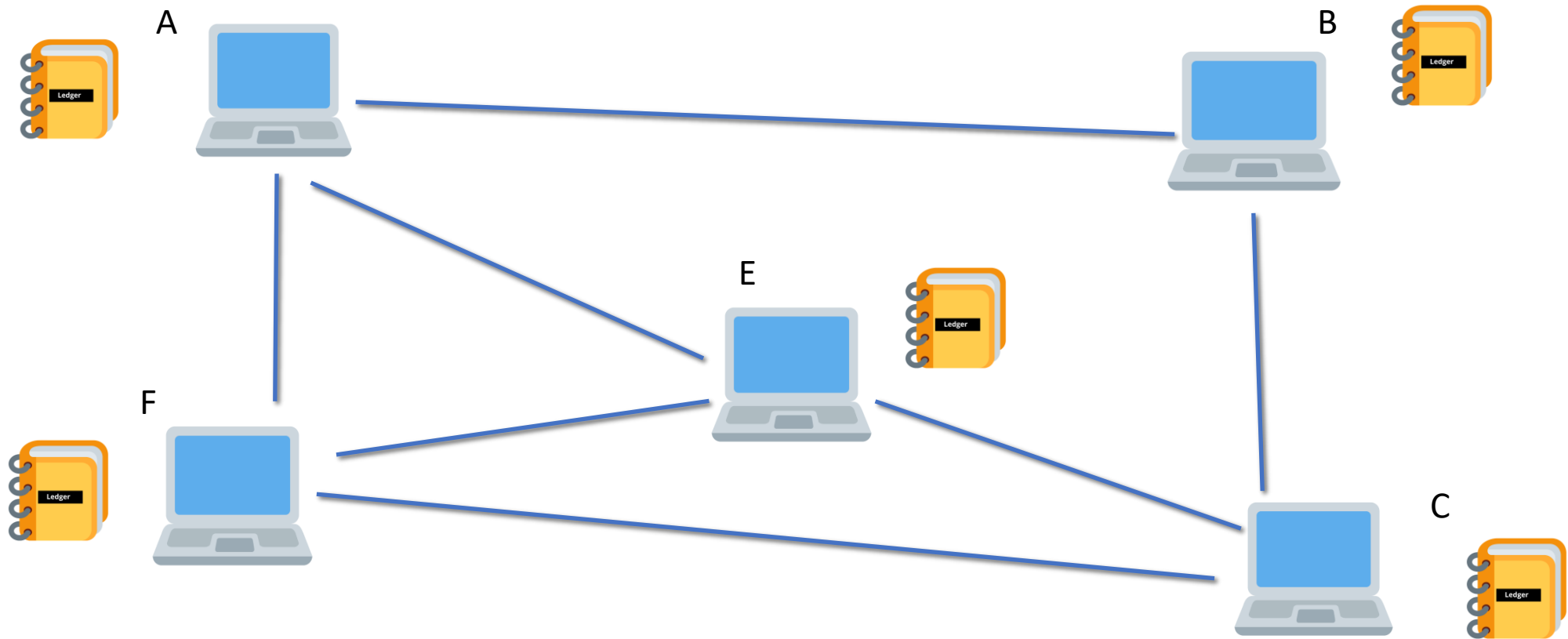
Block

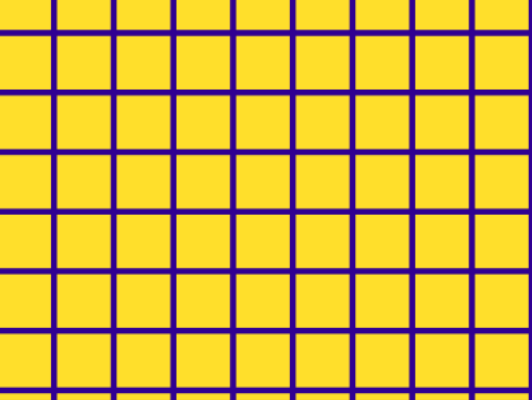












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Applications of Blockchain

Applications of Blockchain

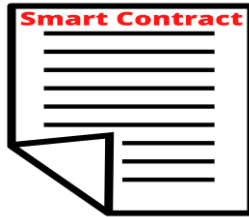
Product Tracking



Healthcare System



Smart Contracts



International Wire Transfer

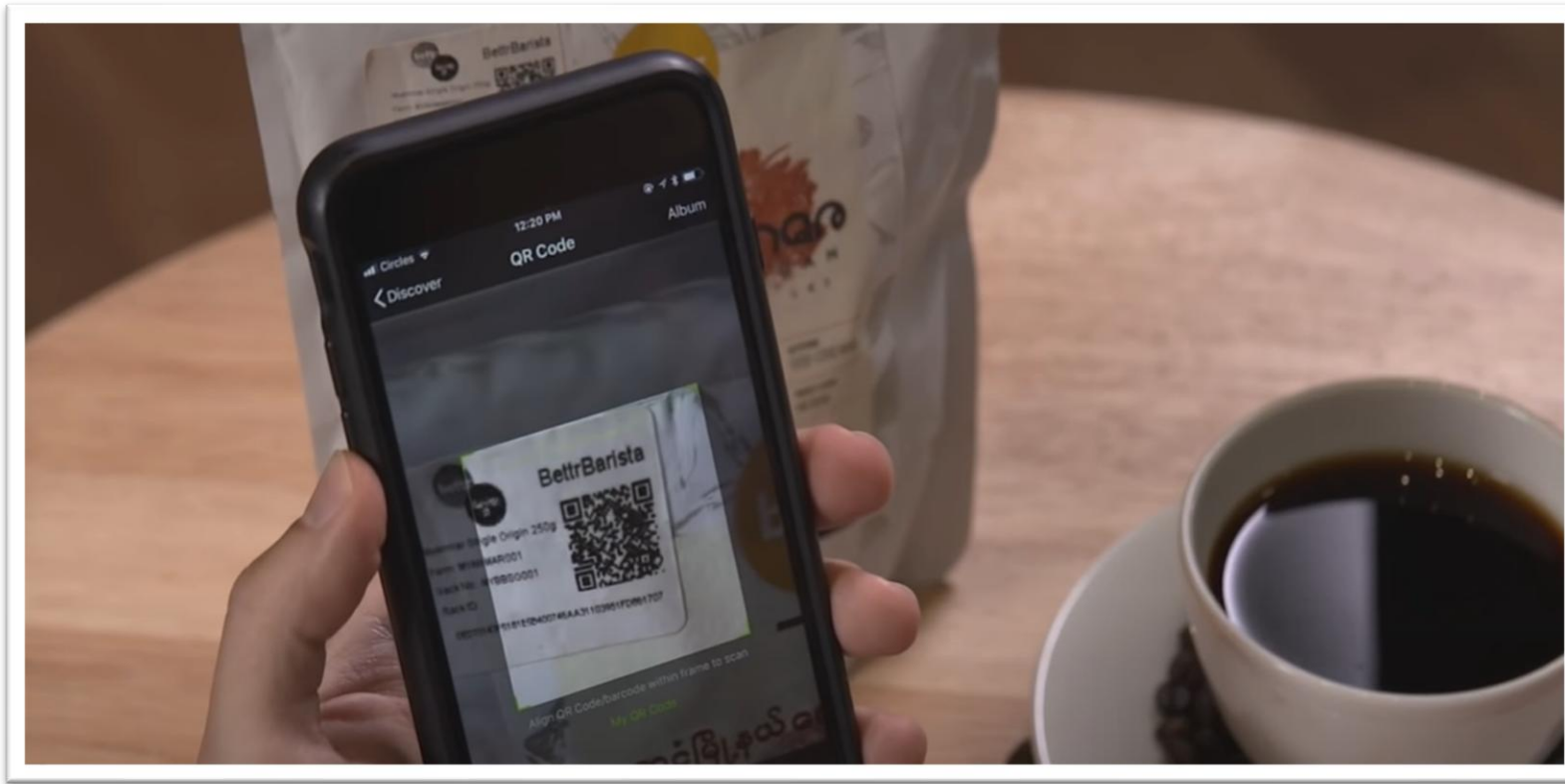


Product Tracking



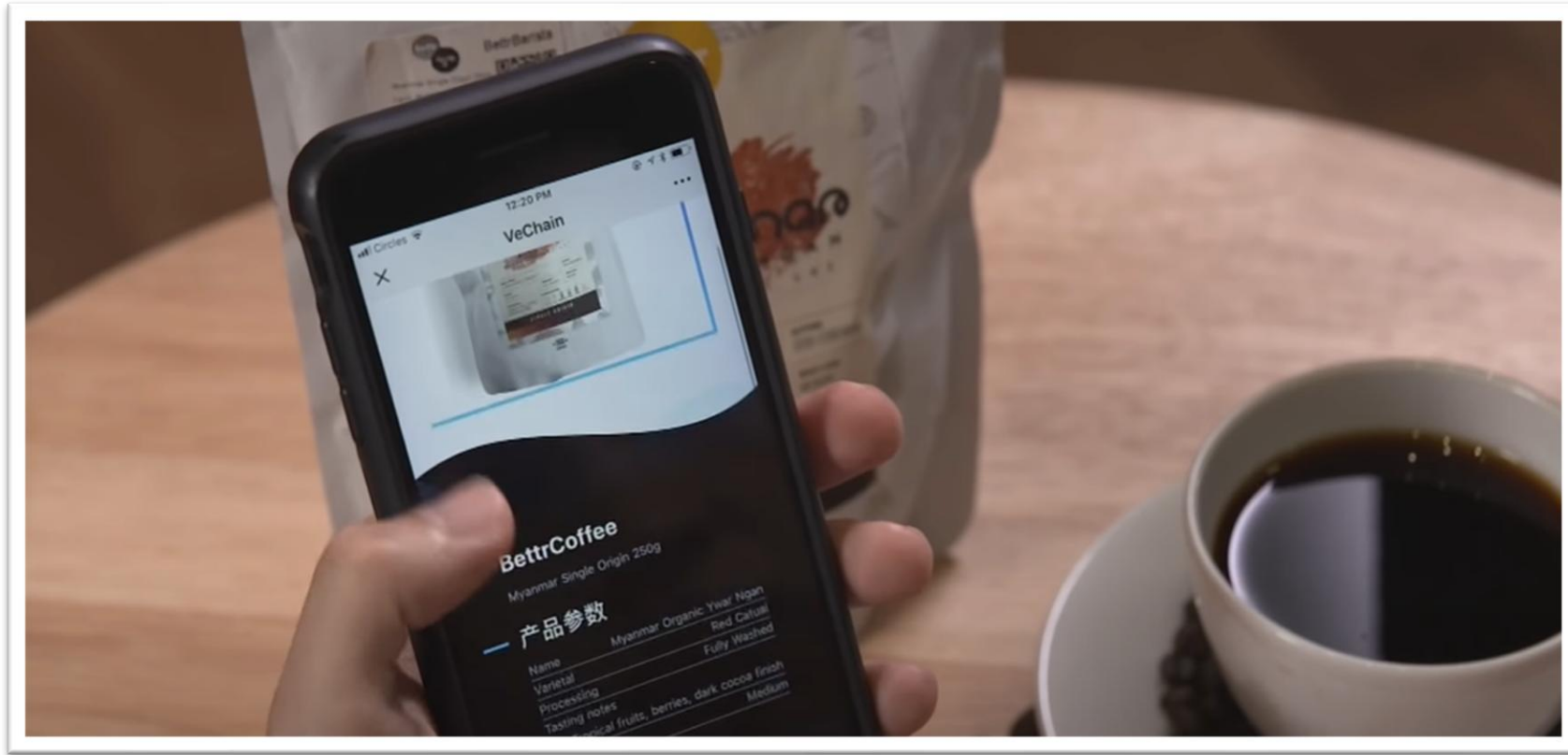
Coffee

Product Tracking



Scan Barcode

Product Tracking



Product Tracking



Applications of Blockchain

Smart Contract

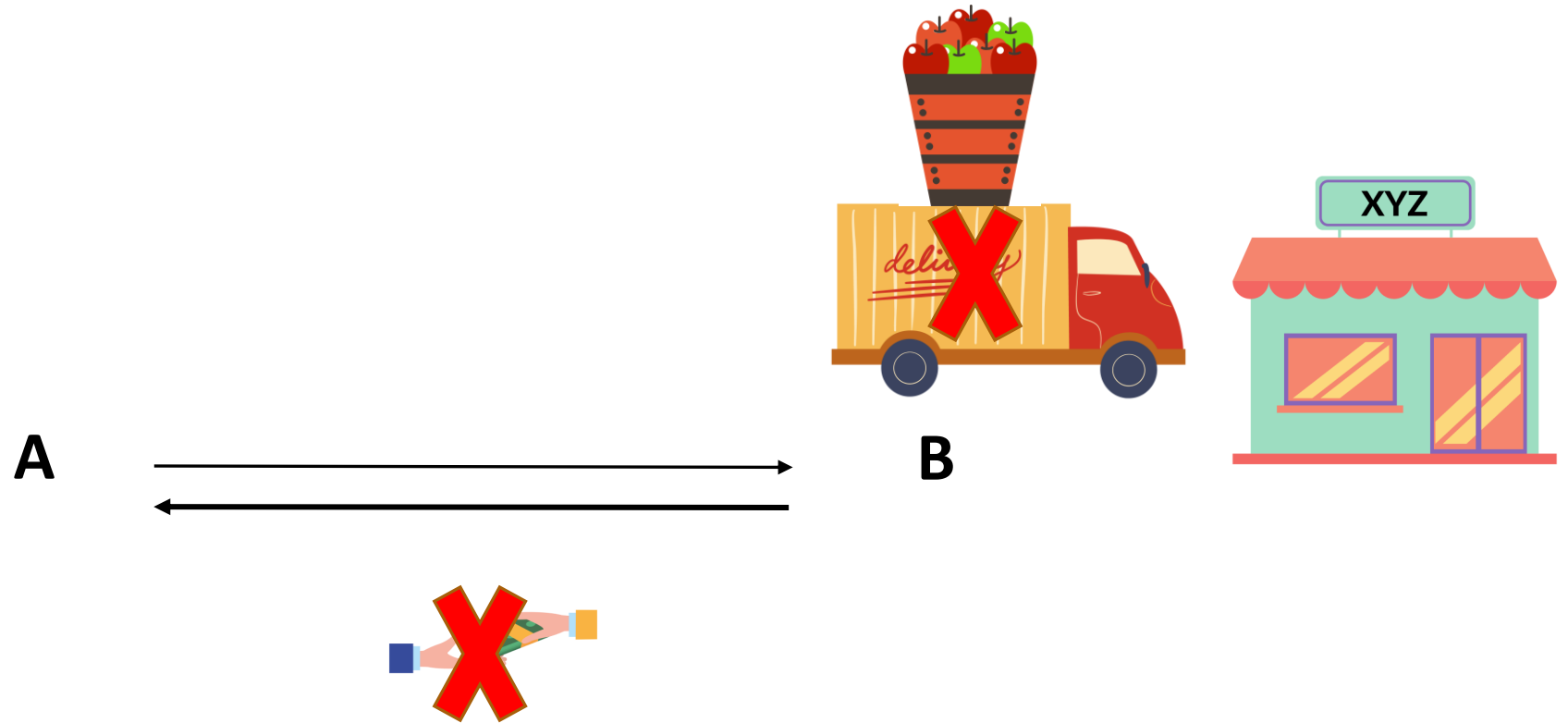


A

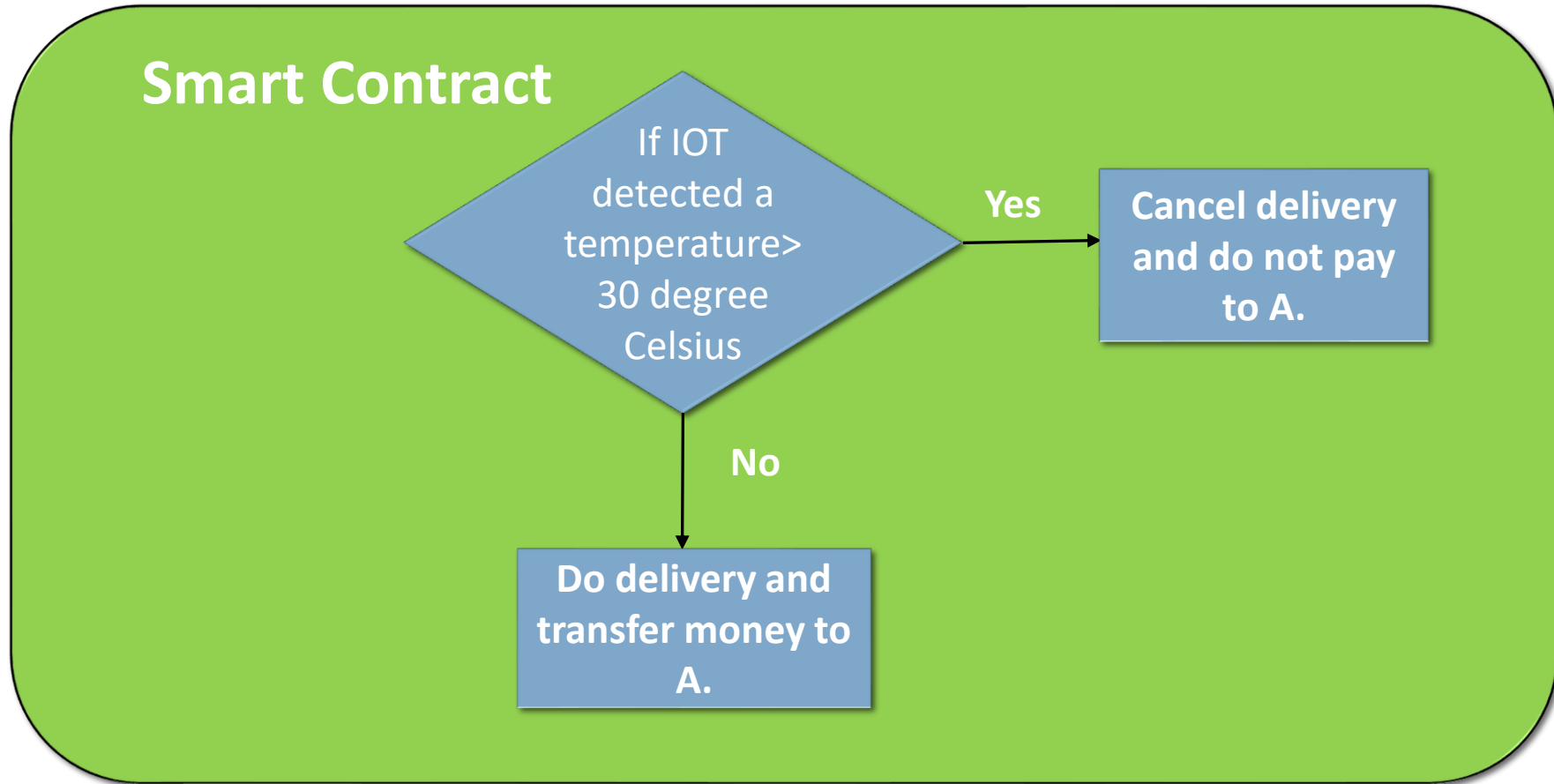


B

Smart Contract

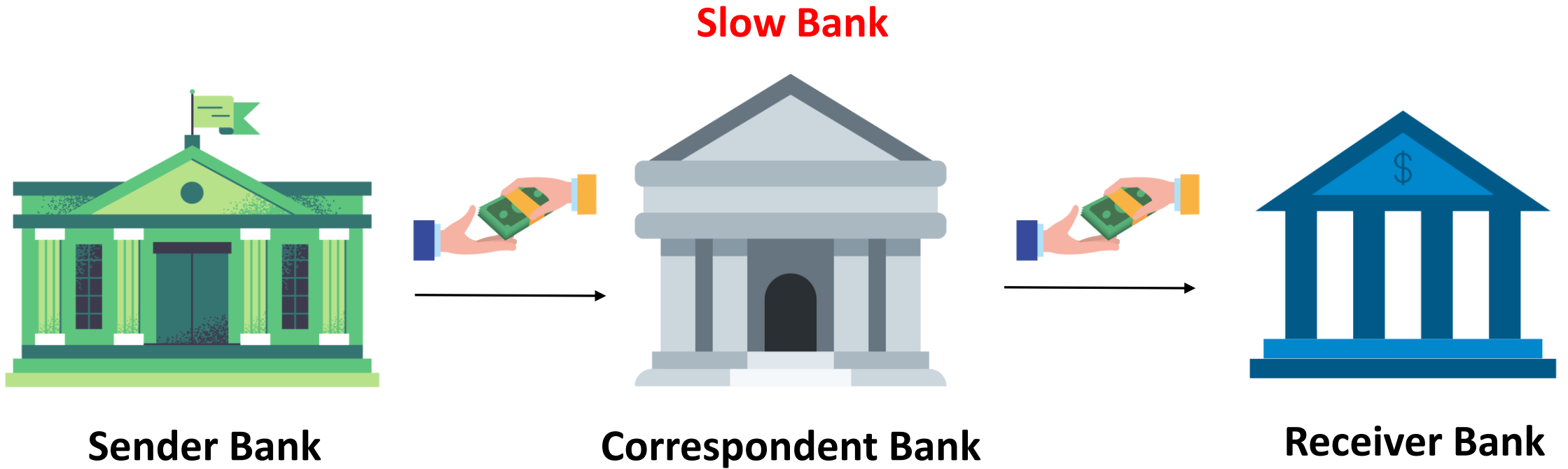


Smart Contract

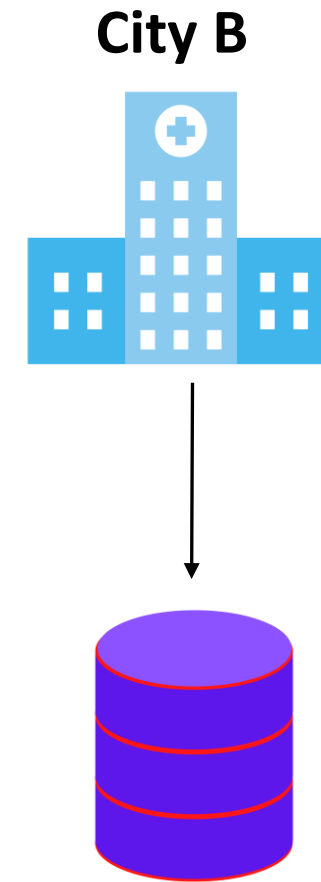
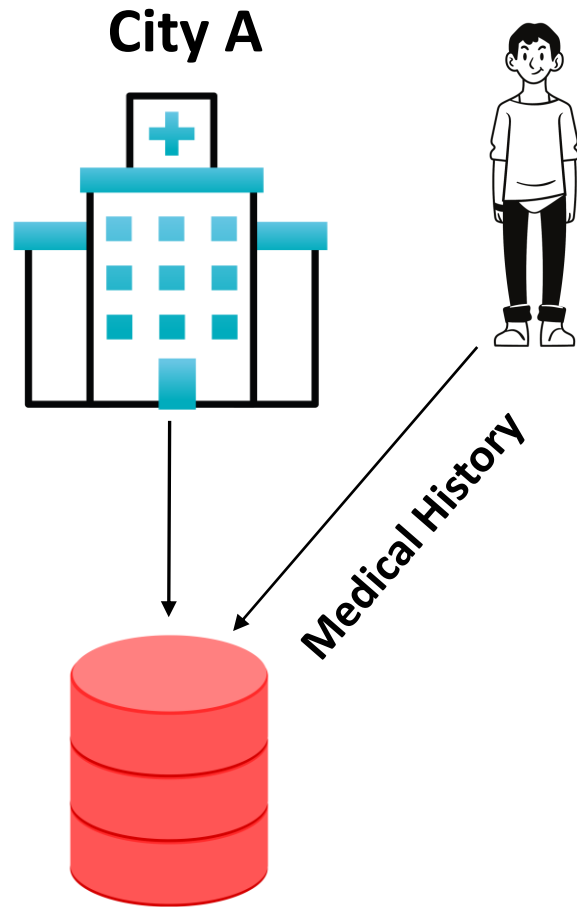


Note- Assuming optimum temperature < 30 degree Celsius.

International Wire Transfer



Healthcare System

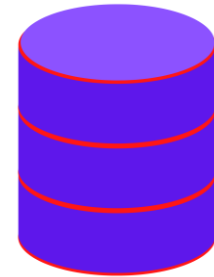
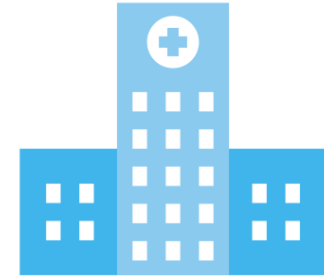


Healthcare System

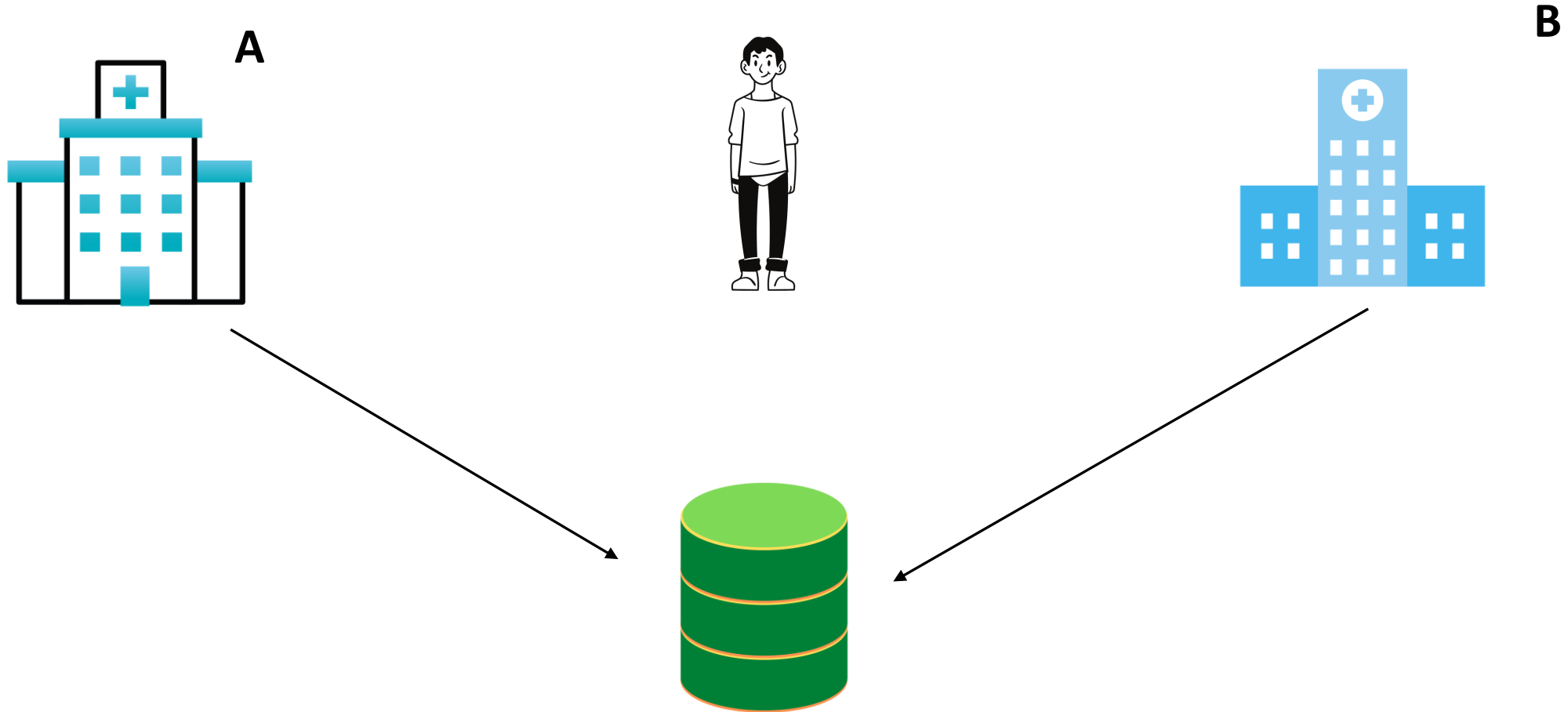
City A



City B

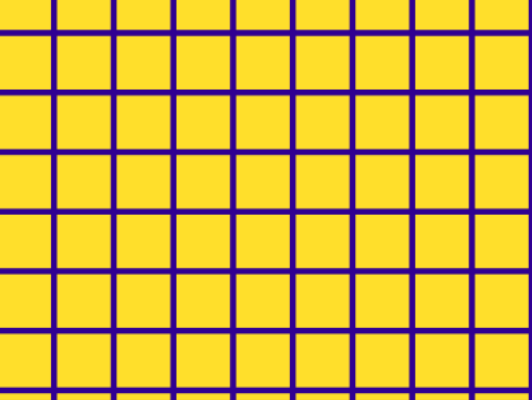


Healthcare System



The background of the slide features a grayscale image of an hourglass with sand flowing from the top bulb to the bottom bulb. To the right of the hourglass, a portion of a calendar is visible, showing dates 22, 23, 24, 29, 30, and 31. The entire background is overlaid with a semi-transparent dark gray filter. The title 'Homework Time' is centered in a large, white, sans-serif font, flanked by two thin white horizontal lines.

Homework Time



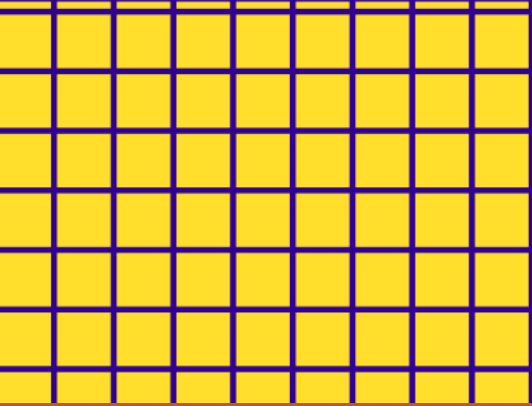
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Hashing Algorithm



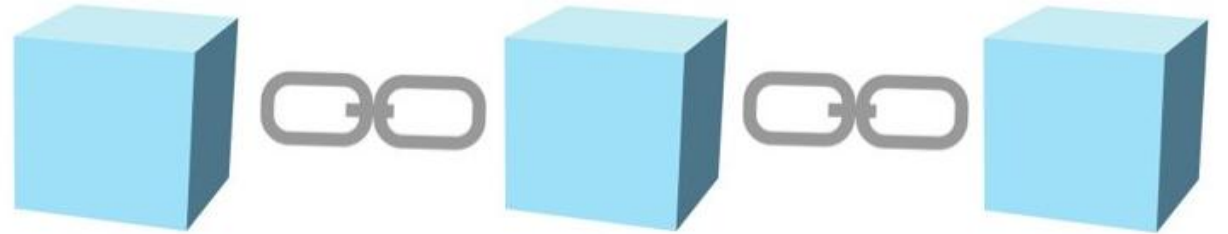
Hashing Algorithm

Hashing Algorithm

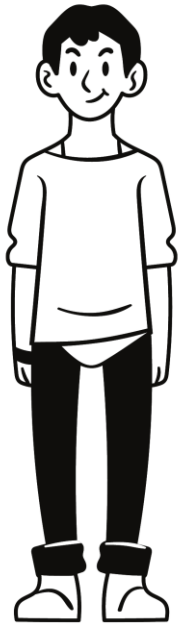
Hashing Algorithm

Block No.-1
Data
Prev Hash:000000000
Hash:0000D8C42

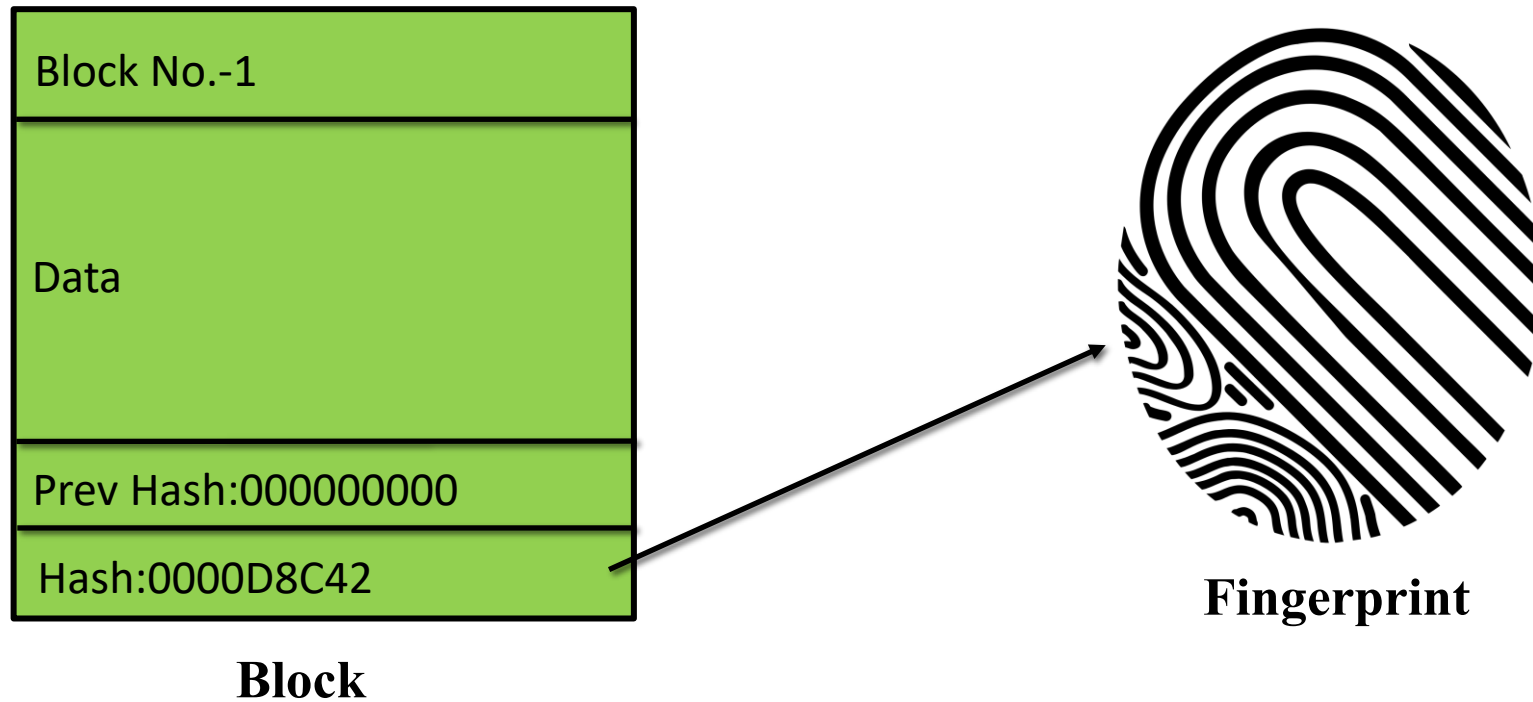
Block



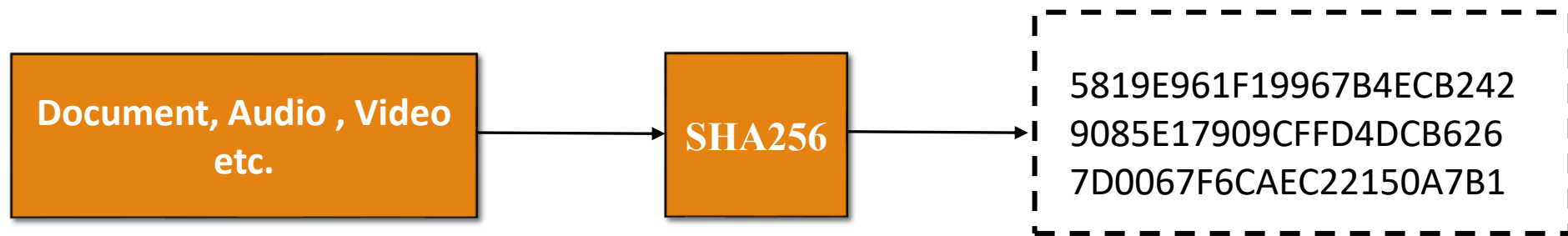
Hashing Algorithm



Hashing Algorithm

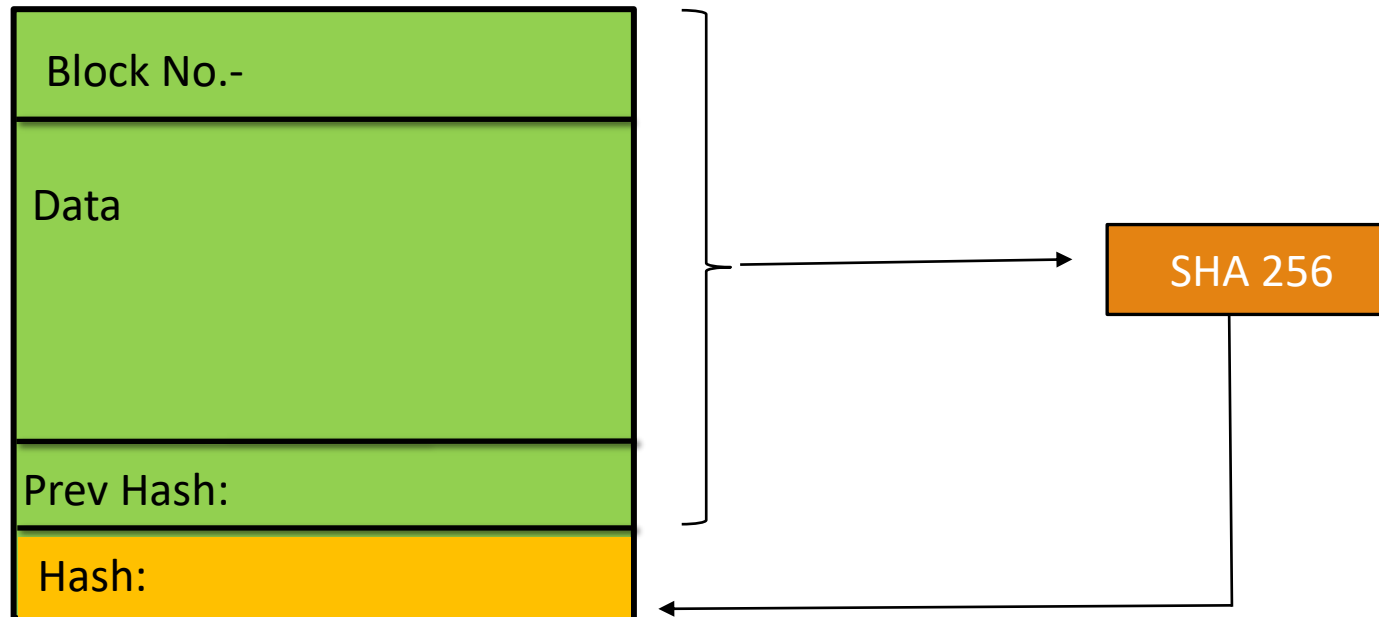


Hashing Algorithm



This has **64 hexadecimal characters**.
Each character is of **4 bits**.
So in total it has $64 * 4$ bits i.e. **256 bits**.

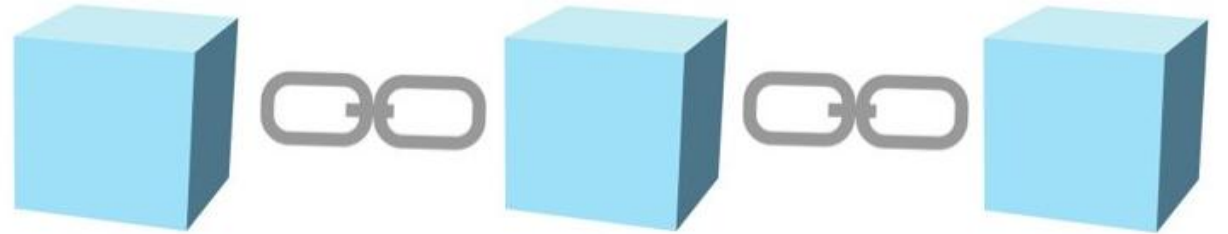
Hashing Algorithm



Hashing Algorithm

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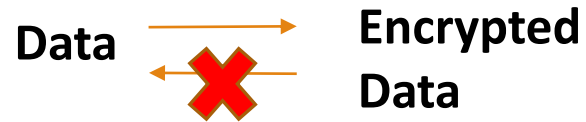
Block



Hashing Algorithm

The five requirements of Hash Algorithm-

One Way



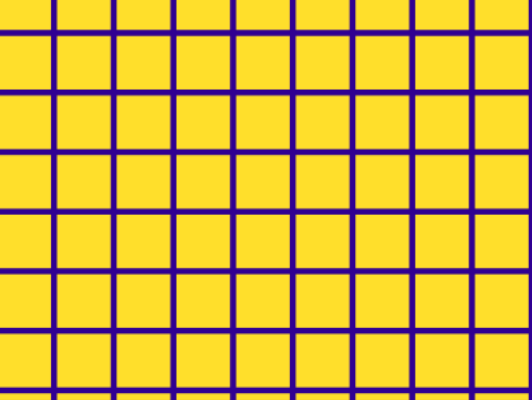
Withstand
Collisions

Deterministic



Avalanche
Effect

Fast
Computation



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Blockchain Formation

Hashing Algorithm

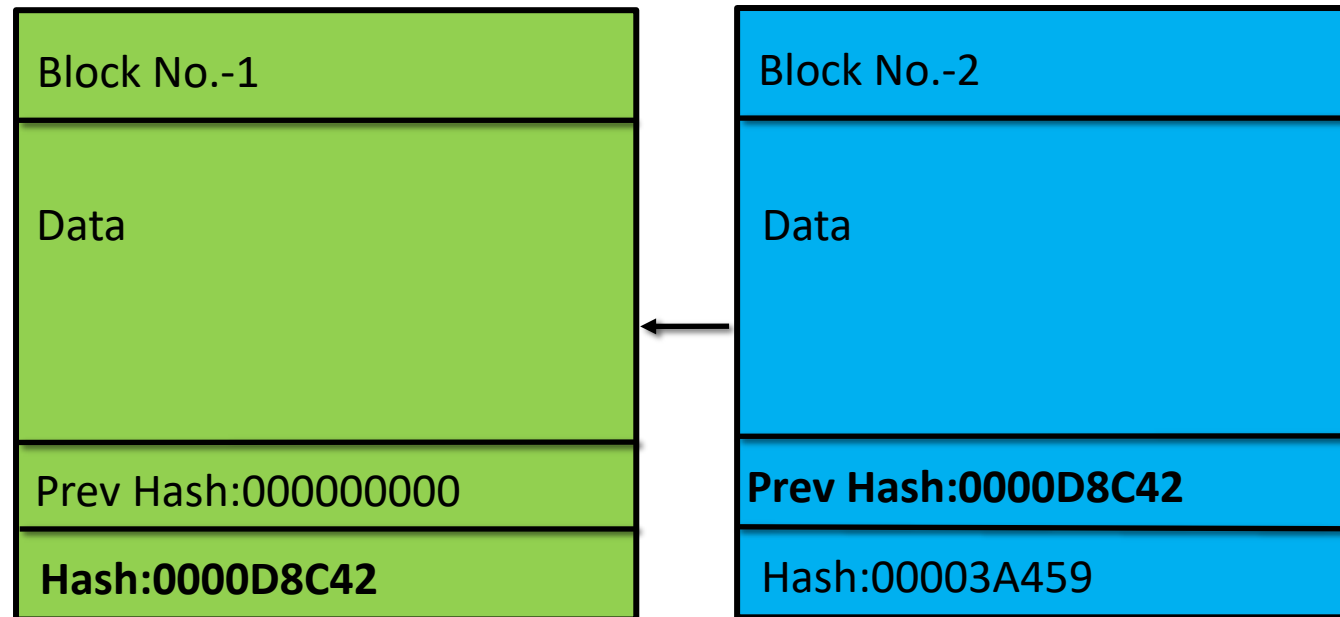
Block No.-1
Data
Prev Hash:000000000
Hash:0000D8C42

Hashing Algorithm

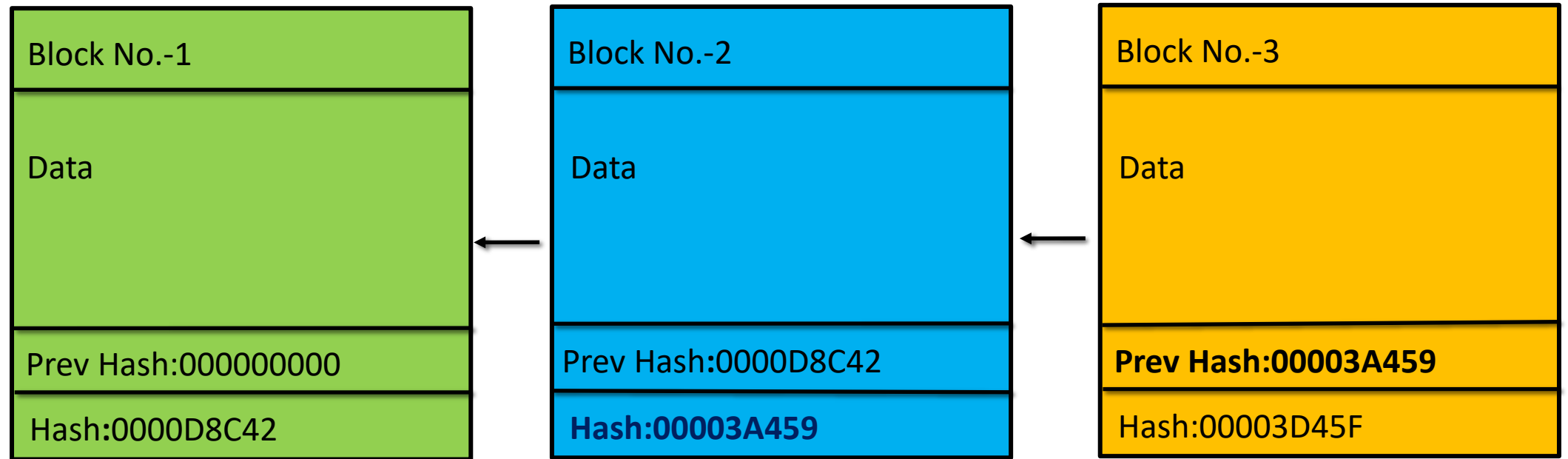
Block No.-1
Data
Prev Hash:000000000
Hash:0000D8C42

Block No.-2
Data
Prev Hash:
Hash:00003A459

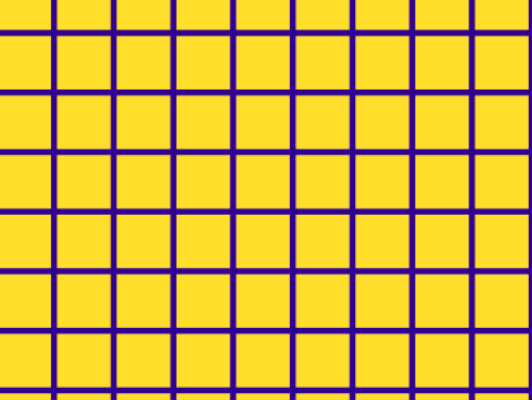
Hashing Algorithm



Hashing Algorithm



Genesis Block



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Immutable Ledger

Immutable Ledger



You



Money



Sales Deed

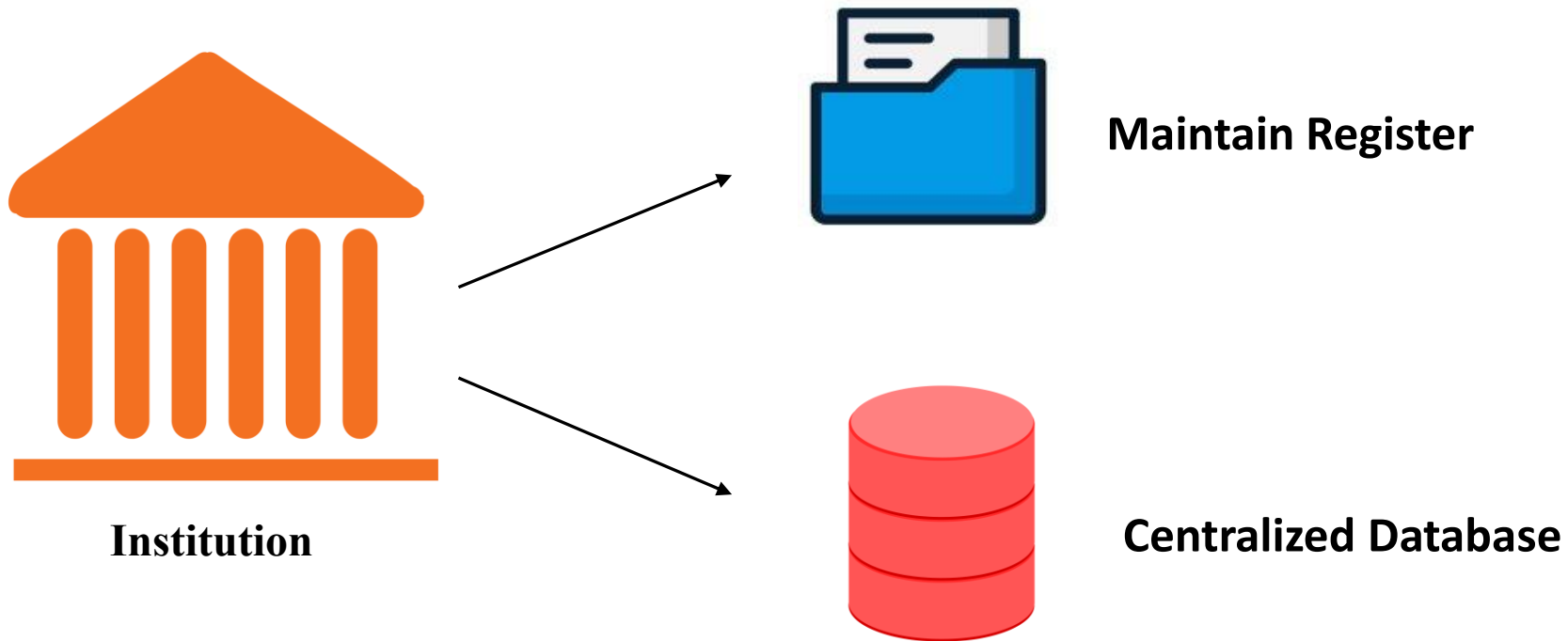


Institution

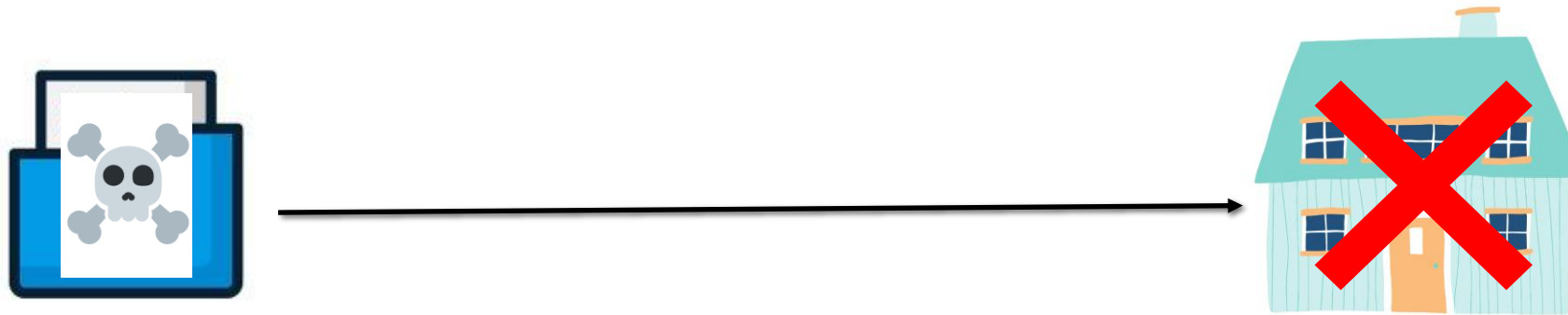


House

Immutable Ledger

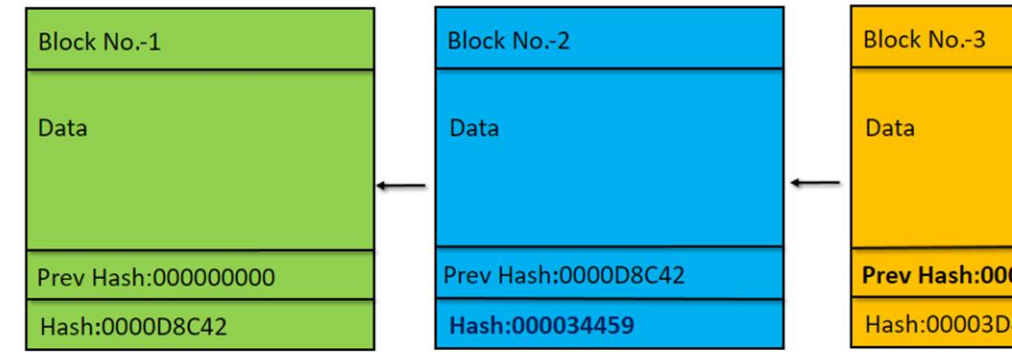


Immutable Ledger



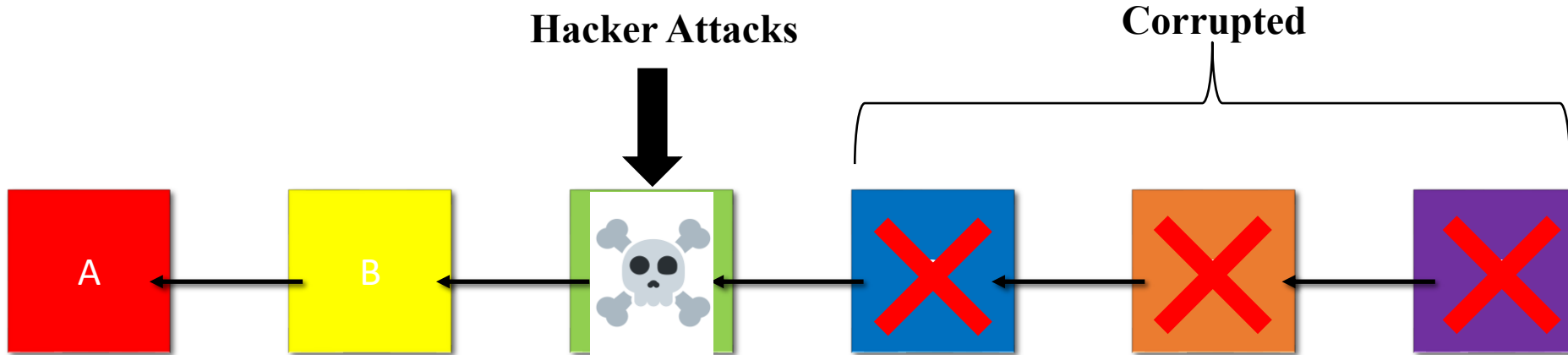


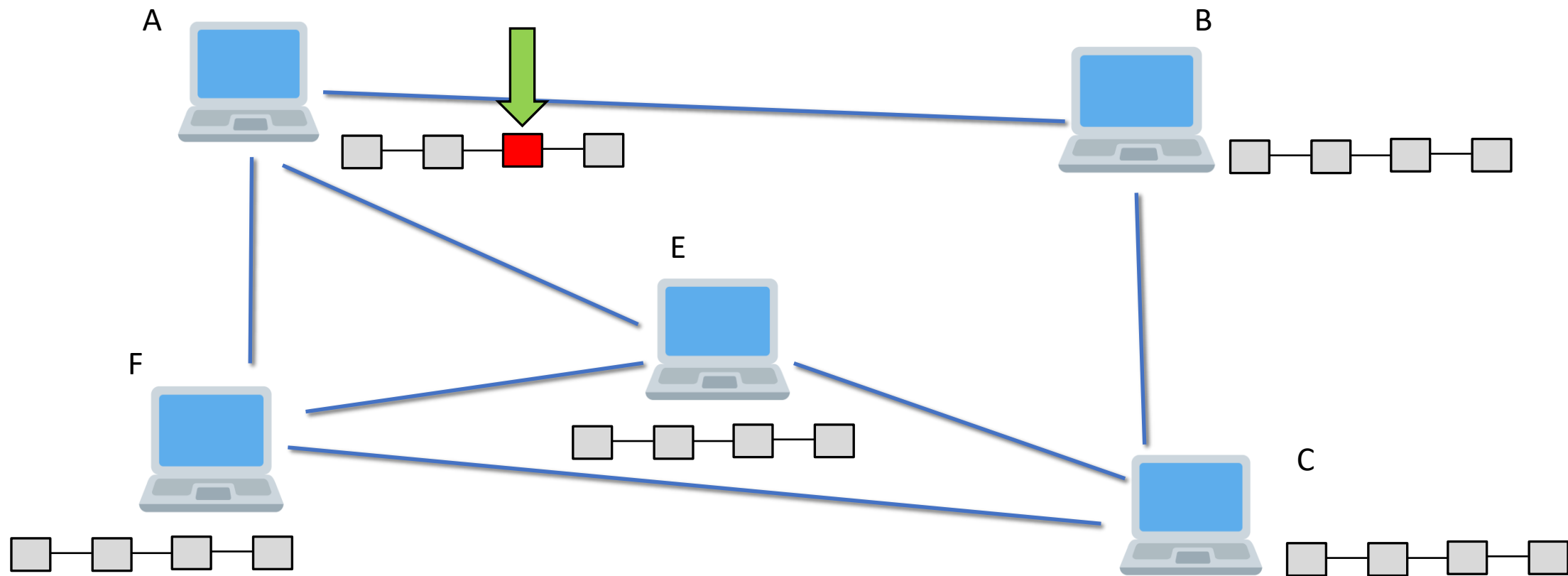
Immutable Ledger

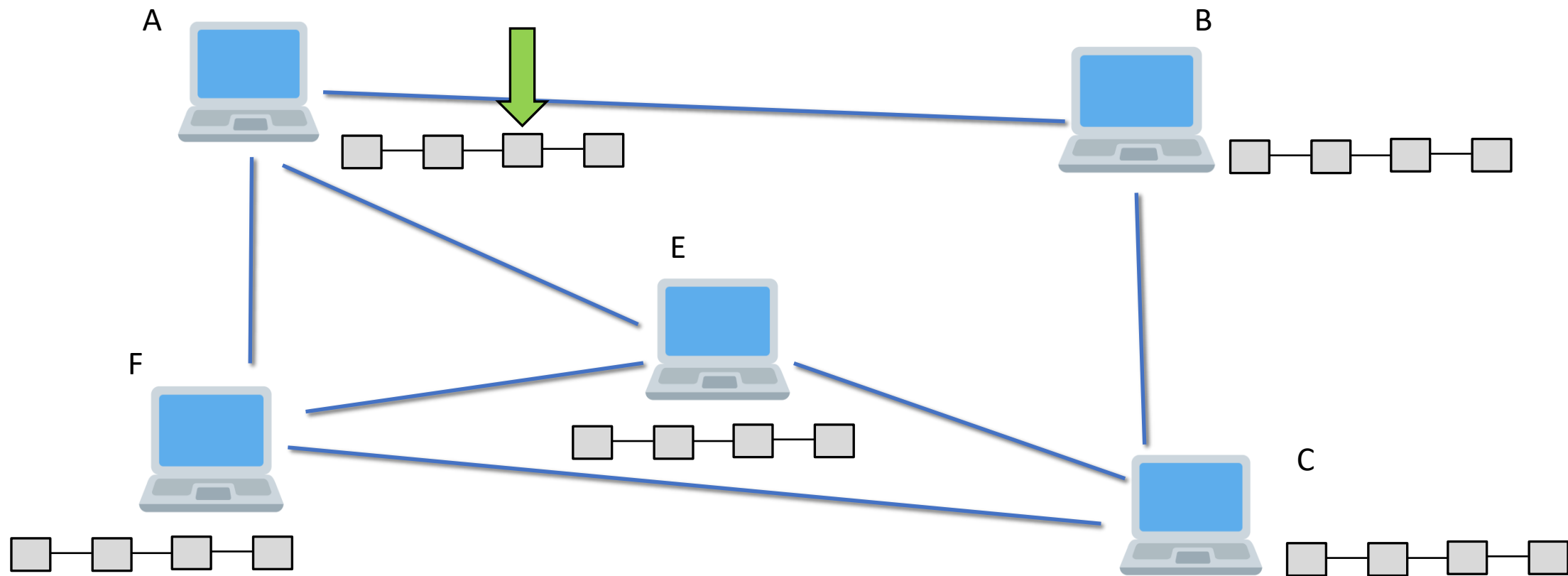


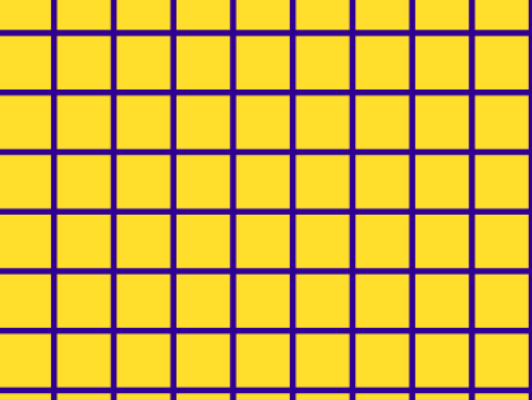
Genesis Block

From previous lecture..









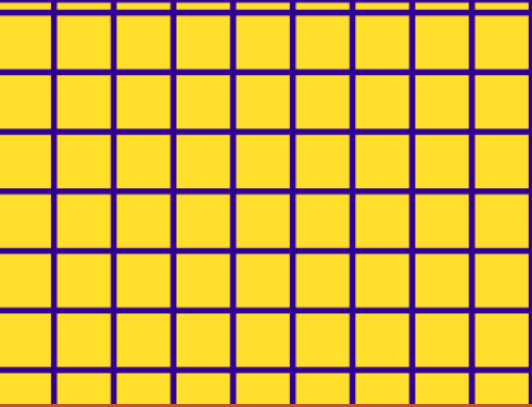
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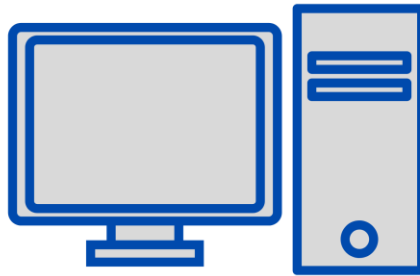


The background of the slide features a conceptual image. In the center, a hand is shown holding a Bitcoin coin. Radiating from the coin and hand are numerous yellow lines that connect to various circular nodes, some of which are highlighted in yellow. This network of nodes and lines is superimposed on a dark blue background that has a faint grid pattern, resembling a globe or a technical diagram. The overall aesthetic is high-tech and digital.

What is a P2P network?

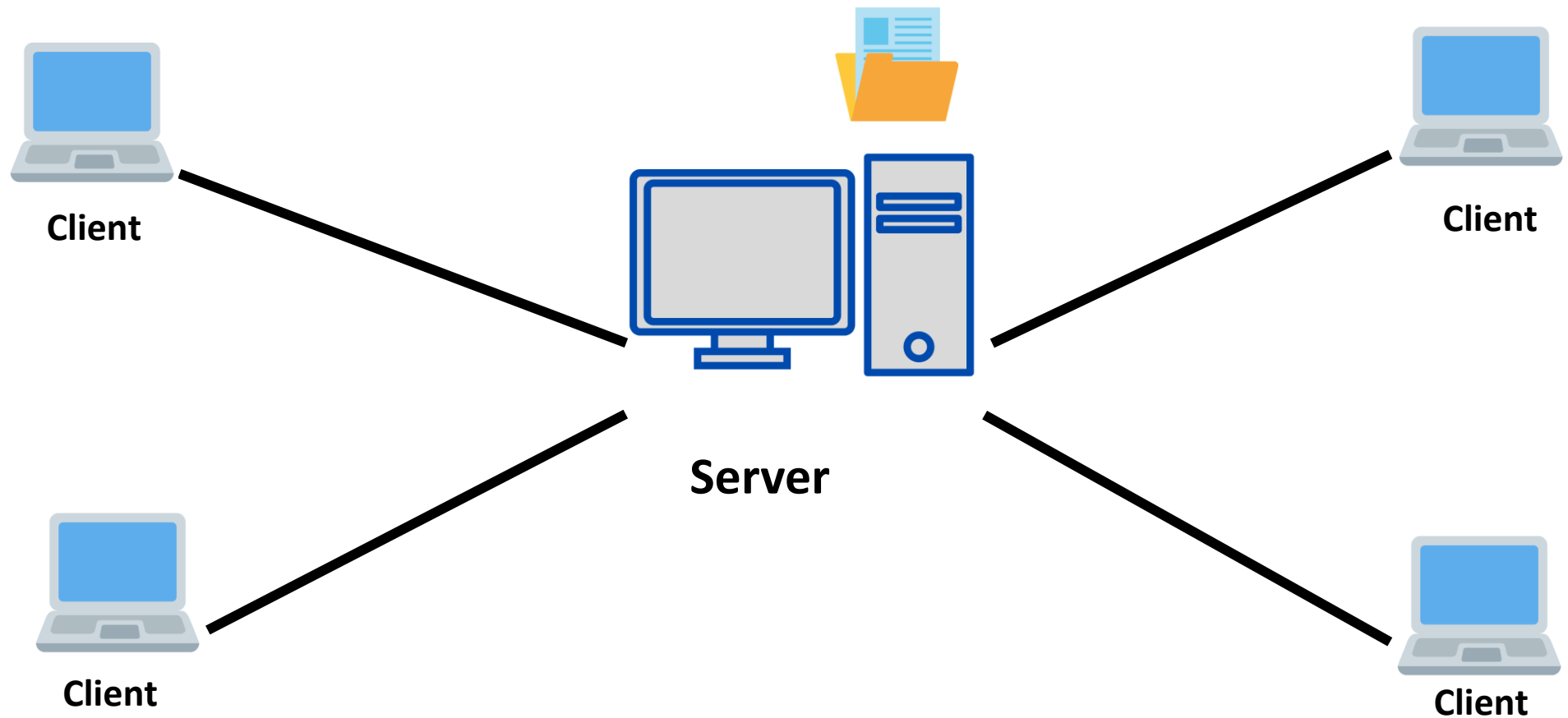
What is a centralized network?

What is a centralized network?

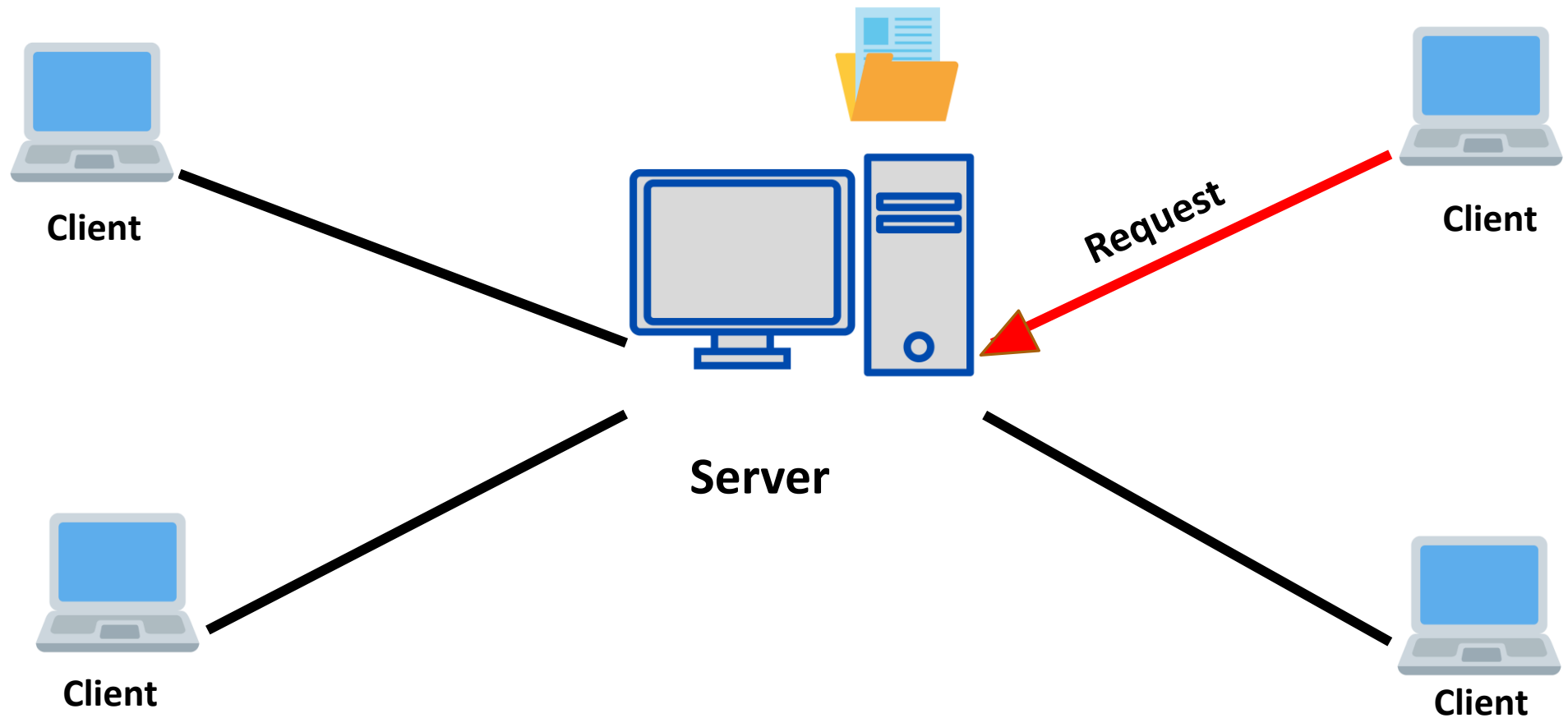


Server

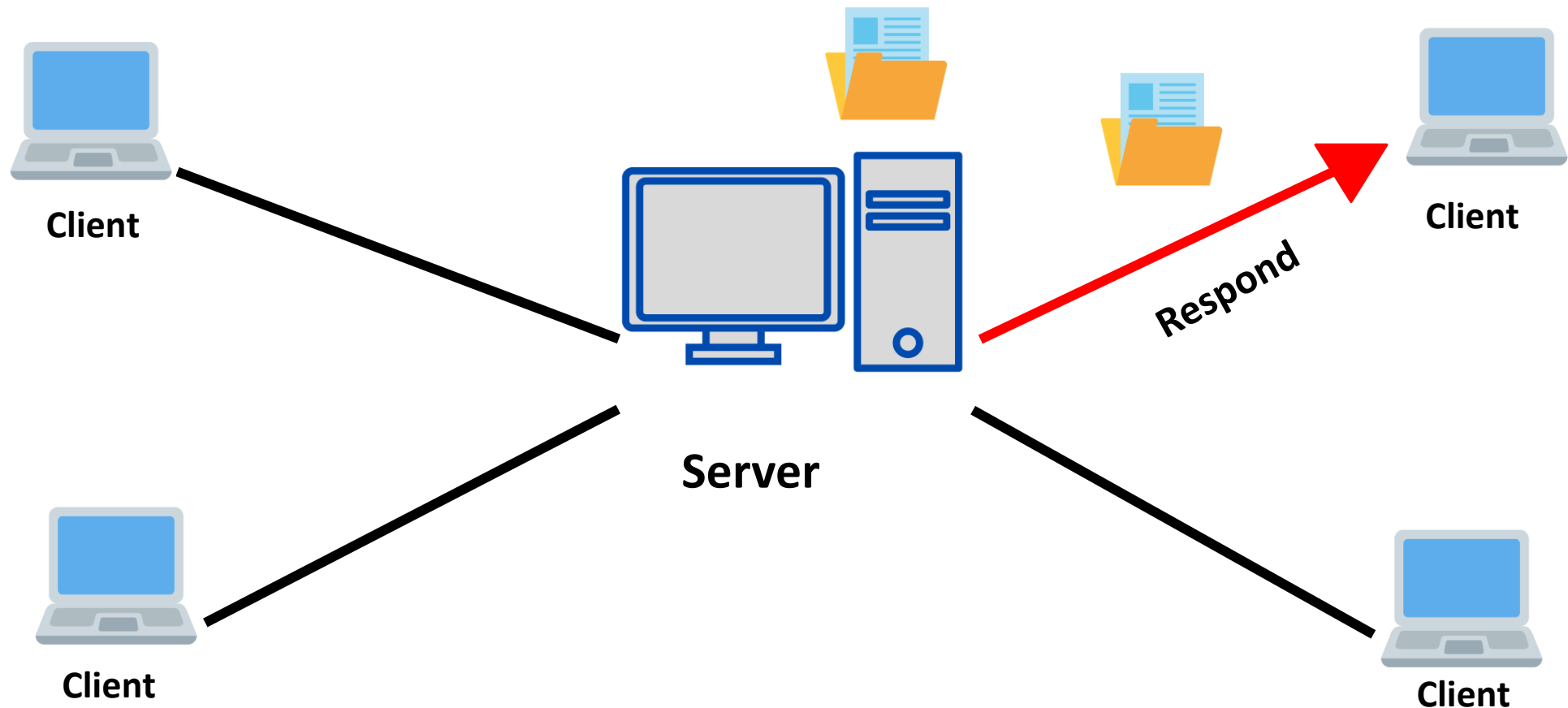
What is a centralized network?



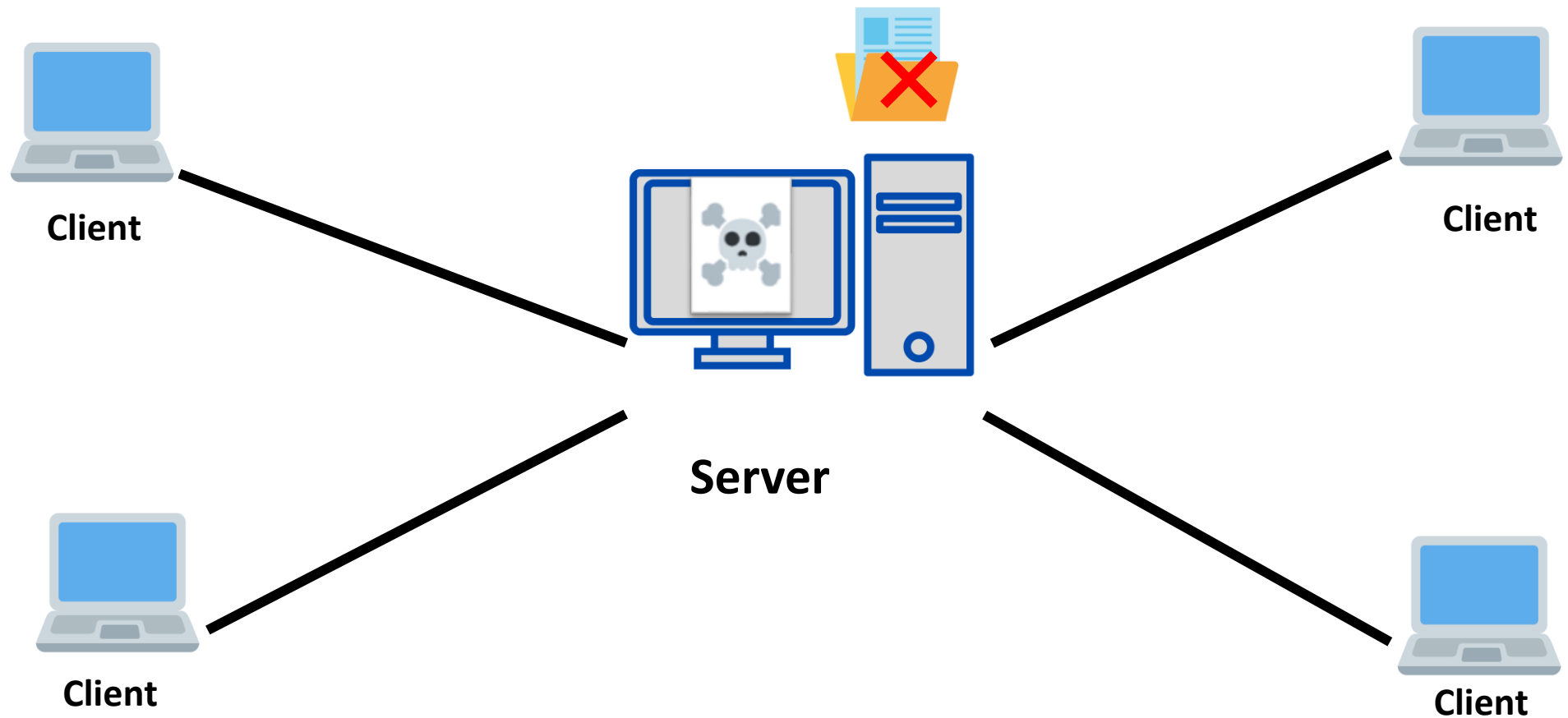
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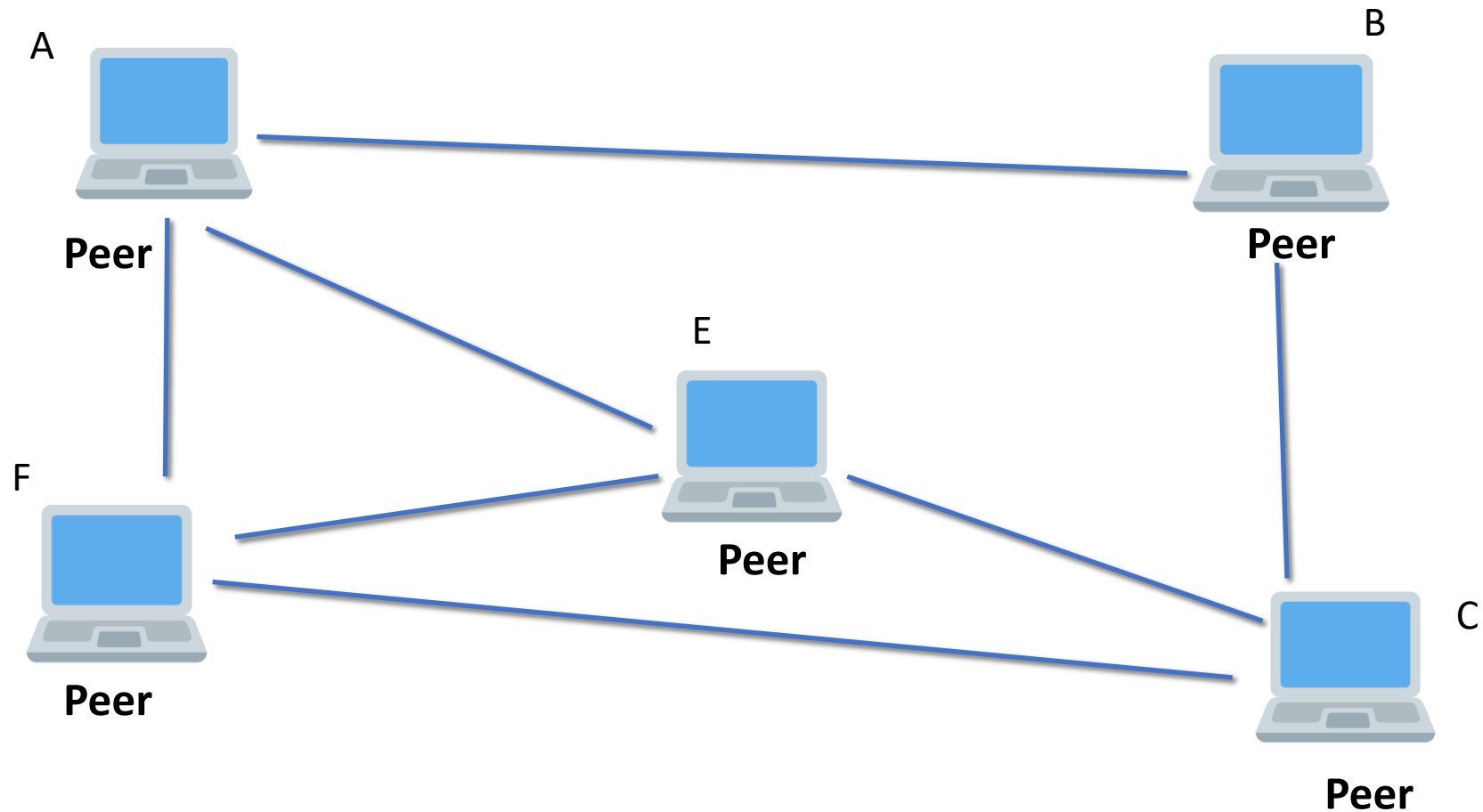
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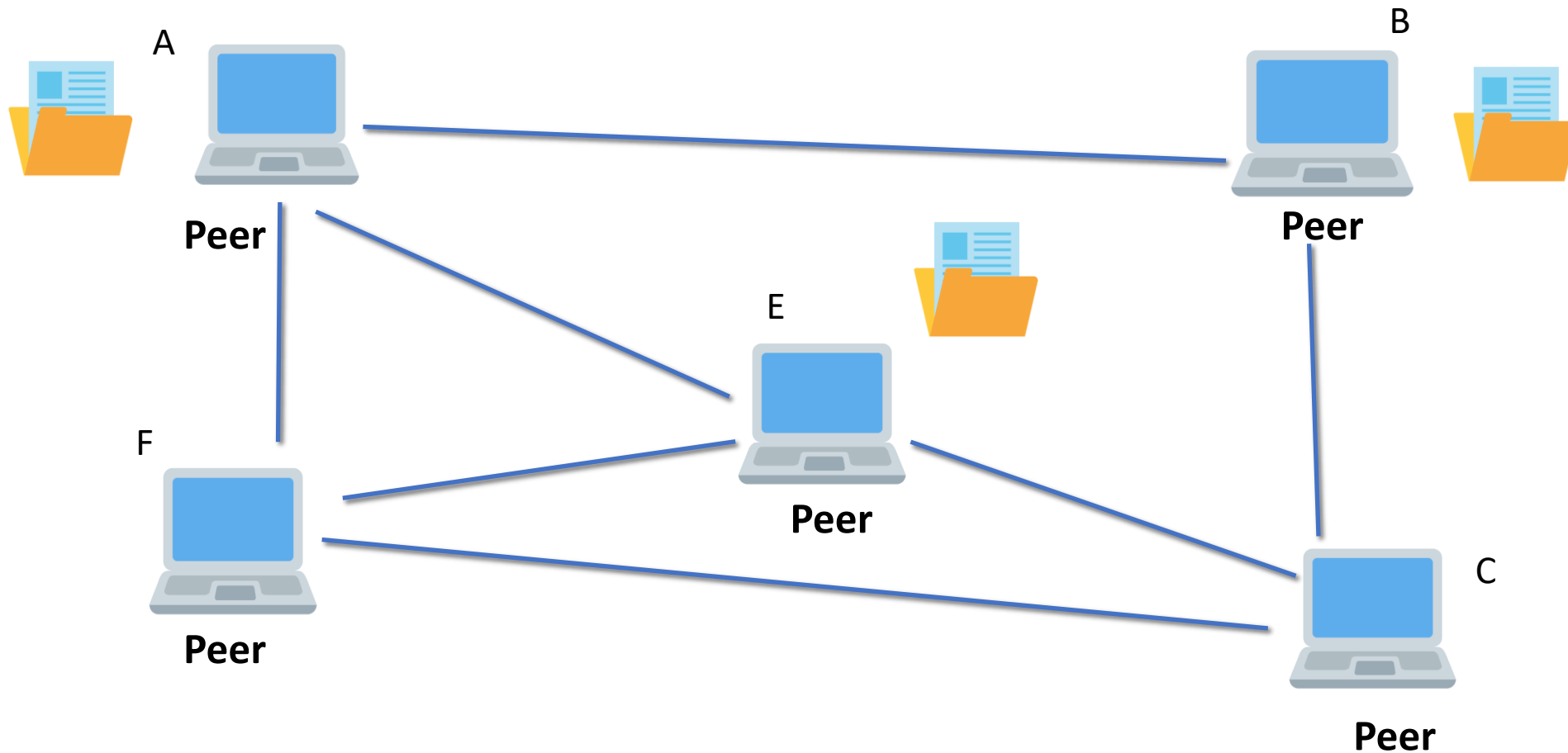
What is a centralized network?

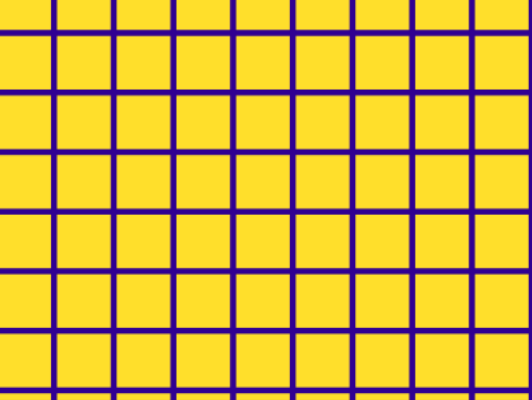


Distributed P2P network



Distributed P2P network





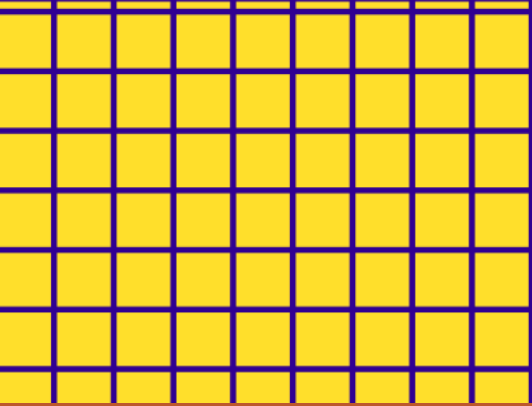
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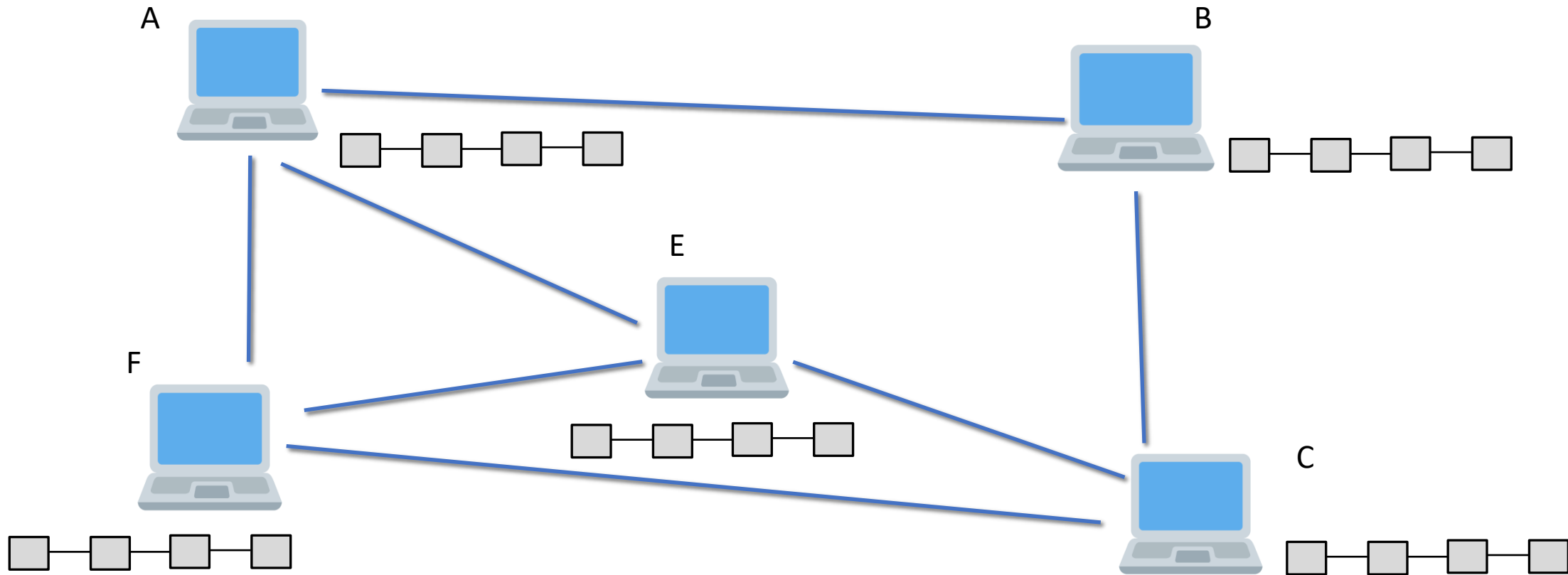
Discord – Link in description



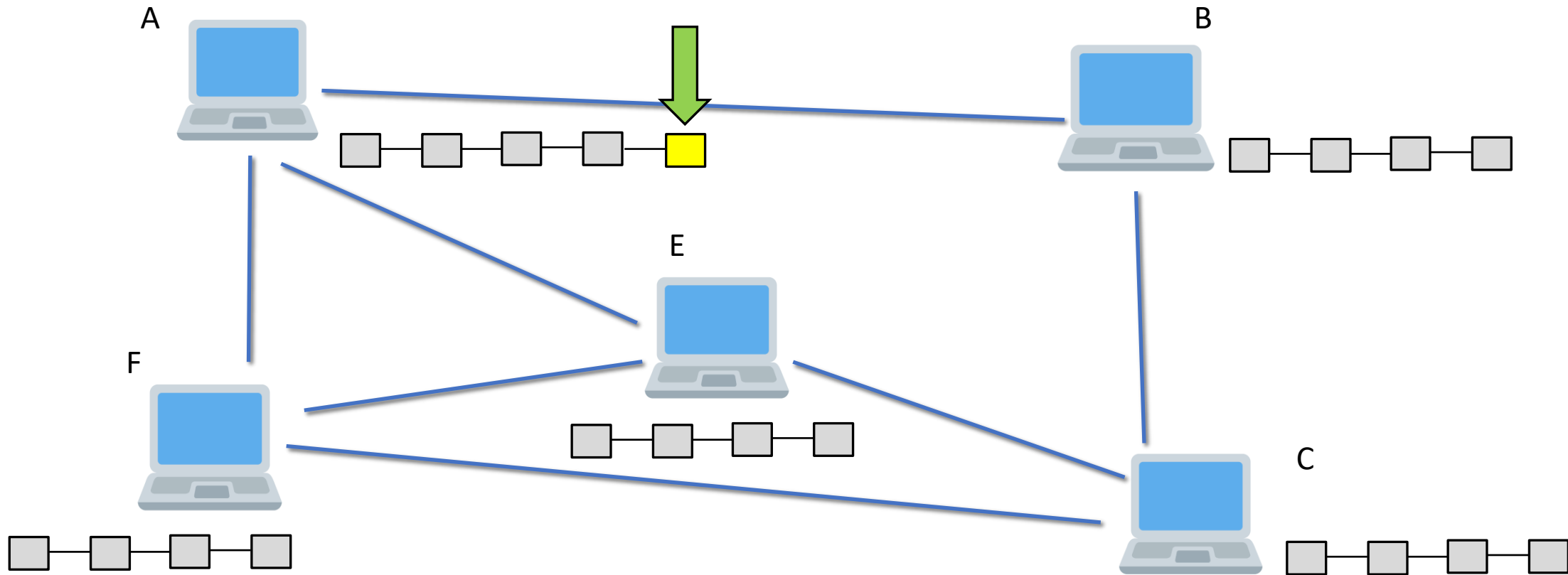


Distributed P2P network in Blockchain

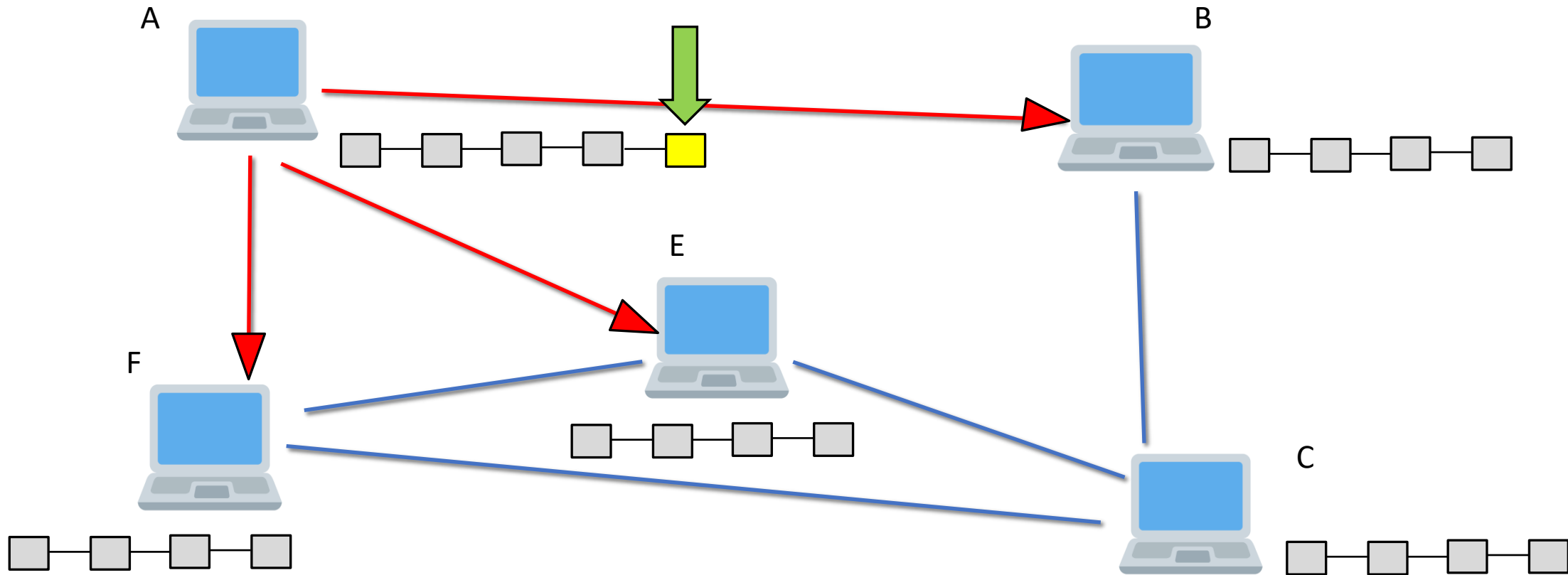
Distributed P2P network



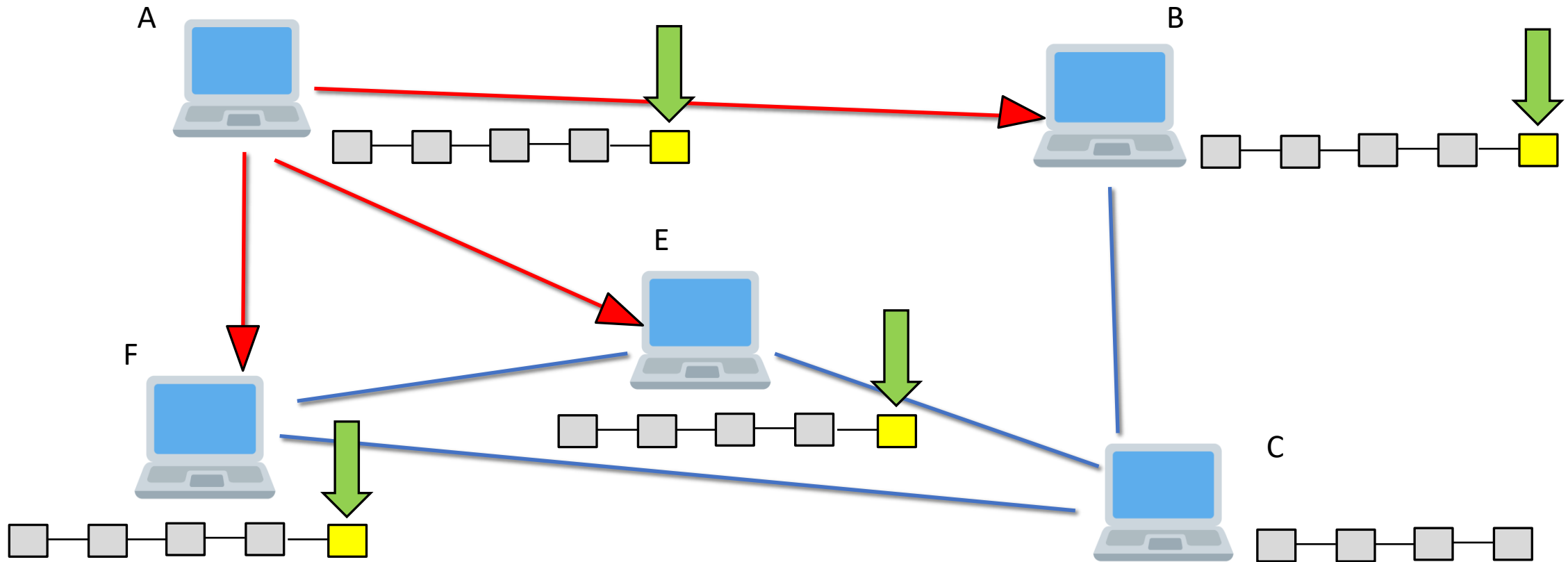
Distributed P2P network



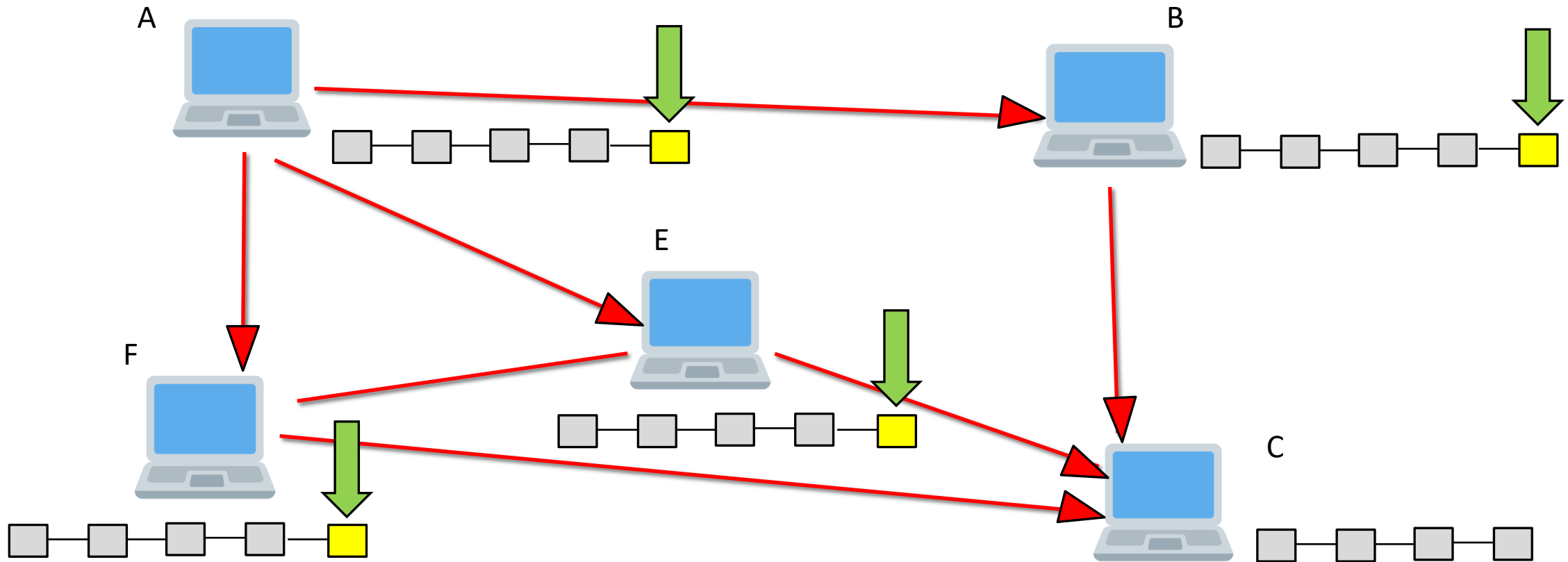
Distributed P2P network



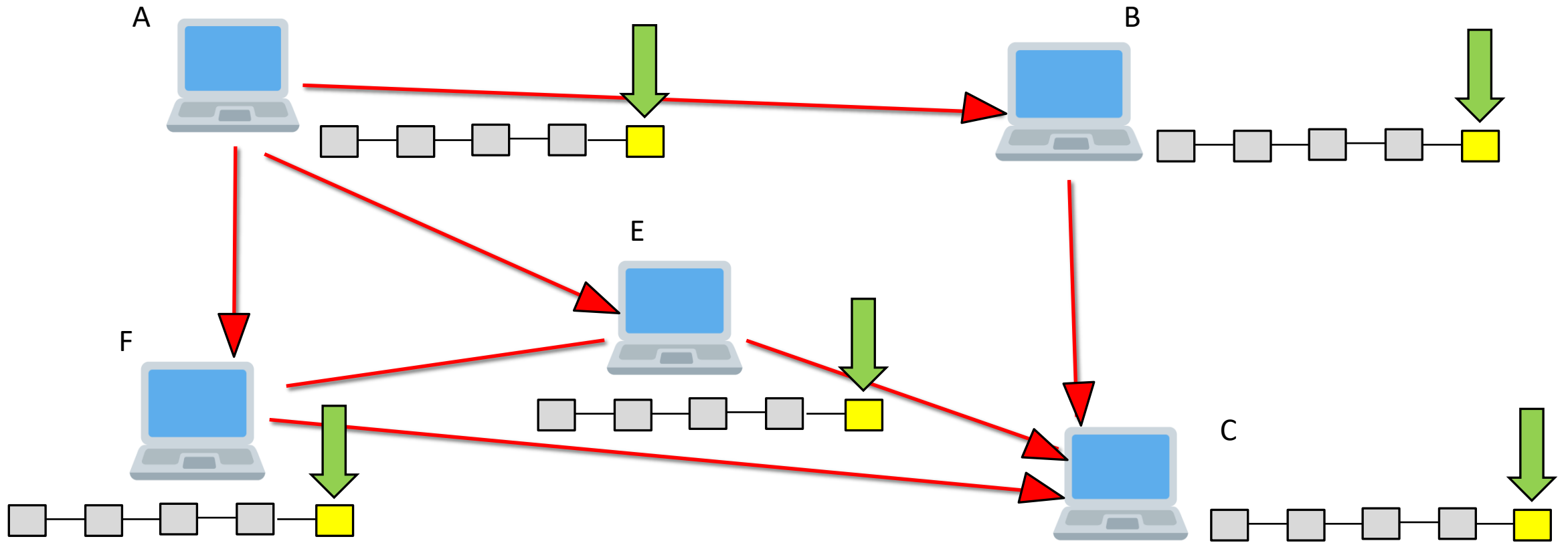
Distributed P2P network



Distributed P2P network



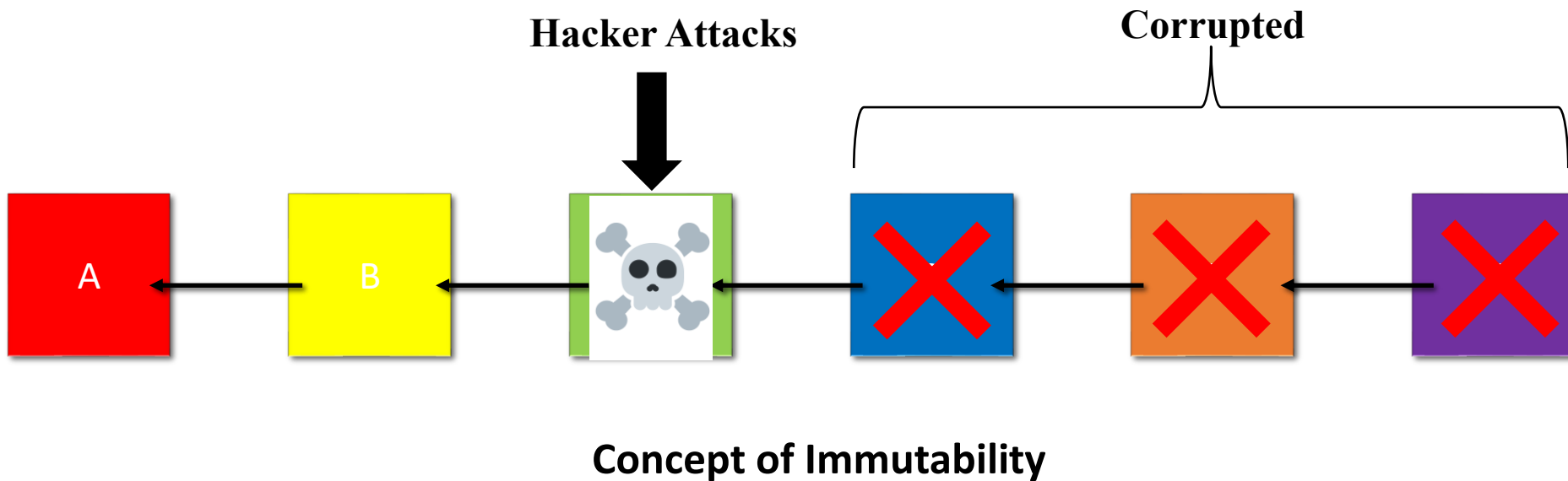
Distributed P2P network



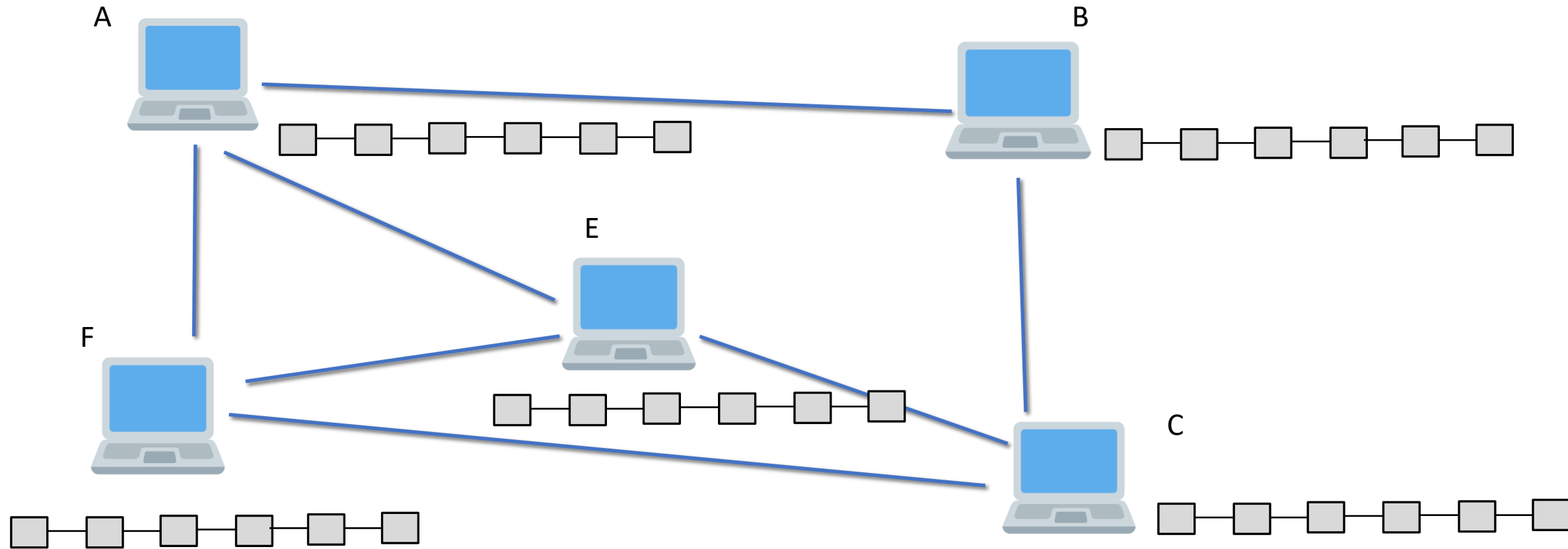
Distributed P2P network

Q)Why we need Distributed P2P network in Blockchain?

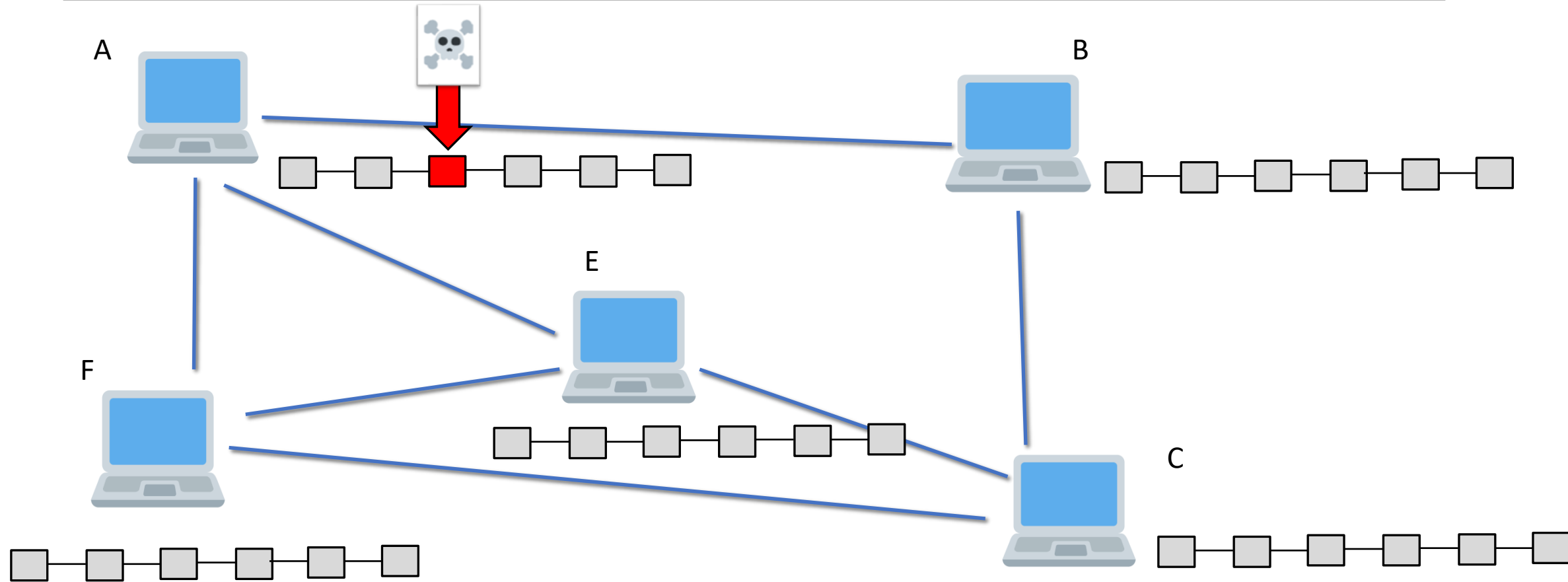
Distributed P2P network



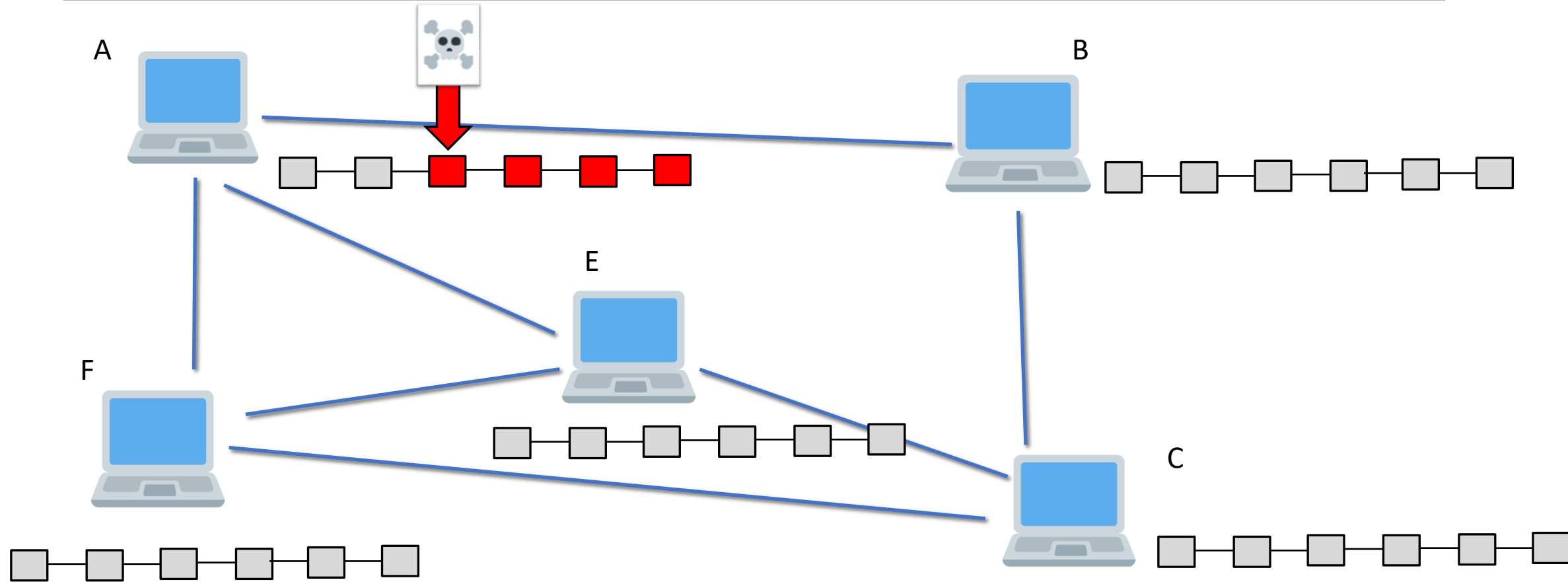
Distributed P2P network



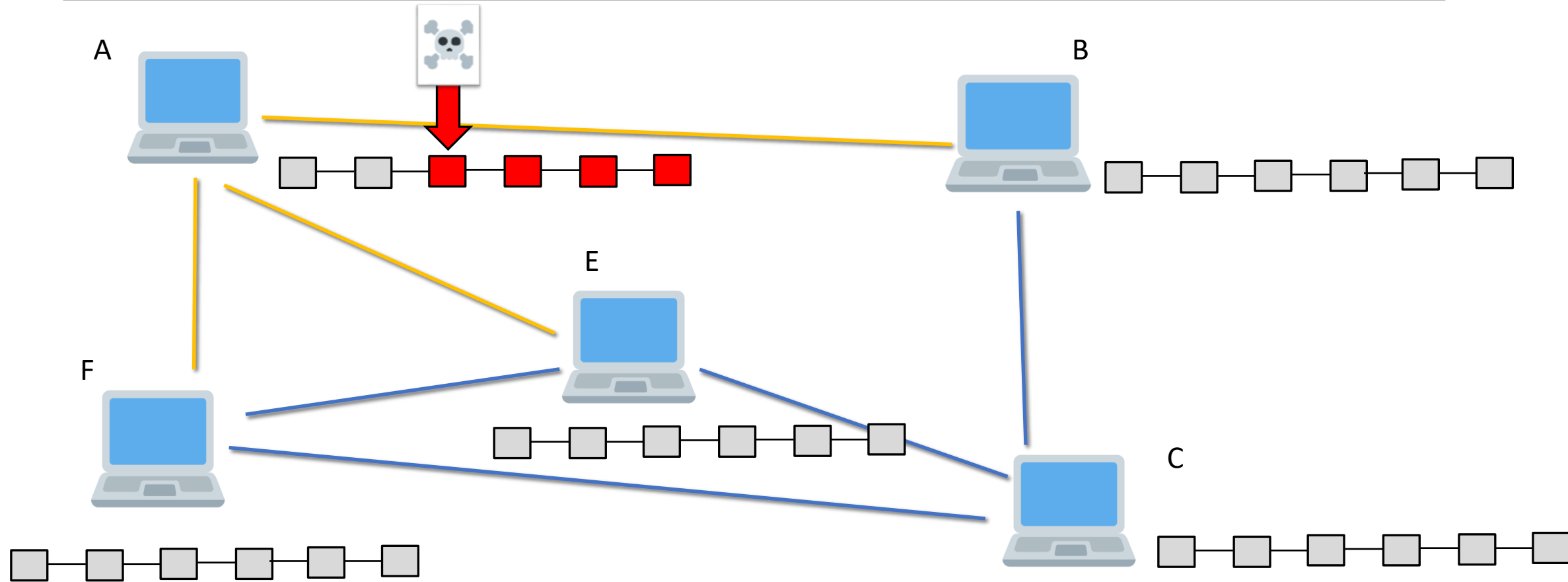
Distributed P2P network



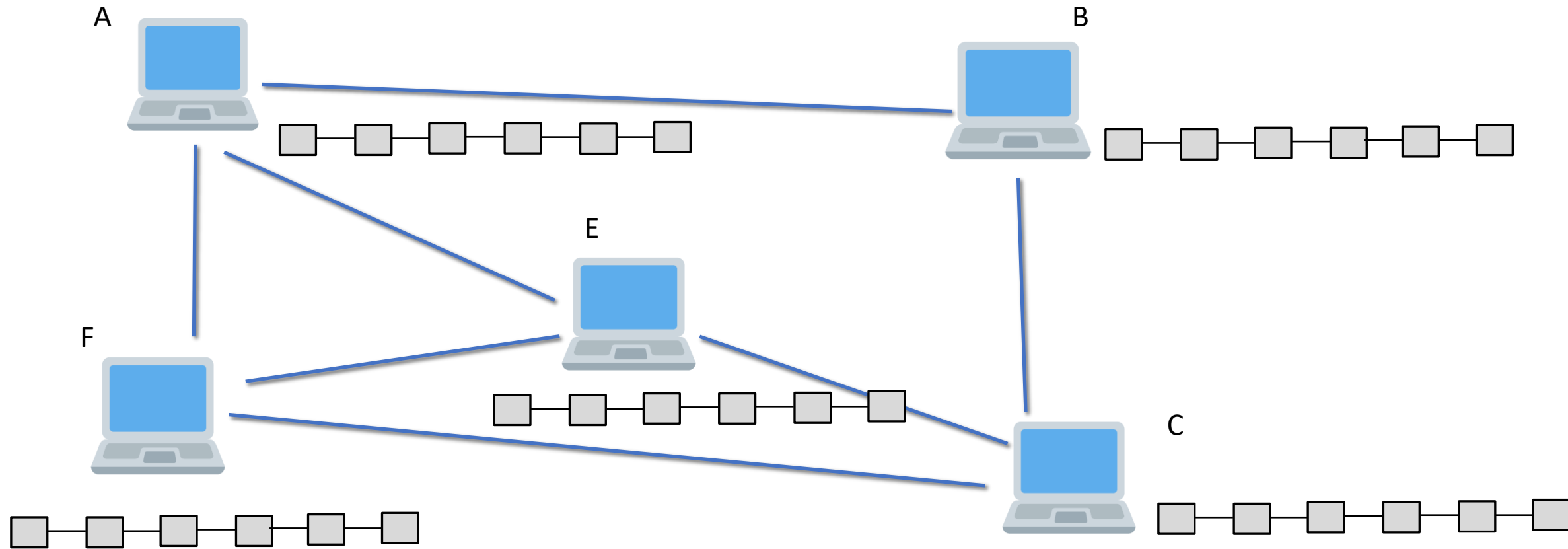
Distributed P2P network

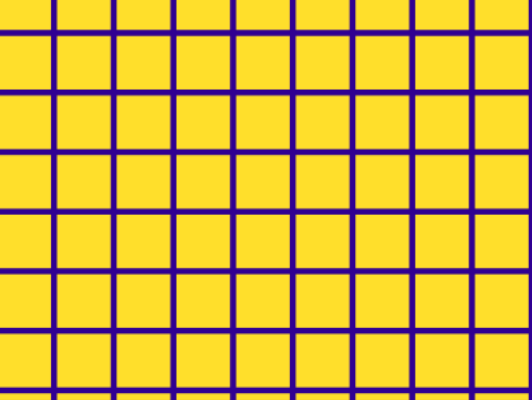


Distributed P2P network



Distributed P2P network





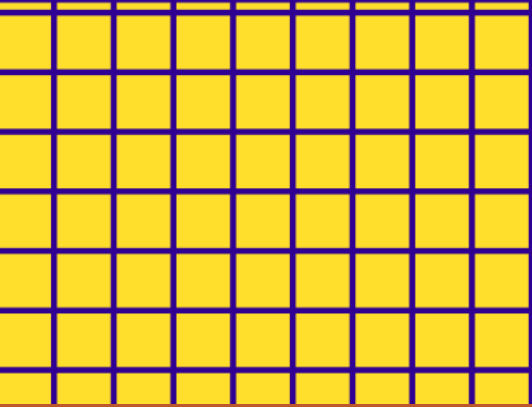
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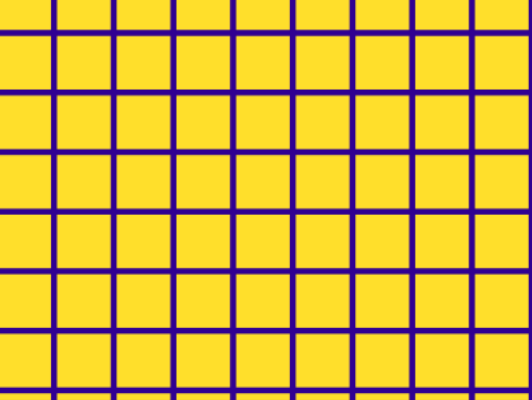
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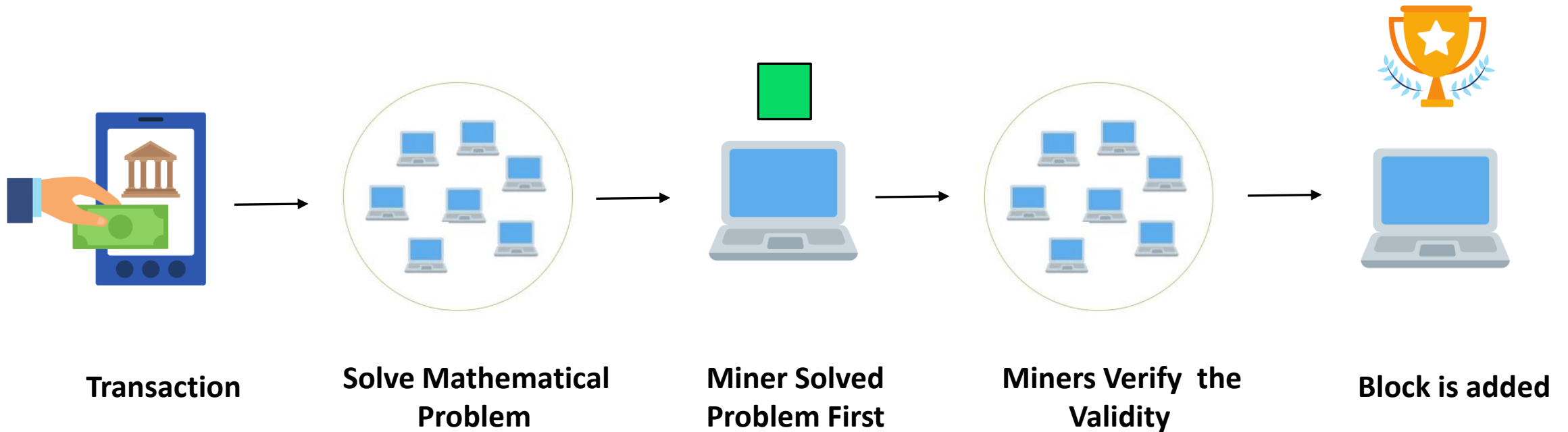
CODE EATER

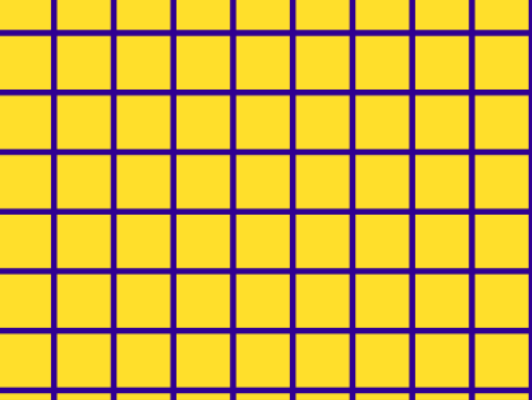




Blockchain Mining

Blockchain Mining





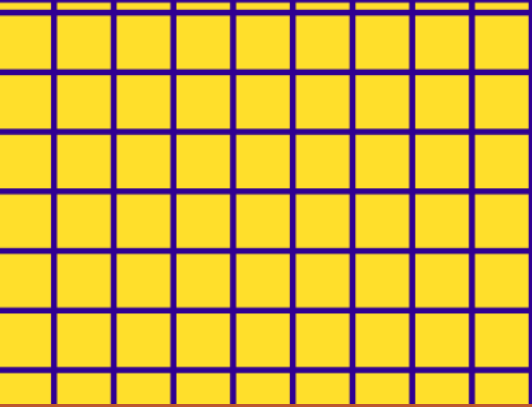
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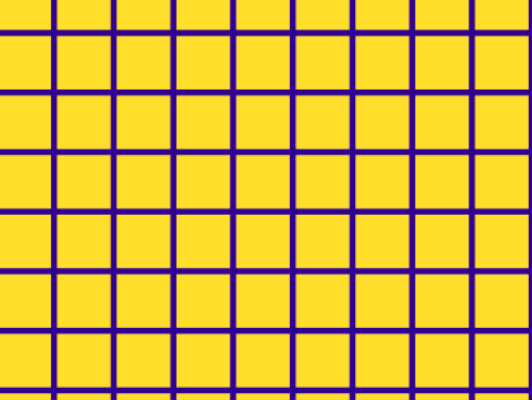


Discord – Link in description



Blockchain Mining

Trust, secure,



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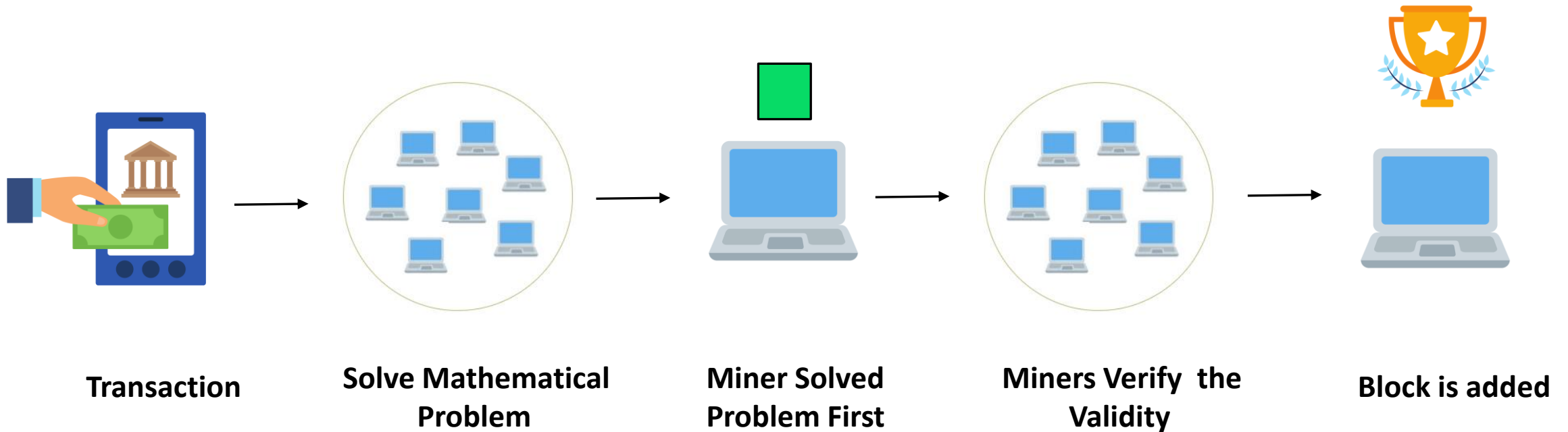
CODE EATER



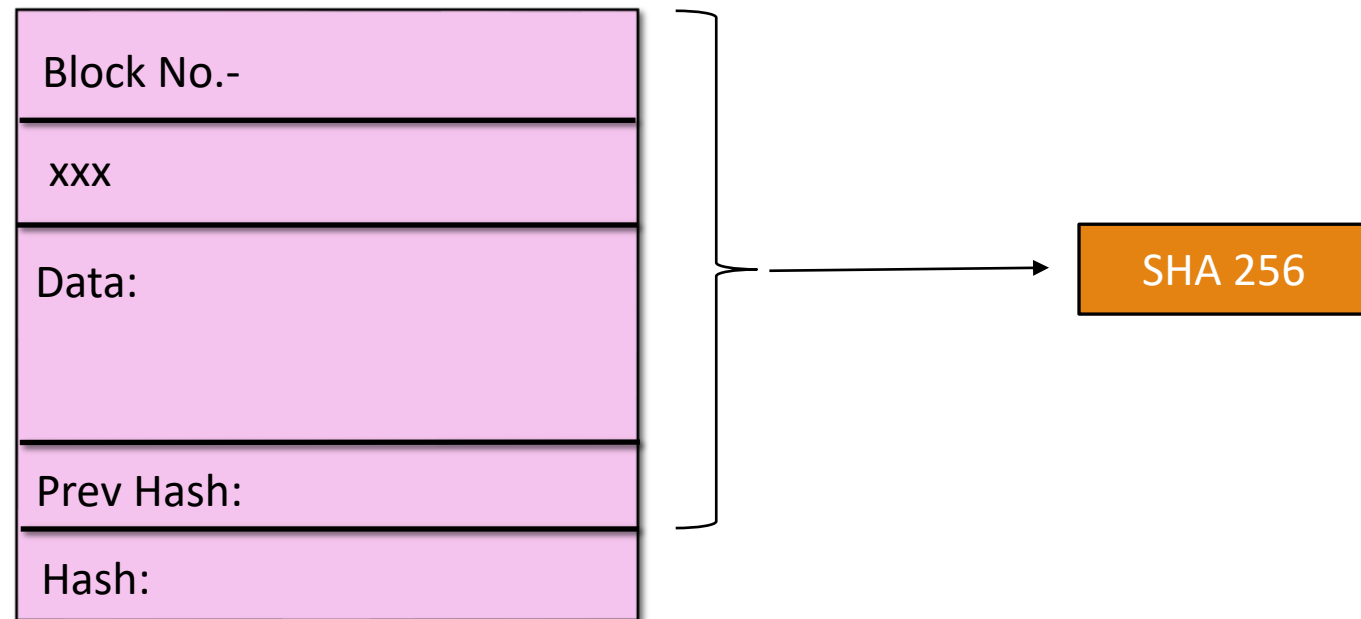


How Mining works: The Nonce

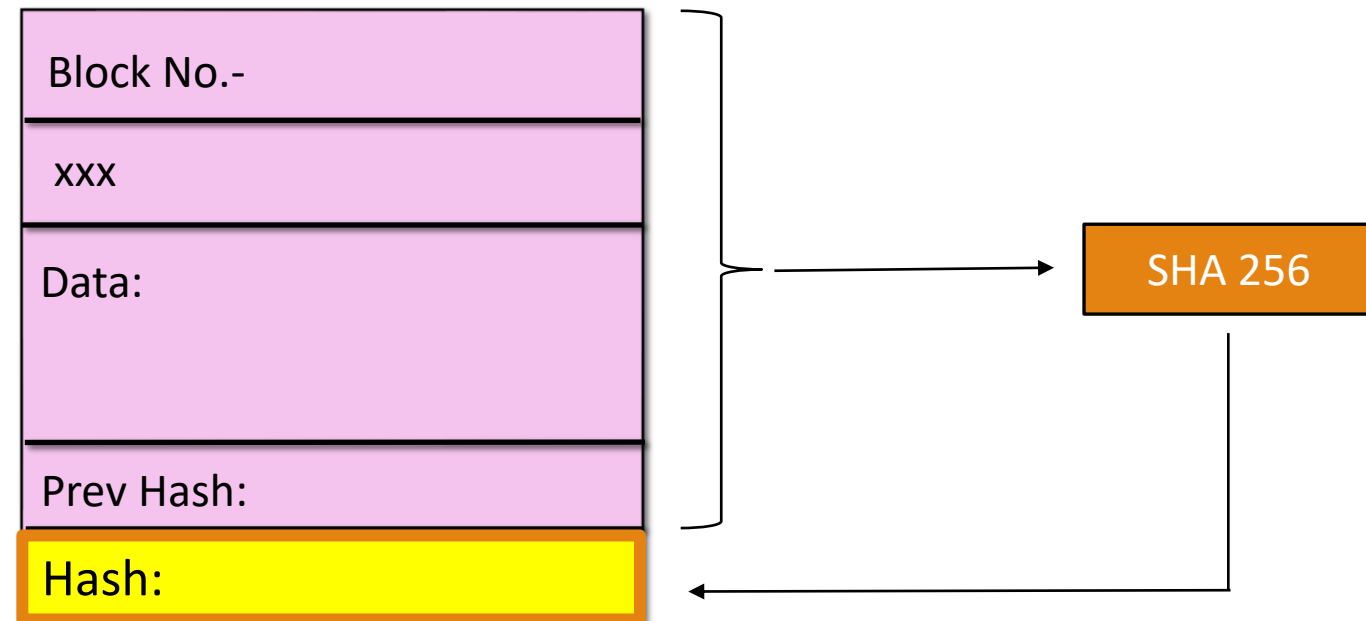
Blockchain Mining



The Nonce



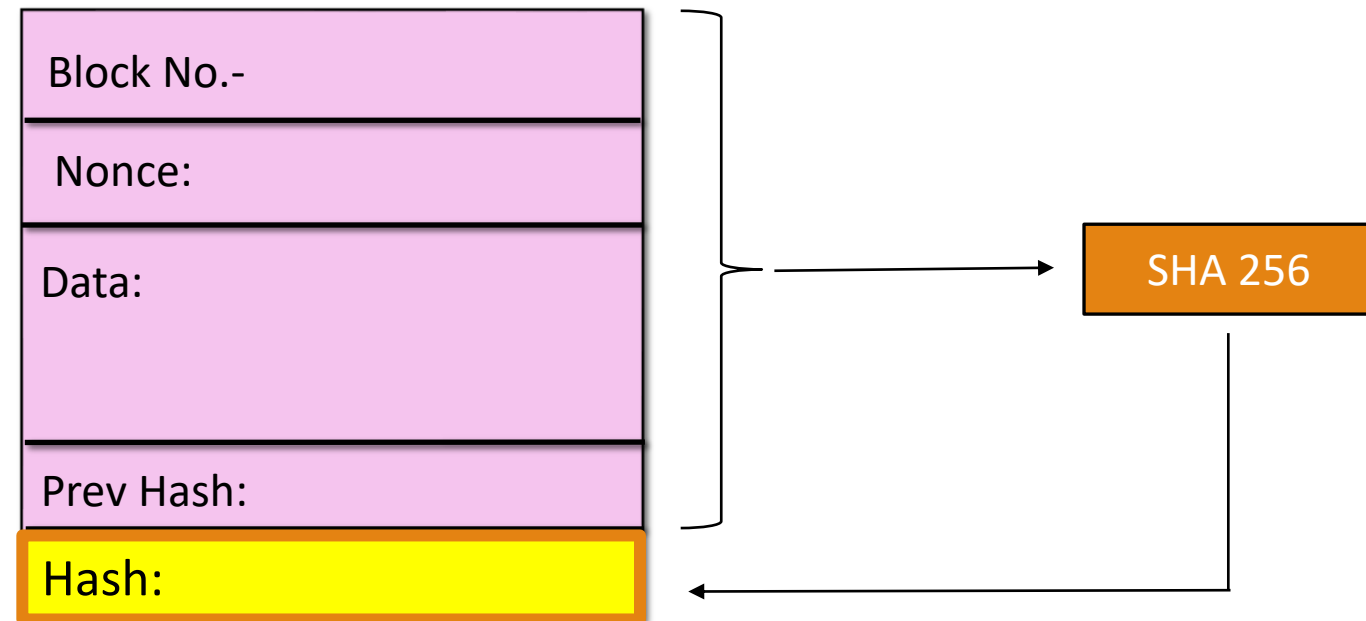
The Nonce



The Nonce

Block No.-
Nonce:
Data:
Prev Hash:
Hash:

The Nonce



The Nonce

Block No.- 6
Nonce:
Data: Kshitij->Rakesh 500 coins Raj->Bella 200 coins
Prev Hash: 0000AB23
Hash:

The Nonce

Block No.- 6
Nonce: 23
Data: Kshitij->Rakesh 500 coins Raj->Bella 200 coins
Prev Hash: 0000AB23
Hash: 00001ba1

The Nonce

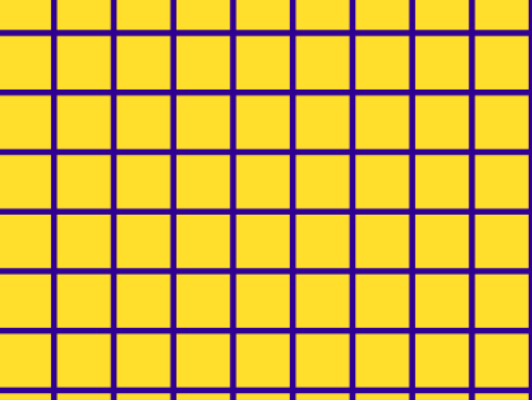
Block No.- 6
Nonce: 50
Data: Kshitij->Rakesh 500 coins Raj->Bella 200 coins
Prev Hash: 0000AB23
Hash: 0000fb12

The Nonce

Block No.- 6
Nonce: 3
Data: Kshitij->Rakesh 500 coins Raj->Bella 200 coins
Prev Hash: 0000AB23
Hash: 0000acc12

The Nonce

Block No.- 6
Nonce: 1001
Data: Kshitij->Rakesh 500 coins Raj->Bella 200 coins
Prev Hash: 0000AB23
Hash: 0000ef23



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How Mining works: The Nonce

How Mining Works ?

Nonce

Target

How Mining Works ?

Nonce:

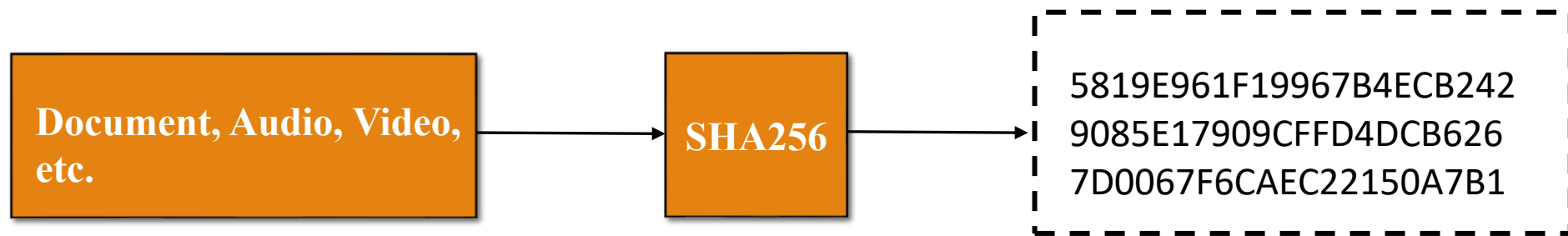
- The nonce is a number that blockchain miners are solving for.

How Mining Works ?

Target:

- Target is a number used in mining.
- It is a number that a block hash must be below for the block to be added on to the blockchain.
- The target adjusts every 2016 blocks (roughly two weeks) to try and ensure that blocks are mined **once every 10 minutes** on average.

Hashing Algorithm



This has **64 hexadecimal characters**.
Each character is of **4 bits**.
So in total it has $64 * 4$ bits i.e. **256 bits**.

How Mining Works ?

Hexadecimal Numbers

Decimal	Hexadecimal
0	0
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	A

Decimal	Hexadecimal
11	B
12	C
13	D
14	E
15	F

How Mining Works ?

- d2fd3930d274b202fe8e7cb431e38a8b64ec396e15f5717e60493234b0de210a
- 52d095795c1dc87ff2f6b4d9b005a1fe2cfed01103763c9443f6d4496df8e800
- 0000005432d9f64f6e05c019f9302162100163b6cdba06bd72eee35cd19aebf

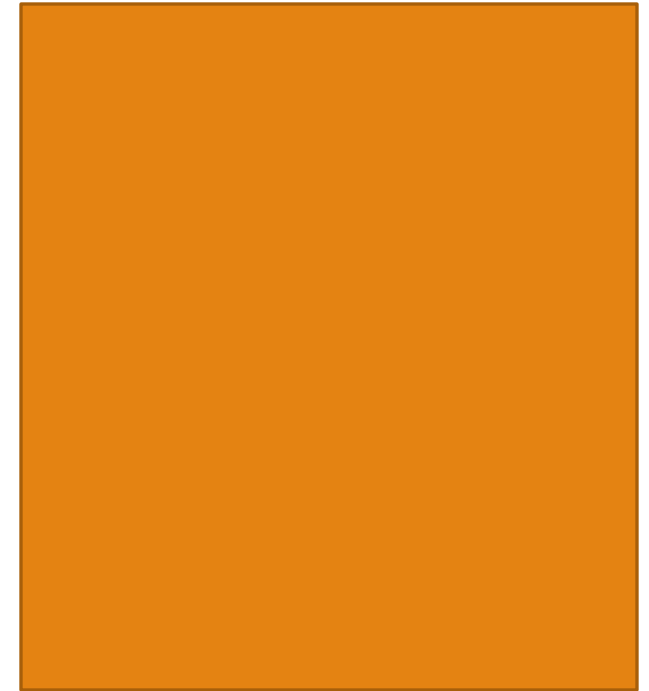
Smallest- 0000000.....0

Largest- ffffffffff.....f

How Mining Works ?

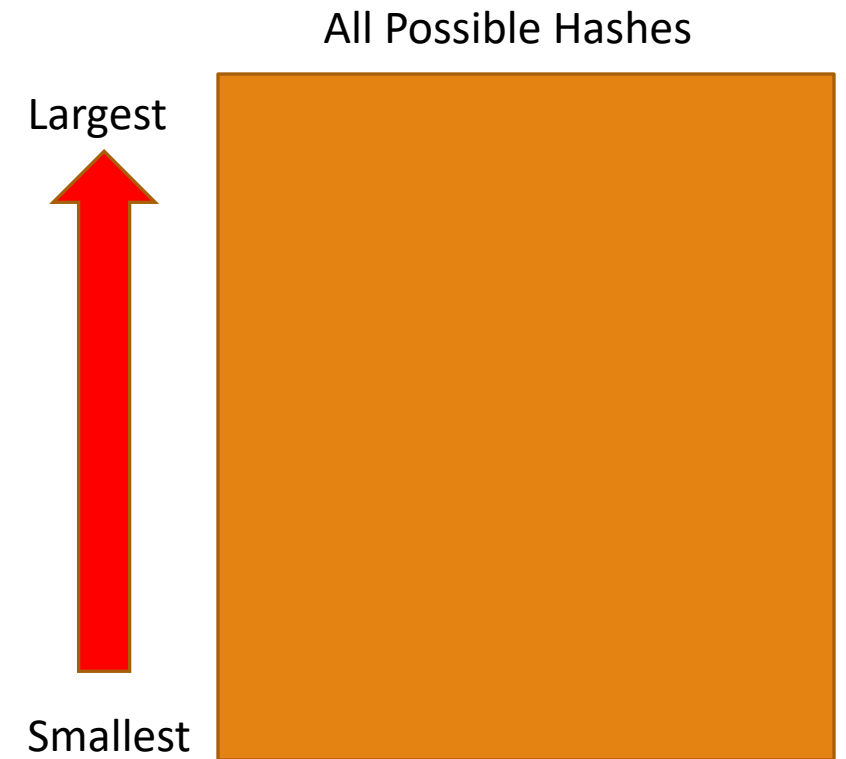
Block No.-6
Nonce:
Data: Kshitij->Rakesh 500 coins Raj->Bella 200 coins
Prev Hash: 0000AB23
Hash:

All Possible Hashes



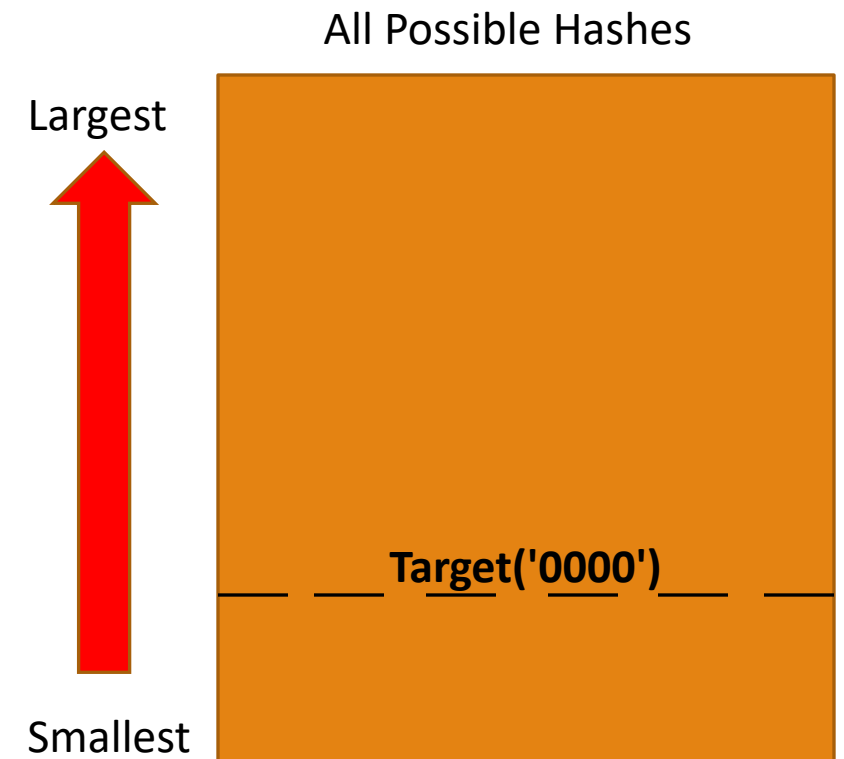
How Mining Works ?

Block No.-6
Nonce:
Data: Kshitij->Rakesh 500 coins Raj->Bella 200 coins
Prev Hash: 0000AB23
Hash:

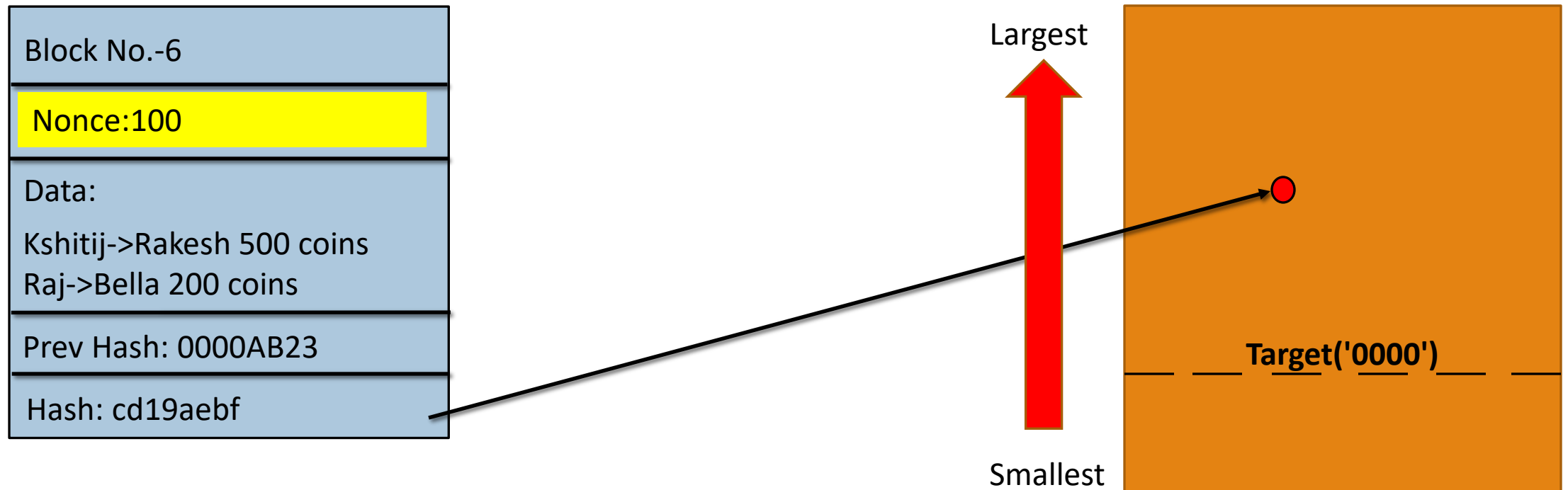


How Mining Works ?

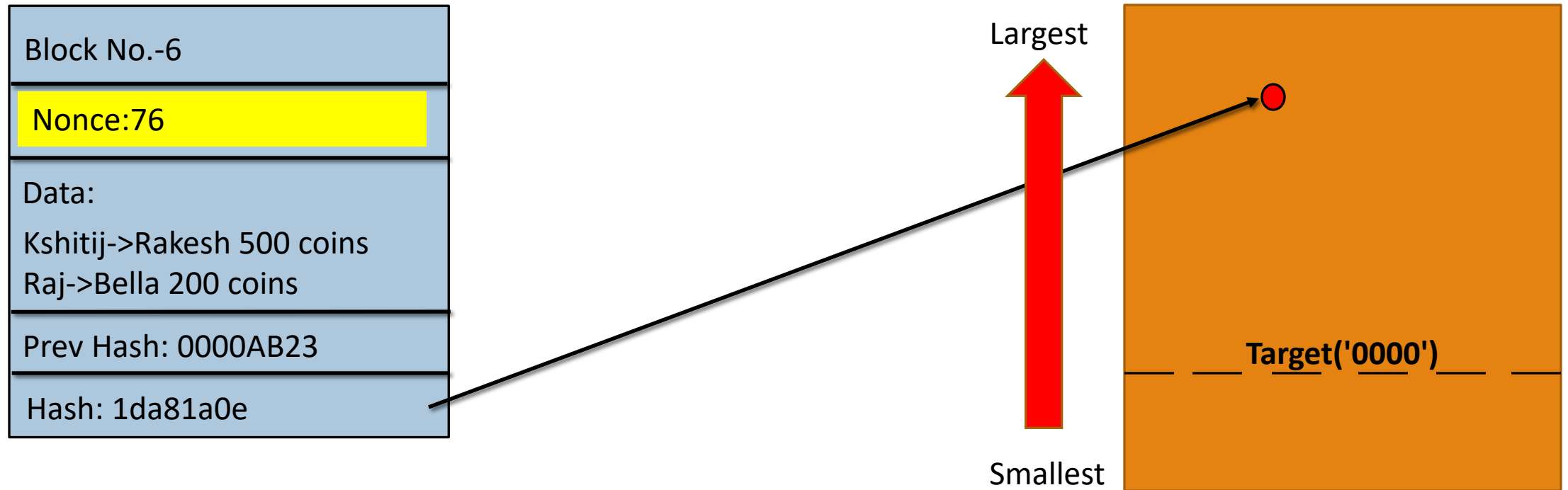
Block No.-6
Nonce:
Data: Kshitij->Rakesh 500 coins Raj->Bella 200 coins
Prev Hash: 0000AB23
Hash:



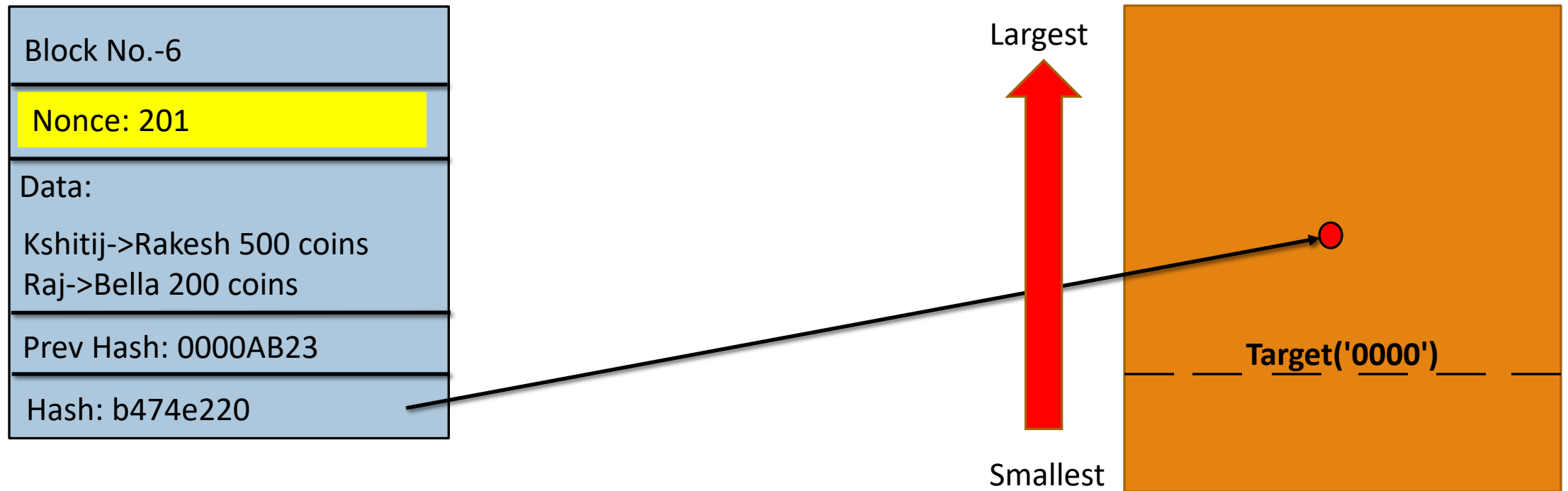
How Mining Works ?



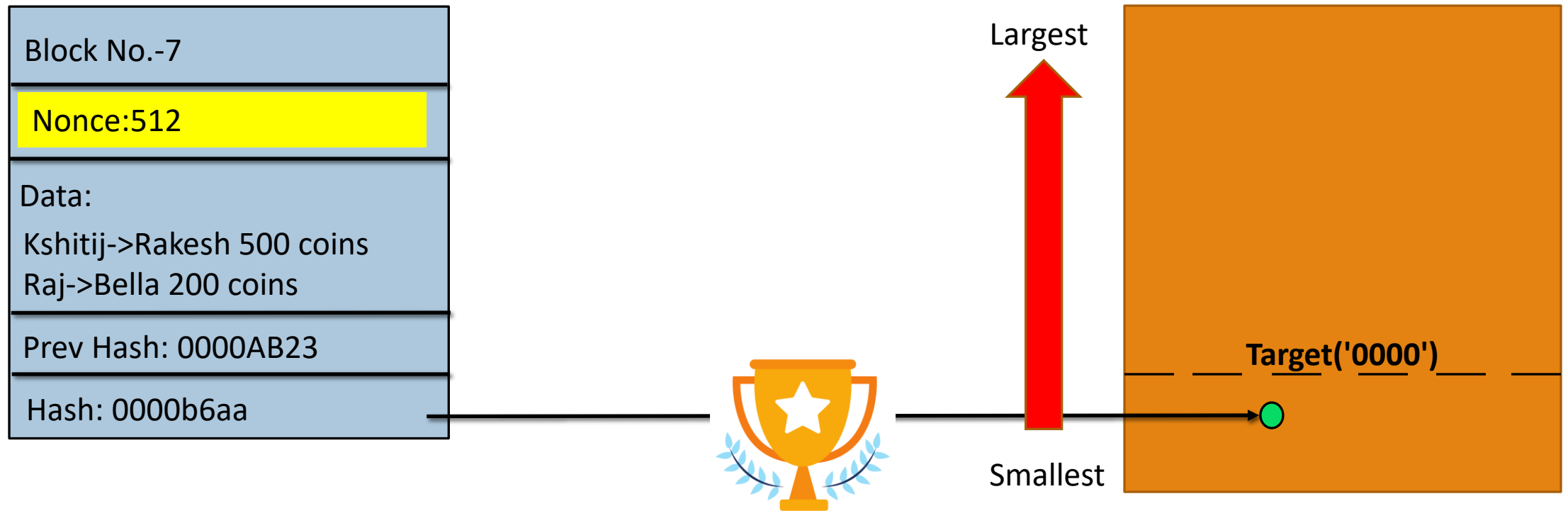
How Mining Works ?

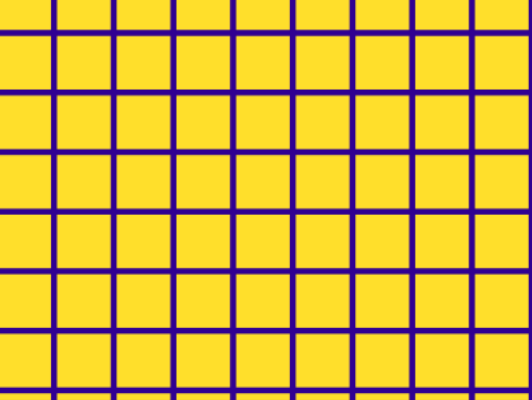


How Mining Works ?



How Mining Works ?





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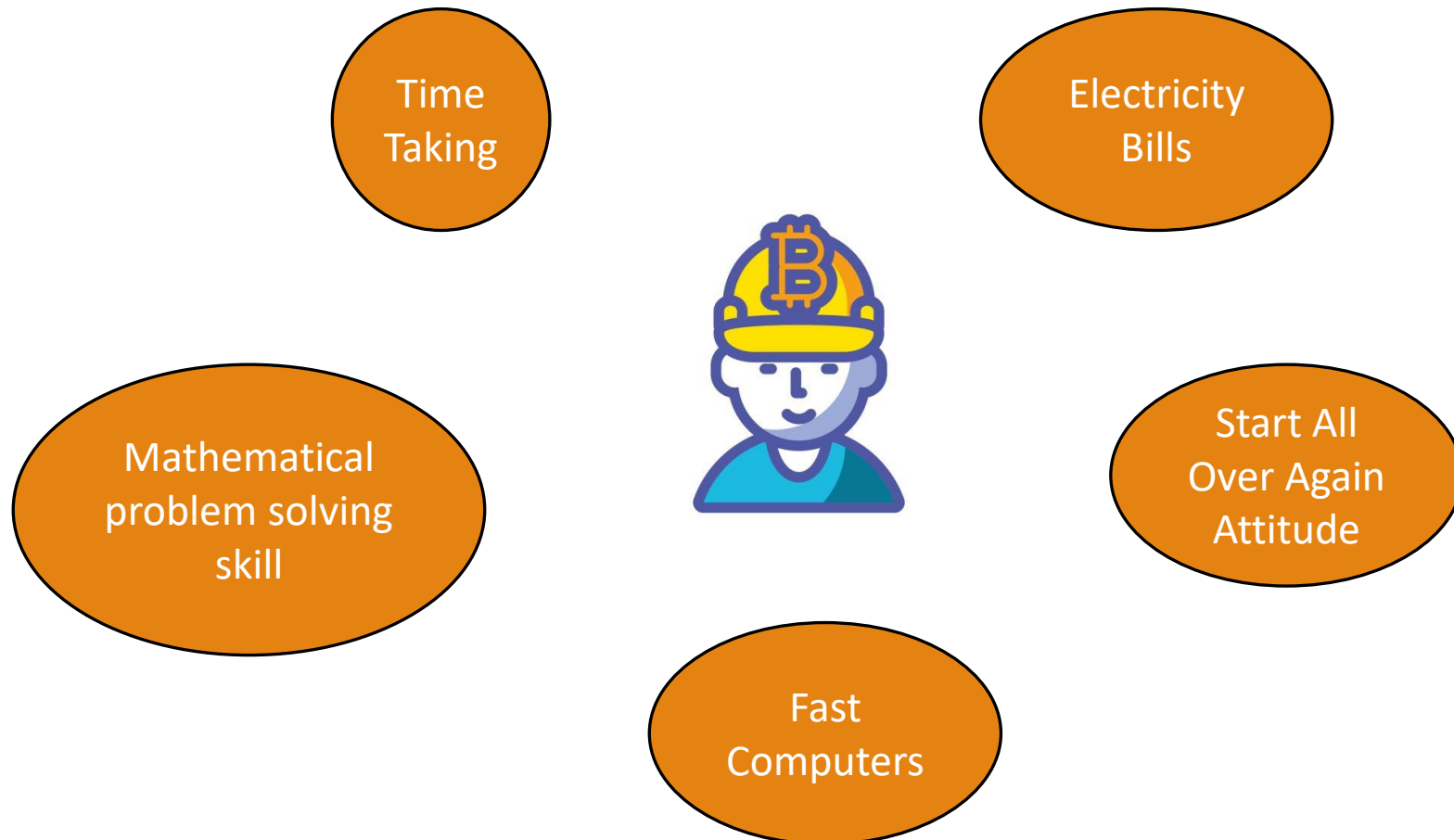


Blockchain Demo



Is Mining that easy?

Challenges faced by Miners



What is Mining?

What is mining? Why mining is required? Nonce basics ? None advanced

Why Mining is required?

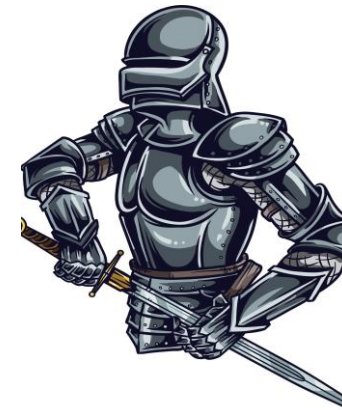
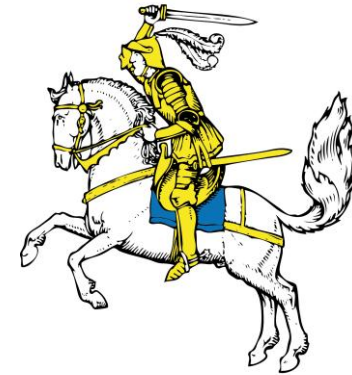


Byzantine Generals Problem

Byzantine Generals Problem

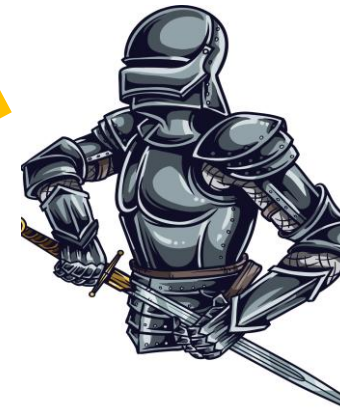


Byzantine Generals Problem

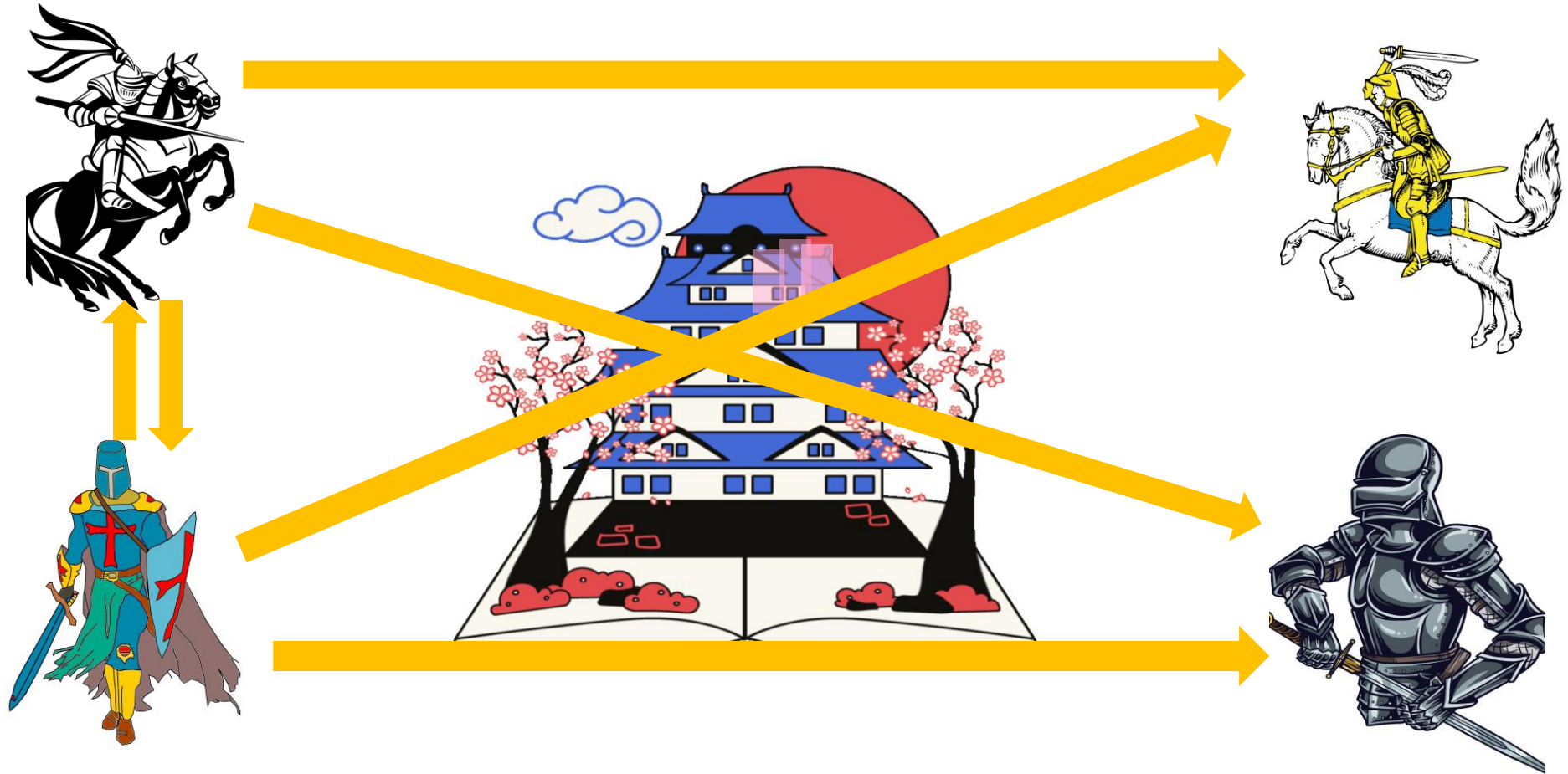


Byzantine Generals Problem

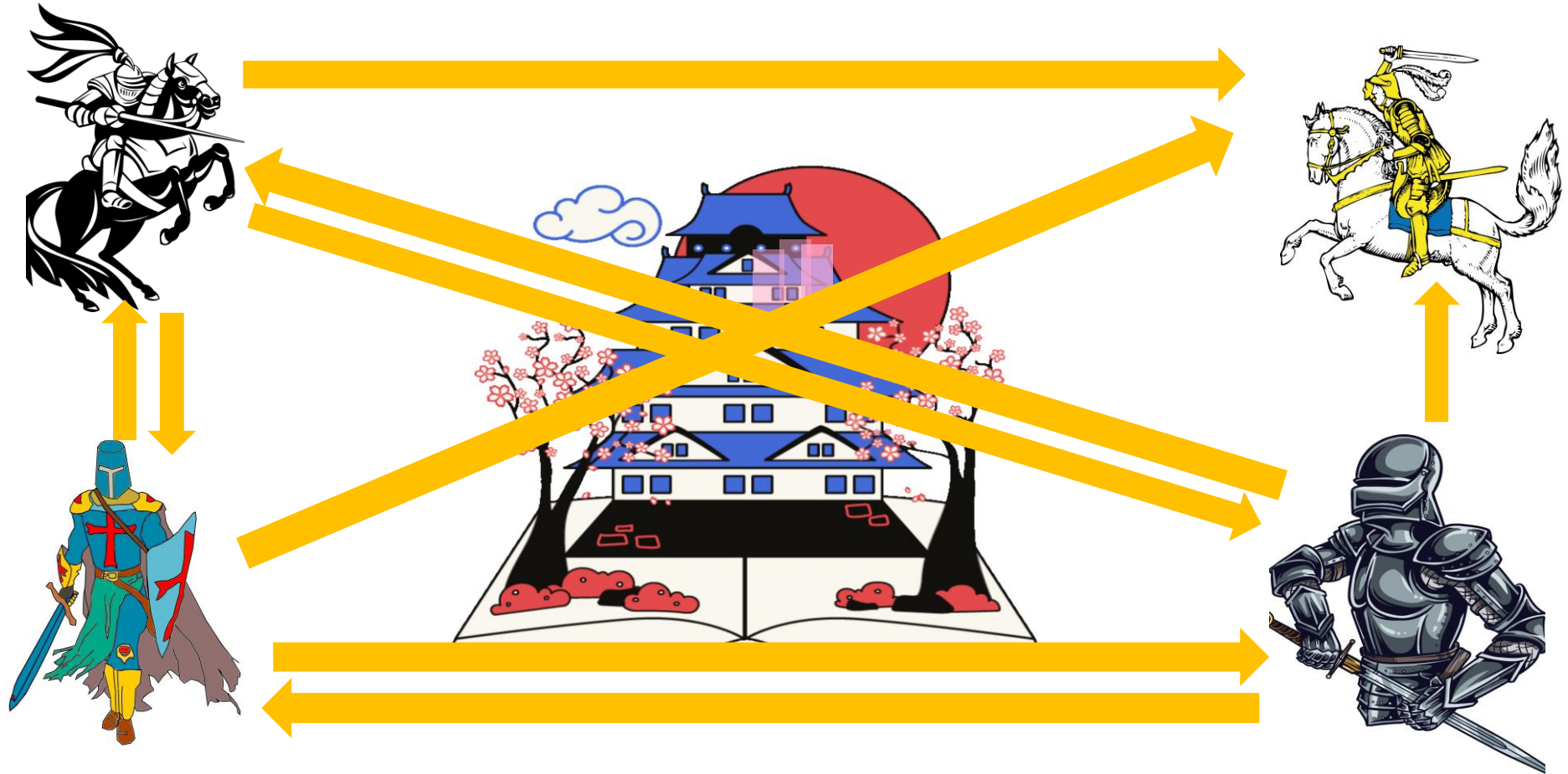
Yellow: Fight



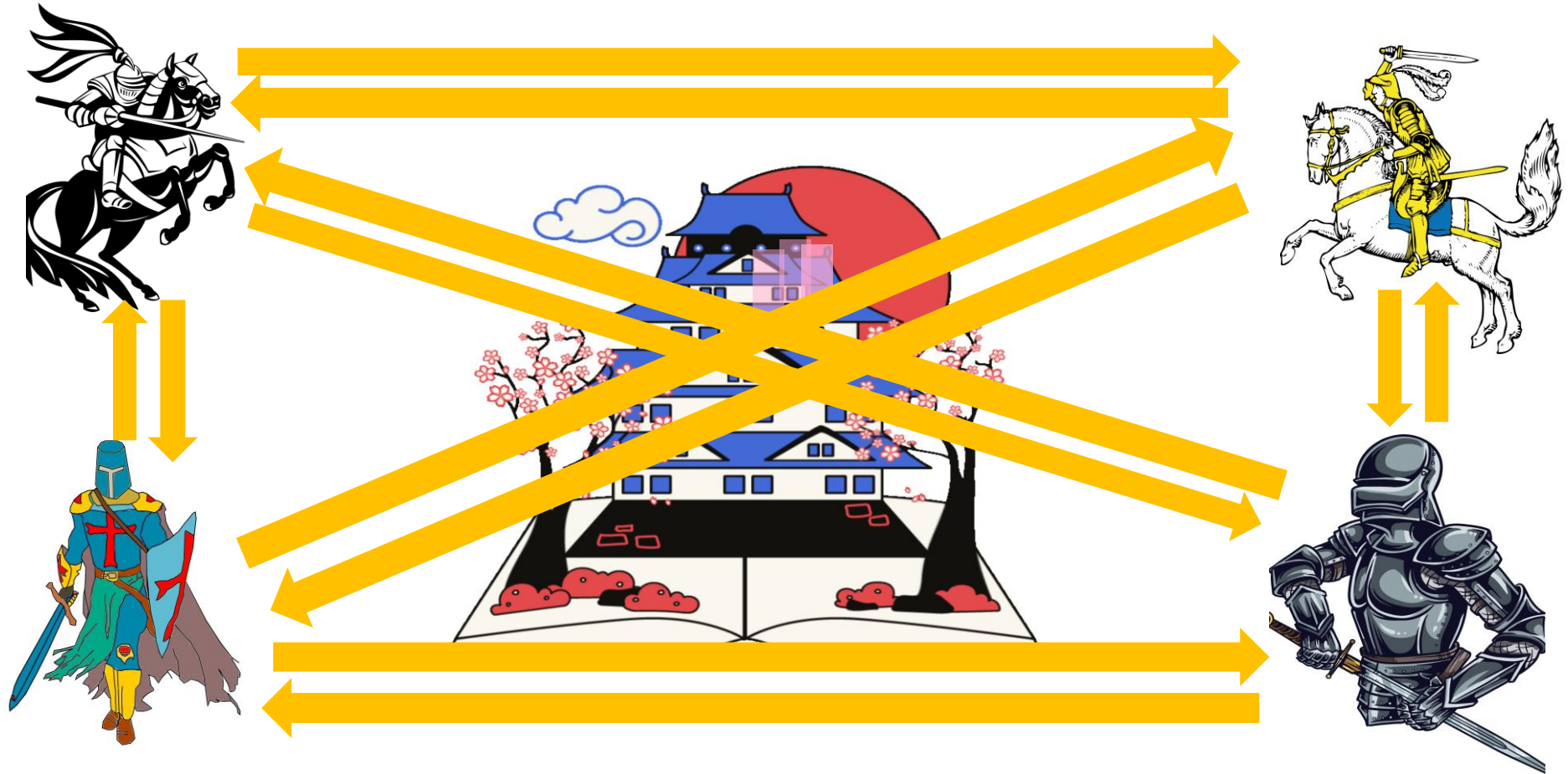
Byzantine Generals Problem



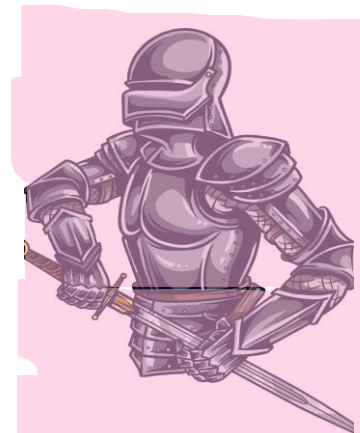
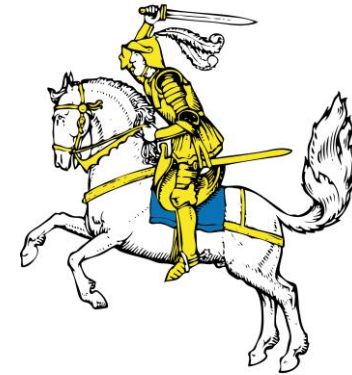
Byzantine Generals Problem



Byzantine Generals Problem

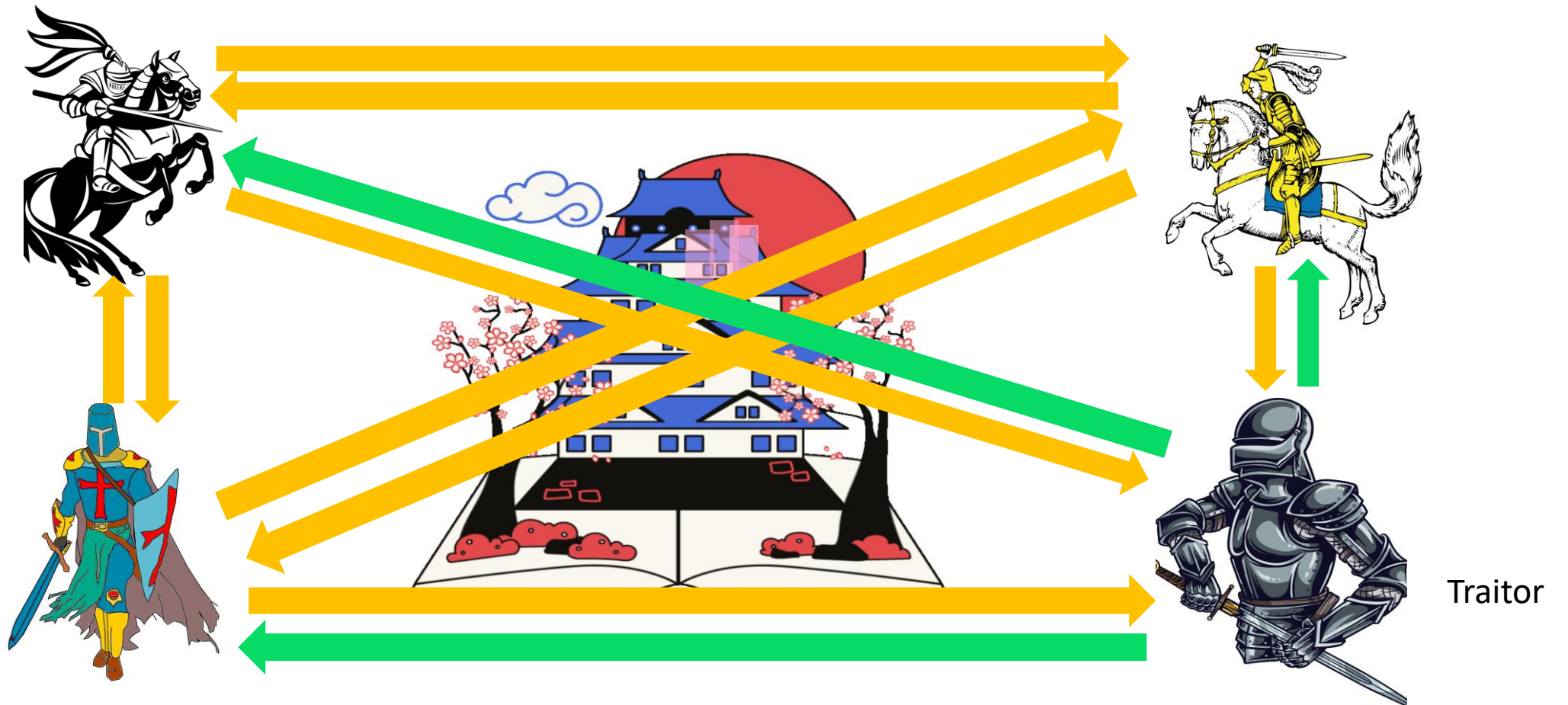


Byzantine Generals Problem



Traitor

Byzantine Generals Problem



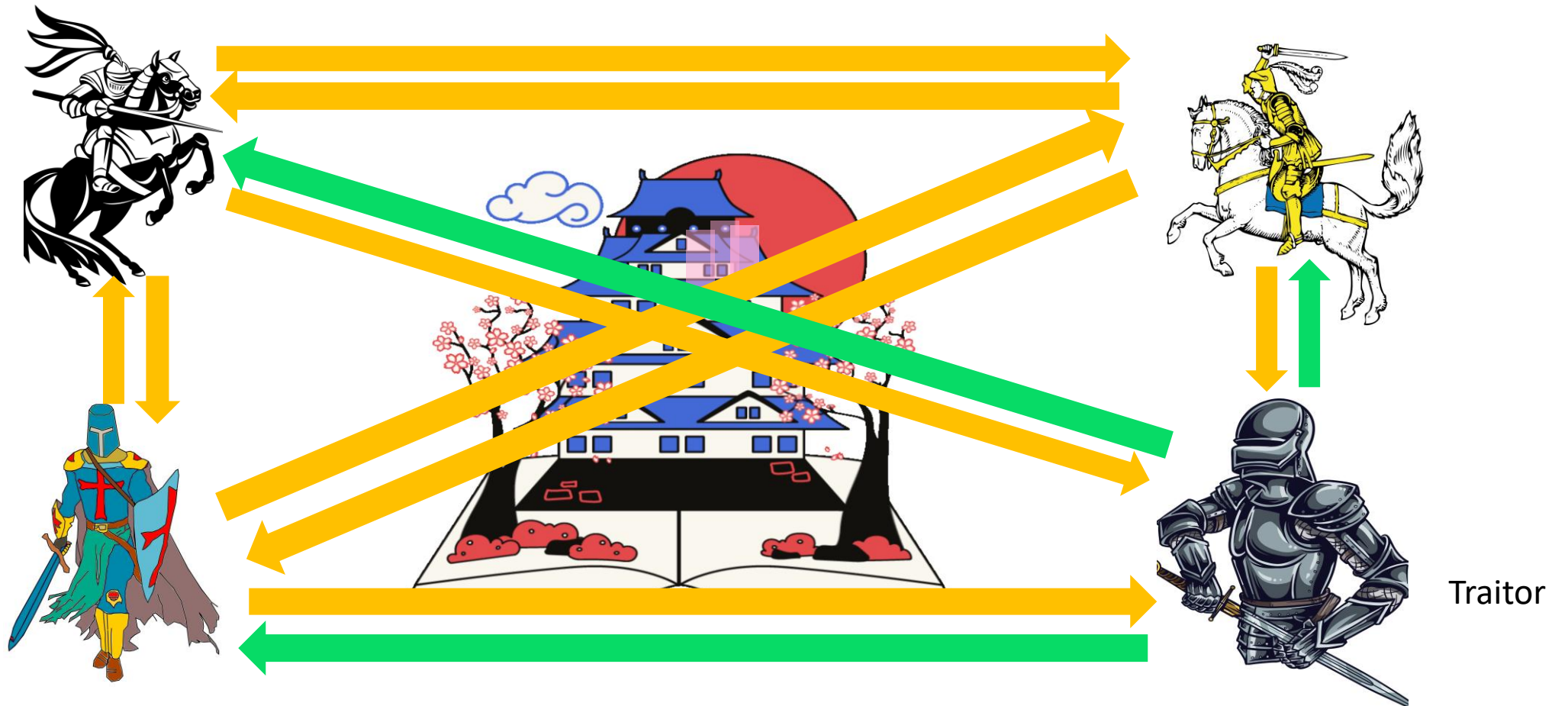
Practical Byzantine Fault Tolerance

by
Miguel Castro

Abstract

Our growing reliance on online services accessible on the Internet demands highly-available systems that provide correct service without interruptions. Byzantine faults such as software bugs, operator mistakes, and malicious attacks are the major cause of service interruptions. This thesis describes a new replication algorithm, BFT, that can be used to build highly-available systems that tolerate Byzantine faults. It shows, for the first time, how to build Byzantine-fault-tolerant systems that can be used in practice to implement real services because they do not rely on unrealistic assumptions and they perform well. BFT works in asynchronous environments like the Internet, it incorporates mechanisms to defend against Byzantine-faulty clients, and it recovers replicas proactively. The recovery mechanism allows the algorithm to tolerate any number of faults over the lifetime of the system provided fewer than $1/3$ of the replicas become faulty within a small window of vulnerability.

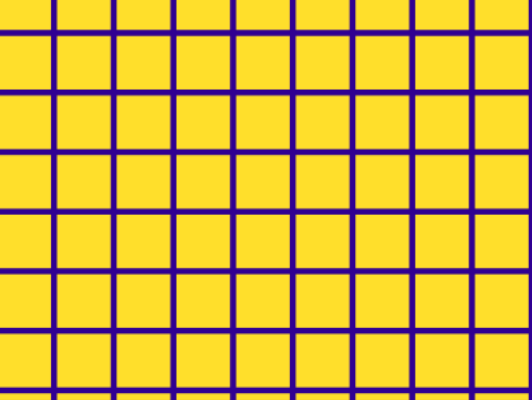
Byzantine Generals Problem





Byzantine Generals Problem

Q) How this Byzantine Fault Tolerance works in Blockchain?



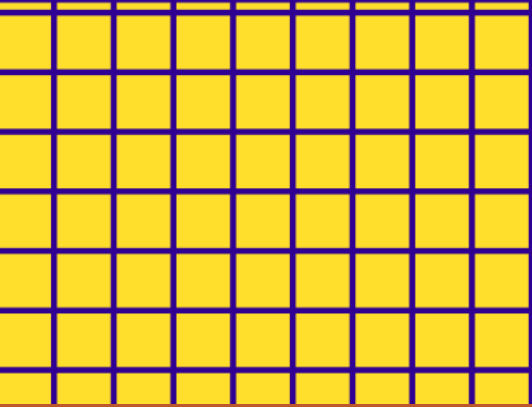
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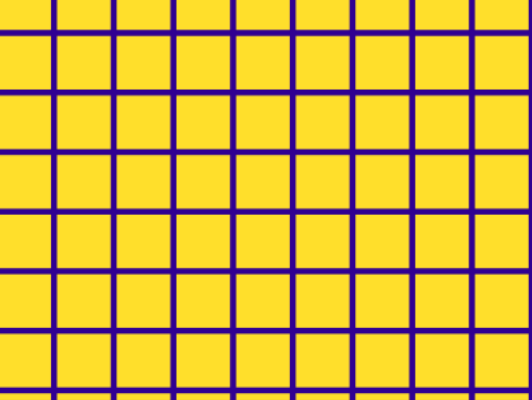
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Consensus Protocol

Consensus Protocol

- **Hashing Algorithm** 
- **Immutable Ledger** 
- **Mining** 
- **Distribute P2P network** 

Consensus Protocol

Prevent Attacks

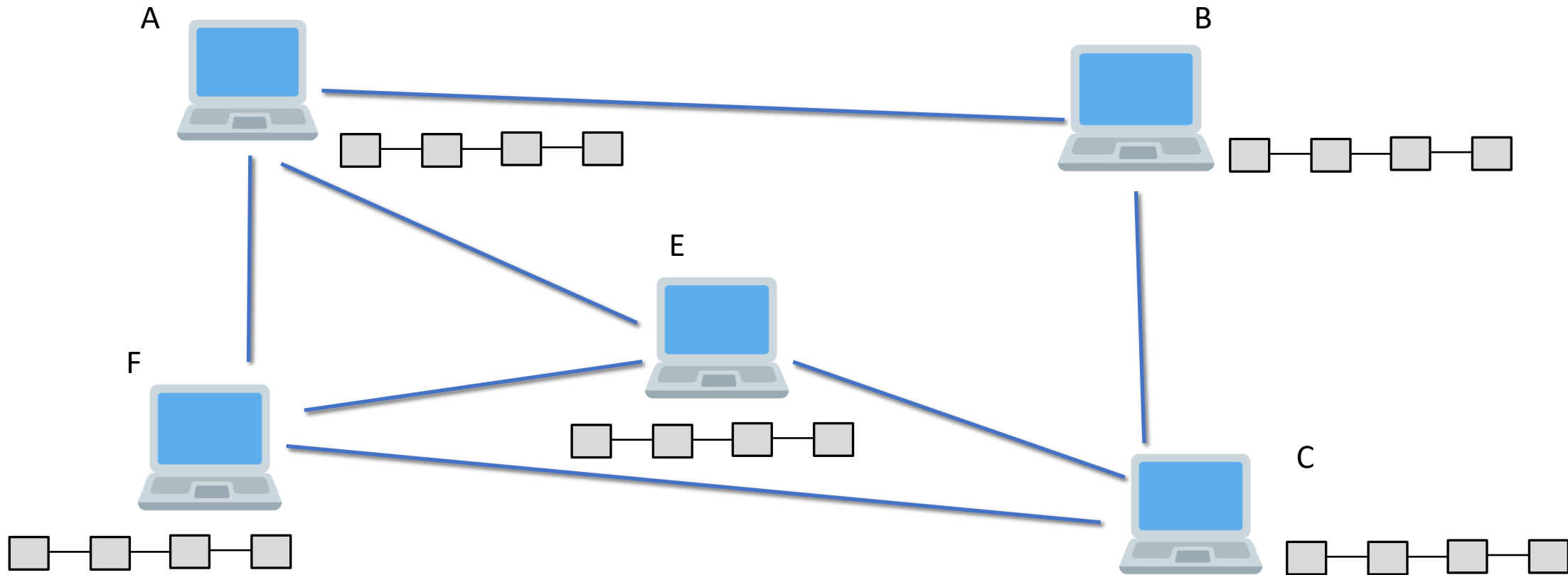
Competing Chain Problem

Consensus Protocol

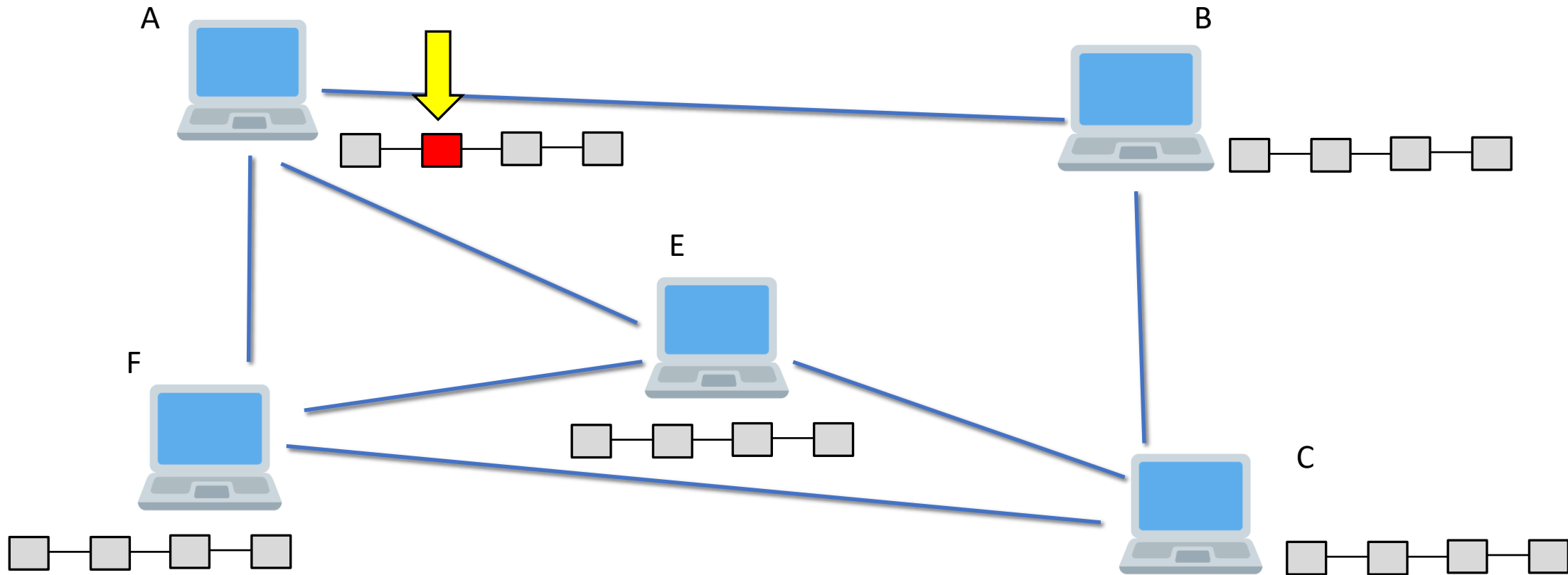
Types of Consensus Protocol-

- **Proof of Work (POW)**
- **Proof of Stake (POS)**
- **Others**

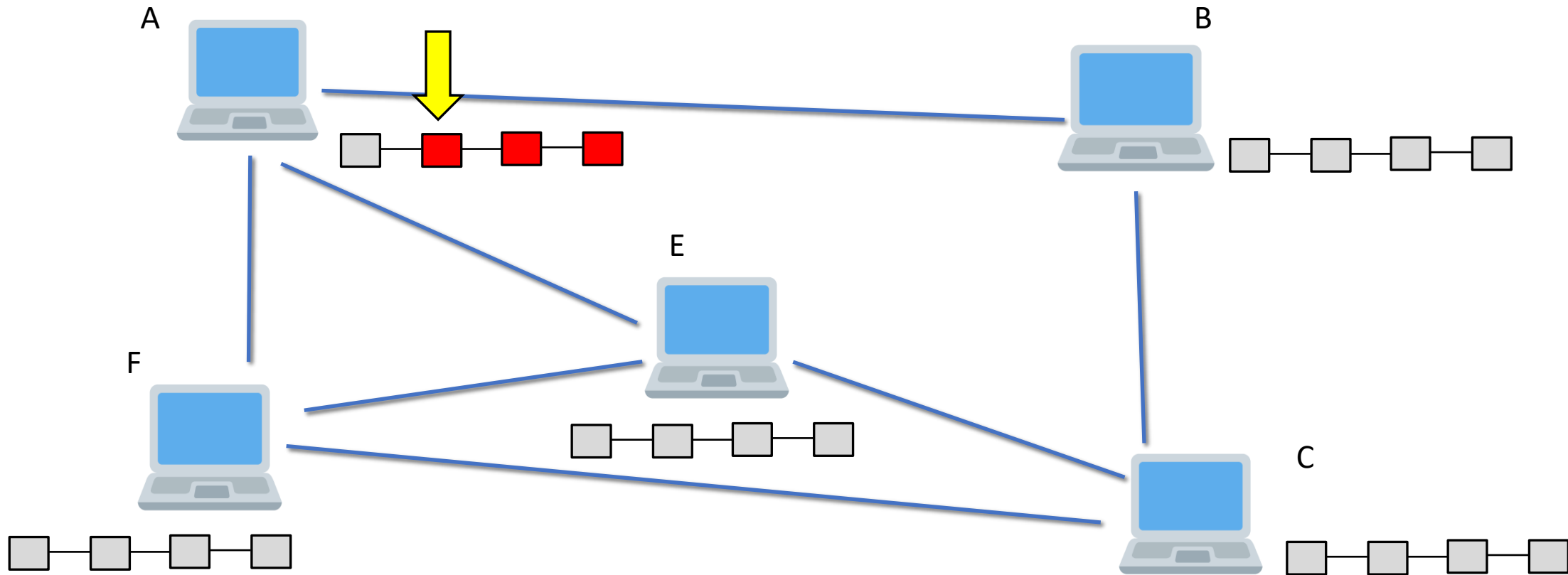
Consensus Protocol



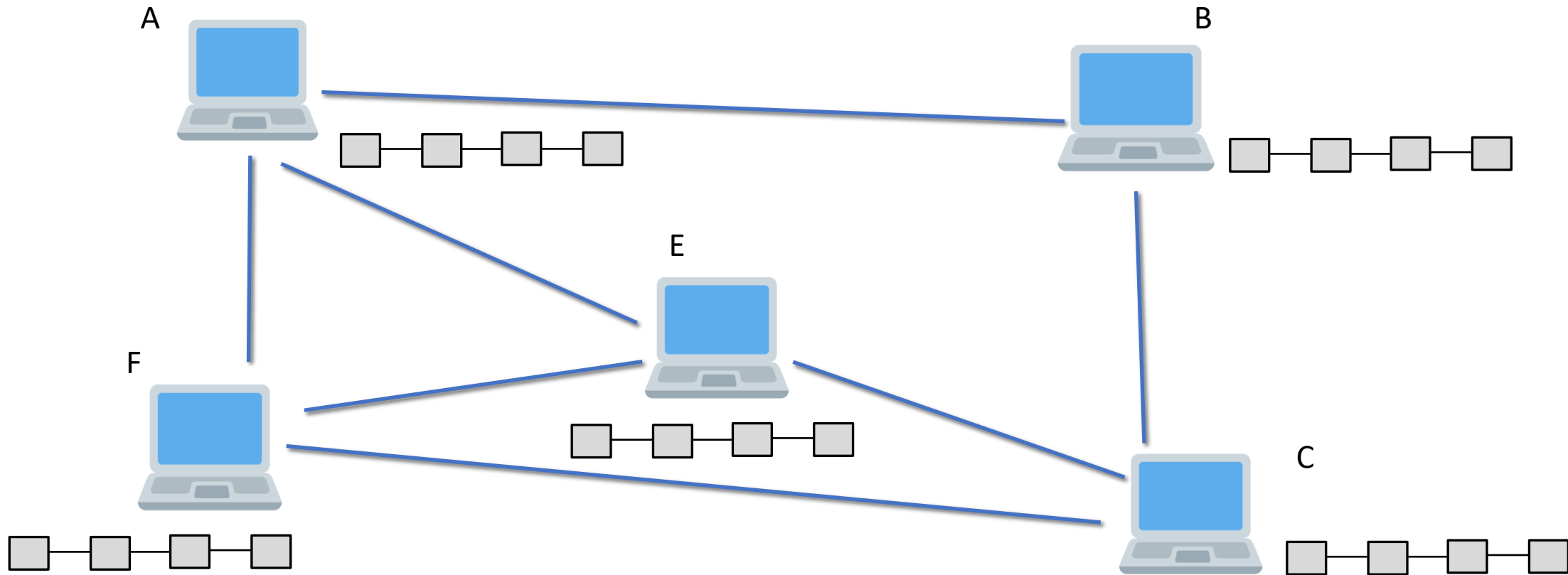
Consensus Protocol



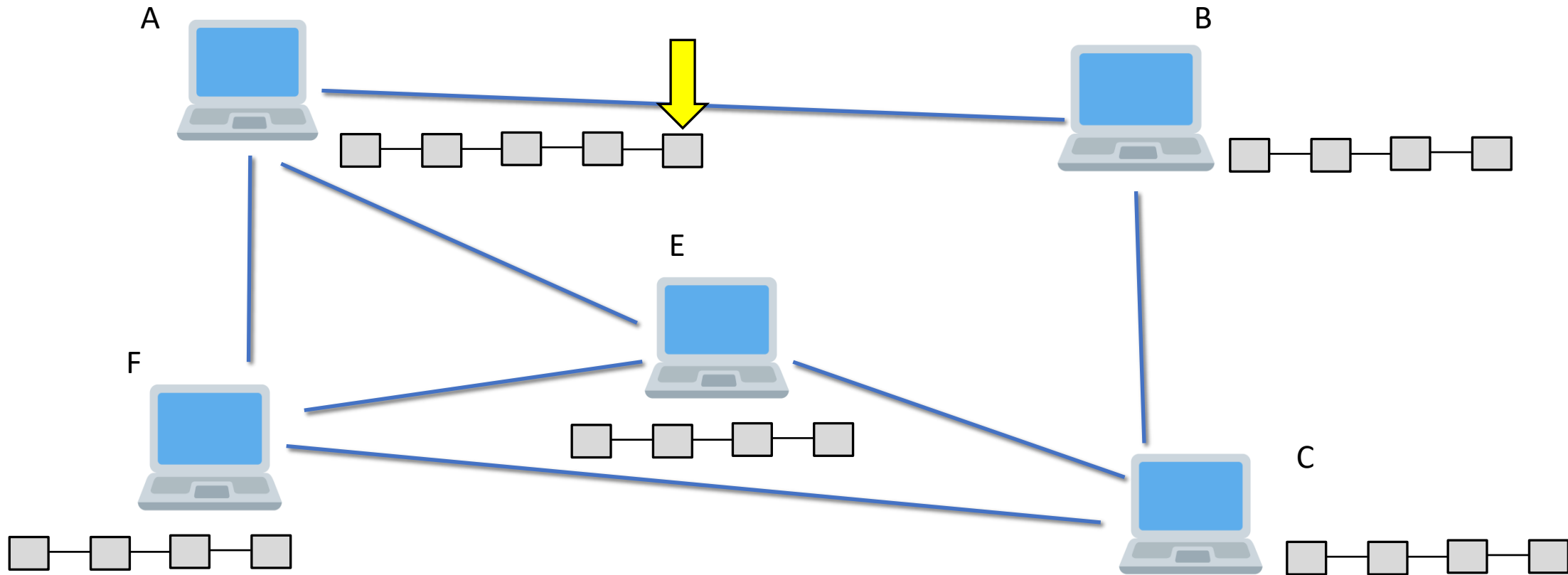
Consensus Protocol



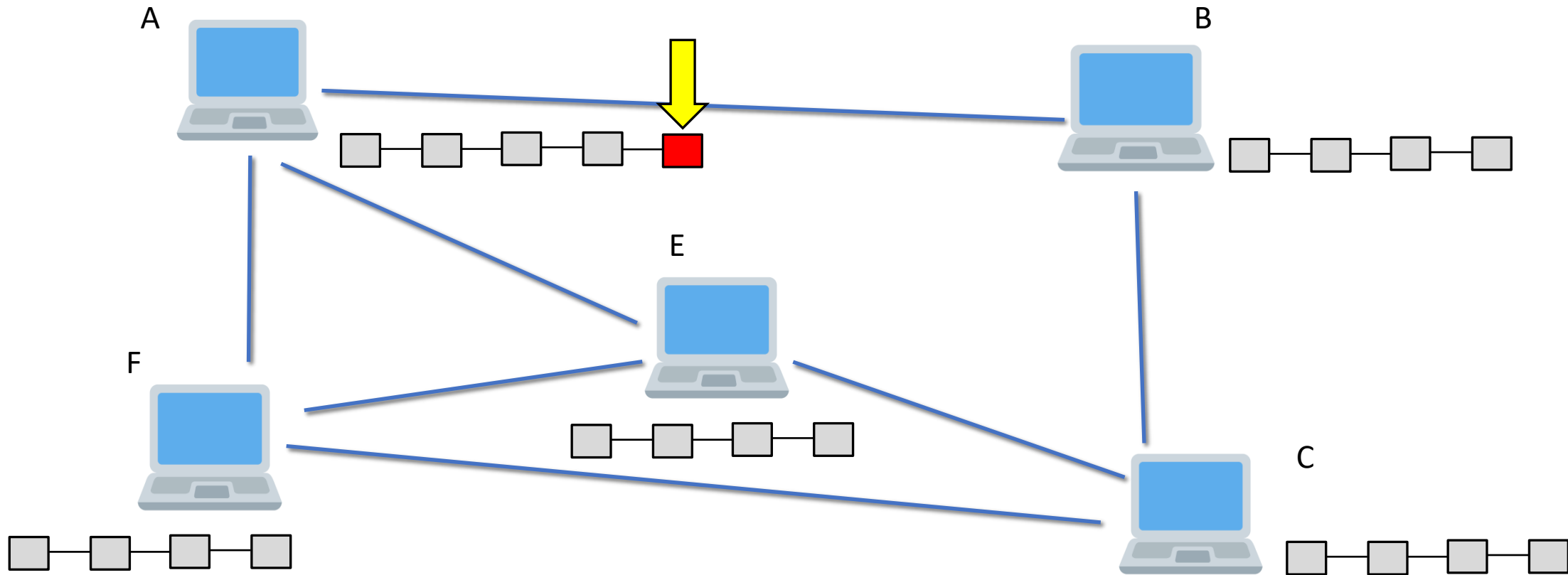
Consensus Protocol



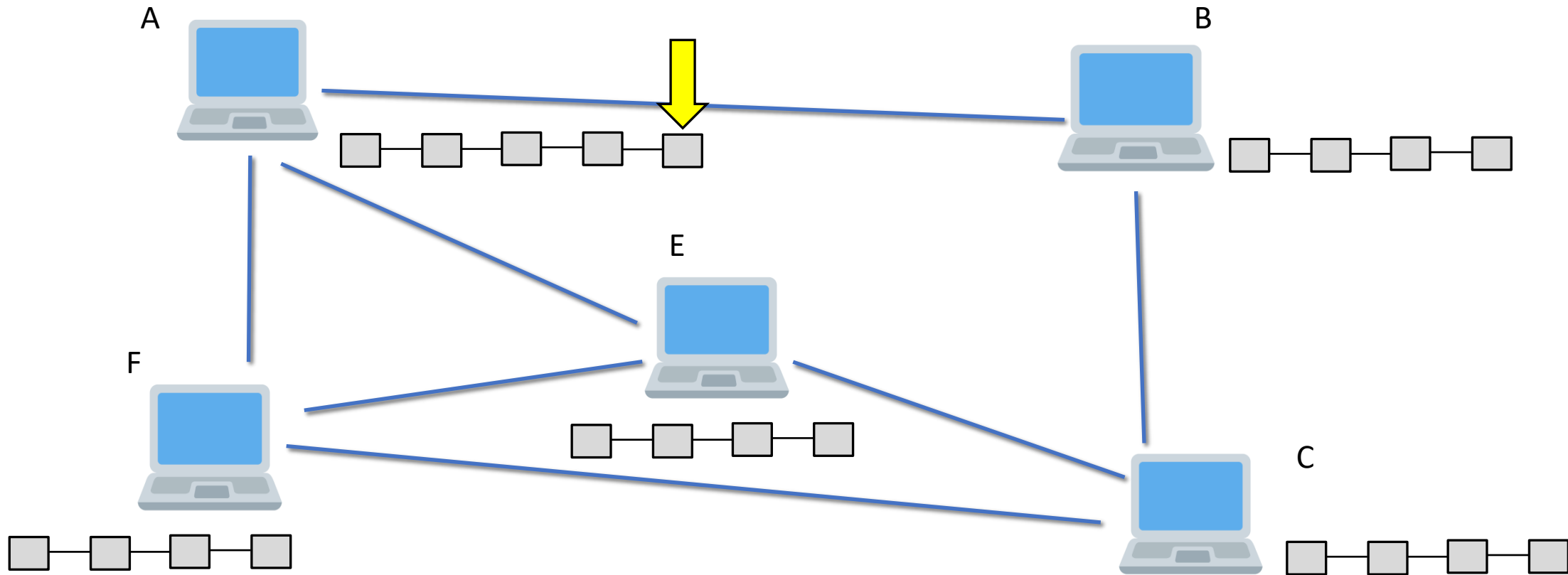
Consensus Protocol



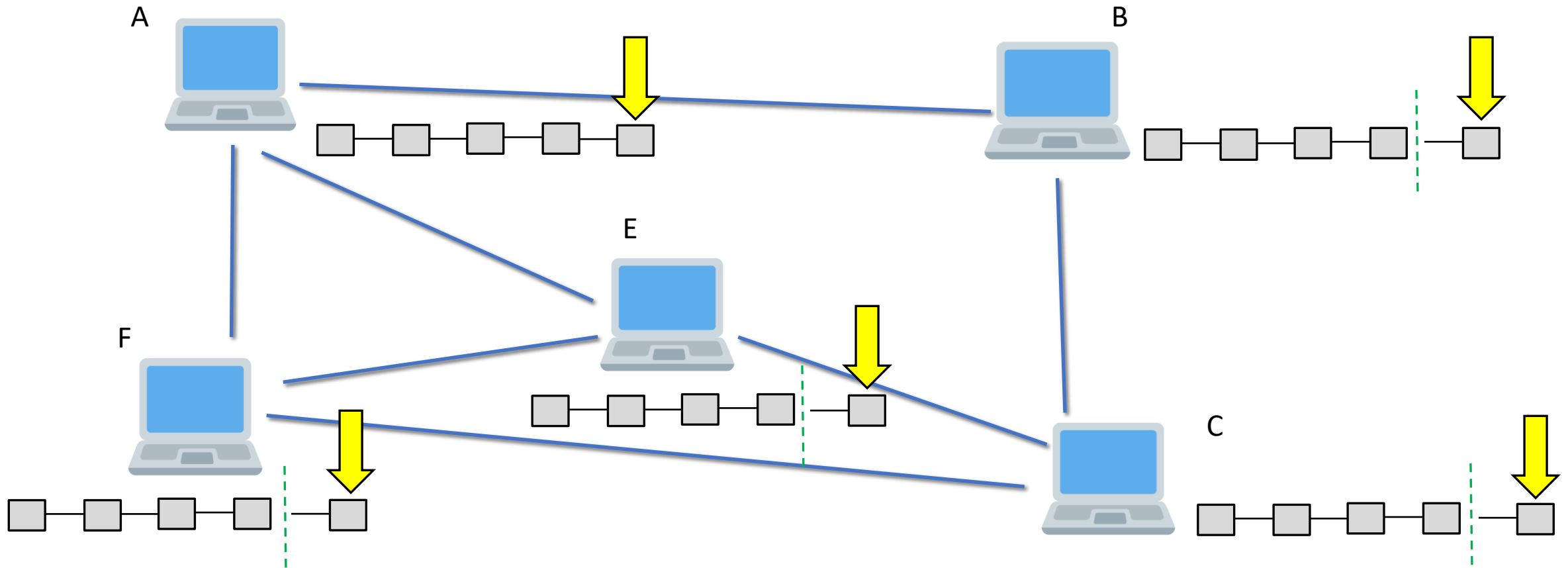
Consensus Protocol



Consensus Protocol

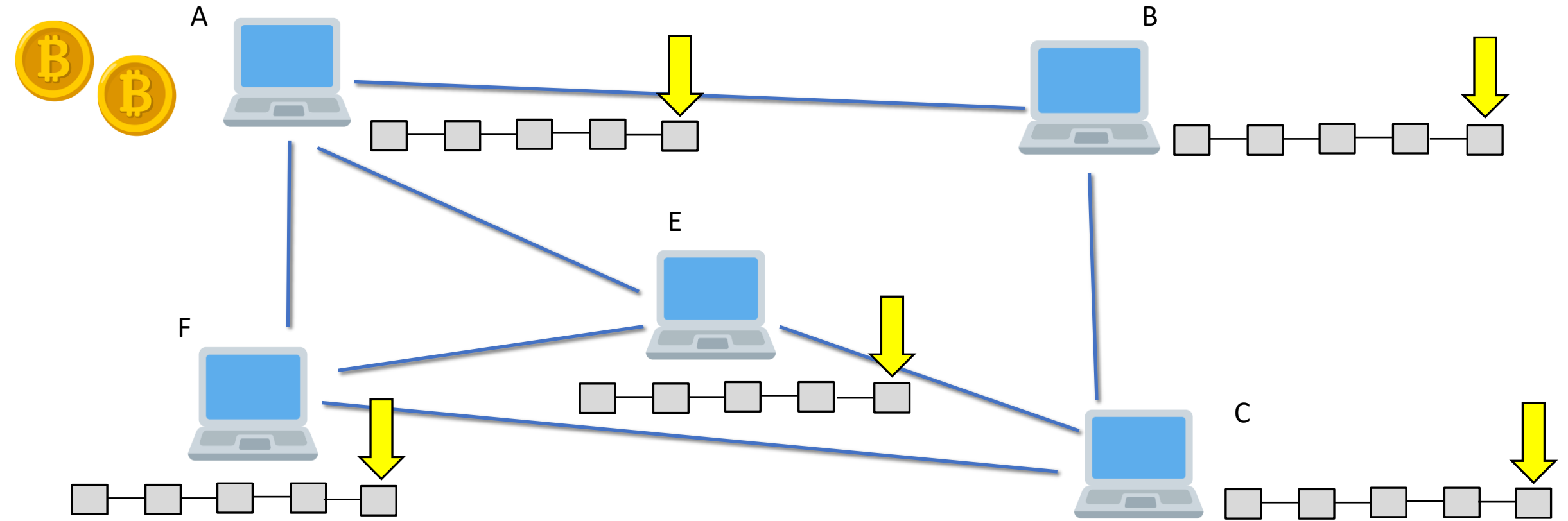


Consensus Protocol



1. Check syntactic correctness
2. Reject if duplicate of block we have in any of the three categories
3. Transaction list must be non-empty
4. Block hash must satisfy claimed nBits proof of work
5. Block timestamp must not be more than two hours in the future
6. First transaction must be coinbase (i.e. only 1 input, with hash=0, n=-1), the rest must not be
7. For each transaction, apply "tx" checks 2-4
8. For the coinbase (first) transaction, scriptSig length must be 2-100
9. Reject if sum of transaction sig opcounts > MAX_BLOCK_SIGOPS
10. Verify Merkle hash
11. Check if prev block (matching prev hash) is in main branch or side branches. If not, add this to orphan block in prev chain; done with block
12. Check that nBits value matches the difficulty rules
13. Reject if timestamp is the median time of the last 11 blocks or before
14. For certain old blocks (i.e. on initial block download) check that hash matches known values
15. Add block into the tree. There are three cases: 1. block further extends the main branch; 2. block make it become the new main branch; 3. block extends a side branch and makes it the new main branch
16. For case 1, adding to main branch:
 1. For all but the coinbase transaction, apply the following:

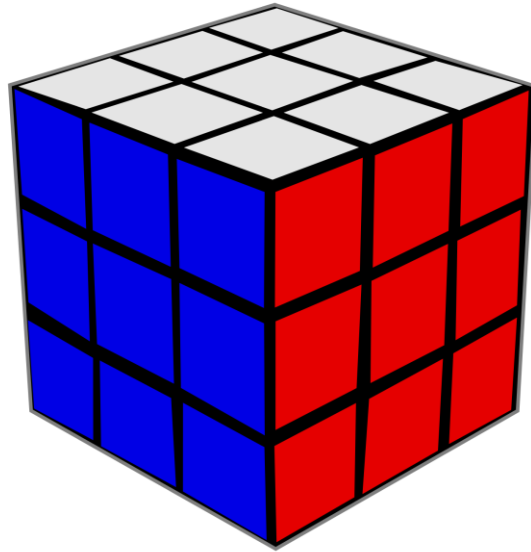
Consensus Protocol



Consensus Protocol

Q)Is this verification and validation a time taking process ?

Consensus Protocol



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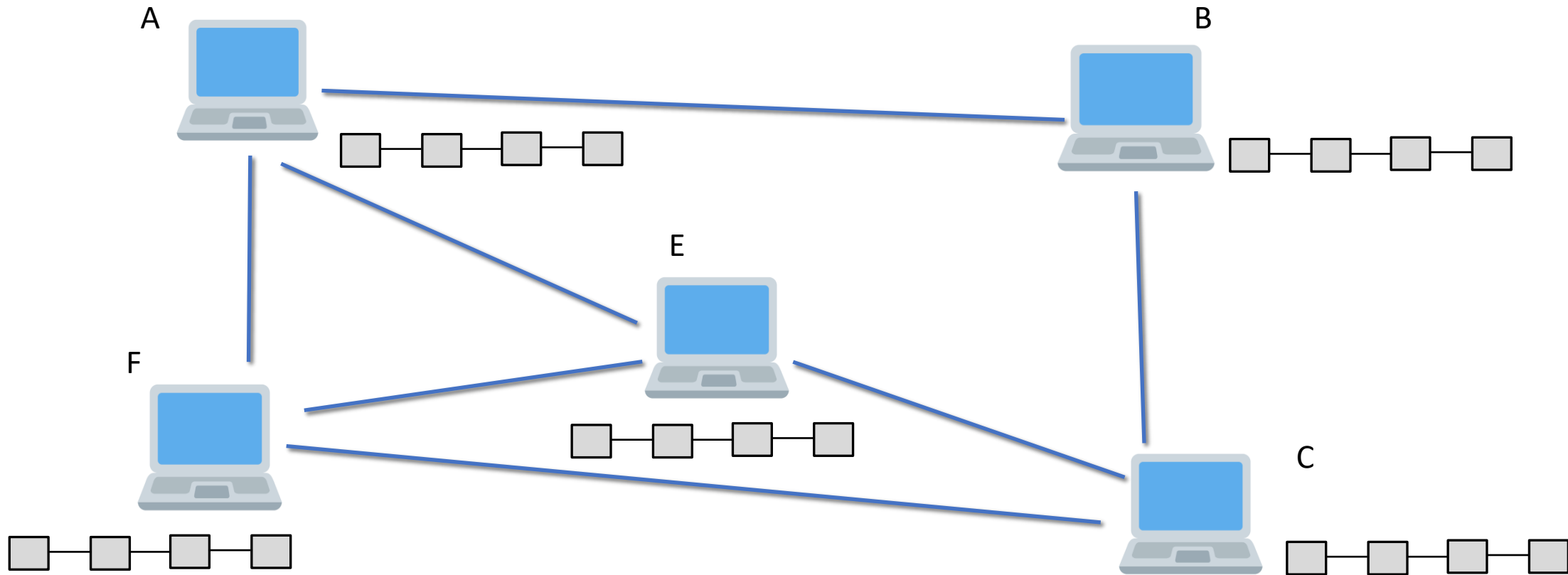
Consensus Protocol

Consensus Protocol

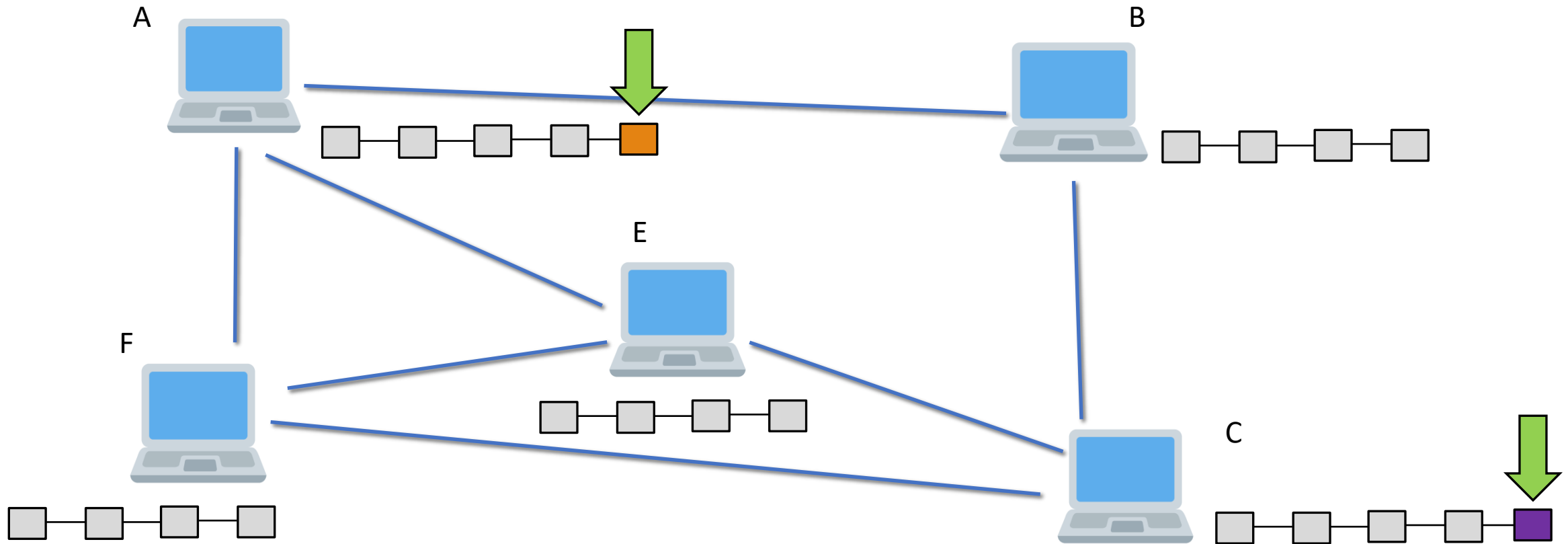
Prevent Attacks

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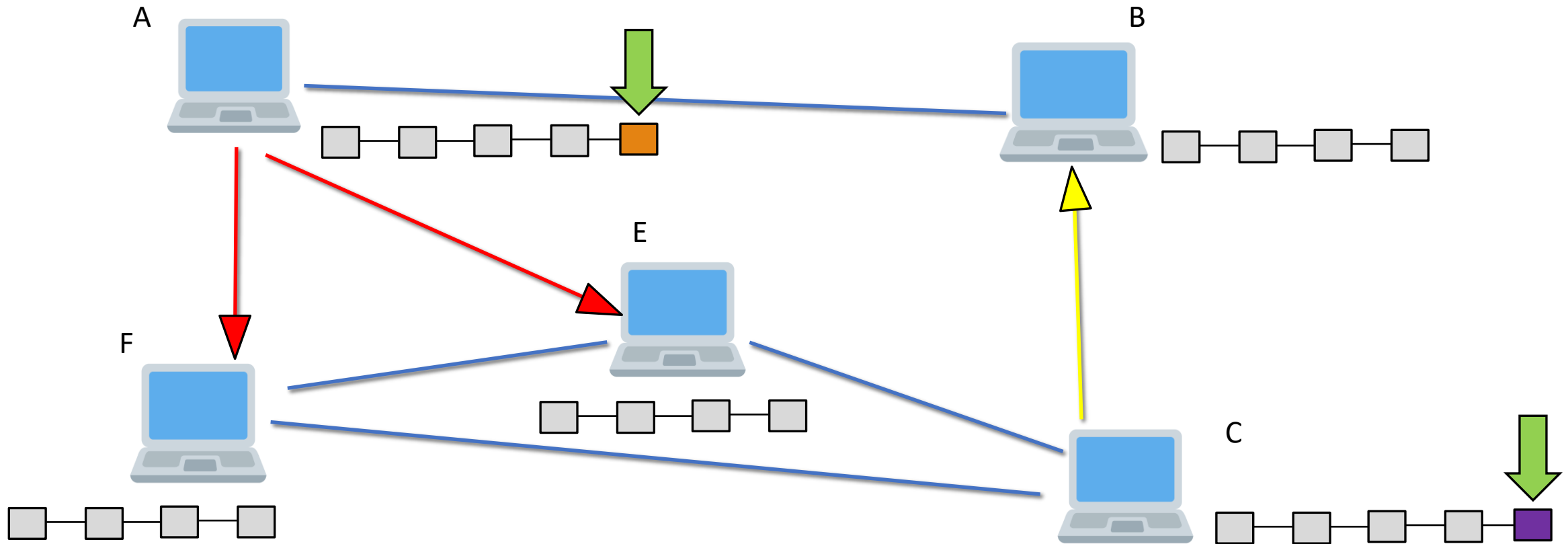
Consensus Protocol



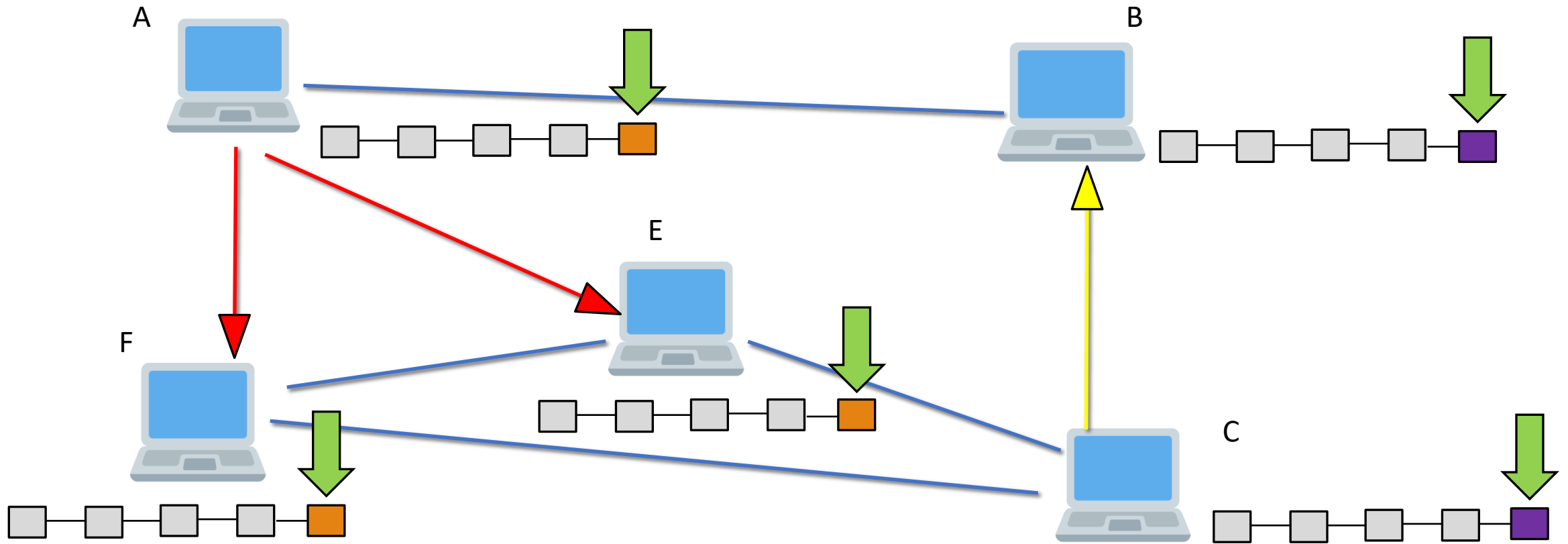
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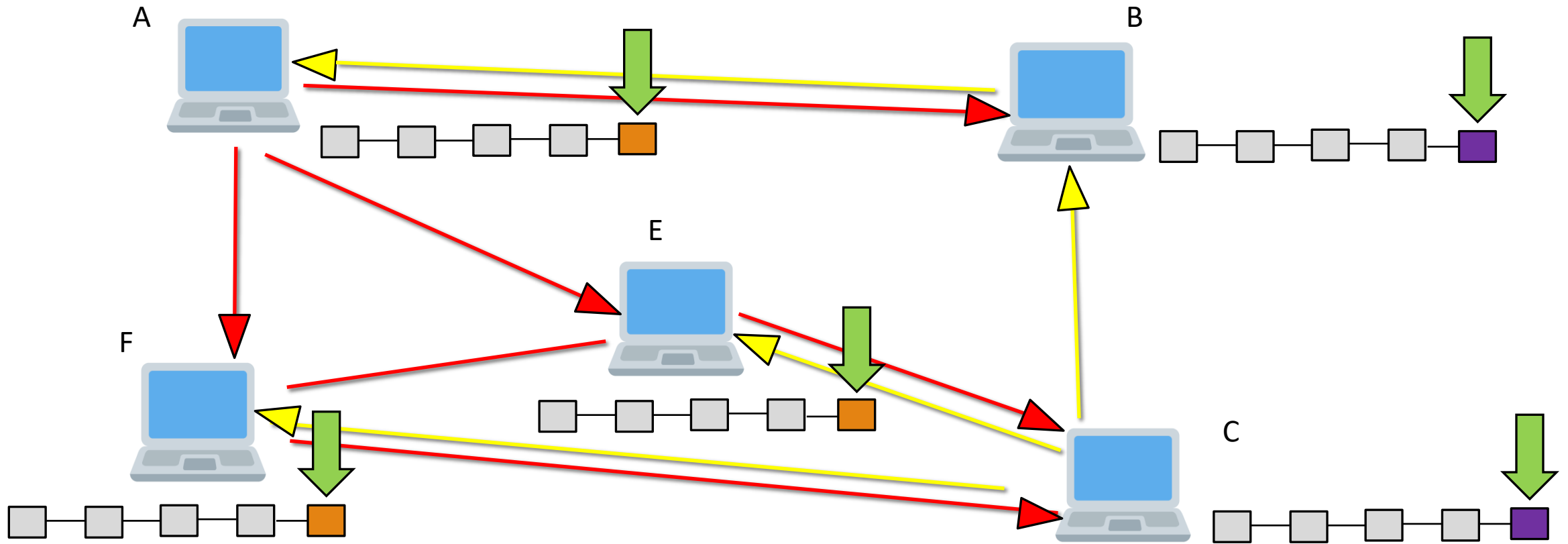
Consensus Protocol



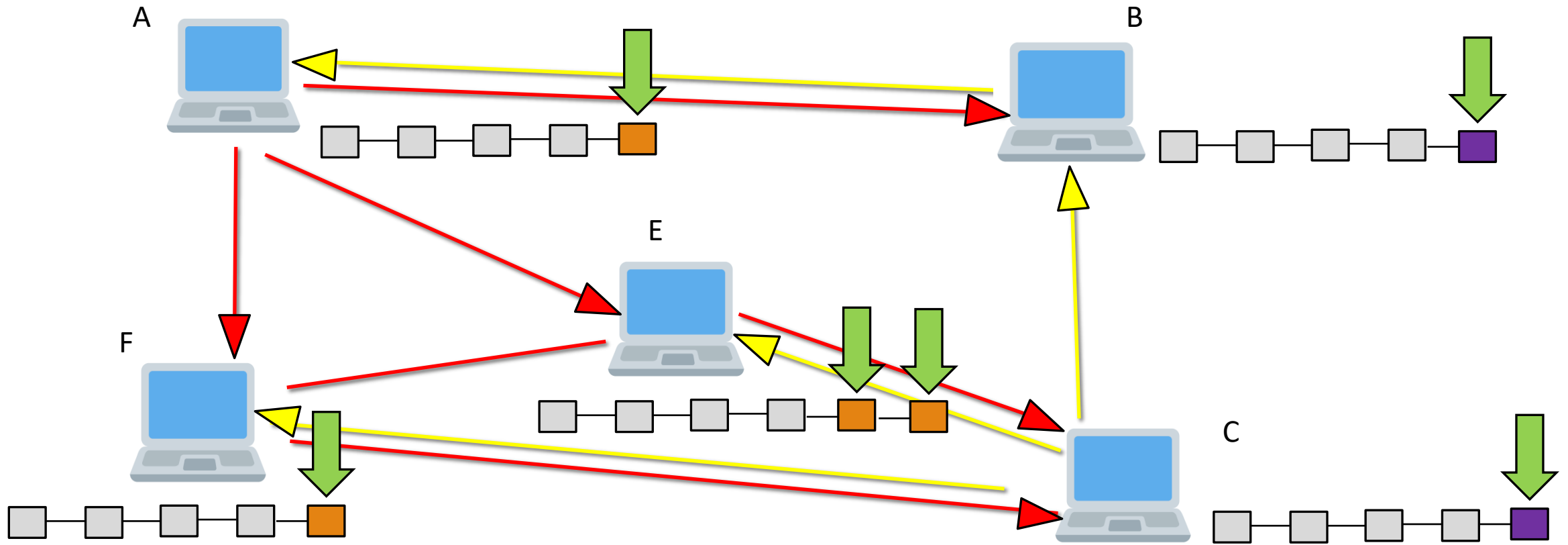
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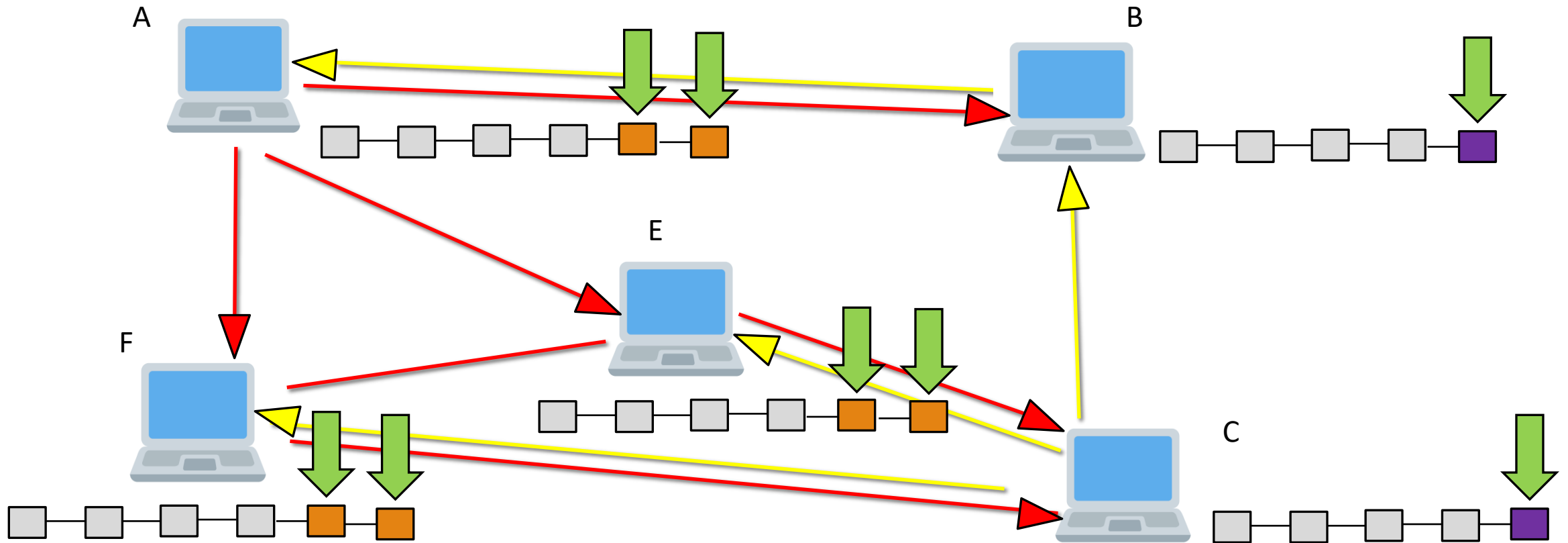
Consensus Protocol



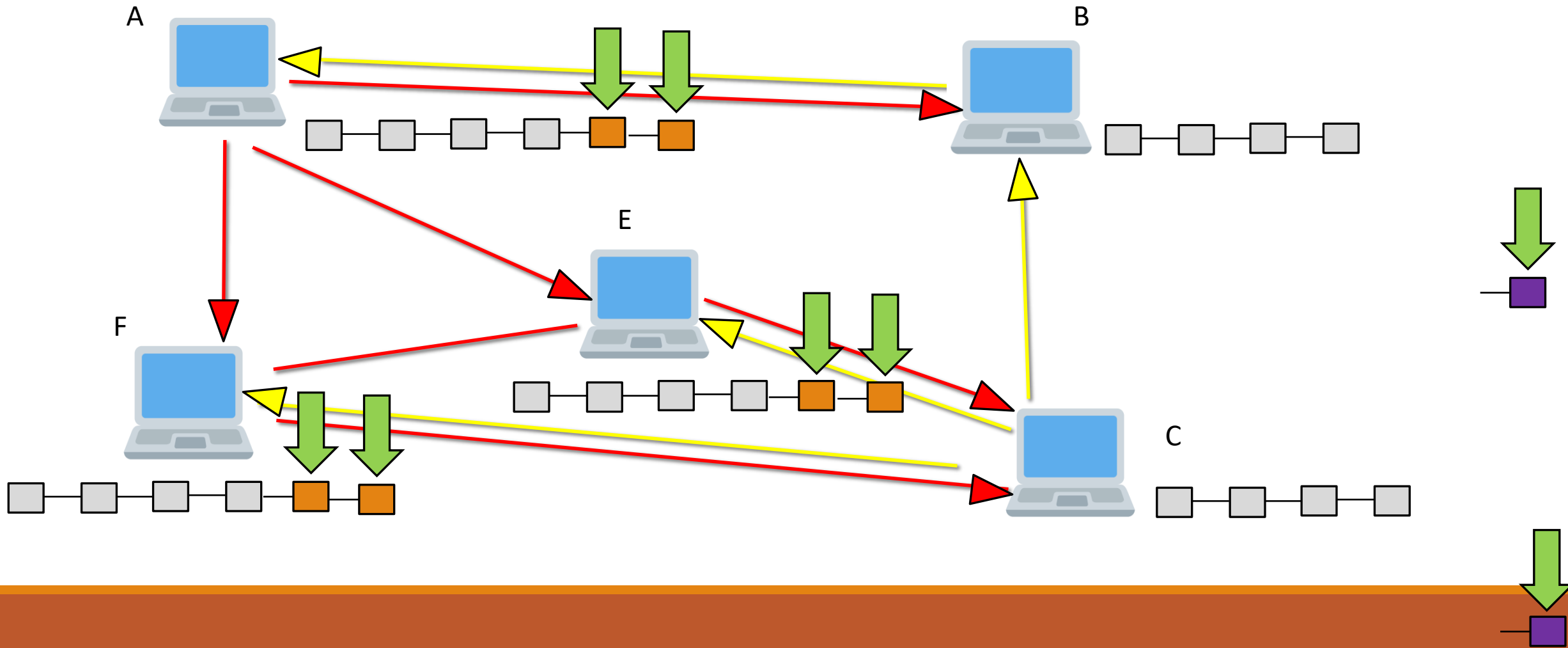
Consensus Protocol



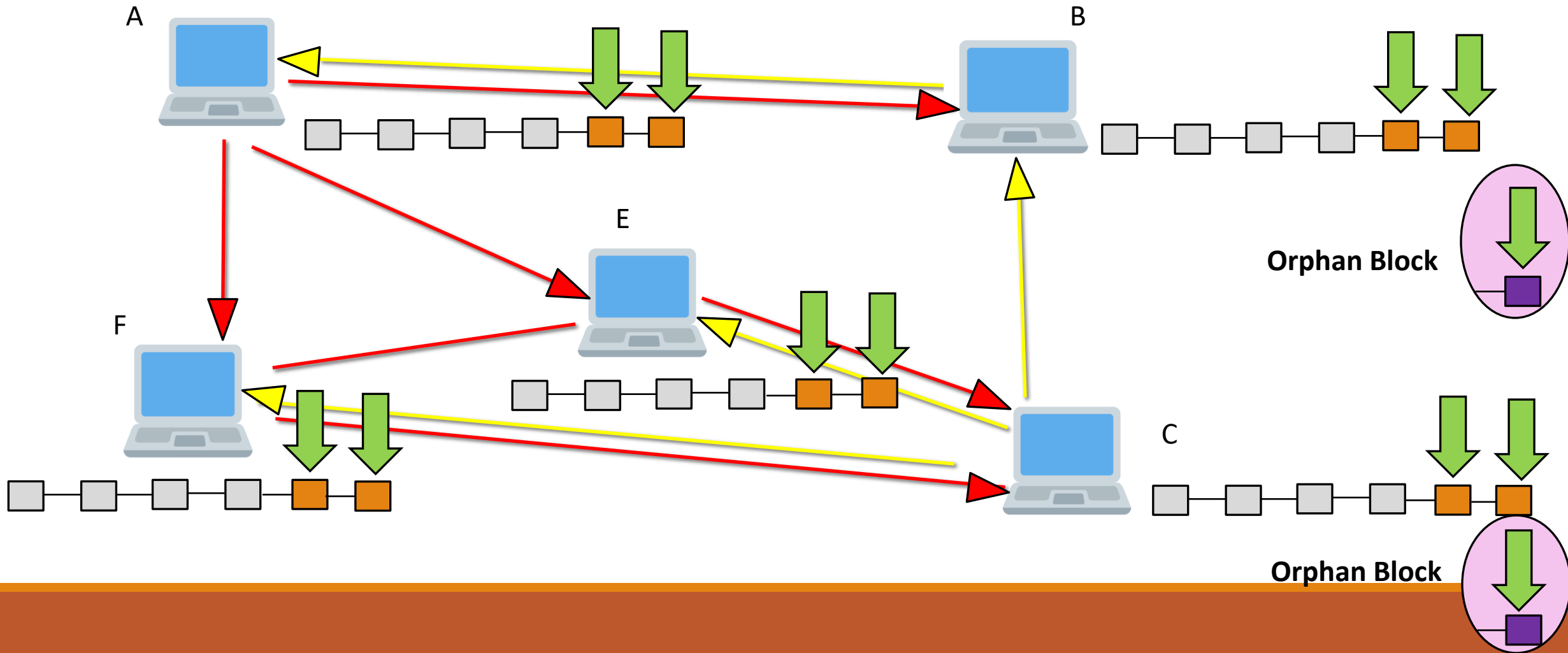
Consensus Protocol



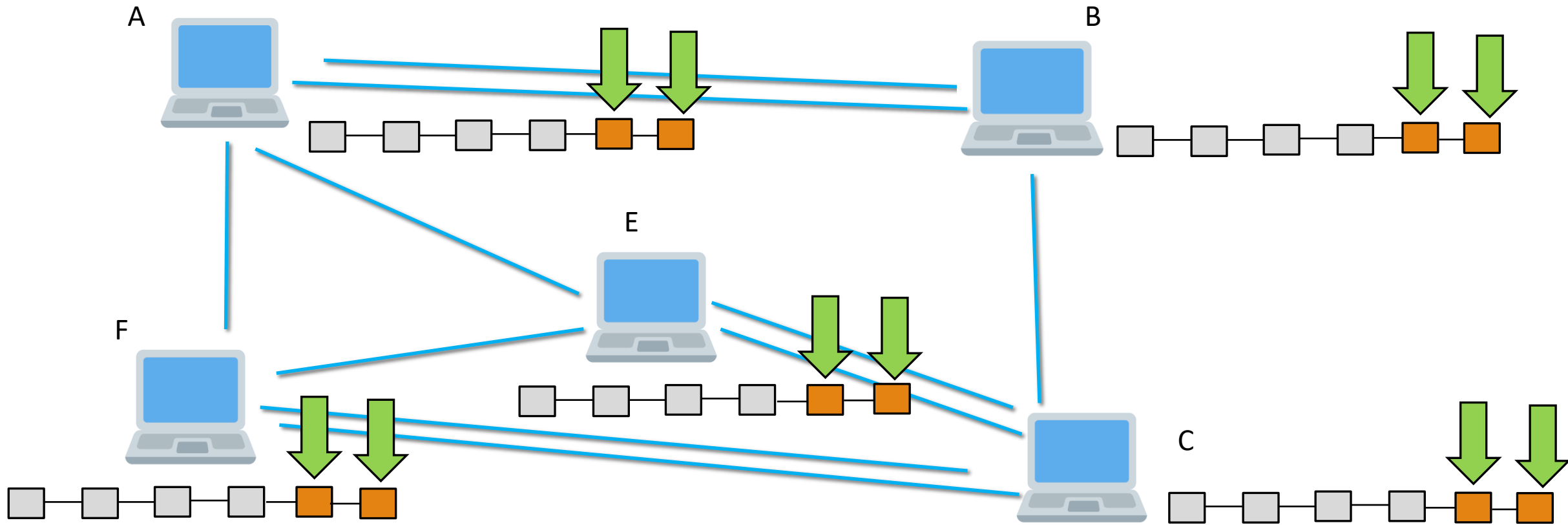
Consensus Protocol



Consensus Protocol



Consensus Protocol



Consensus Protocol

Note-

- The Consensus Protocol of Blockchain is much better than the Byzantine Fault Tolerance as Consensus Protocol only needs a **51%** majority while Byzantine Fault Tolerance needs **approximately 66%**.
- All the transaction in the **Orphan Block** will be dropped and the miner that had mined the block will not get any reward.
- So that's why wait for the **6 confirmations** before assuming payment to be successful.

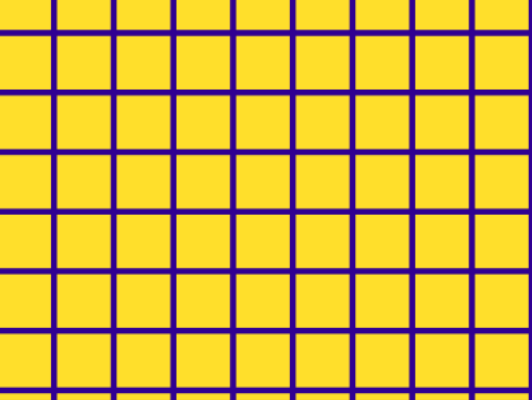
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Contents – Module B

What is Bitcoin?

Nonce

Transactions

**Bitcoin's Monetary
Policy**

**CPU's vs GPU's vs
ASICs**

Wallets

Mining

Mempool

**Public Key and Private
Key**



Cryptocurrency

History Of Bitcoin

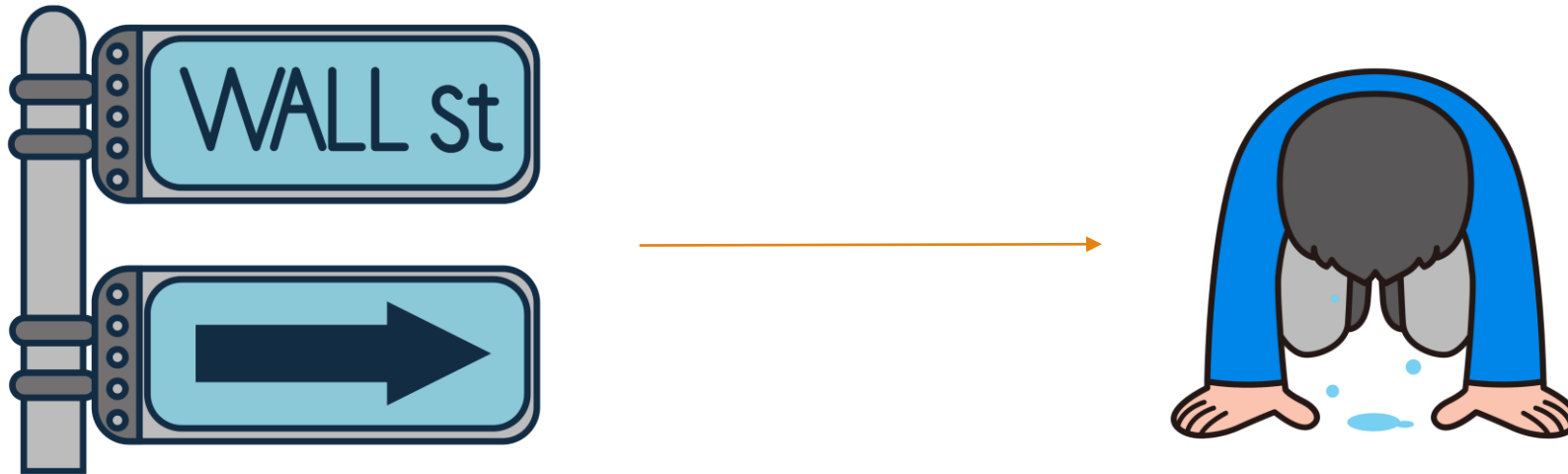


US Economy

Nearly **9 million** – Lost Job

Nearly **1.8 million** – Small business closed

History Of Bitcoin



History Of Bitcoin

Lehman Brothers



HISTORY OF BITCOIN



History Of Bitcoin

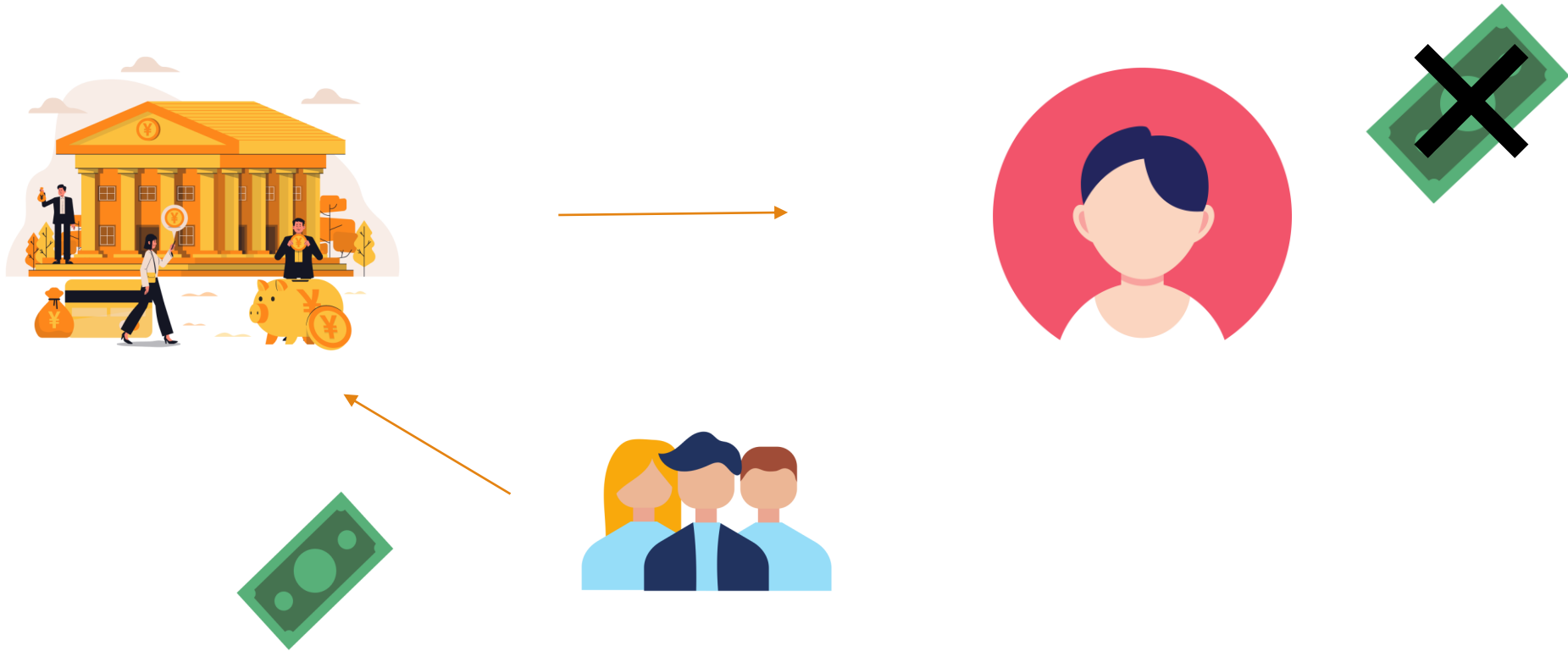


Foolishness of Banks



US Government

History Of Bitcoin



History Of Bitcoin



Foolishness of Banks

US Government

History Of Bitcoin



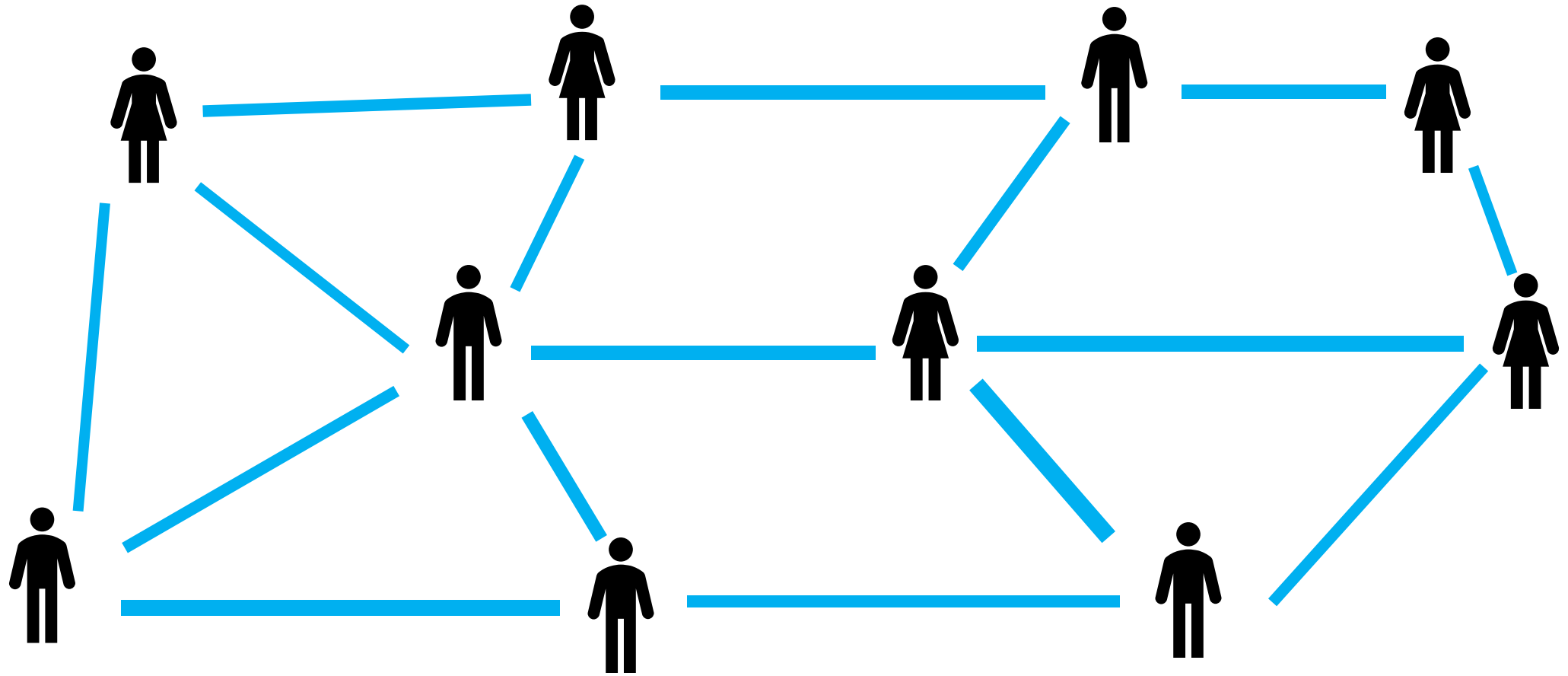
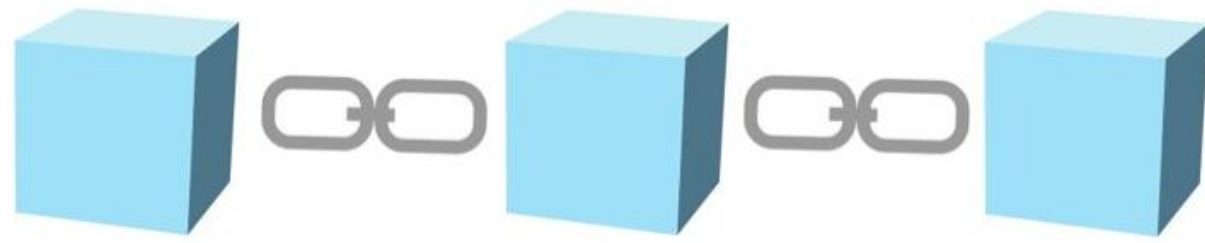
Bitcoin



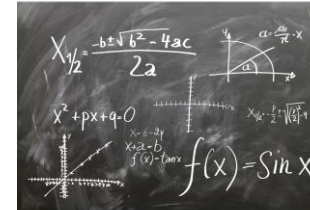
Satoshi Nakamoto

Bitcoin

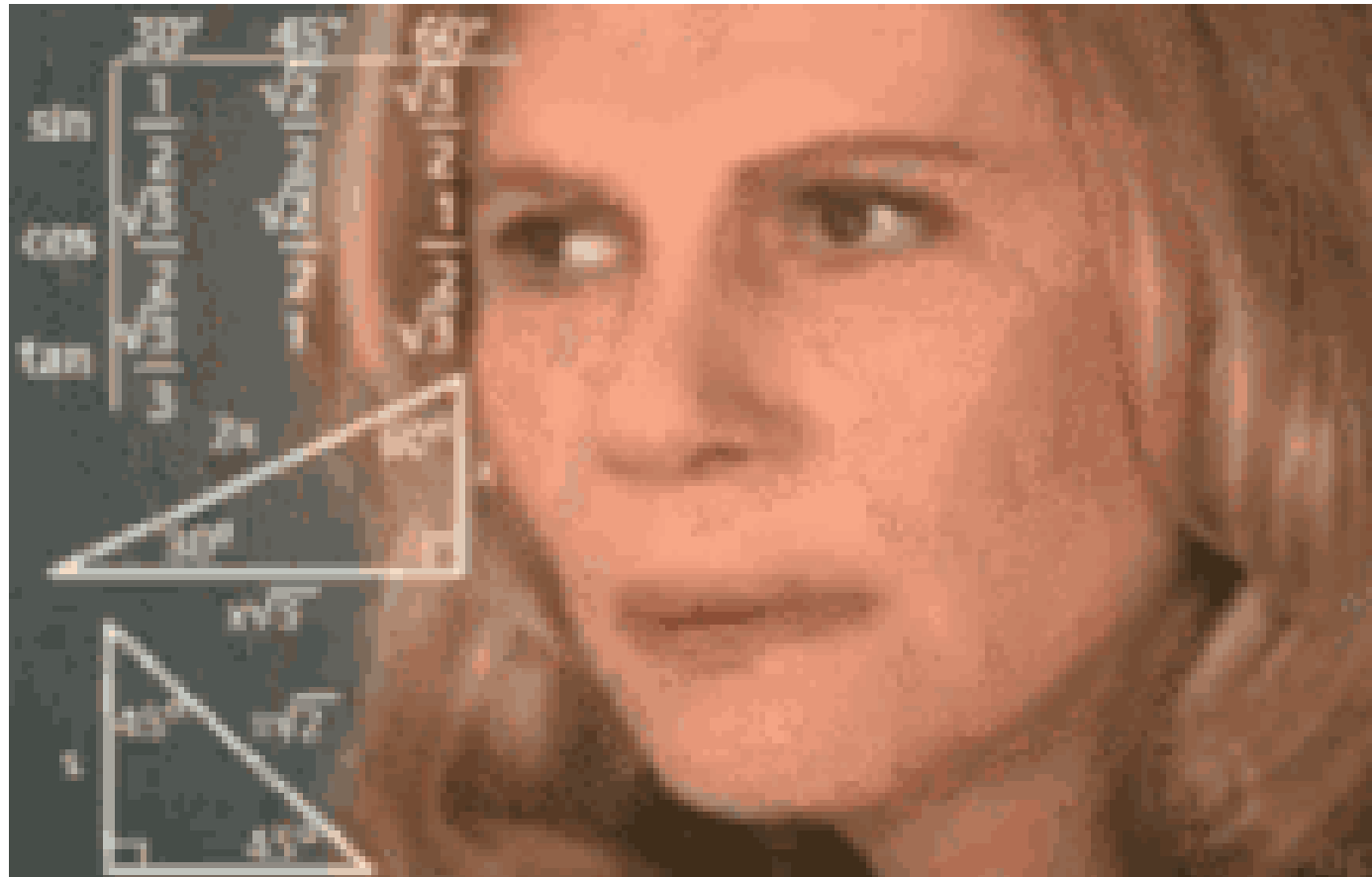
- Bitcoin is a decentralized **digital currency, without a central bank or single administrator**, that can be sent from user to user on the peer-to-peer bitcoin network without the need for **intermediaries**.
- It uses Blockchain Technology.



Trust



Mathematical Problem





CHRISTIAN
BALE
STEVE
CARELL
RYAN
GOSLING
BRAD
PITT

AN ADAM MCKAY FILM

THE BIG SHORT

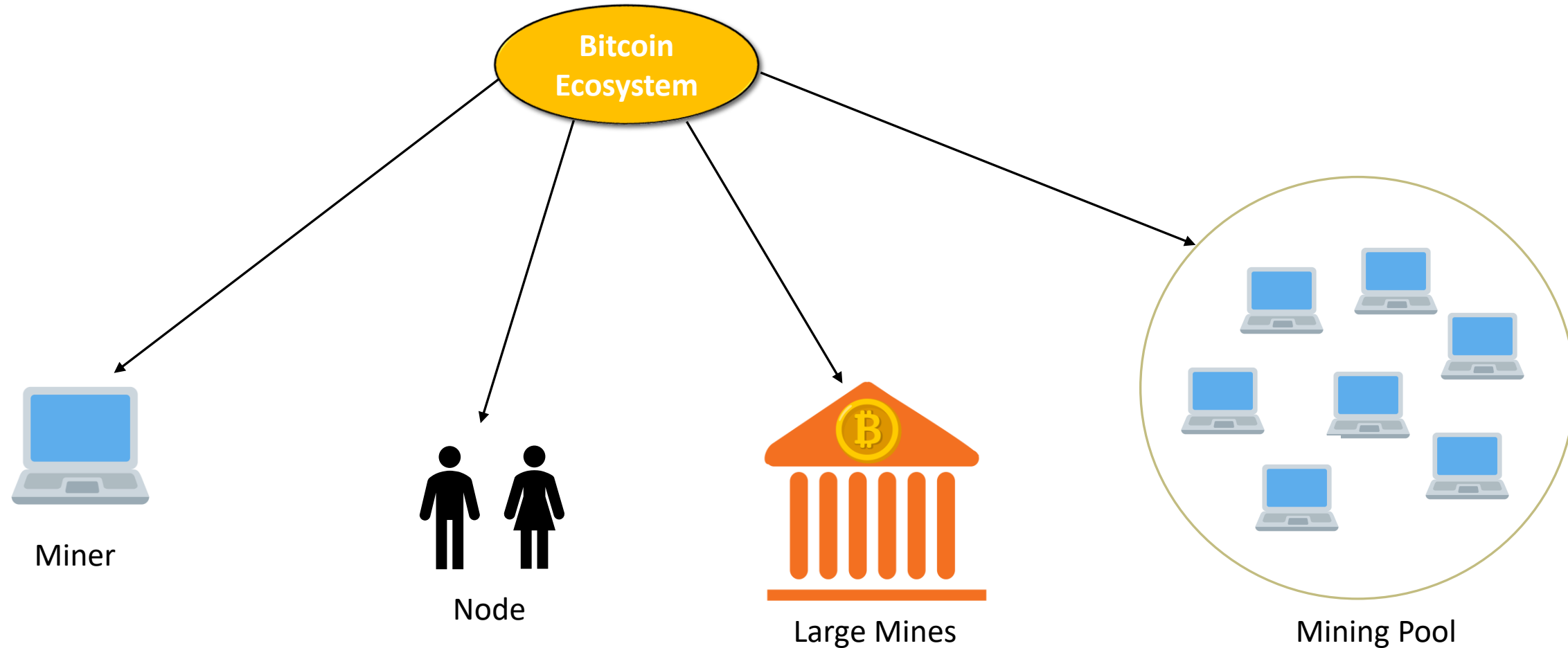
THIS IS A TRUE STORY

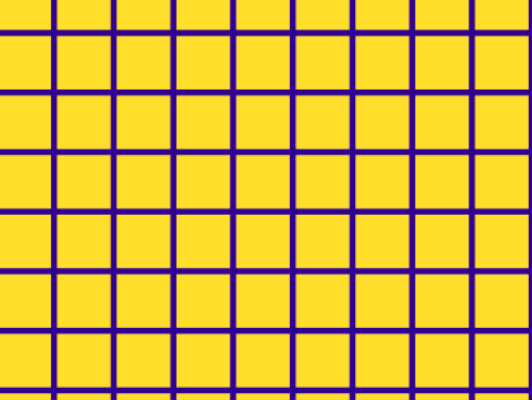
**"FASCINATING
AND FUNNY.
IT'S A TRIUMPH."**

Joe Morgenstern,
THE WALL STREET JOURNAL

The Big Short

Bitcoin Ecosystem





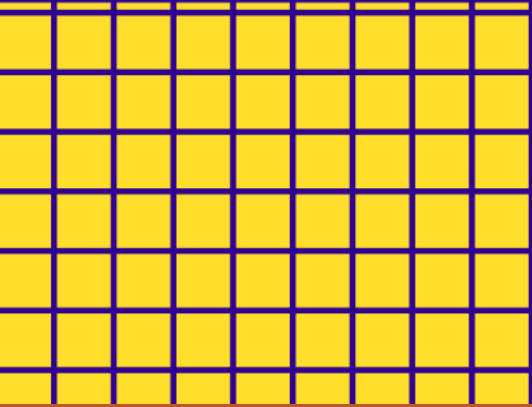
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Blockchain vs Coin

Technology

Blockchain

Protocol/Coin

Waves

Bitcoin

Ethereum

Blockchain vs Coin vs Token

Technology

Blockchain

Protocol/Coin

Waves

Bitcoin

Ethereum

Token

WGB

BI

INTL

WGR

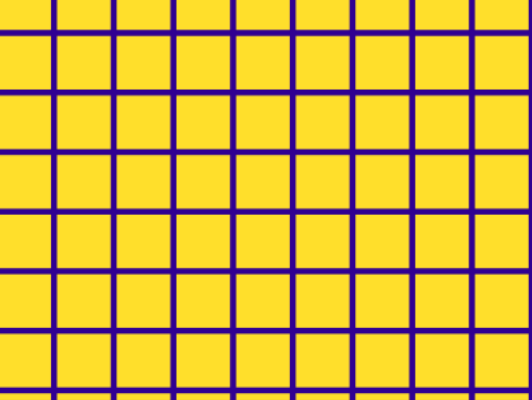


TRX

SNT

REP

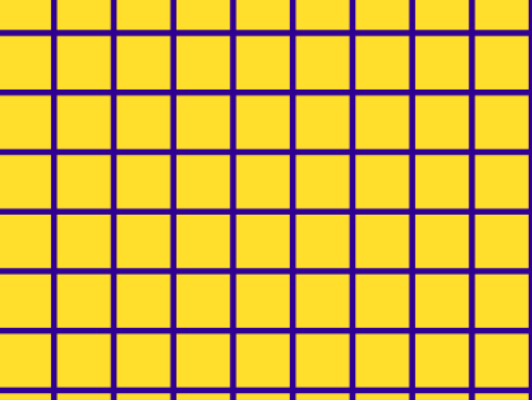
AE



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Bitcoin's Monetary Policy

Bitcoin's Monetary Policy

The Halving

Block Frequency

The Halving

Event	Date	Block number	Reward
Launch of Bitcoin	03 Jan. 2009	0	50 new XBT
1st halving	28 Nov. 2012	210'000	25 new XBT
2nd halving	11 May 2020	630'000	12.5 new XBT
3rd halving	Expected 2024	740'000	3.125 new XBT
4th halving	Expected 2028	850'000	1.5625 new XBT
Maximum supply reached	Expected 2140	6'930'000	0 new XBT

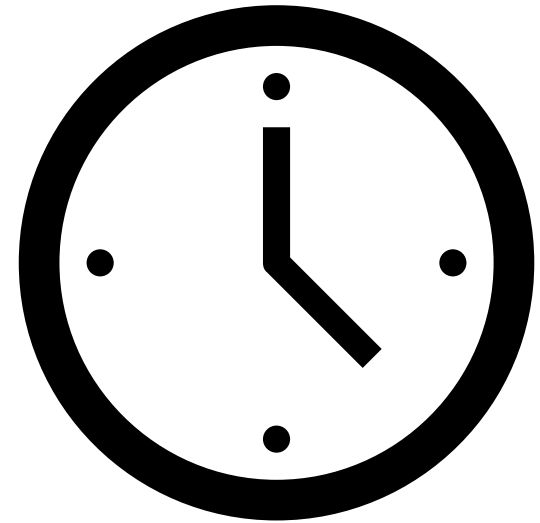
Note- Supply cap of Bitcoin is 21 million.

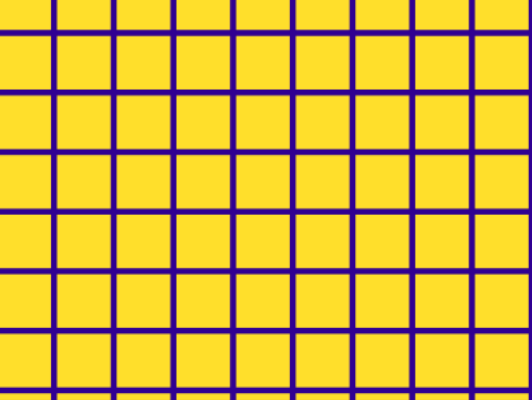
The Halving



Block Frequency

This states that **on an average** it will take 10 minute to create a new block.





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An abstract graphic on the left side of the slide. It features a dark blue background with a grid of lighter blue squares. Overlaid on this is a circular design composed of several concentric rings. The outermost ring contains binary code (0s and 1s) in a light blue color. Inside this, there are rings of small, vertical, light blue bars of varying heights, resembling a bar chart or a stylized representation of data. The center of the graphic is a solid dark blue circle.

Bitcoin's Target History

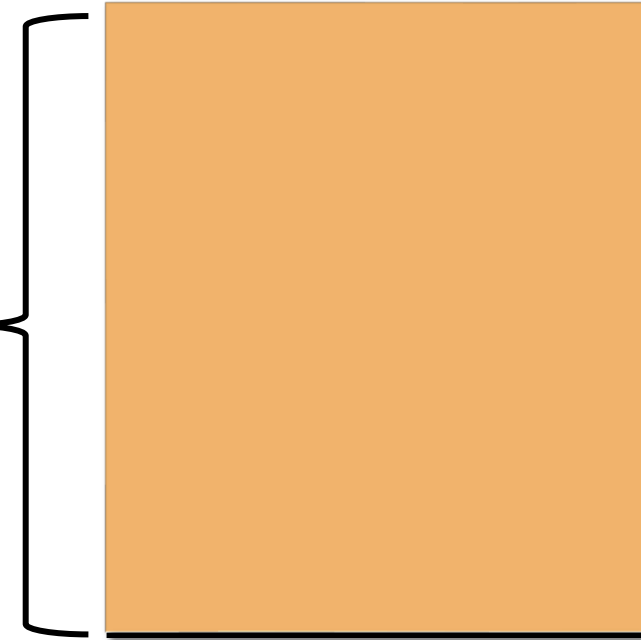


Bitcoin's Monetary Policy

Understanding Mining Difficulty

Let's take a five digit number= XXXXX

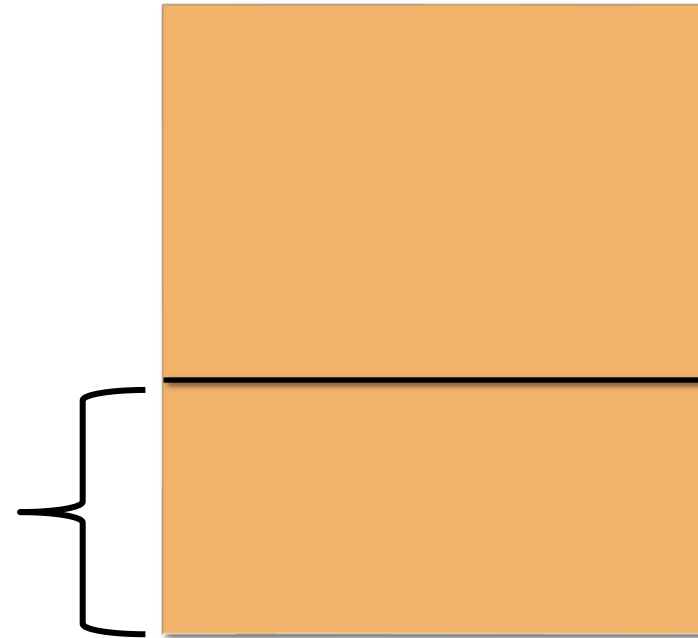
Area that can be covered



Understanding Mining Difficulty

Let's take a five digit number= **0**XXXX

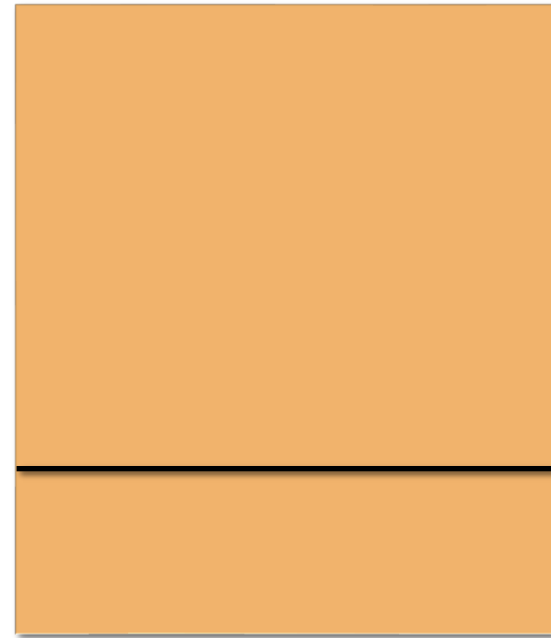
Area that can be covered



Understanding Mining Difficulty

Let's take a five digit number= 00XXX

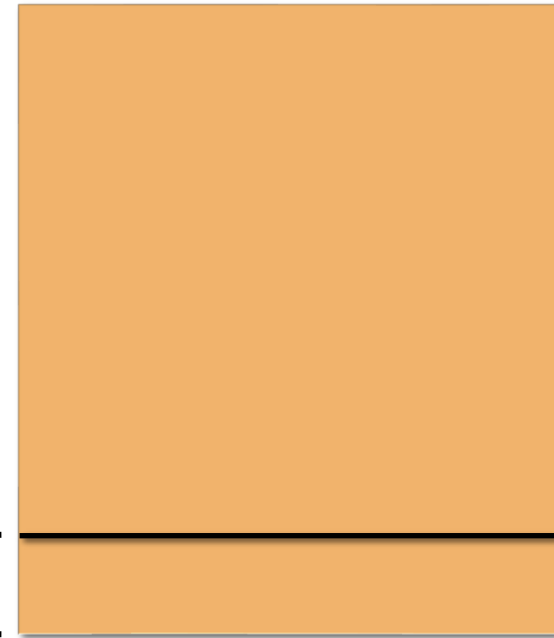
Area that can be covered



Understanding Mining Difficulty

Let's take a five digit number= 000XX

Area that can be covered



Understanding Mining Difficulty

Current Target:

[illegible]

19 leading 0's

Understanding Mining Difficulty

Total Possible 64-digits hexadecimal numbers = $16^{64} \simeq 10^{77}$

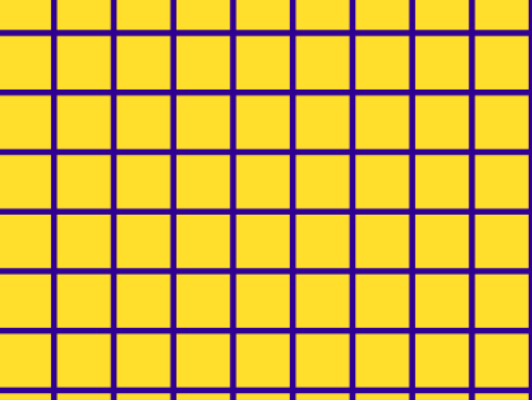
Total valid hashed(with 19 leading 0's) = $16^{(64-19)} \simeq 10^{54}$

Probability that a randomly picked hash is valid = $(10^{54}/10^{77}) \simeq 10^{-23}$

Understanding Mining Difficulty

Q) Who adjusts the difficulty ?

To compensate for increasing hardware speed and varying interest in running nodes over time, the proof-of-work difficulty is determined by a moving average targeting an average number of blocks per hour. If they're generated too fast, the difficulty increases. – [Satoshi Nakamoto](#)



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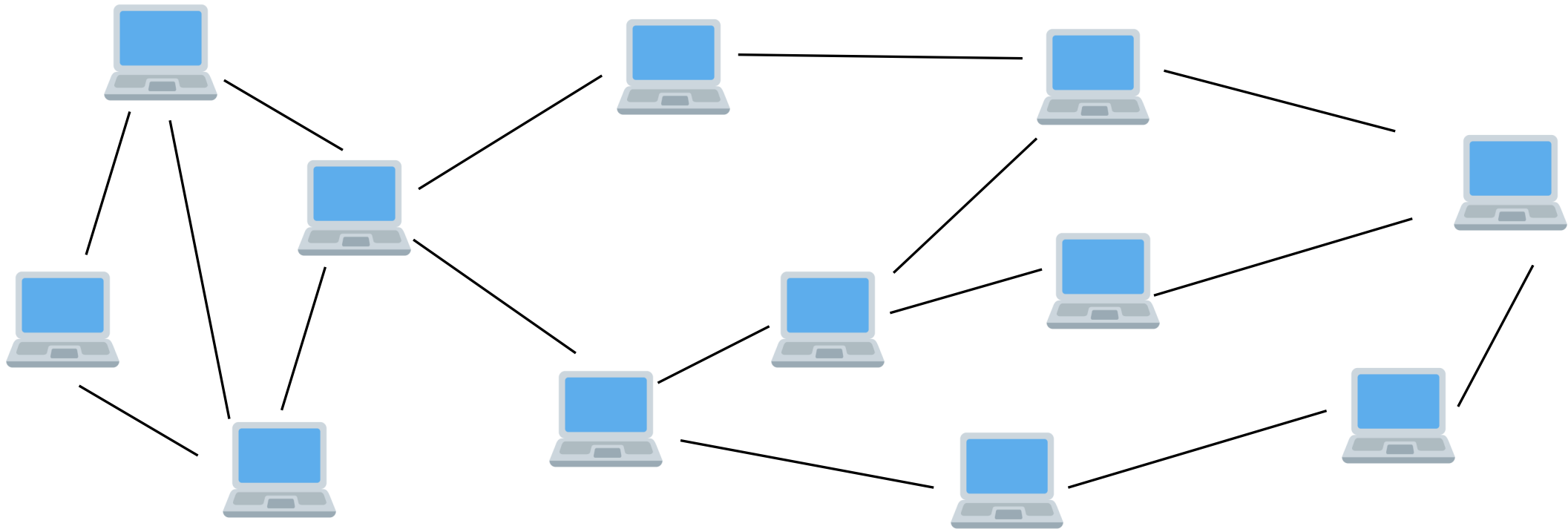
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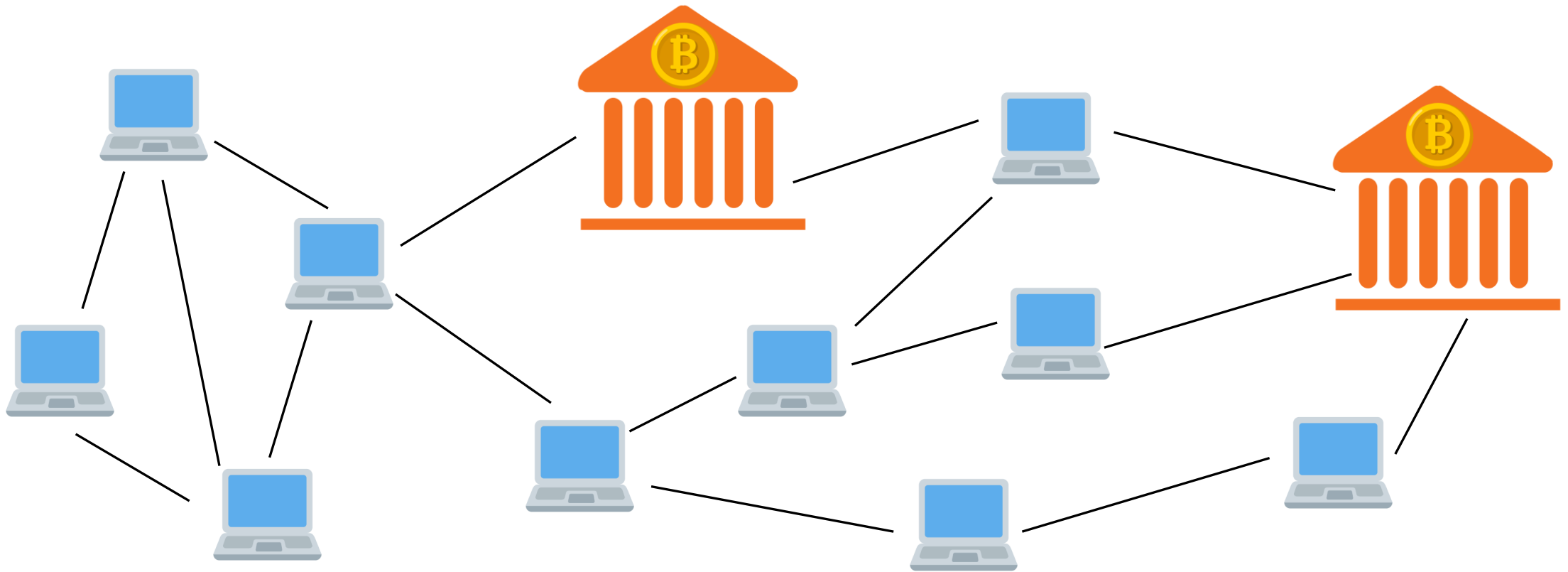


Mining Pool

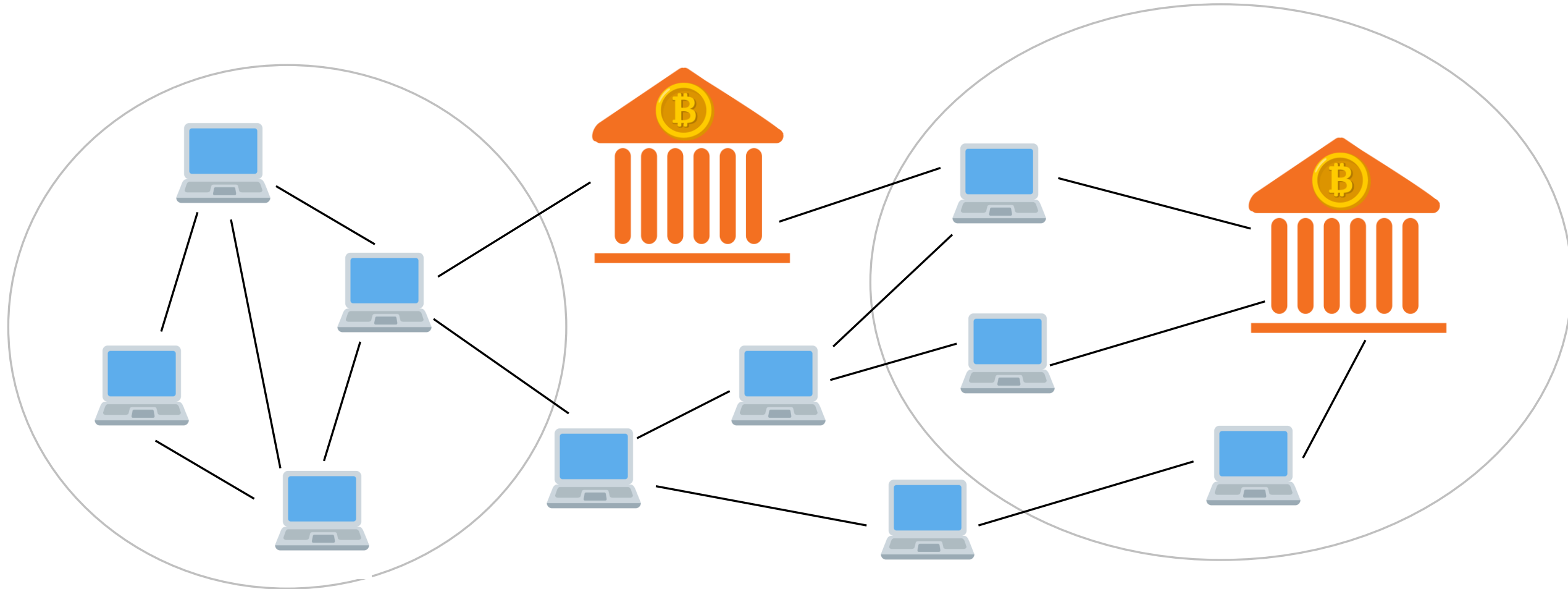
Mining Pools



Mining Pools

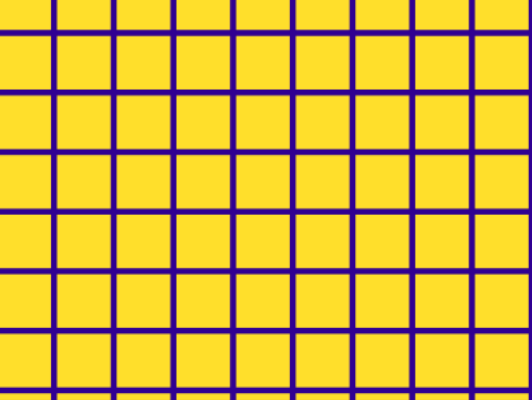


Mining Pools



Mining Pools Benefits

- You don't have to buy costly equipment's.
- You don't have to worry about mining setup.
- Distribution of rewards according to the mining power.
- No wastage of power.



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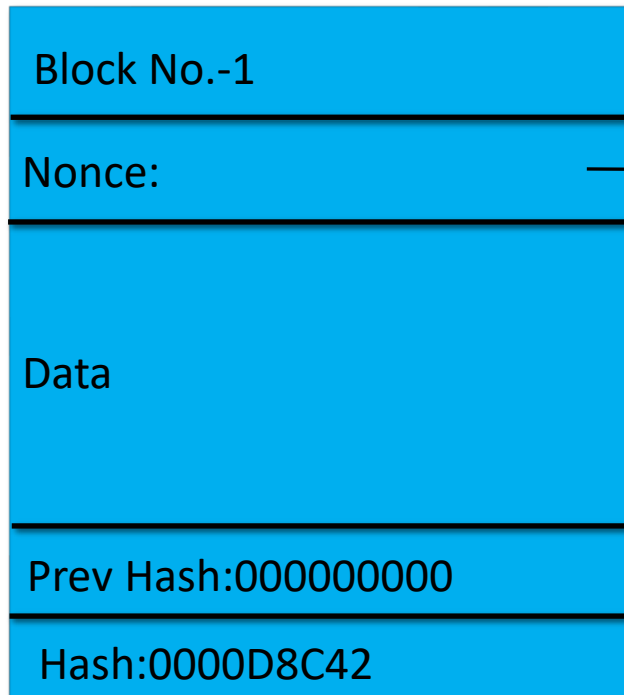
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Nonce Range

Nonce Range



Nonce is a 32 bit number.

Range of Nonce = 0 to $2^{32} - 1 \approx 0$ to 4×10^9

Nonce Range

SHA 256

Total number of possible hashes = $16 \times 16 \times \dots \times 16 = 16^{64} \simeq 10^{77}$

Nonce Range

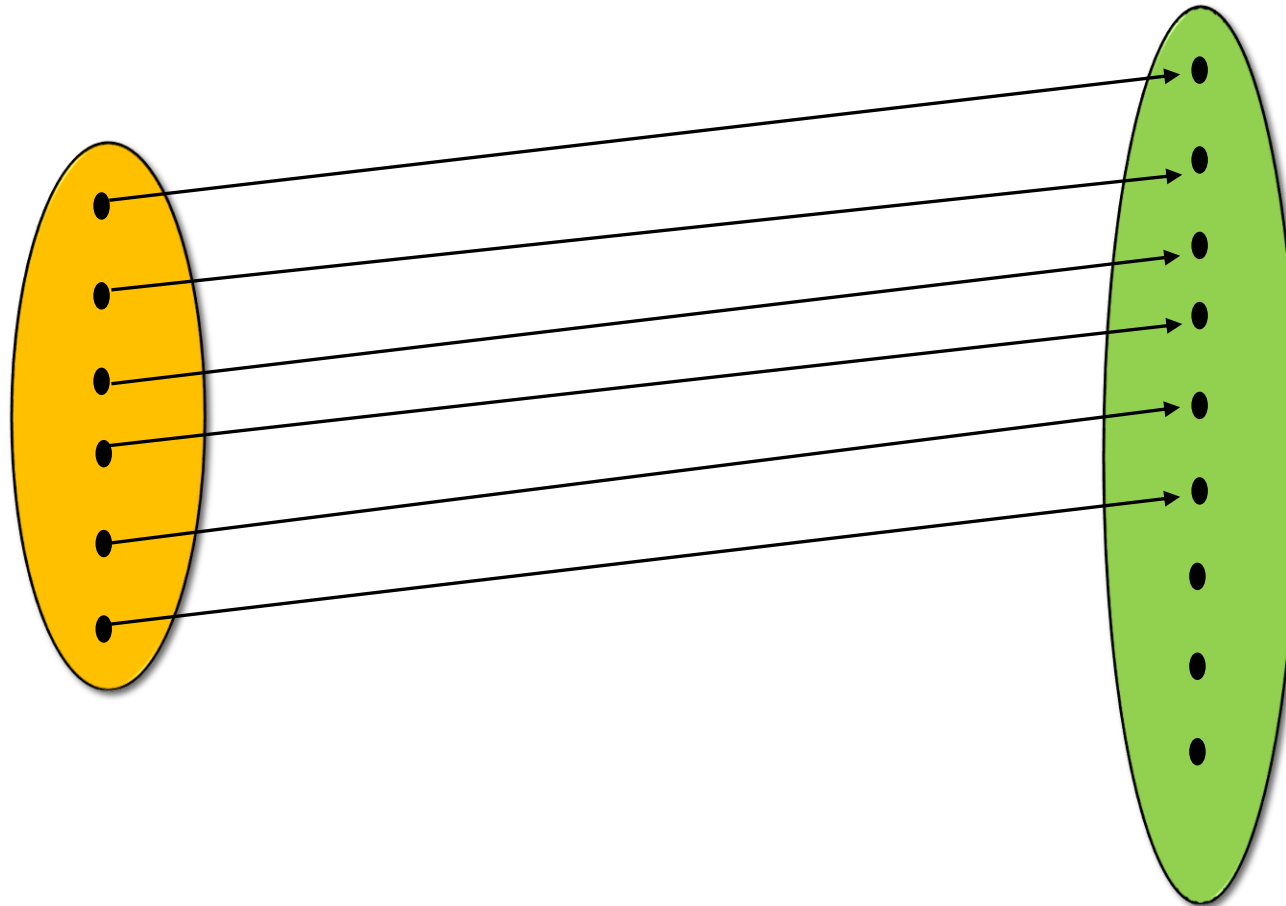
Total valid hashed $\simeq 10^{77}$

Total number of Nonce that we can generate $\simeq 4 \times 10^9$

$$10^{77} \ggggg 4 \times 10^9$$

=> That there are not enough nonce to generate the valid hash.

Nonce Range



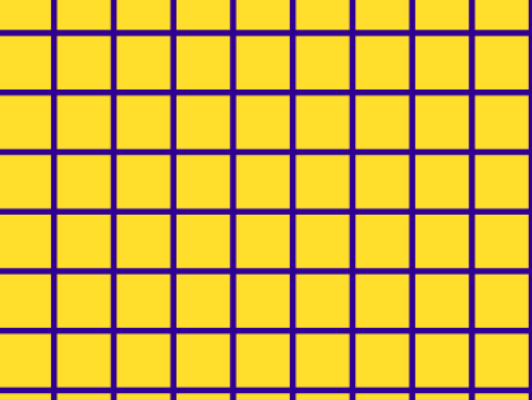
?

Nonce Range

A modest mines does 10^8 hashes/sec.

4×10^9 nonce will be covered in $= (4 \times 10^9)/(10^8) = 40$ seconds.

Q) So what the miners do when all the nonce get exhausted and miners have not hit the target ?



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Timestamp

Recap of Nonce Range

A modest mines does 10^8 hashes/sec.

4×10^9 nonce will be covered in $= (4 \times 10^9)/(10^8) = 40$ seconds.

Q) So what the miners do when all the nonce get exhausted and miners have not hit the target ?

Timestamp

Block No.-1

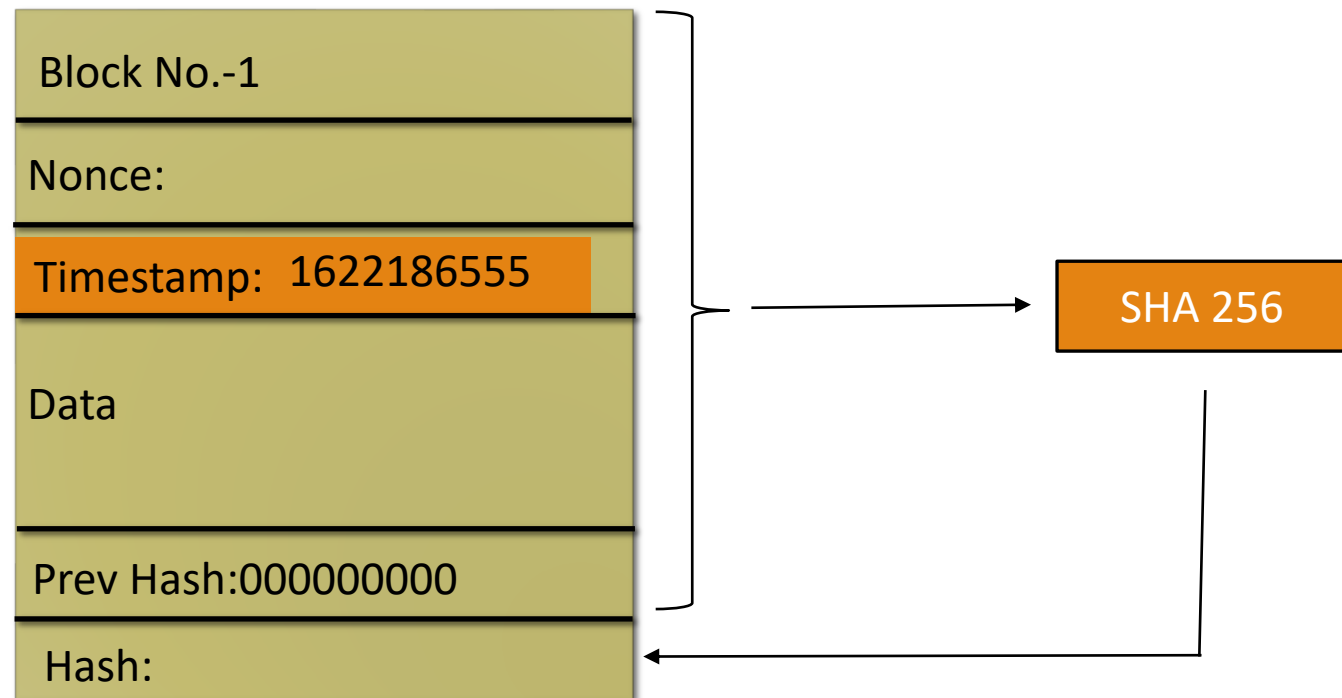
Nonce:

Data

Prev Hash:0000000000

Hash:

Timestamp



Timestamp

Block No.-1
Nonce:
Timestamp: 1622186555
Data
Prev Hash:0000000000
Hash:

A miner exhaust **4 Billion nonce** in **40 sec.**

A miner will exhaust **0.1 Billion nonce** in **1 sec.**

Timestamp

Block No.-1
Nonce:
Timestamp: 1622186555
Data
Prev Hash:0000000000
Hash:

0.5 seconds

Timestamp

Block No.-1
Nonce:
Timestamp: 1622186556
Data
Prev Hash:0000000000
Hash:



Timestamp

Block No.-1
Nonce:
Timestamp: 1622186557
Data
Prev Hash:0000000000
Hash:



Timestamp

Block No.-1
Nonce:
Timestamp: 1622186558
Data
Prev Hash:0000000000
Hash:



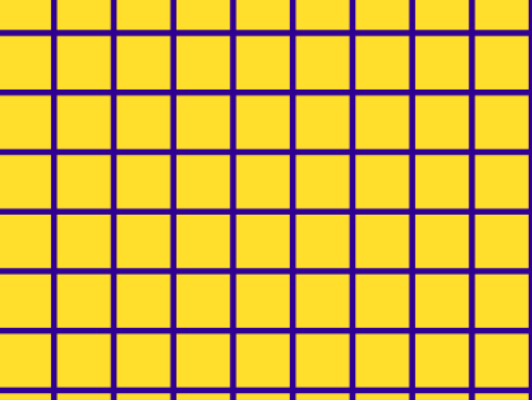
Timestamp

Current hashing rate is equal to **234 million trillion hashes/sec.**

4×10^9 nonce will be covered in $= (4 \times 10^9) / (10^6 \times 10^{12}) = 4 \times 10^{-9}$ seconds.

4×10^{-9} sec <<<<< 1 sec

Q)What should the miners do in idle time? Should they wait for timestamp to change?



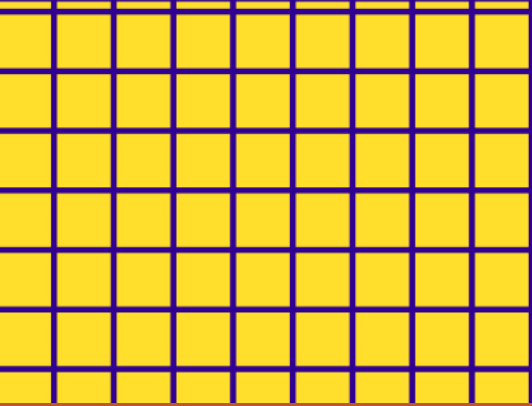
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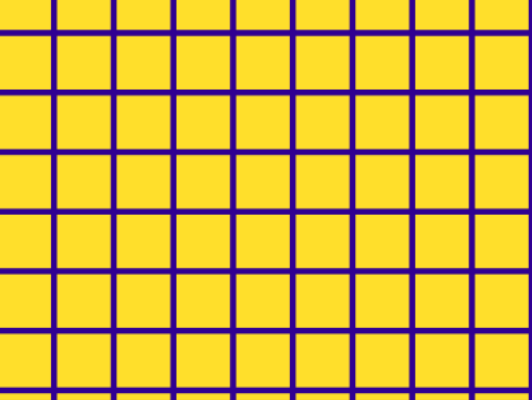
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Discord – Link in description





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Mempool

Recap Timestamp...

Current hashing rate is equal to **234 million trillion hashes/sec.**

4×10^9 nonce will be covered in $= (4 \times 10^9) / (10^6 \times 10^{12}) = 4 \times 10^{-9}$ seconds.

4×10^{-9} sec <<<<< 1 sec

Q)What should the miners do in idle time? Should they wait for timestamp to change?

Mempool

Block No.-1
Nonce:
Timestamp:
Transactions:
Prev Hash:0000000000
Hash:

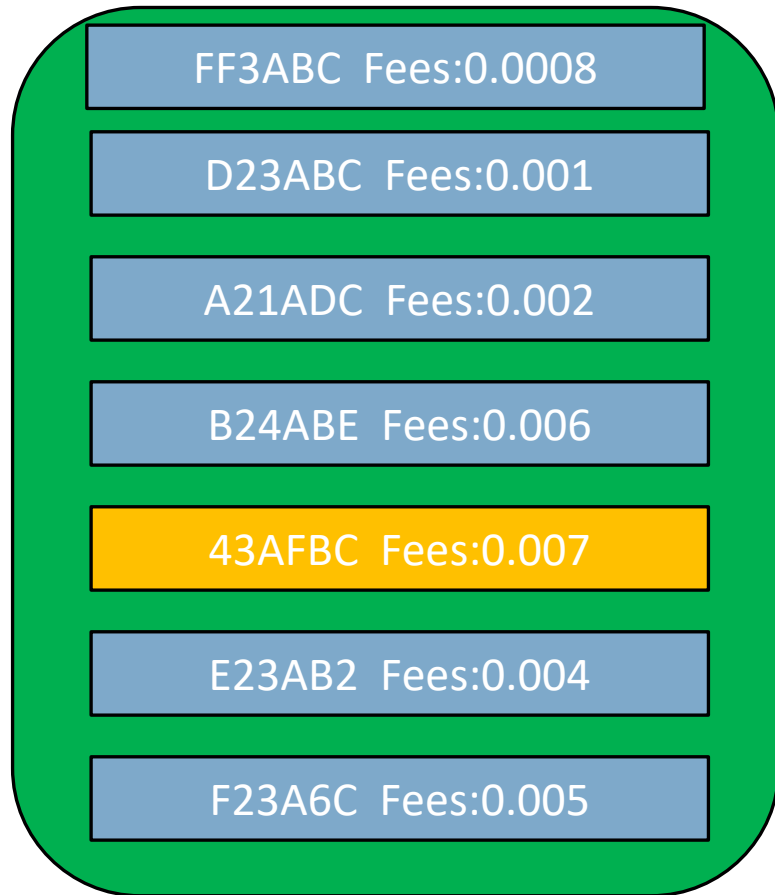
Mempool



Mempool

Block No.-1
Nonce:
Timestamp:
Transactions:
Prev Hash:000000000
Hash:

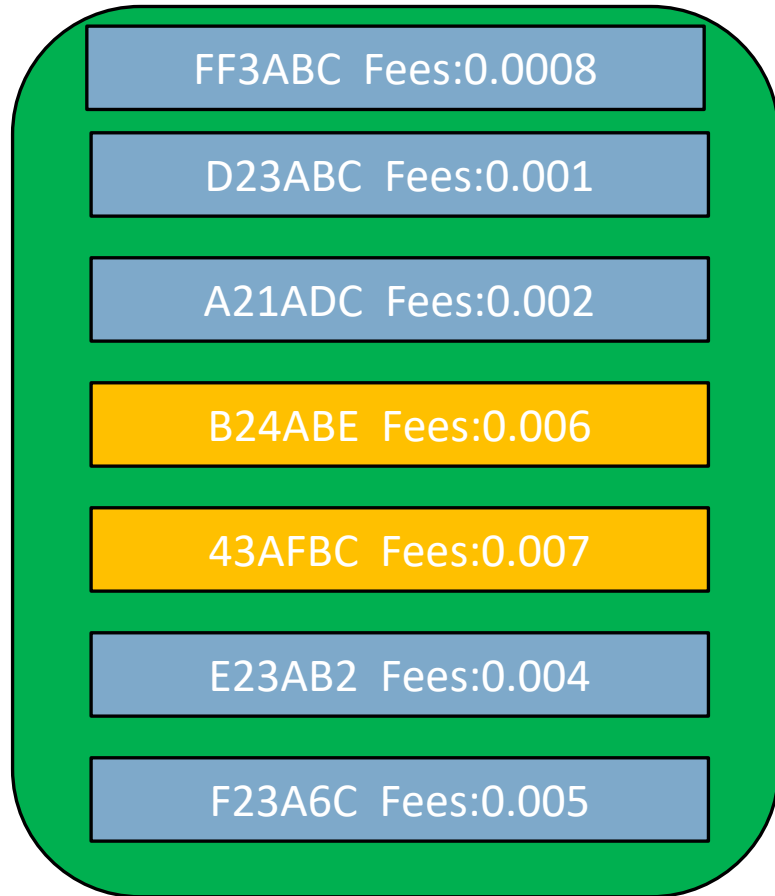
How actually mining of transaction takes place?



Mempool

Block No.-1
Nonce:
Timestamp:
Transactions:
Prev Hash:0000000000
Hash:

How actually mining of transaction takes place?



Mempool

Block No.-1
Nonce:
Timestamp:
Transactions:
Prev Hash:0000000000
Hash:

How actually mining of transaction takes place?



Mempool

Block No.-1
Nonce:
Timestamp:
Transactions:
Prev Hash:0000000000
Hash:

How actually mining of transaction takes place?



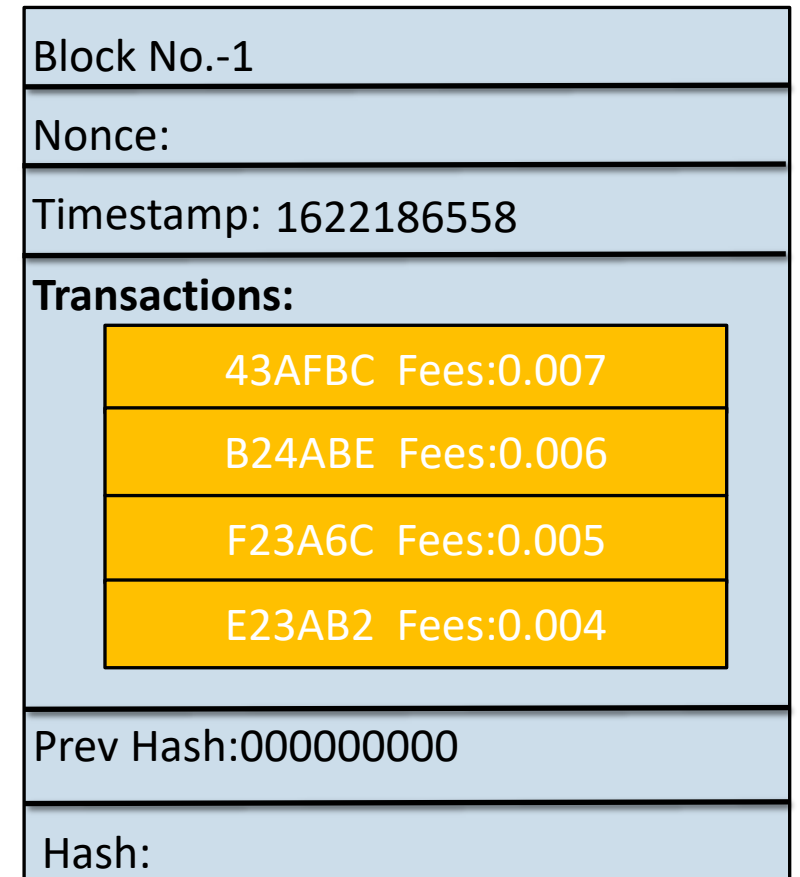
Mempool

Block No.-1
Nonce:
Timestamp:
Transactions:
Prev Hash:0000000000
Hash:

How actually mining of transaction takes place?



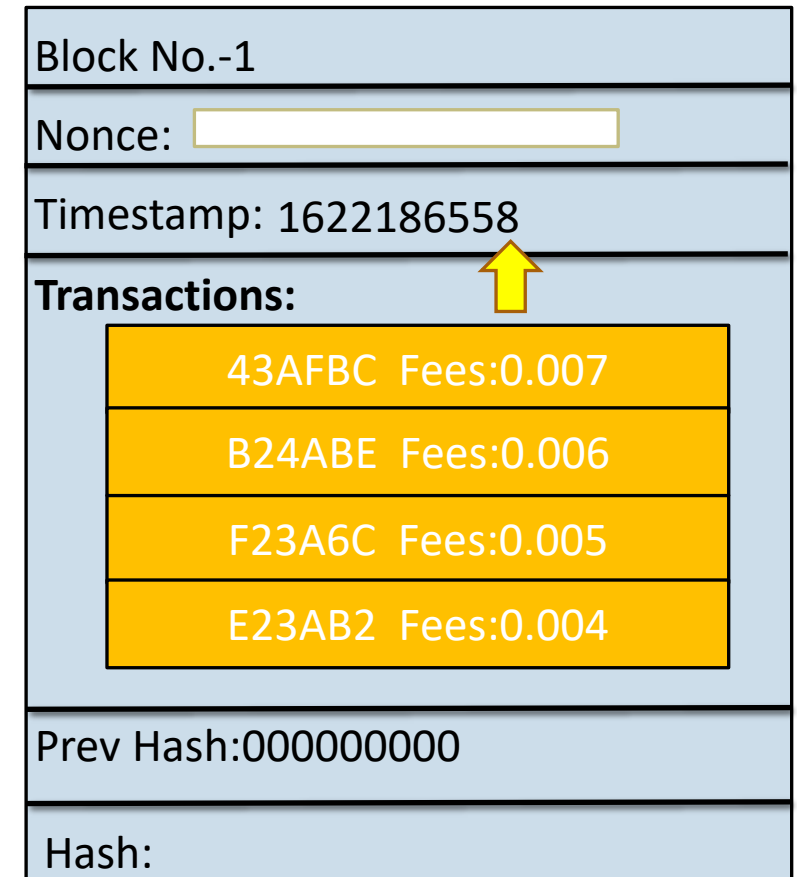
Mempool



How actually mining of transaction takes place?



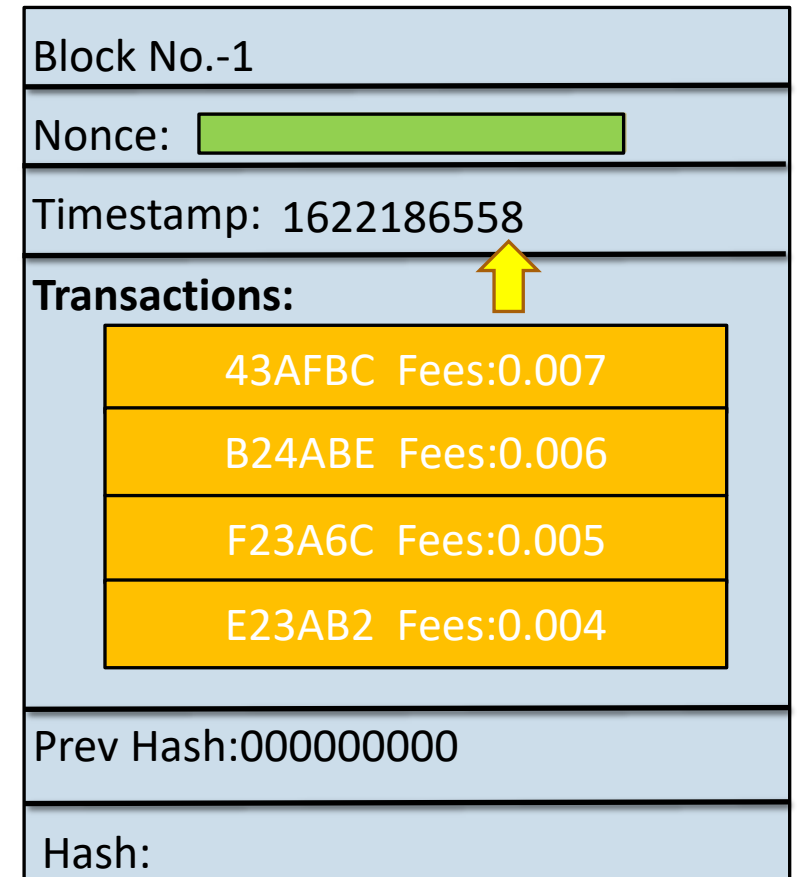
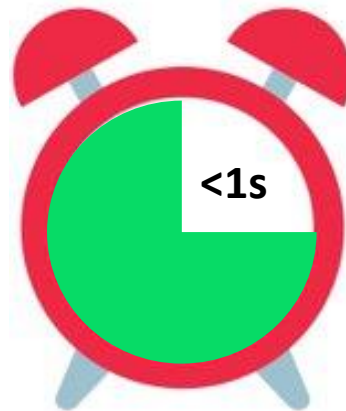
Mempool



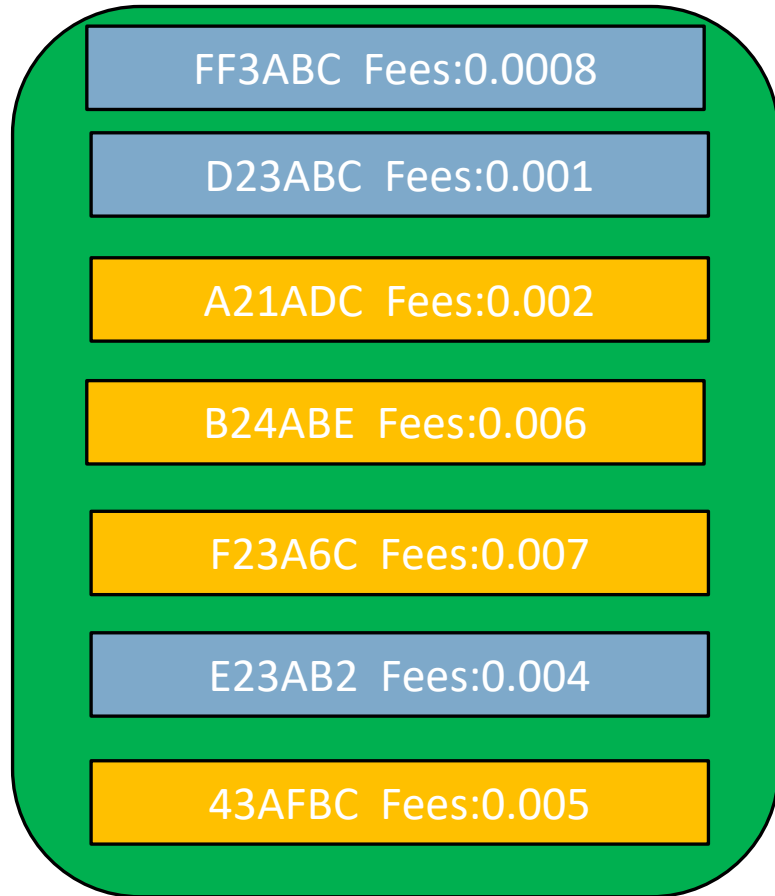
How actually mining of transaction takes place?



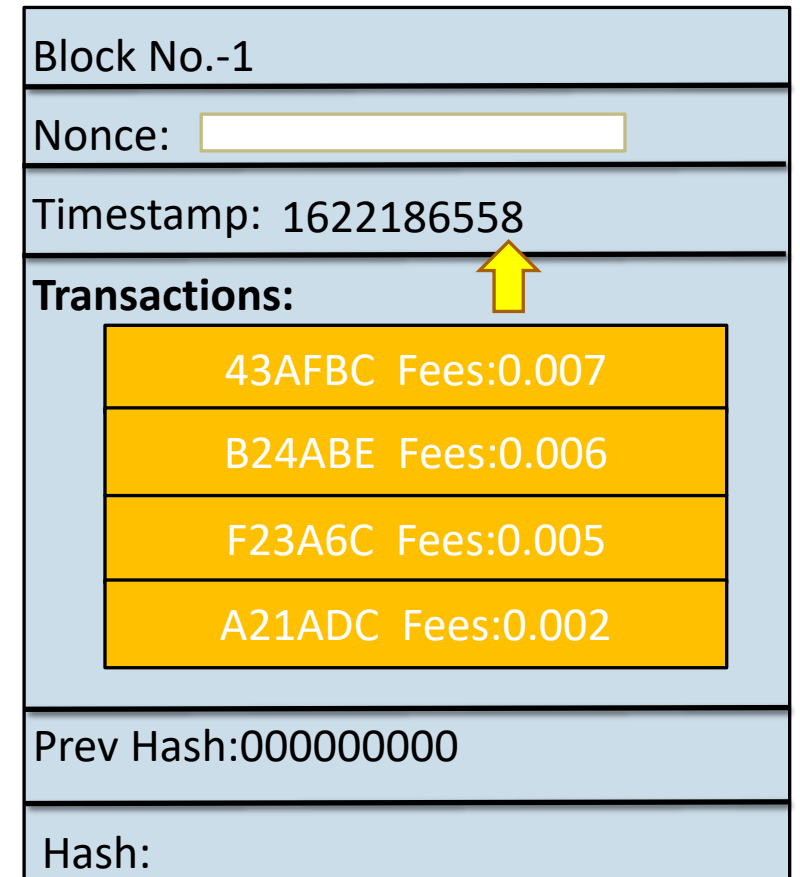
Mempool



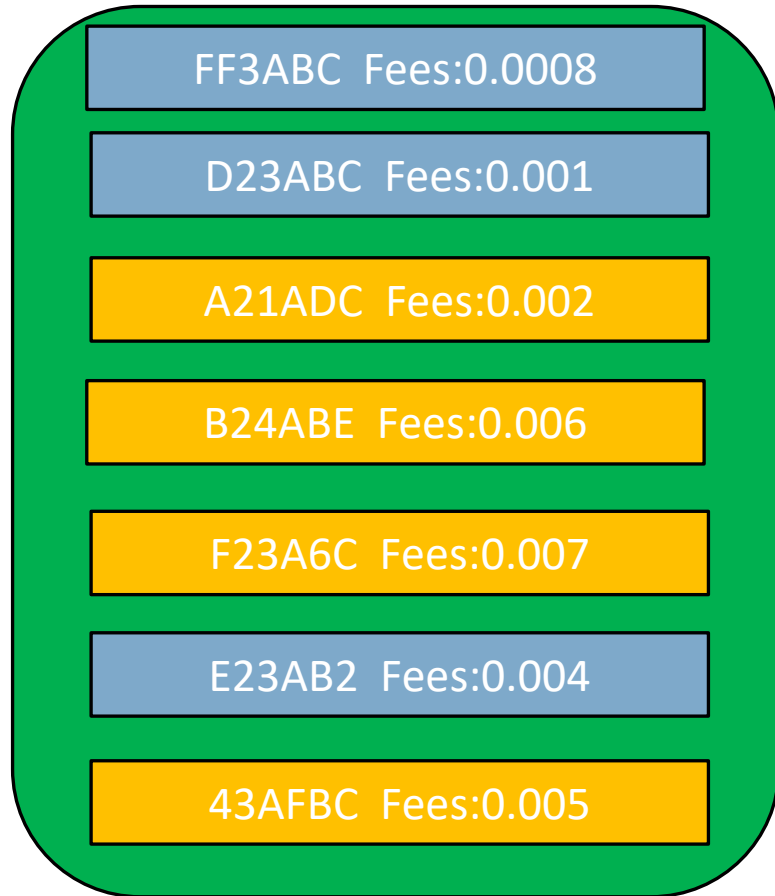
How actually mining of transaction takes place?



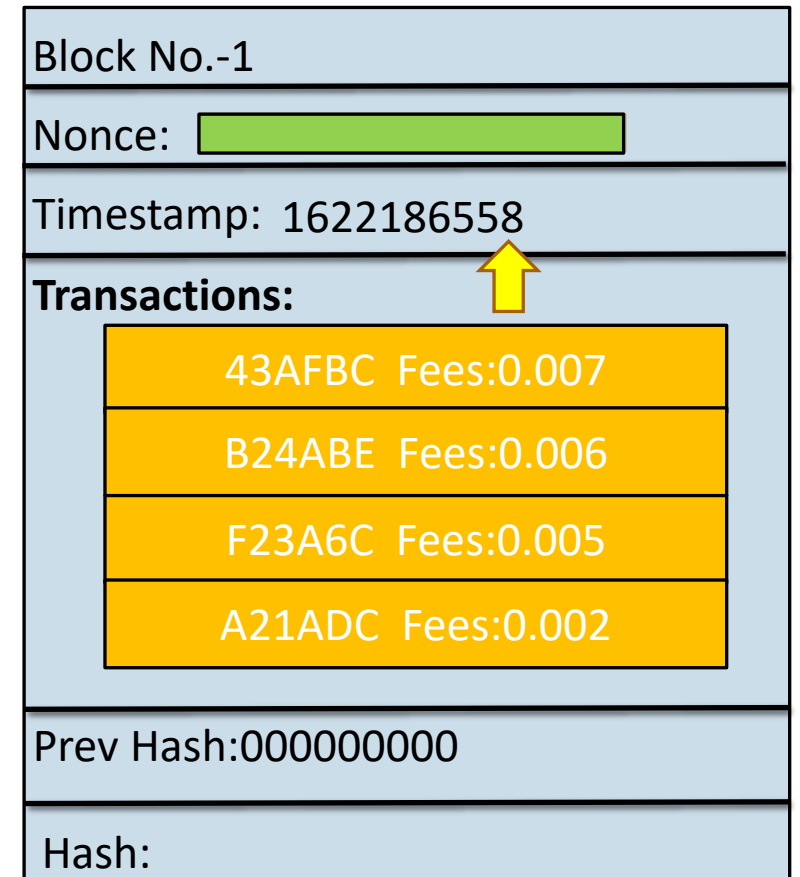
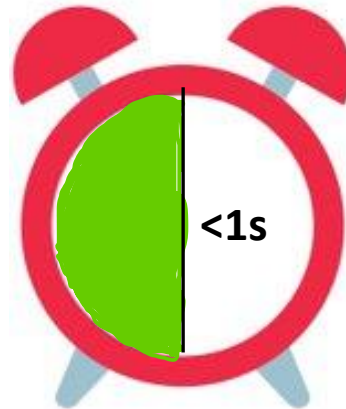
Mempool



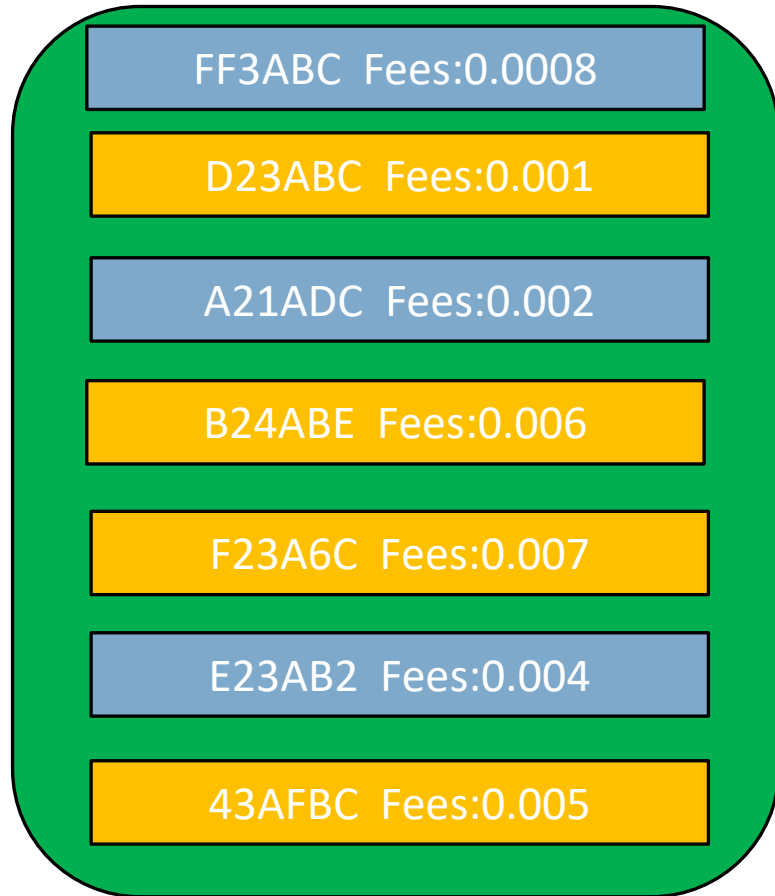
How actually mining of transaction takes place?



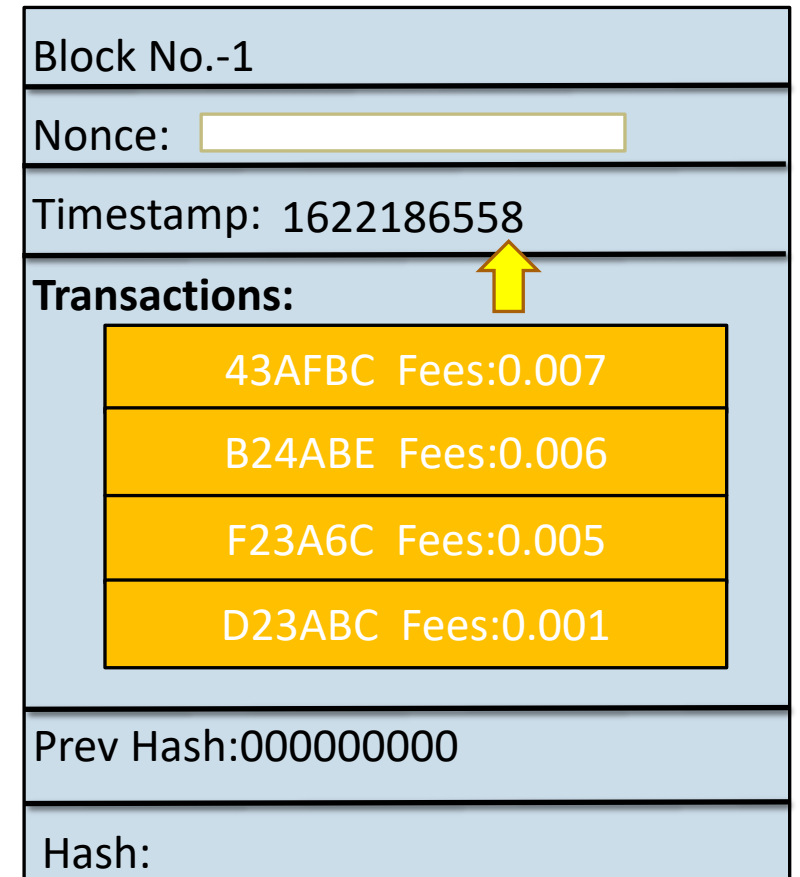
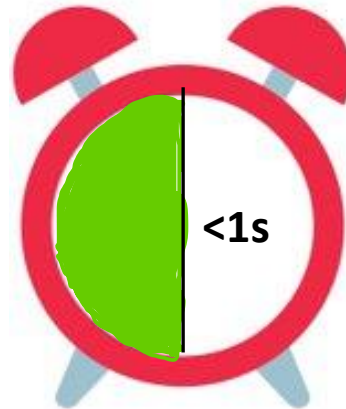
Mempool



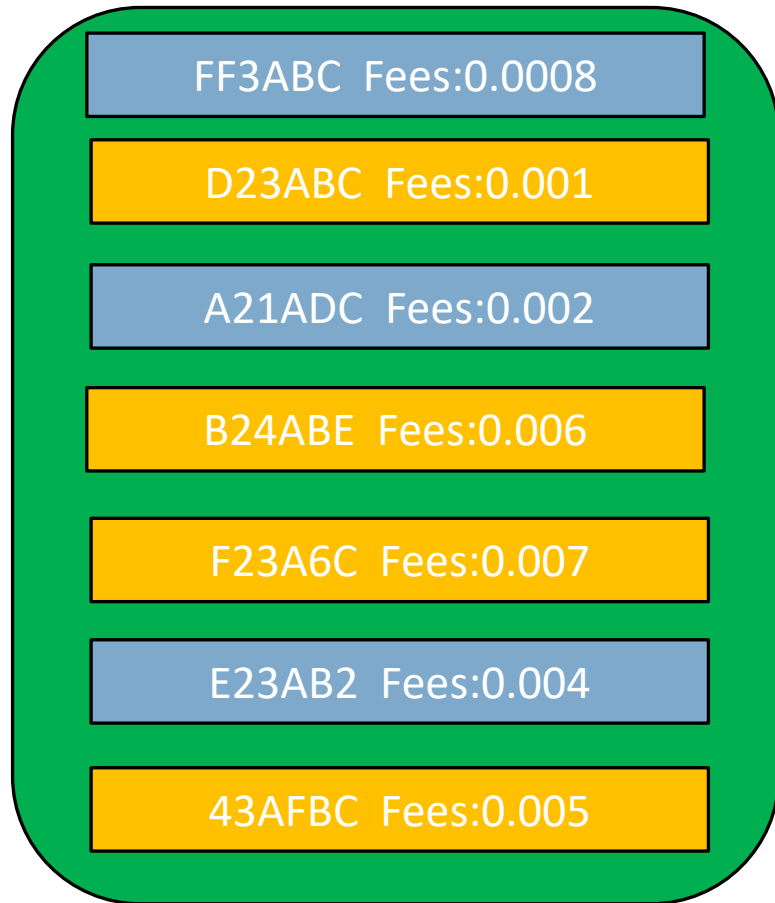
How actually mining of transaction takes place?



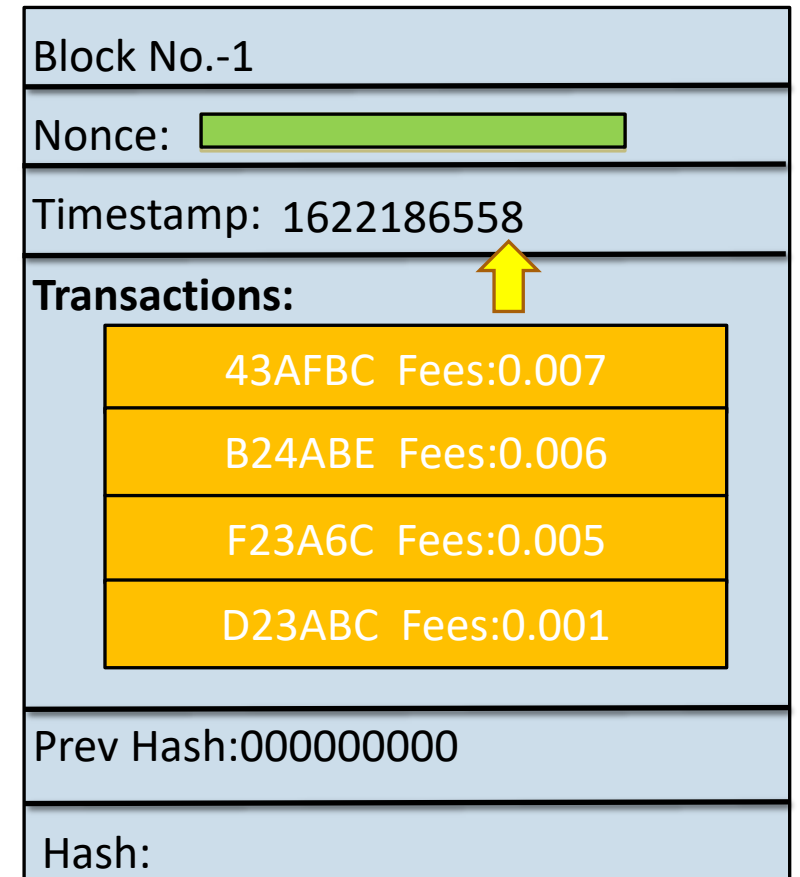
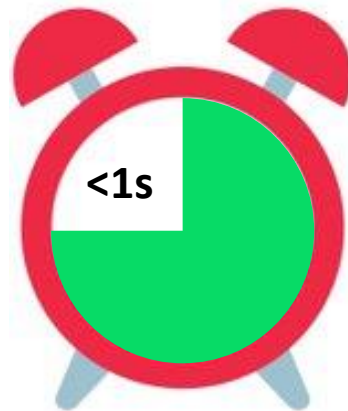
Mempool



How actually mining of transaction takes place?



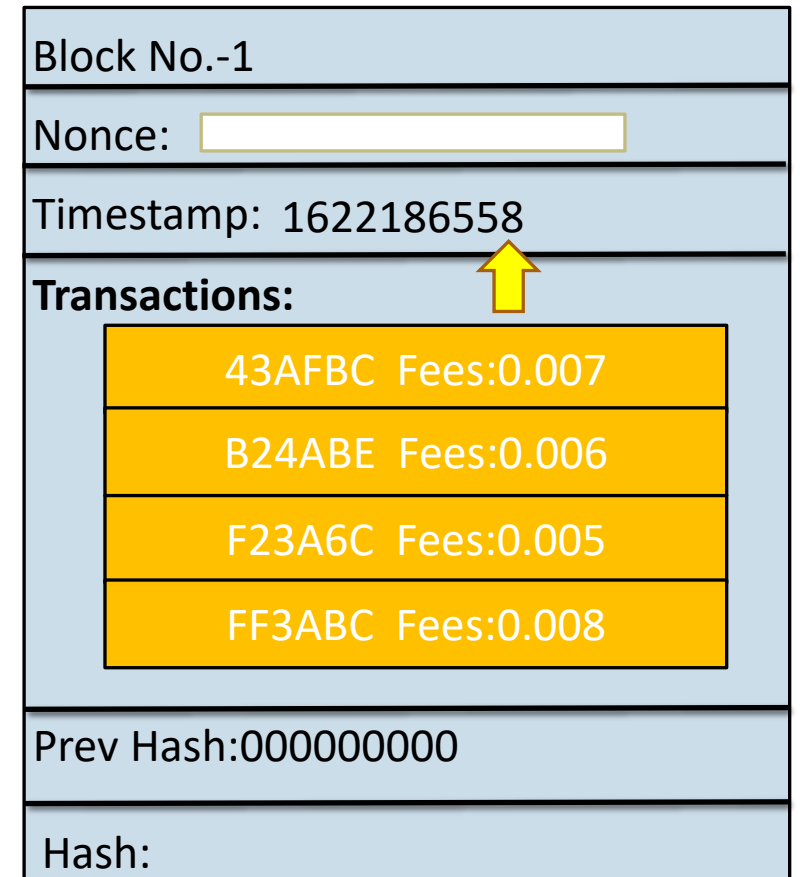
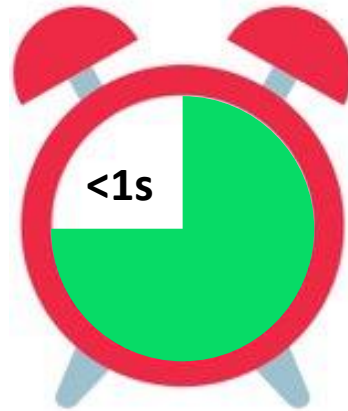
Mempool



How actually mining of transaction takes place?



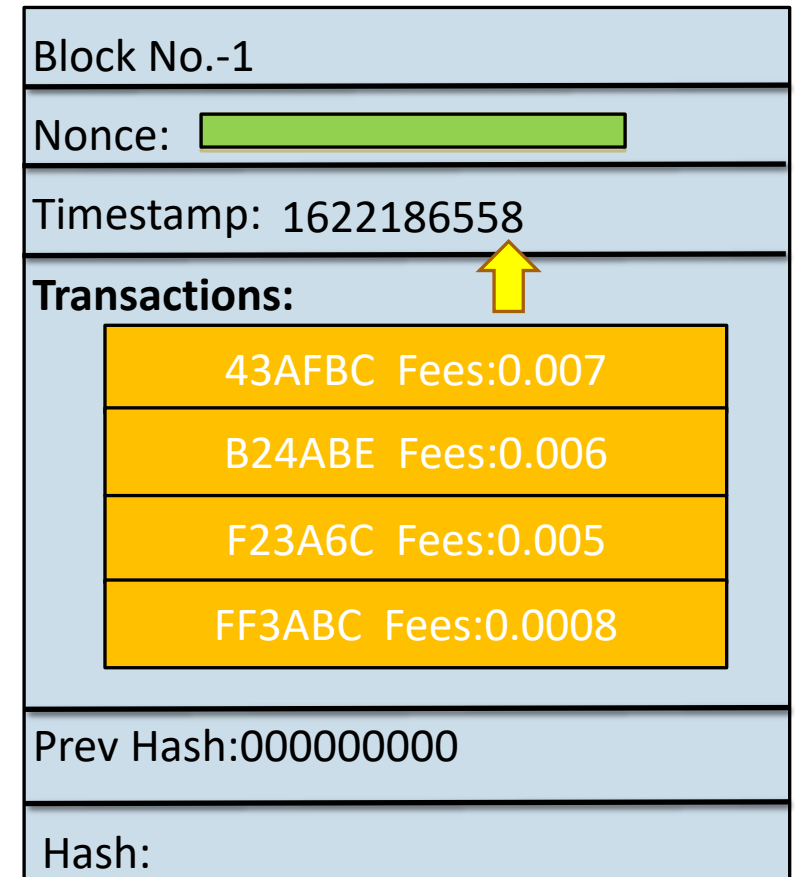
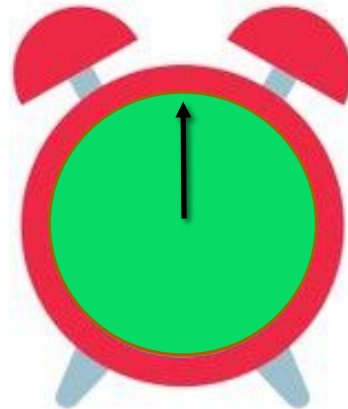
Mempool



How actually mining of transaction takes place?



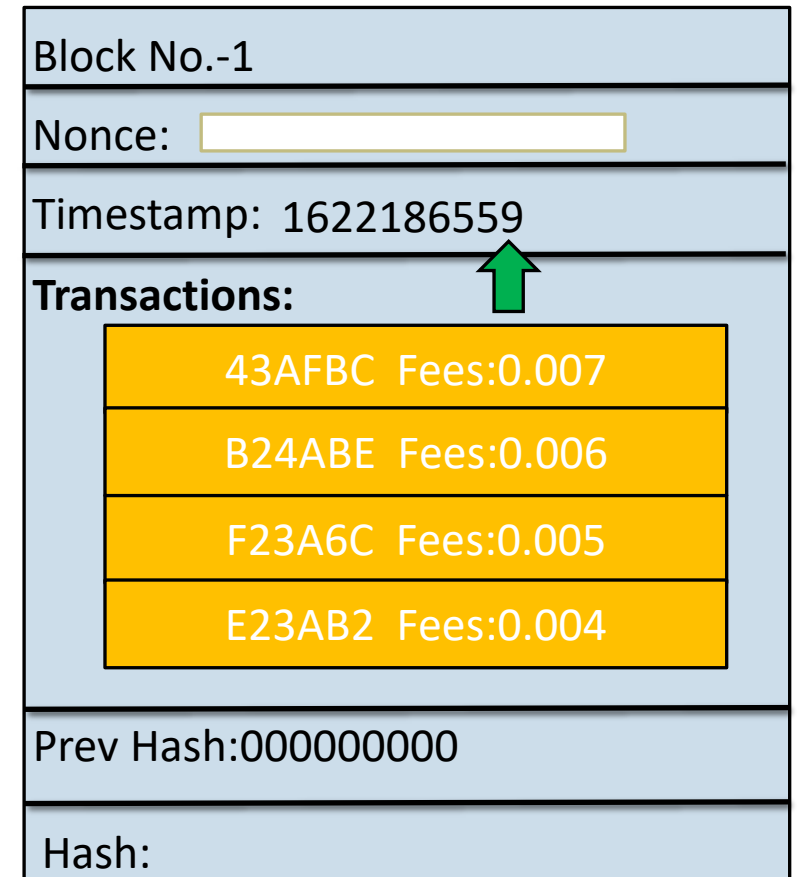
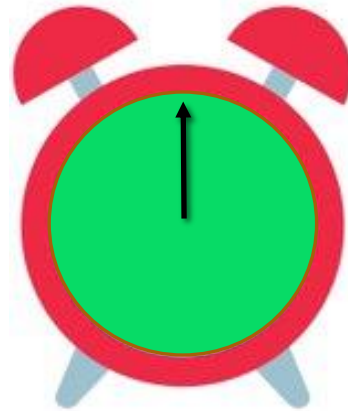
Mempool



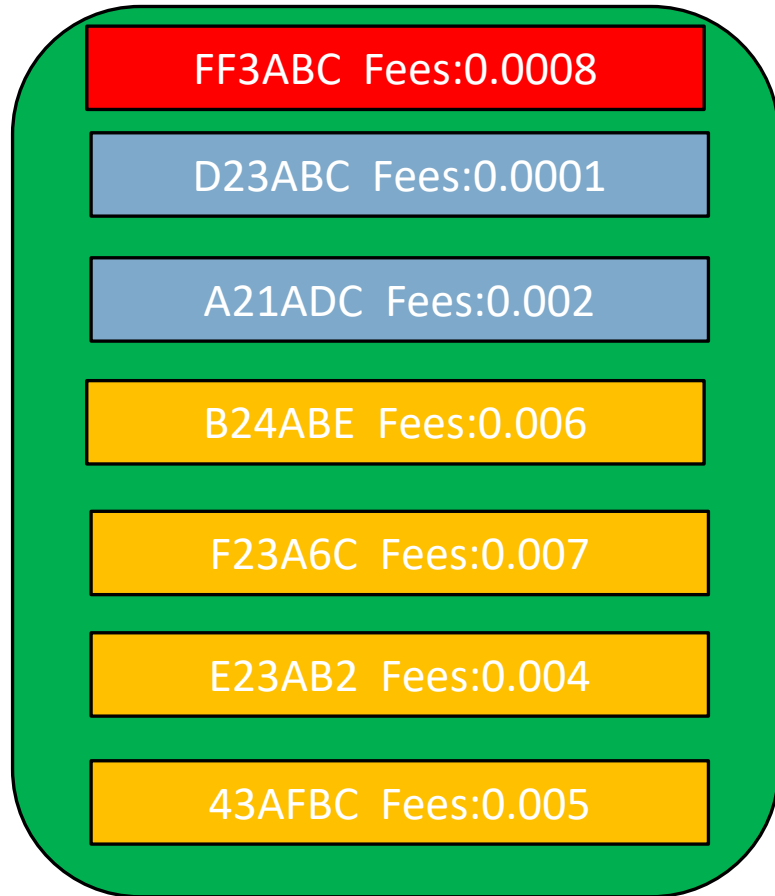
How actually mining of transaction takes place?



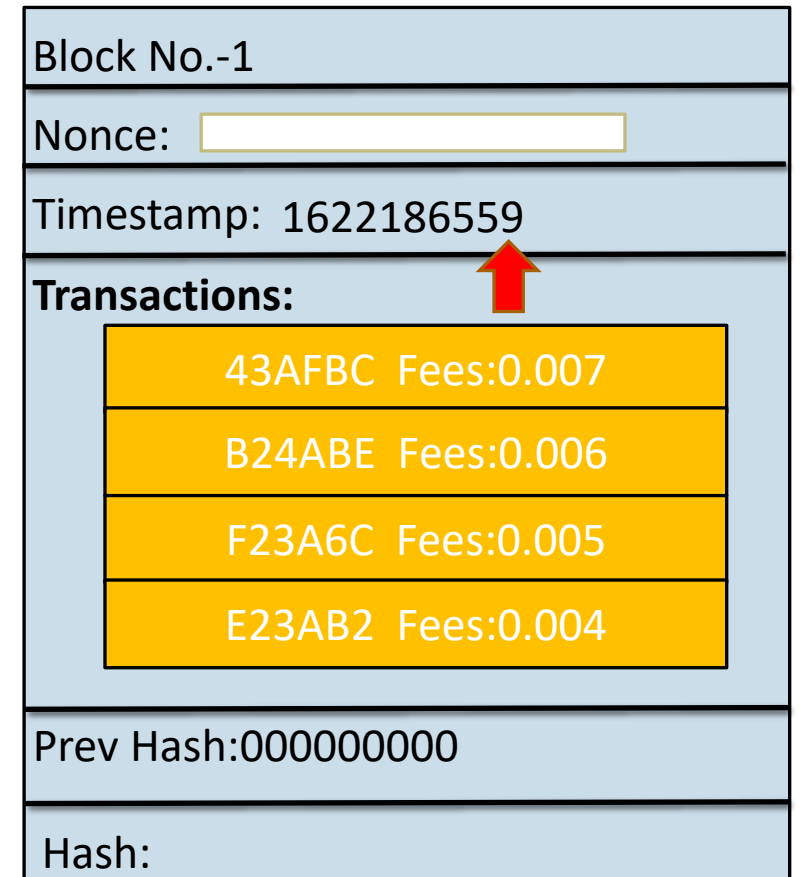
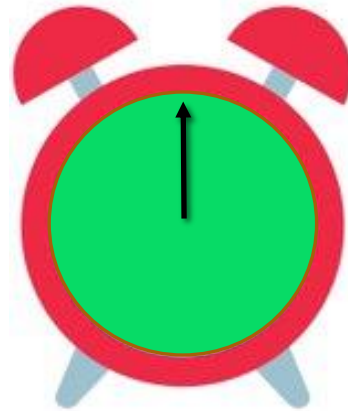
Mempool

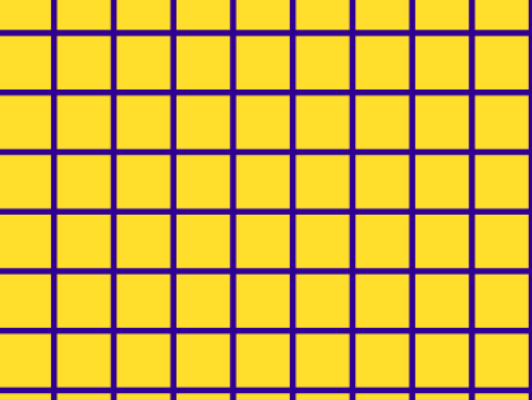


How actually mining of transaction takes place?



Mempool





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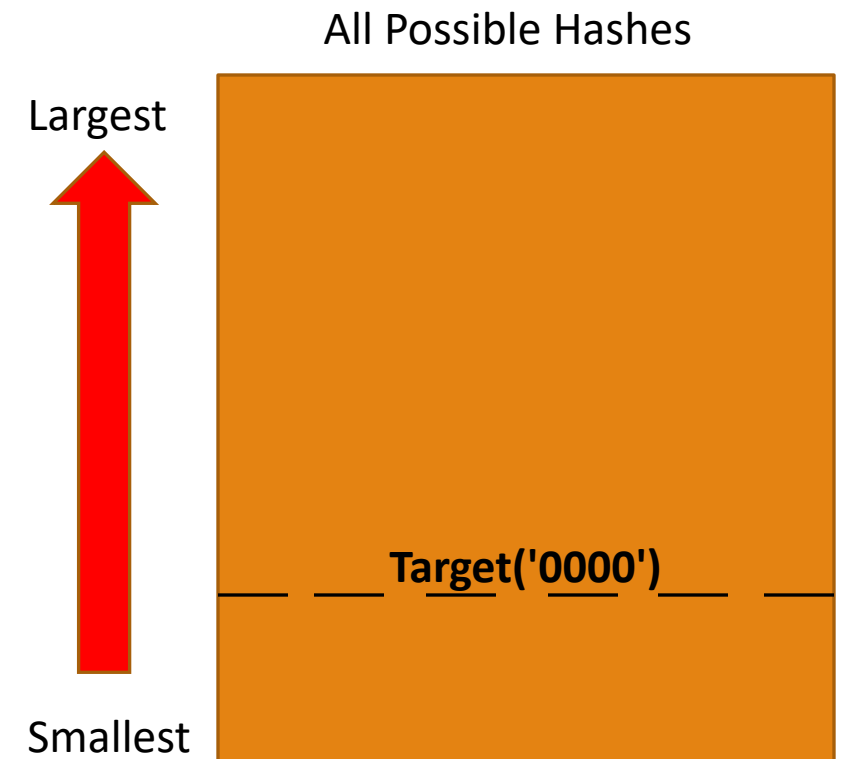


An abstract digital graphic on the left side of the slide. It features concentric circles and rings of binary code (0s and 1s) in various shades of blue and teal. The background has a grid-like pattern. The overall aesthetic is high-tech and futuristic.

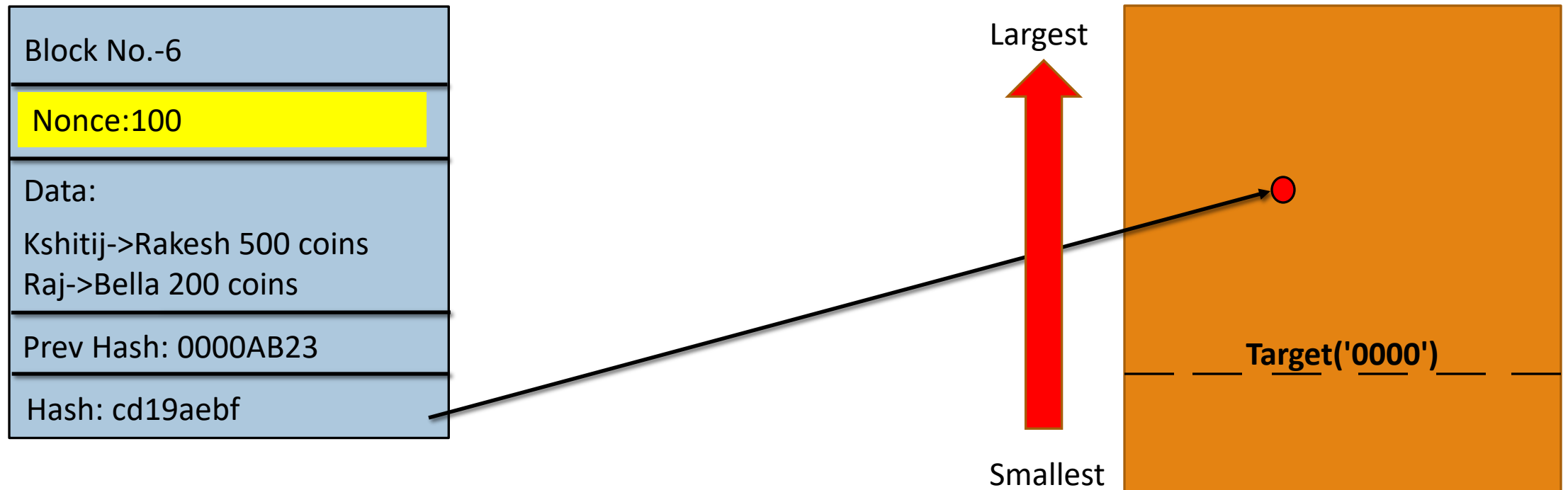
CPU Vs GPU Vs ASIC

How Mining Works ?

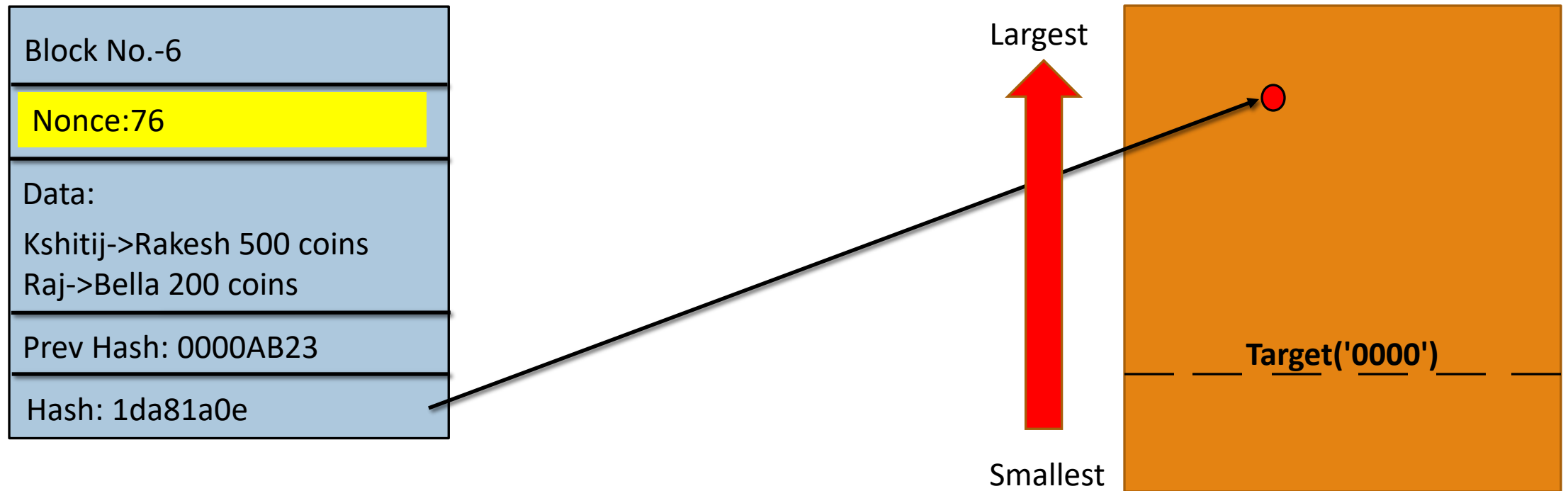
Block No.-6
Nonce:
Data: Kshitij->Rakesh 500 coins Raj->Bella 200 coins
Prev Hash: 0000AB23
Hash: cd19aebf



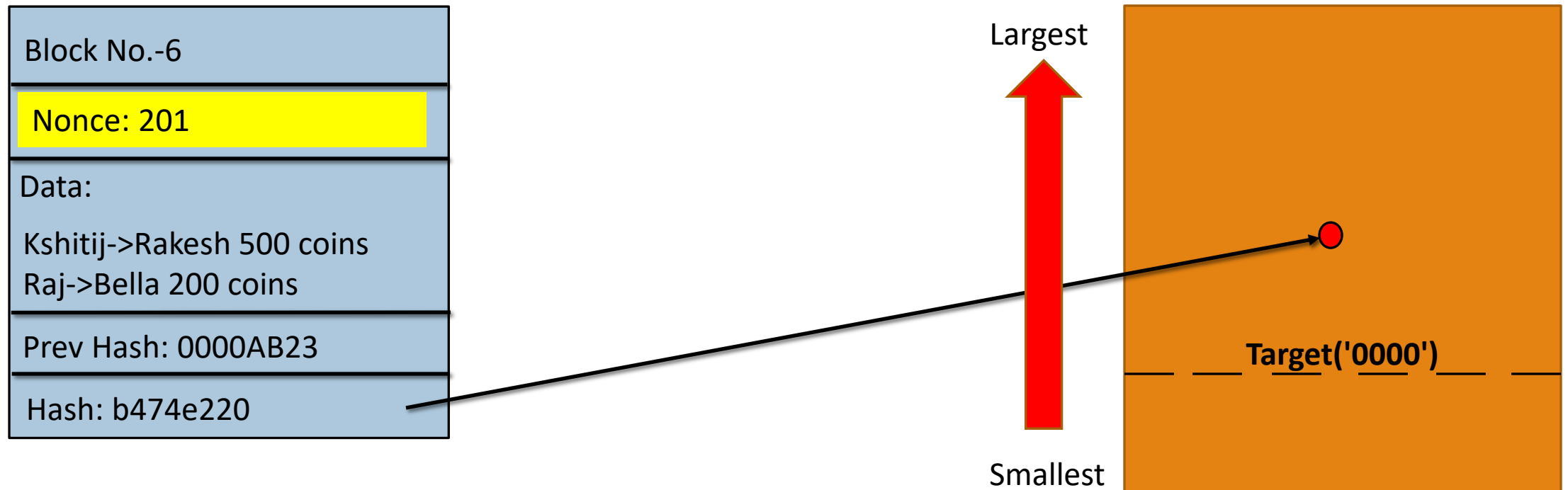
How Mining Works ?



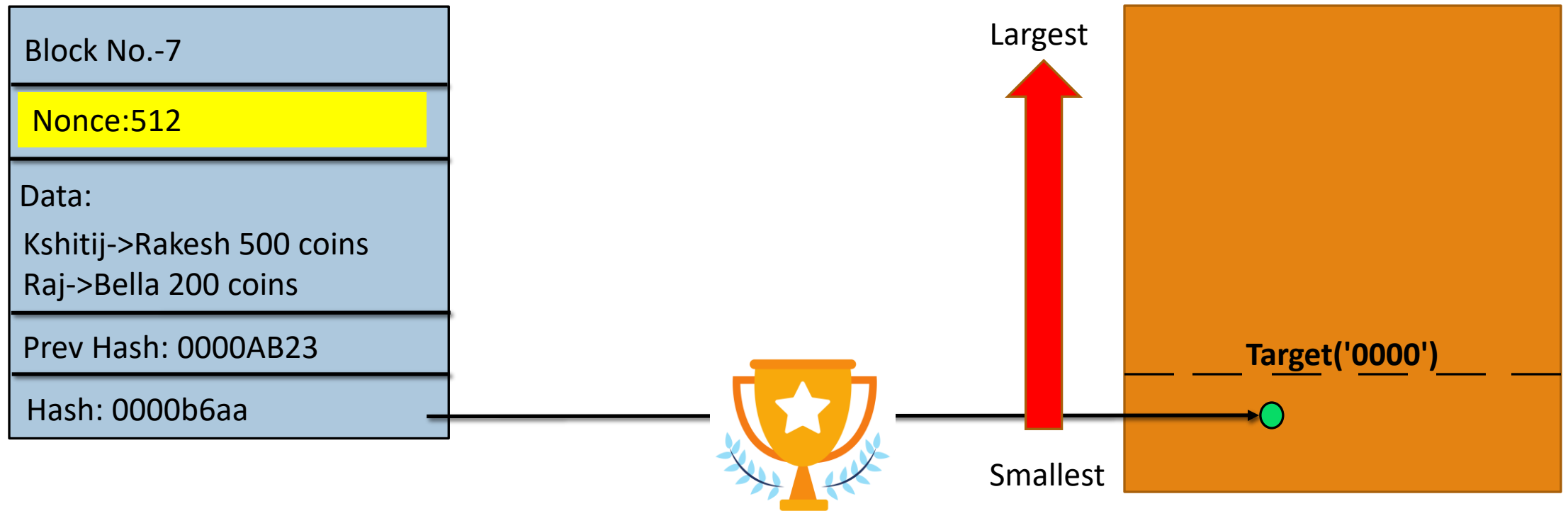
How Mining Works ?



How Mining Works ?



How Mining Works ?

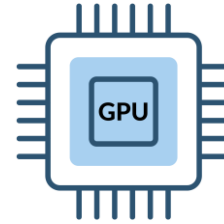


CPU_s Vs GPU_s Vs ASIC_s

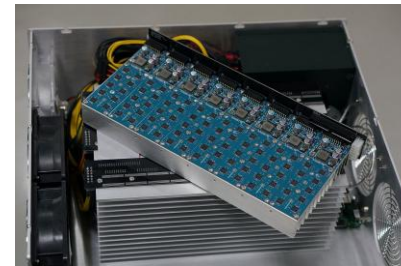
CPU < 10 MH/s

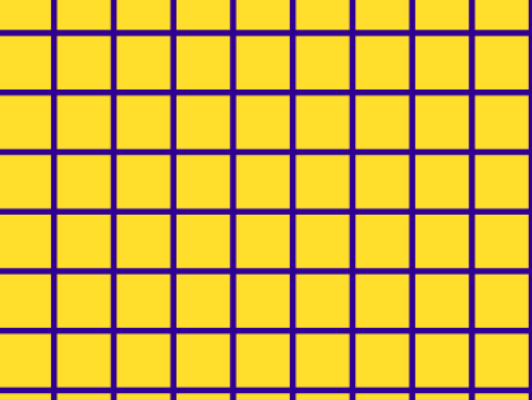


GPU < 1 GH/s



ASIC > 110 TH/s





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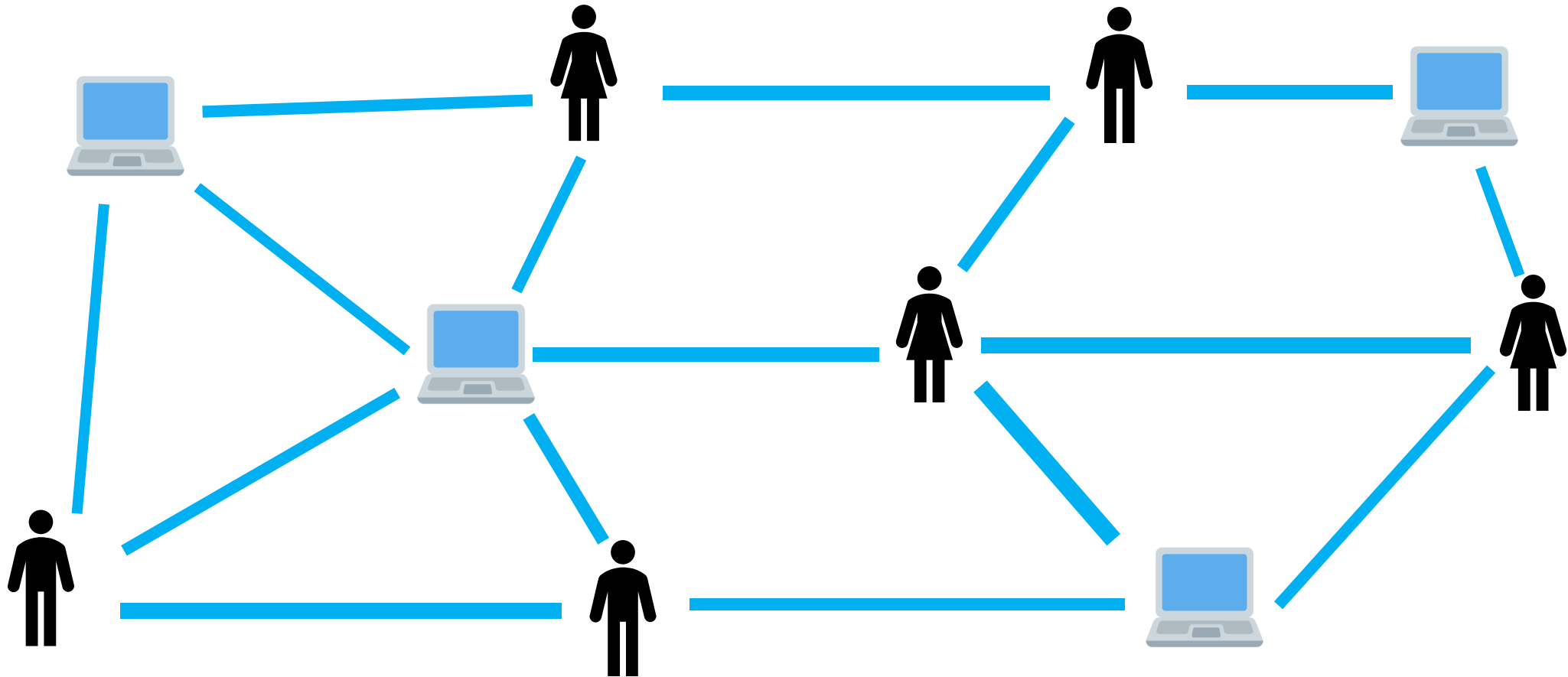
CODE EATER



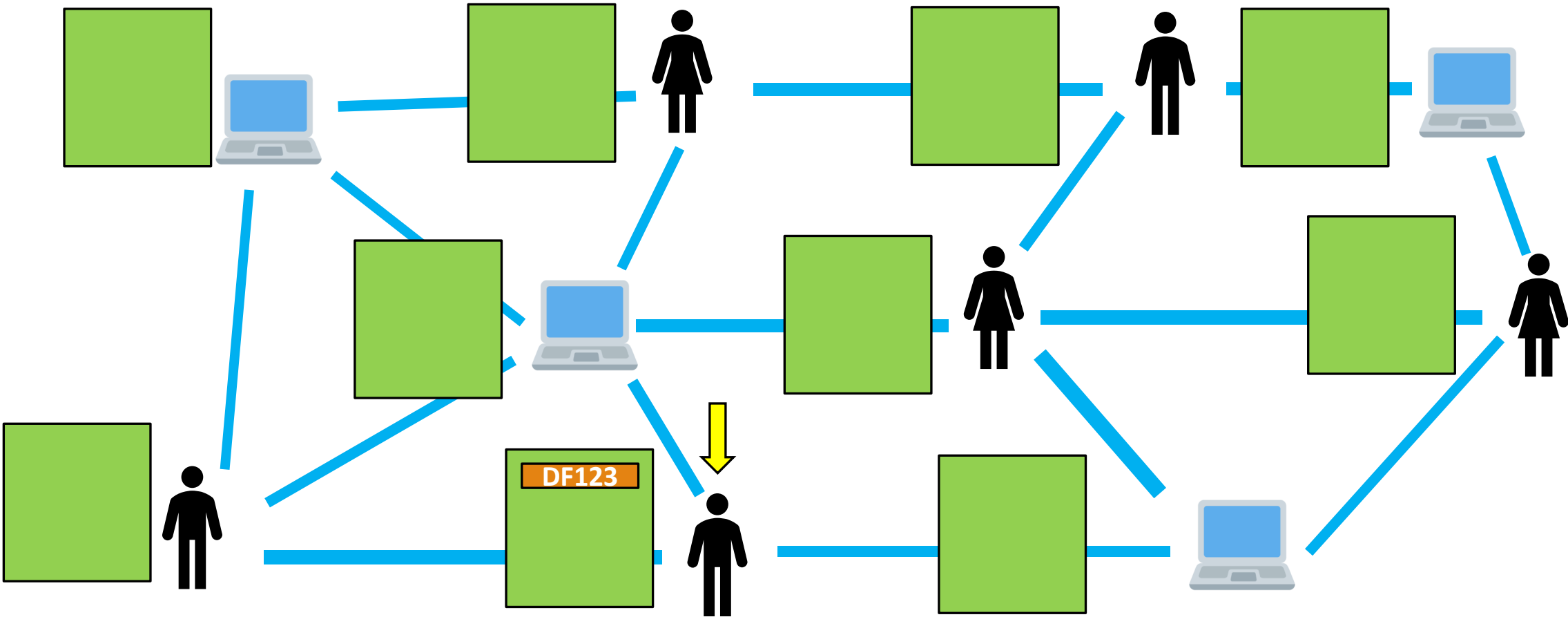
An abstract graphic on the left side of the slide. It features several concentric circles and arcs on a dark blue background. The arcs are composed of small, light blue rectangular segments, creating a sense of motion or a digital interface. Binary code (0s and 1s) is scattered along the outer edges of these arcs. The overall effect is a futuristic, technological aesthetic.

How mempool works?

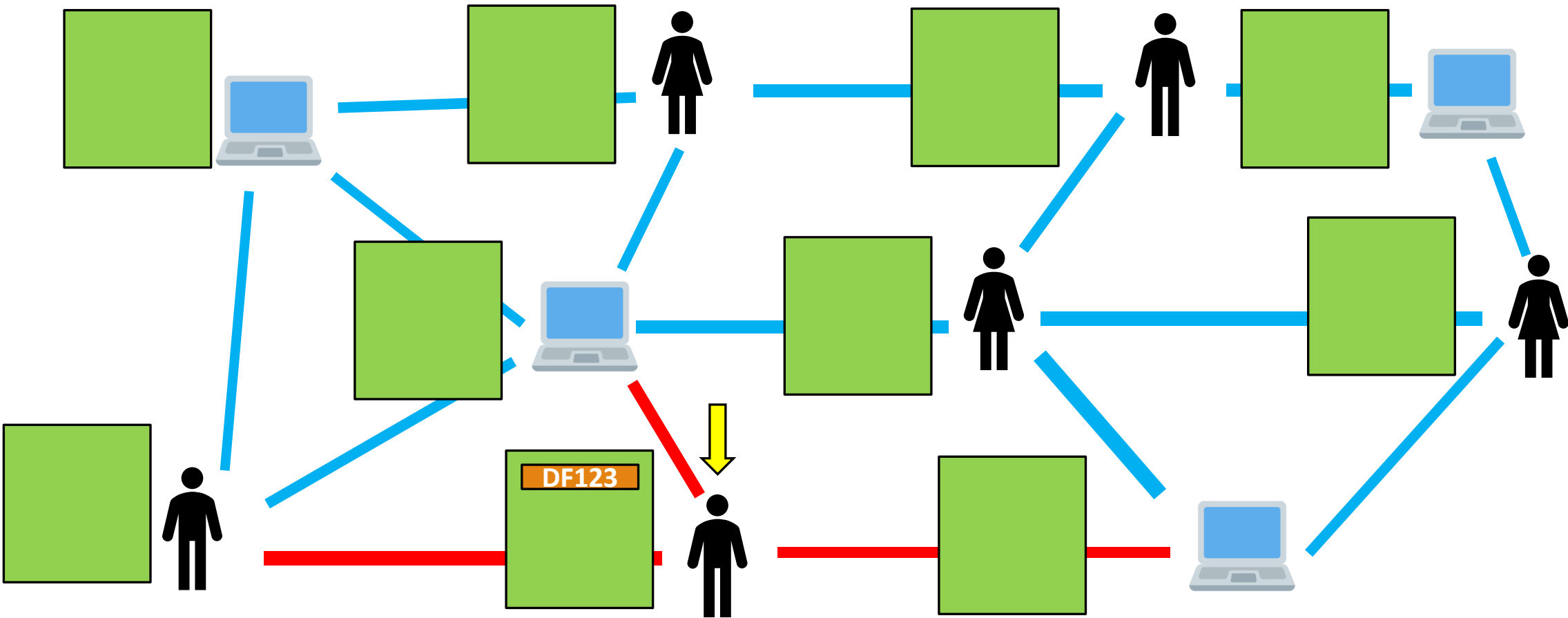
How do Mempool works?(Behind the scenes)



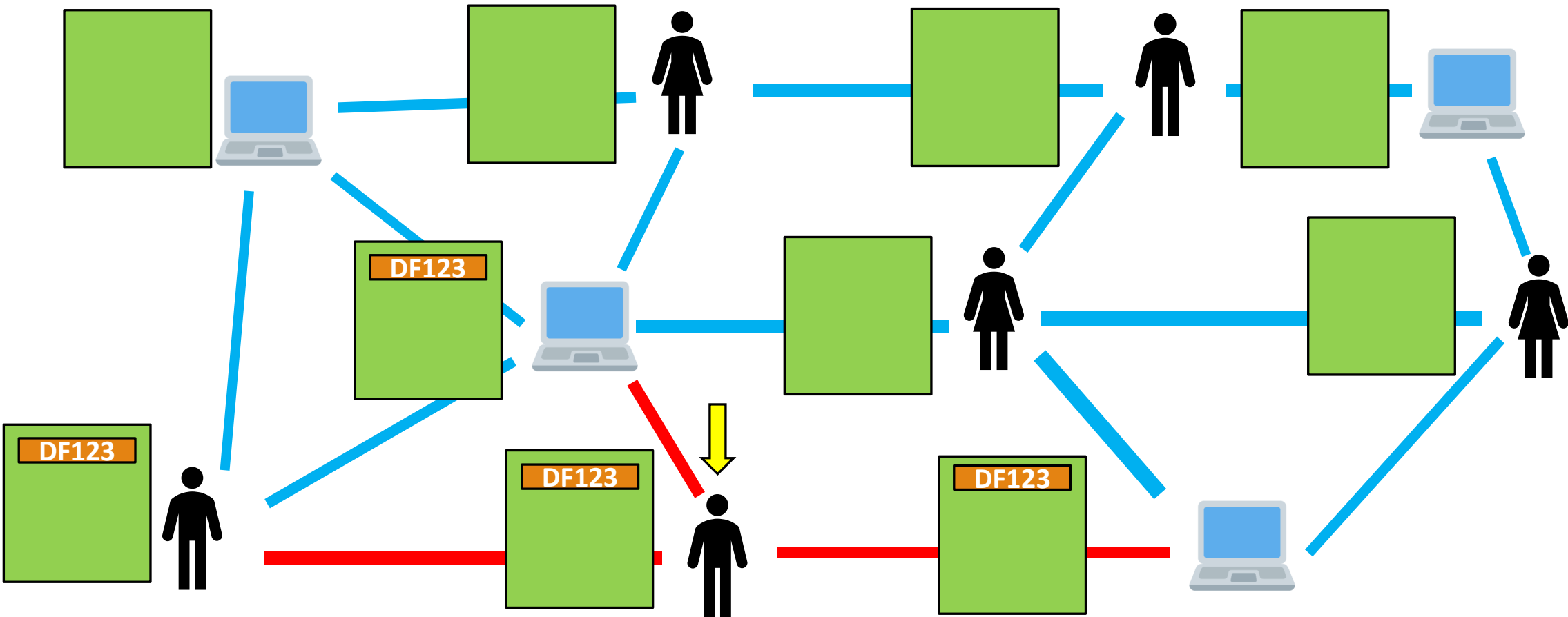
How do Mempool works?(Behind the scenes)



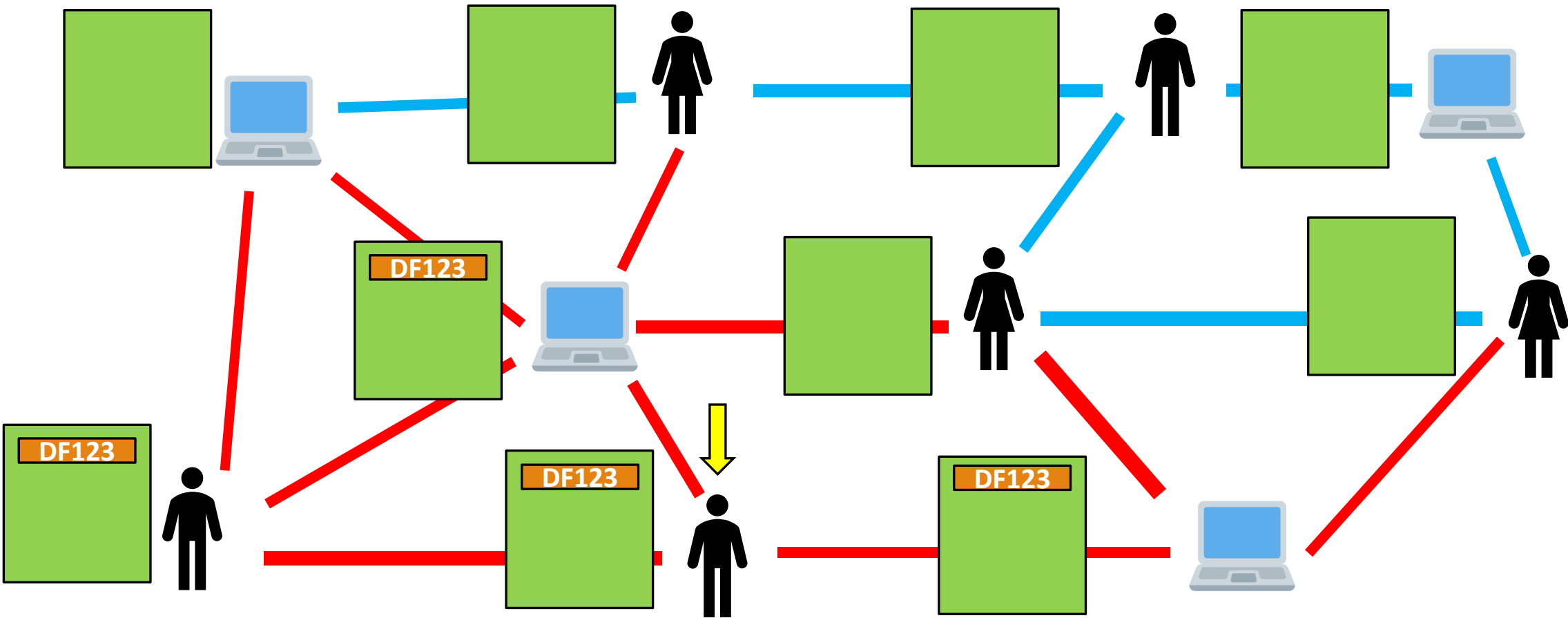
How do Mempool works?(Behind the scenes)



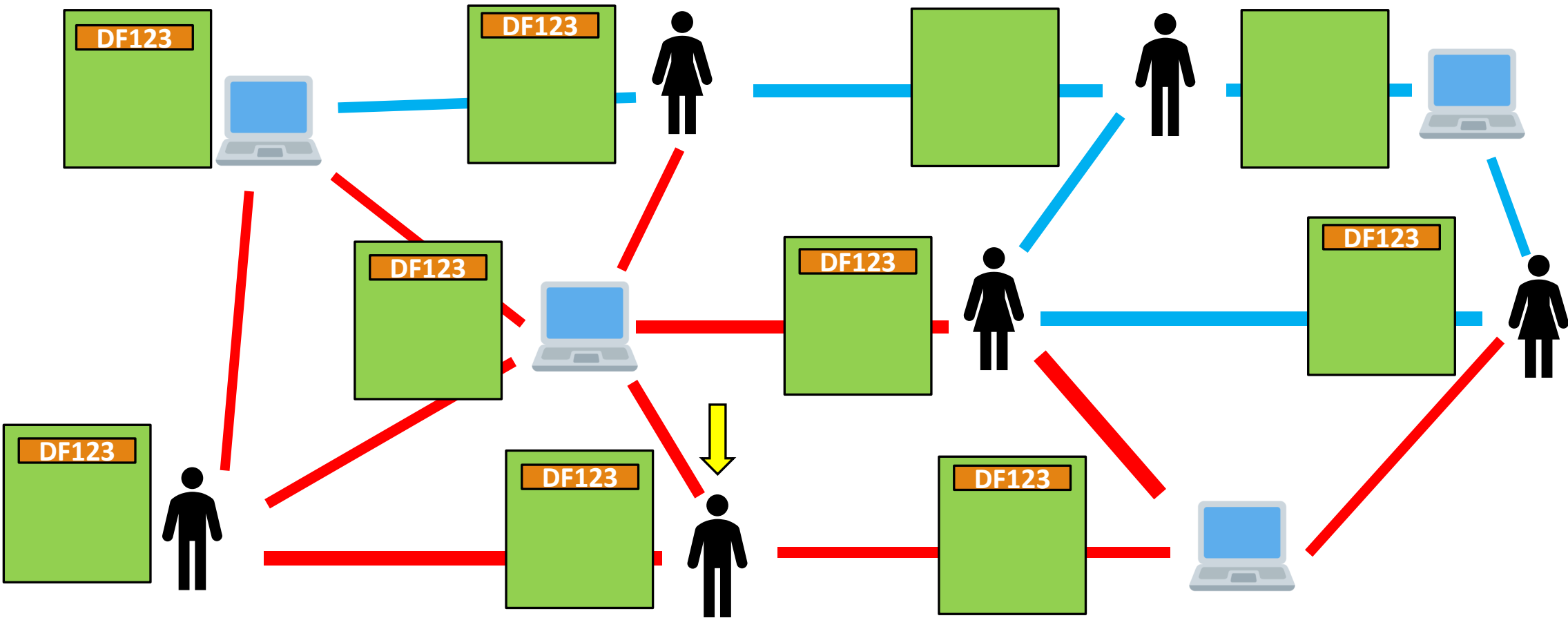
How do Mempool works?(Behind the scenes)



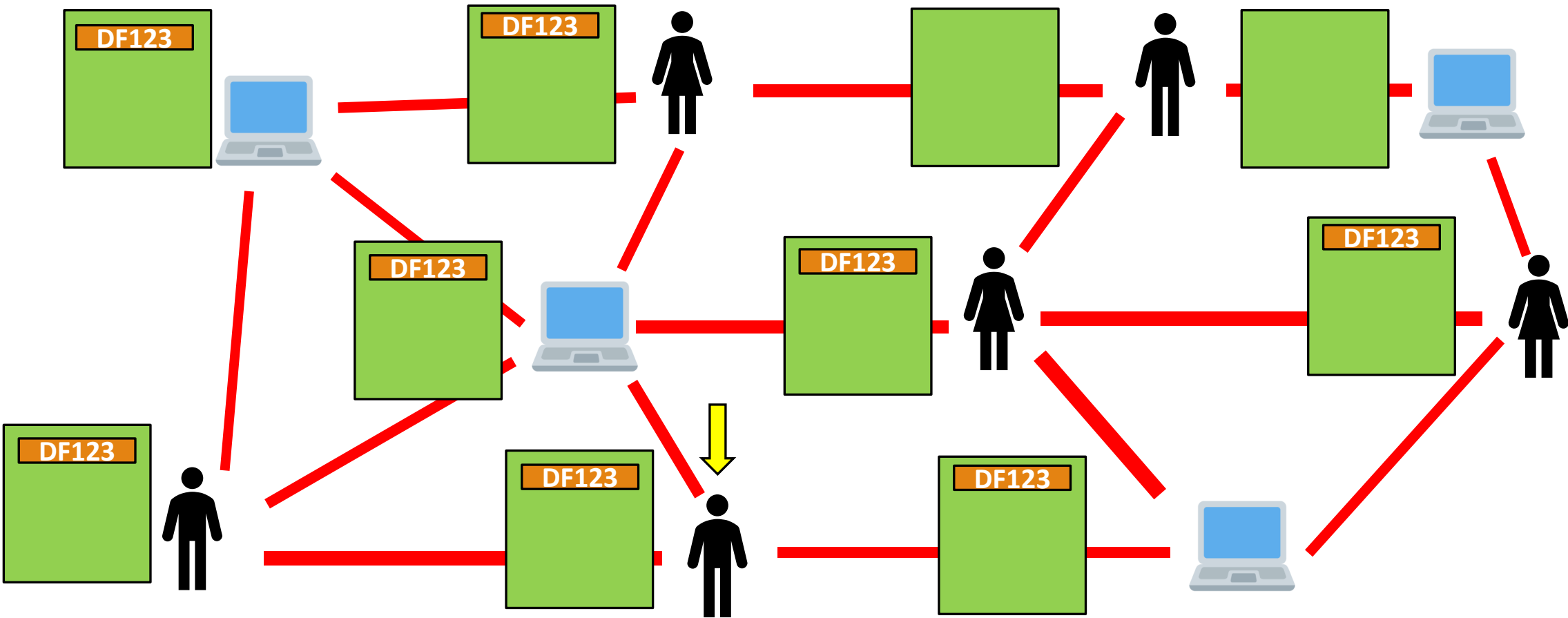
How do Mempool works?(Behind the scenes)



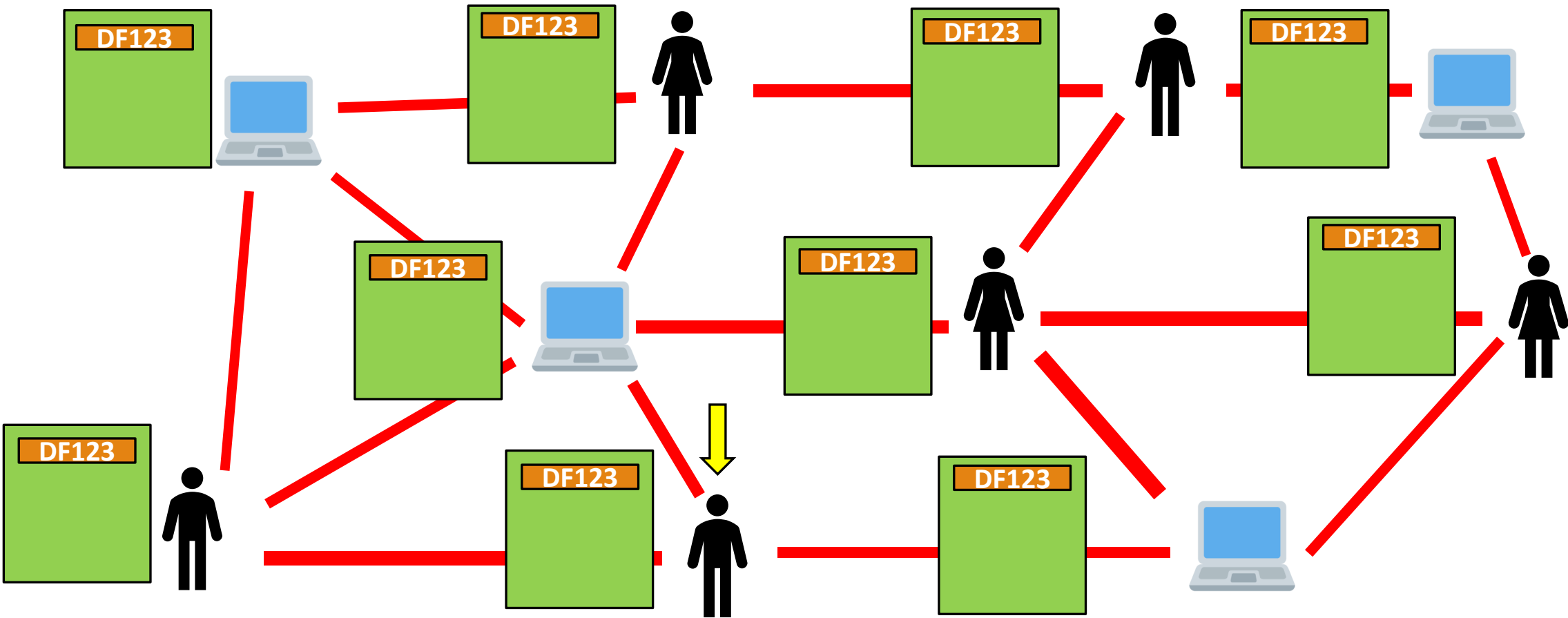
How do Mempool works?(Behind the scenes)



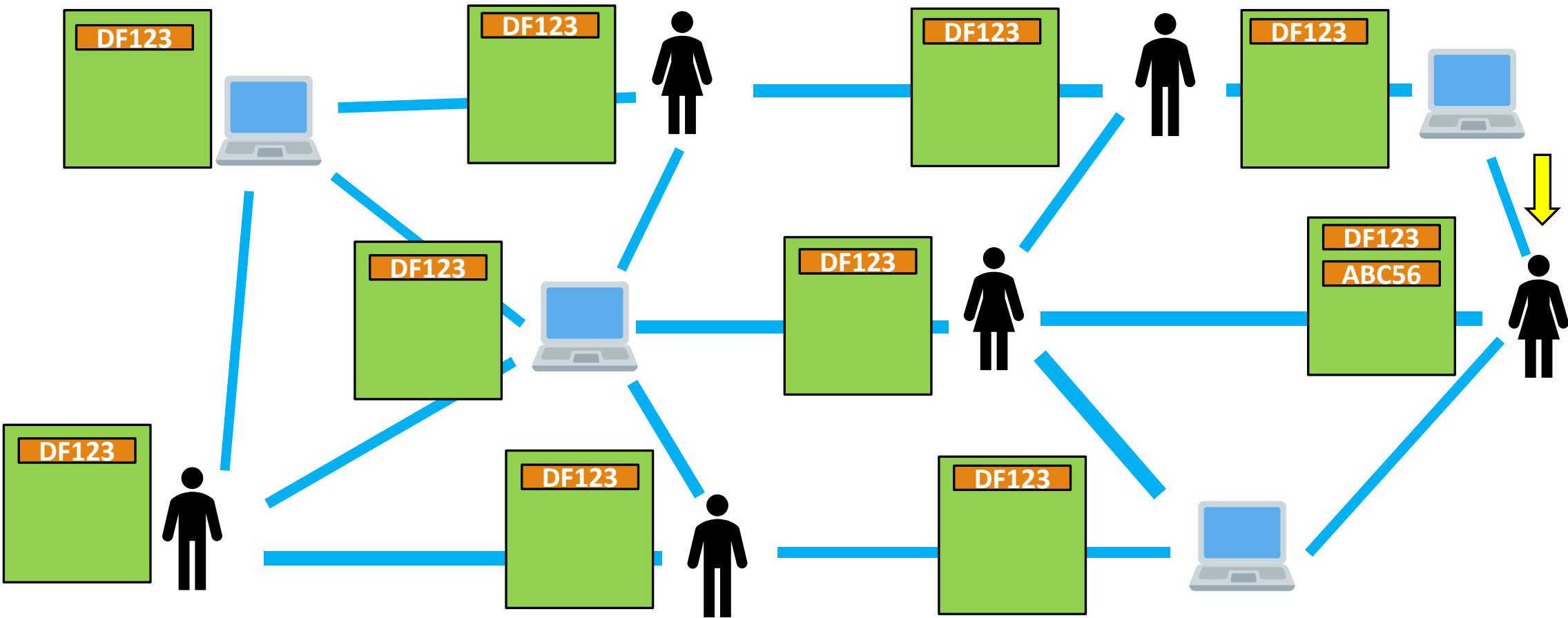
How do Mempool works?(Behind the scenes)



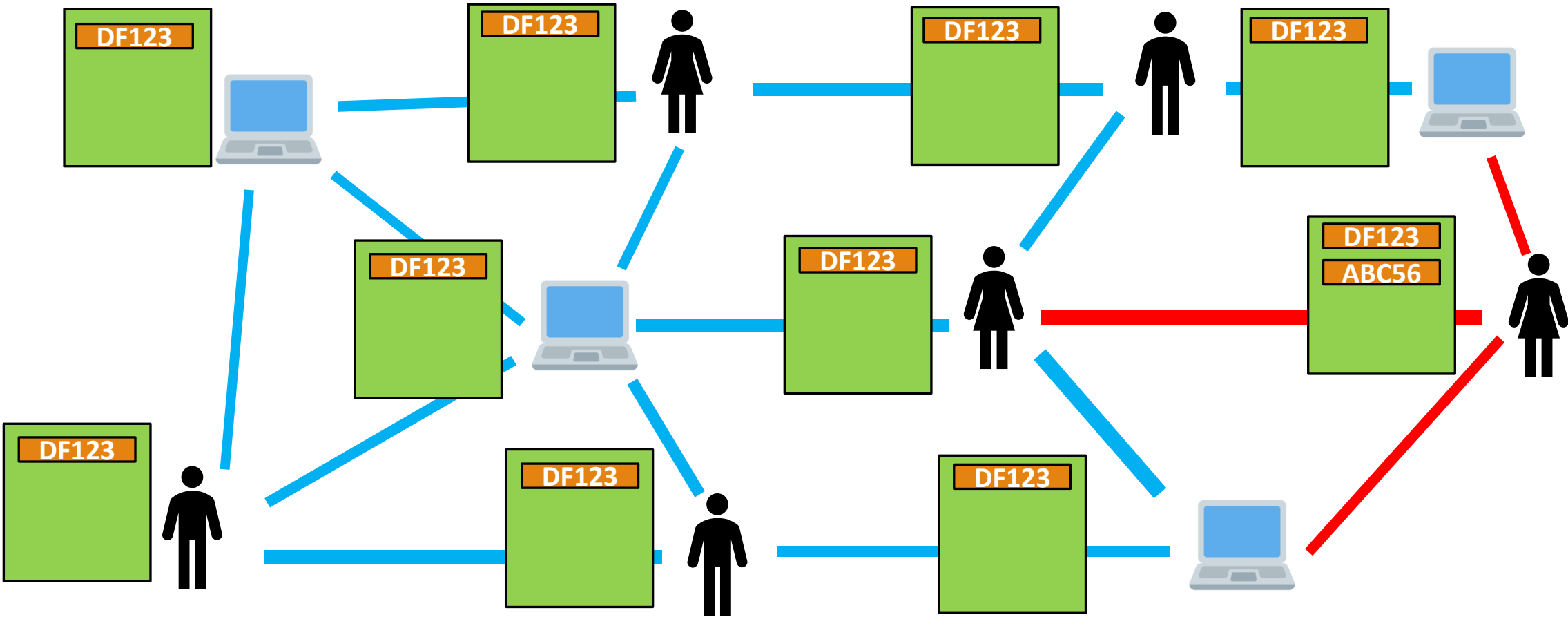
How do Mempool works?(Behind the scenes)



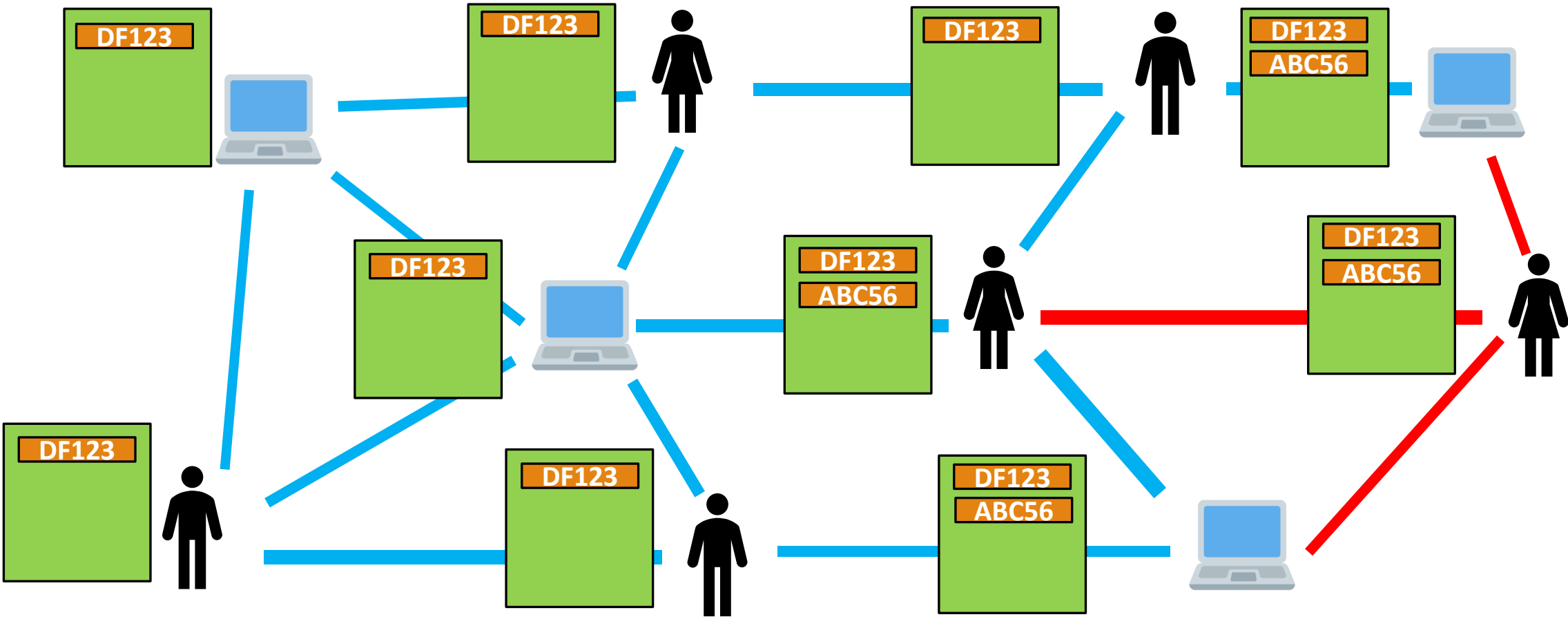
How do Mempool works?(Behind the scenes)



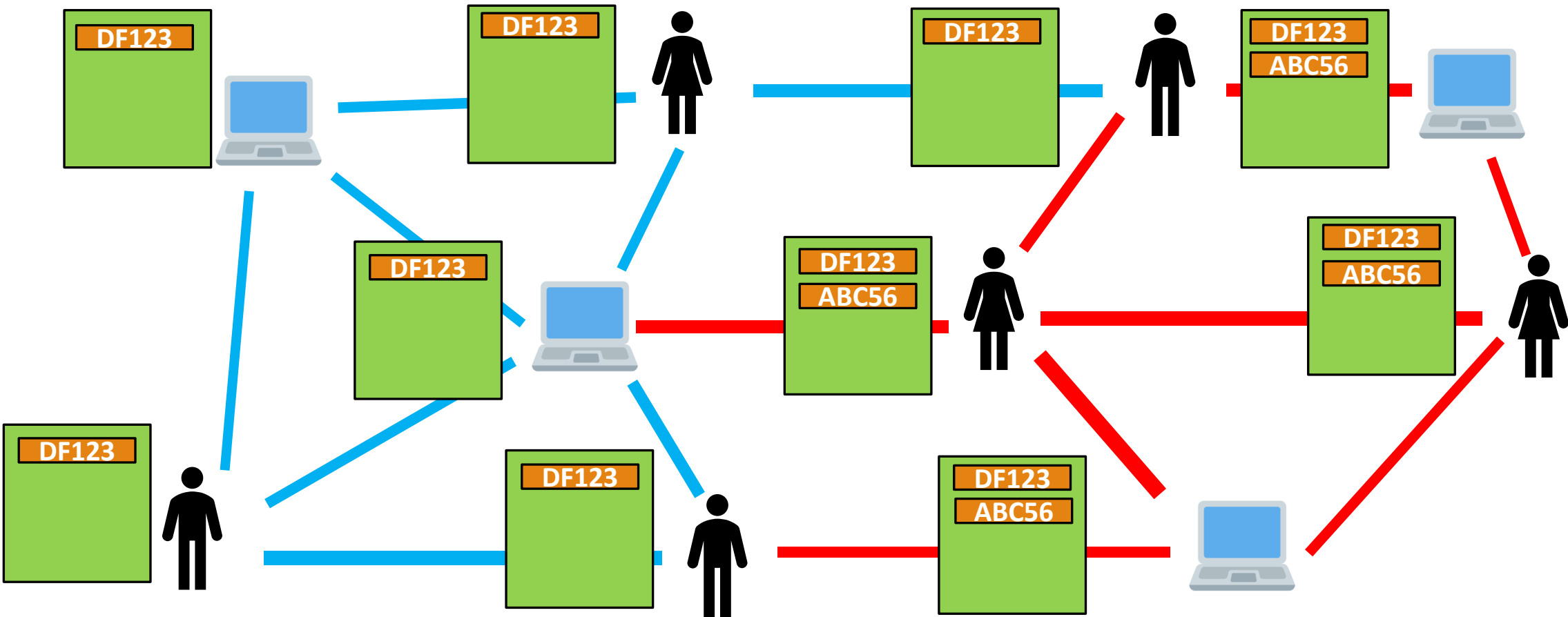
How do Mempool works?(Behind the scenes)



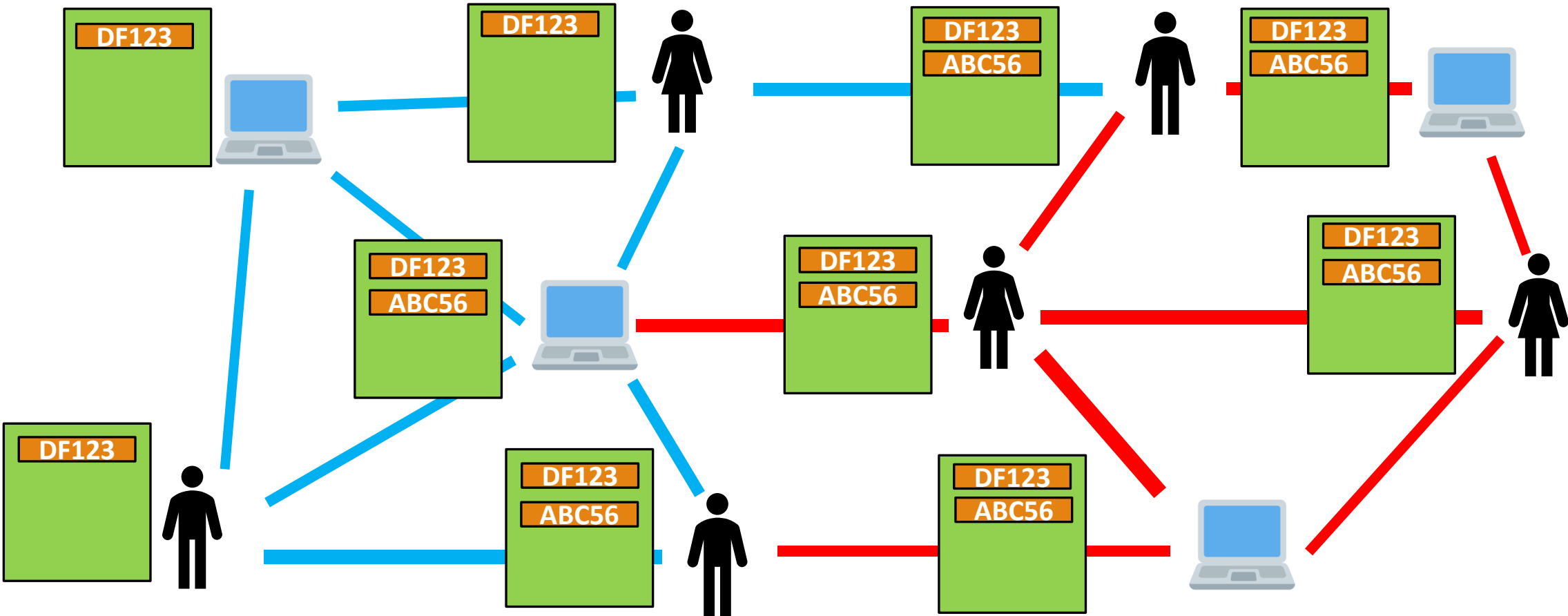
How do Mempool works?(Behind the scenes)



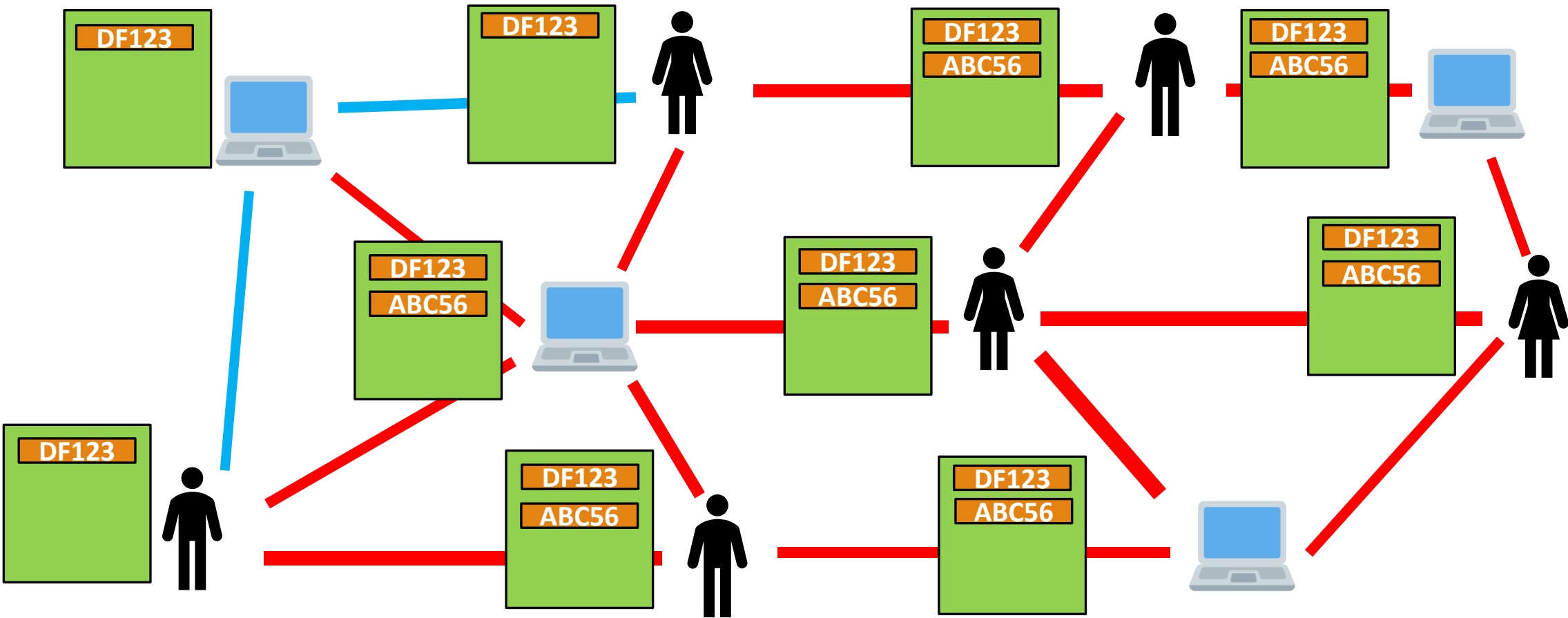
How do Mempool works?(Behind the scenes)

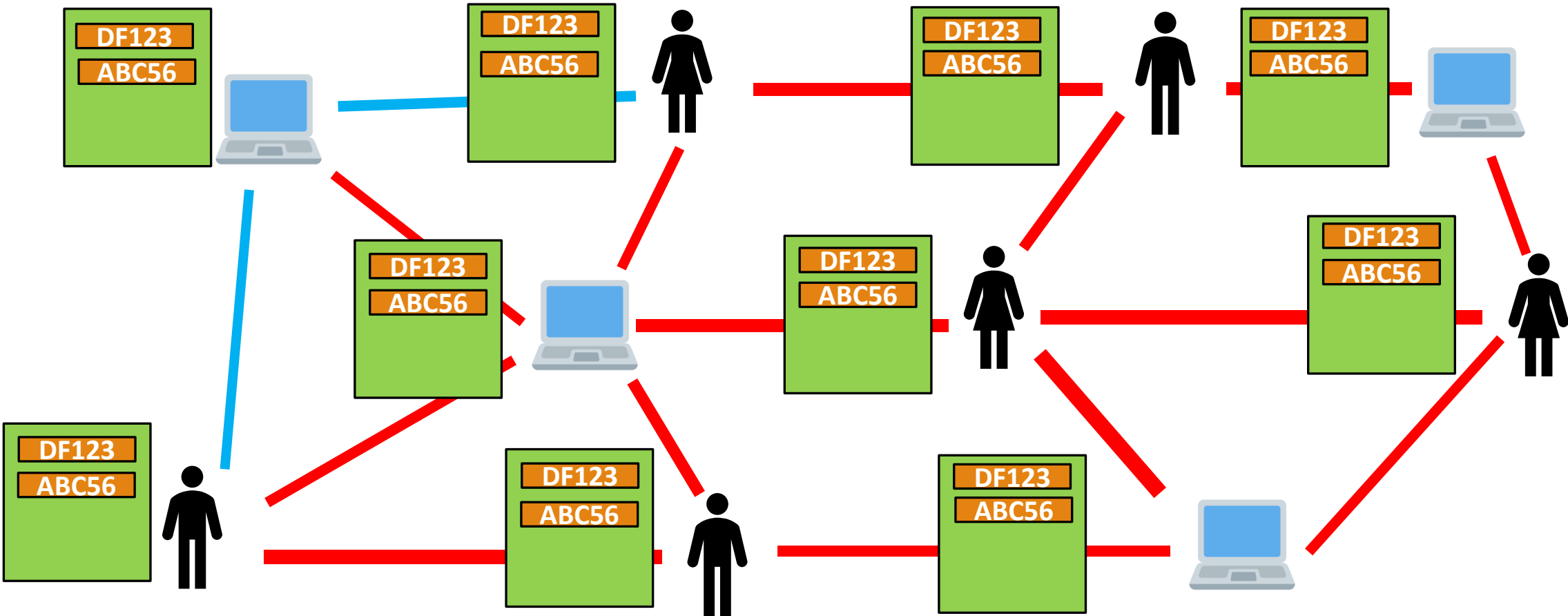


How do Mempool works?(Behind the scenes)

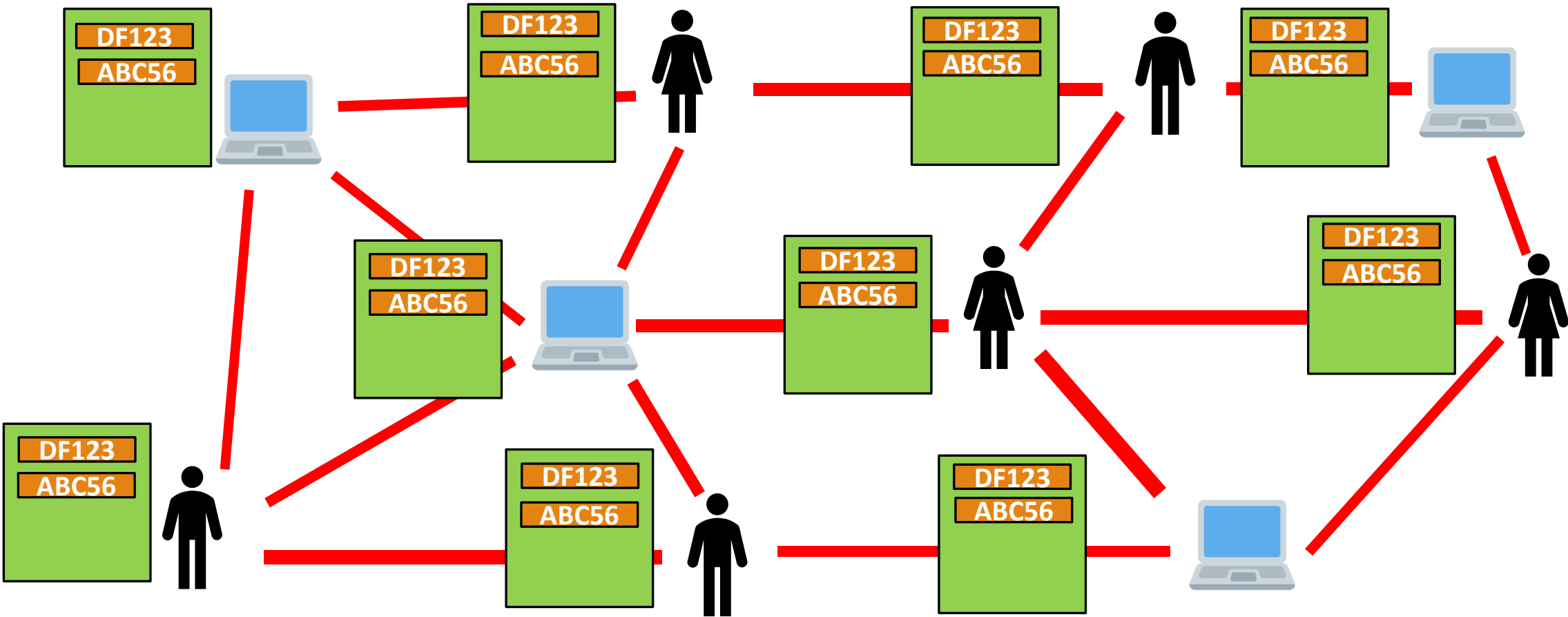


How do Mempool works?(Behind the scenes)

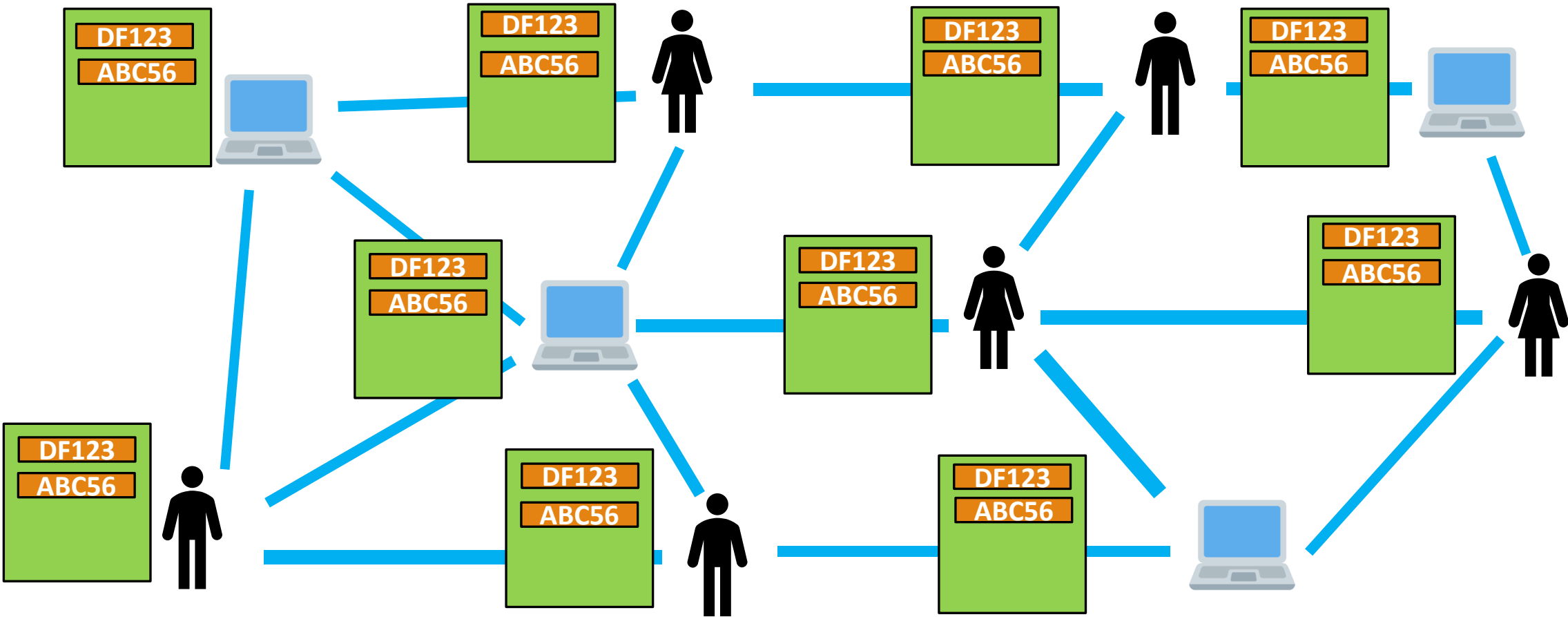


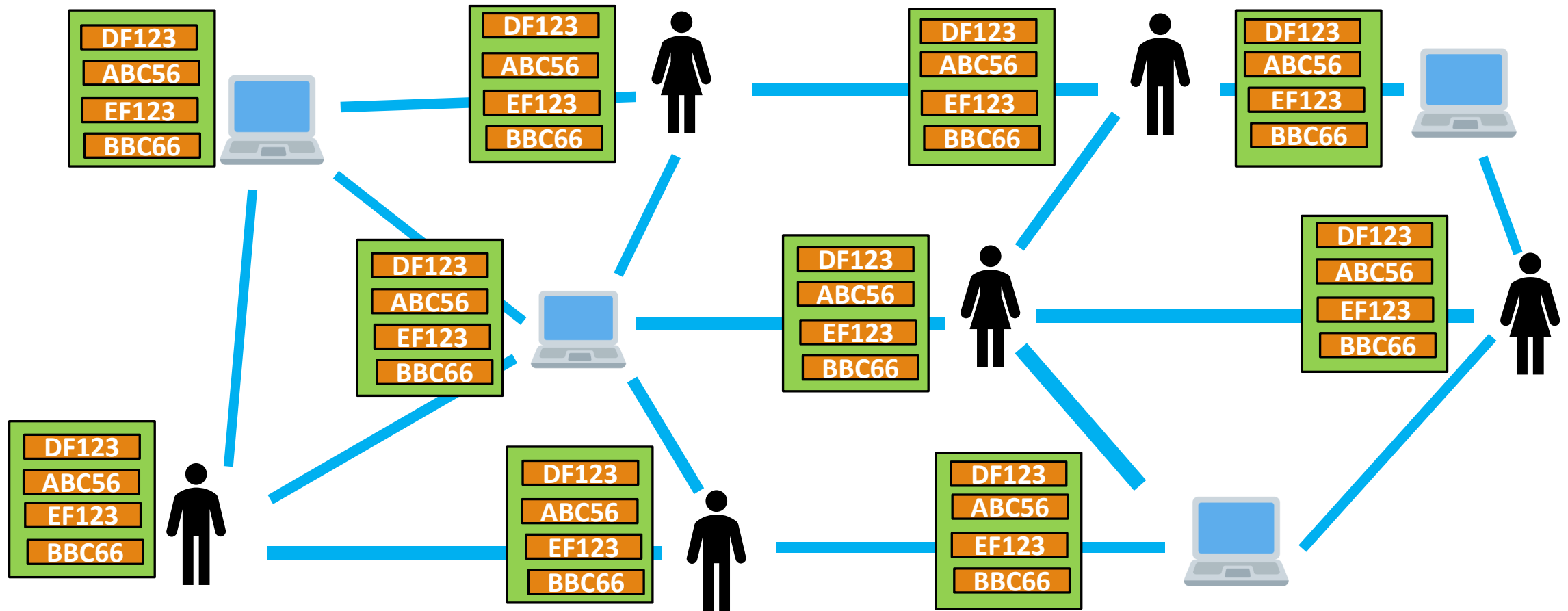


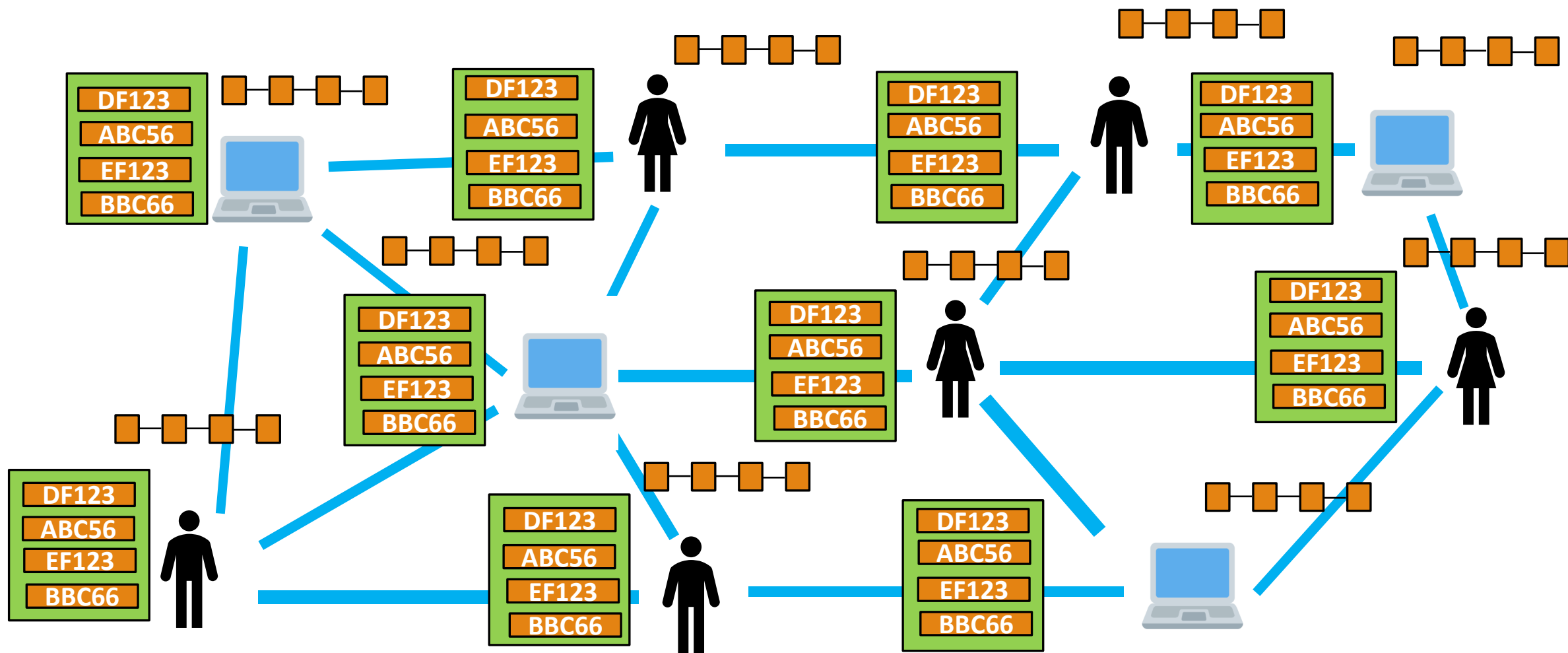
How do Mempool works?(Behind the scenes)

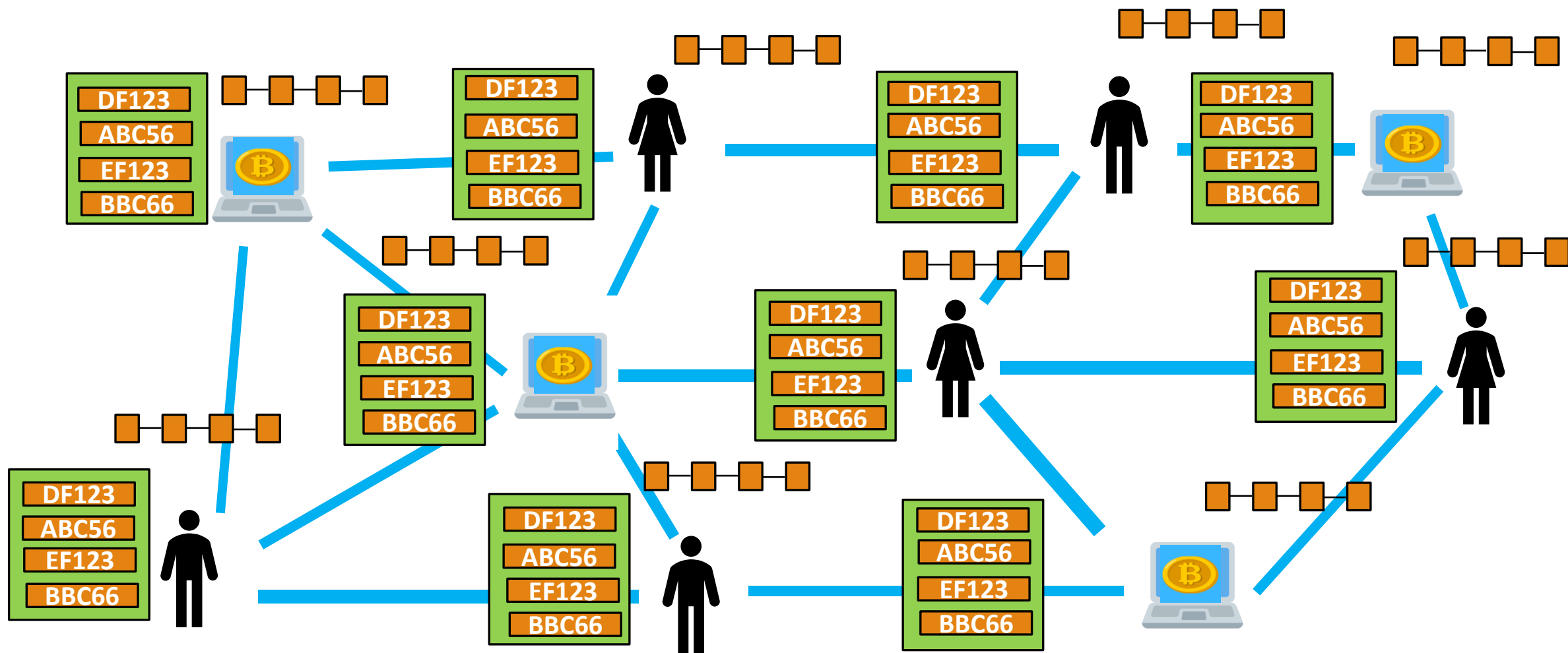


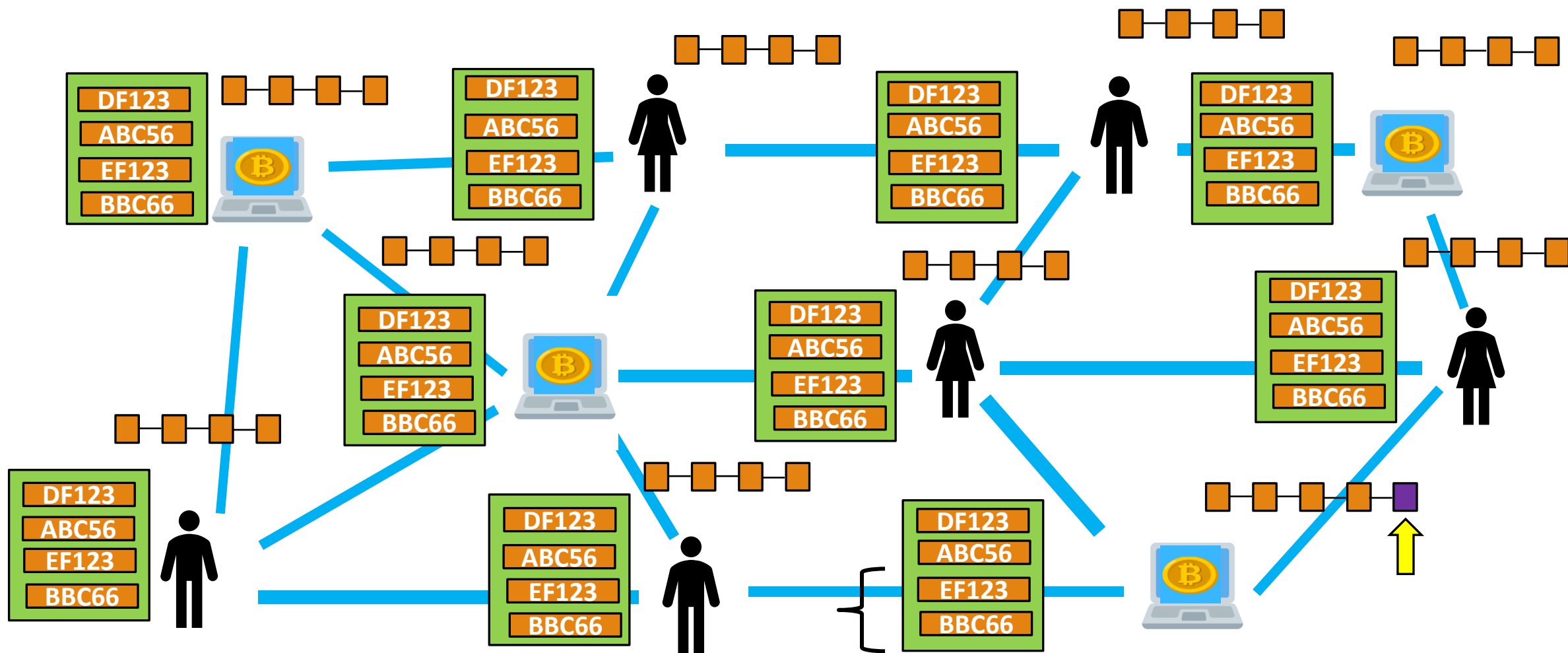
How do Mempool works?(Behind the scenes)

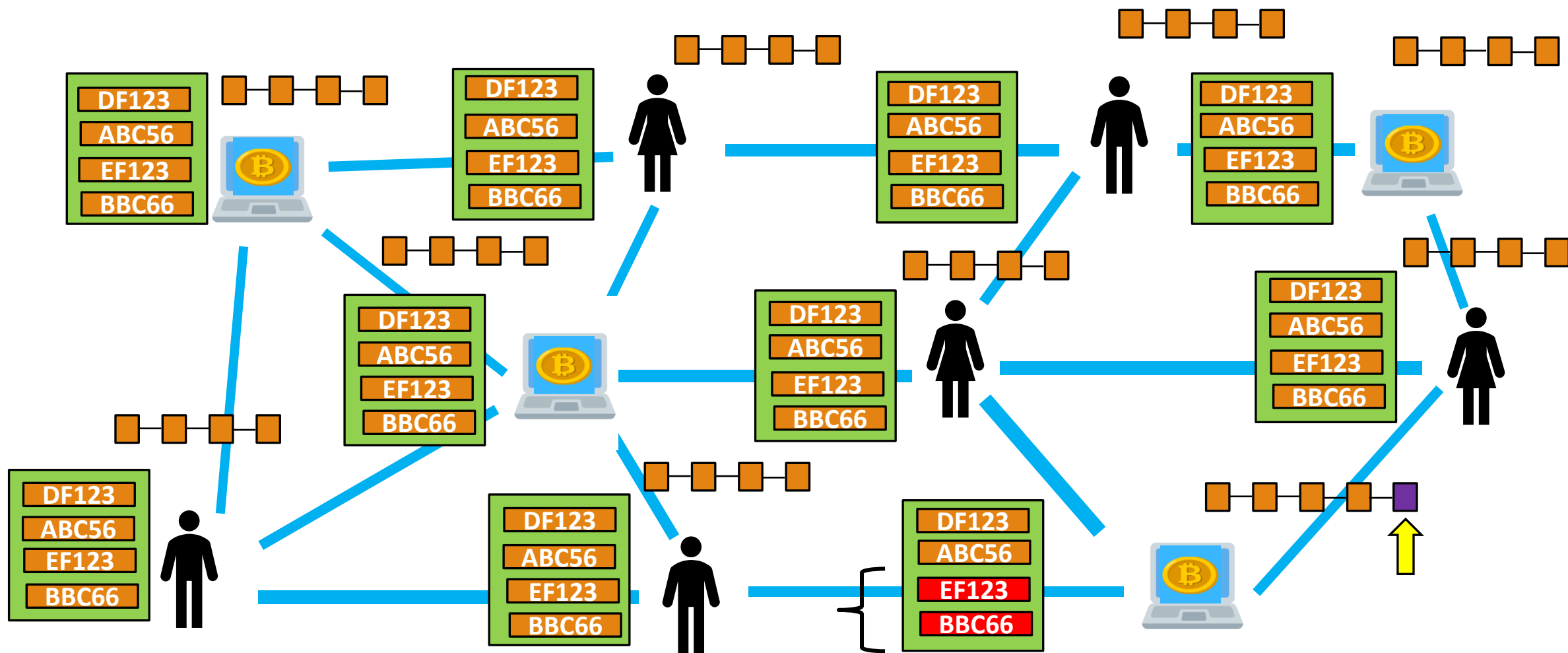


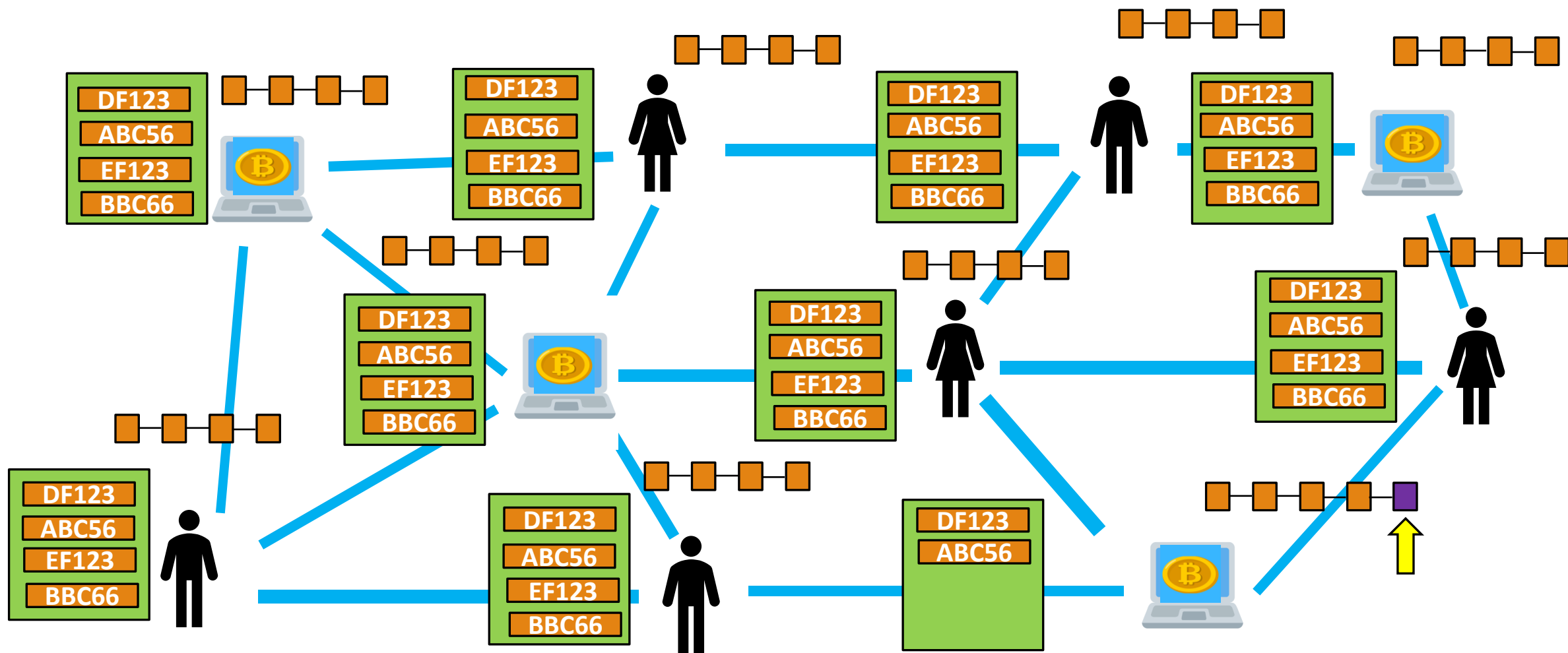


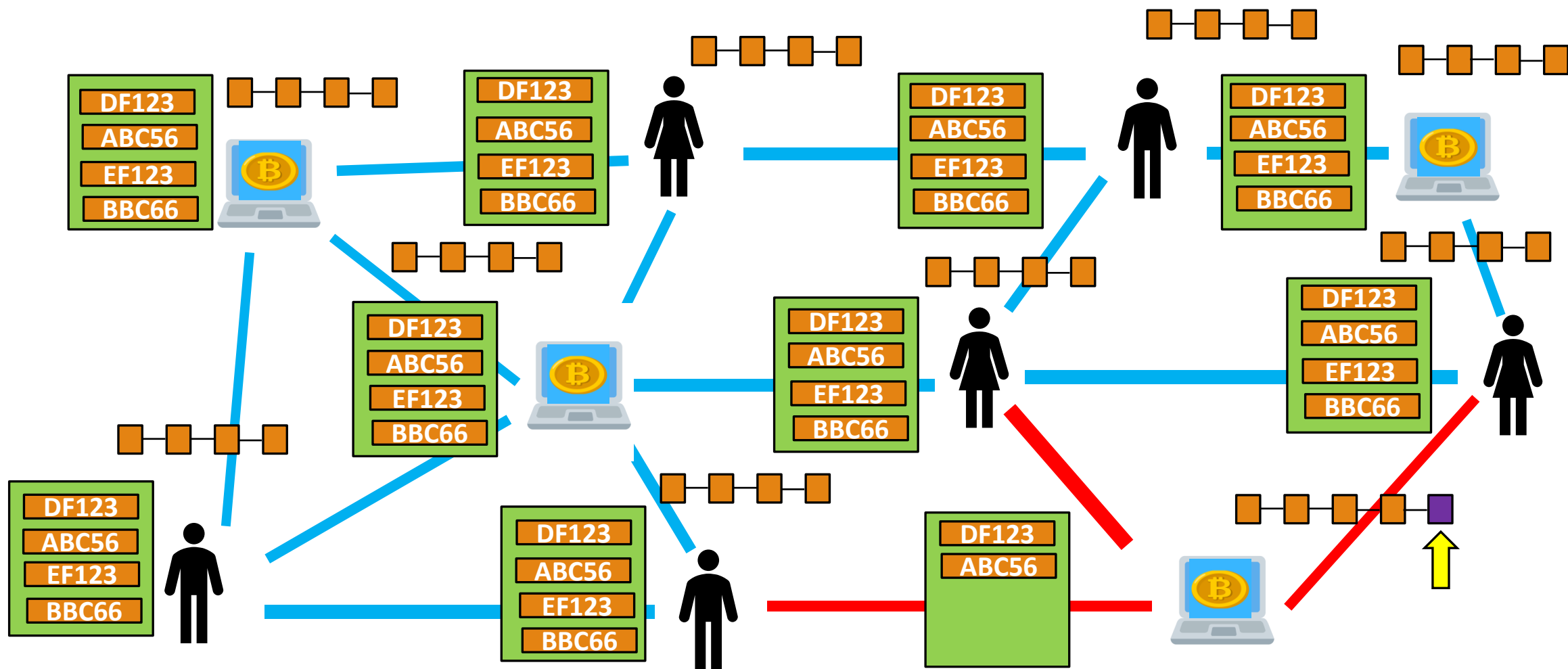


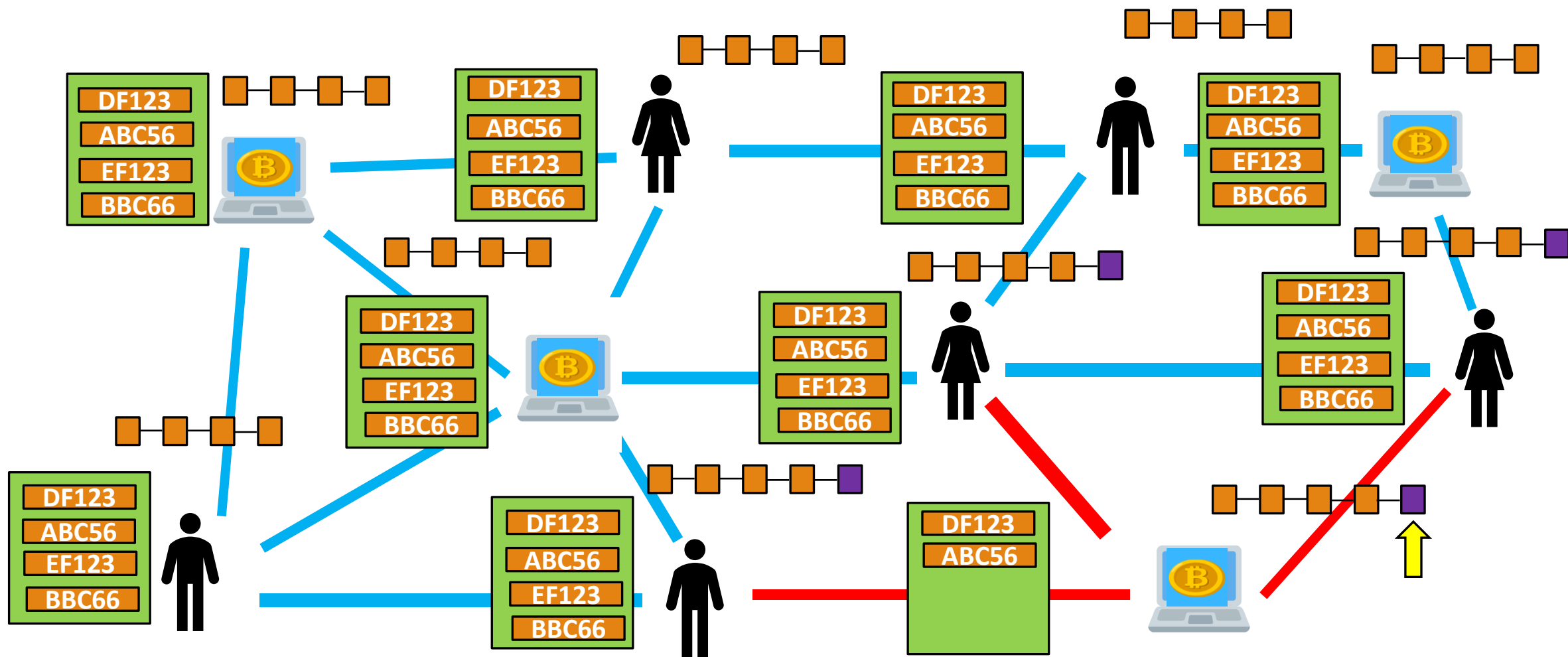


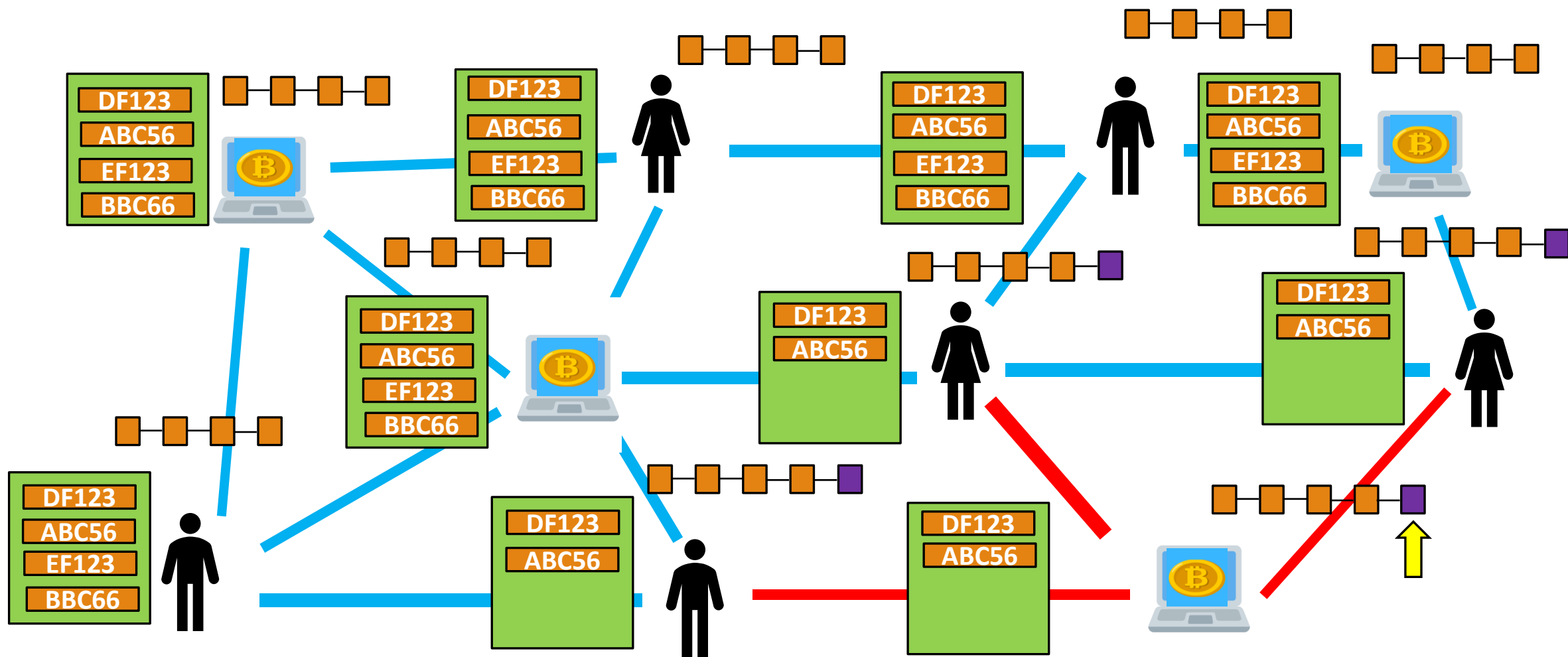


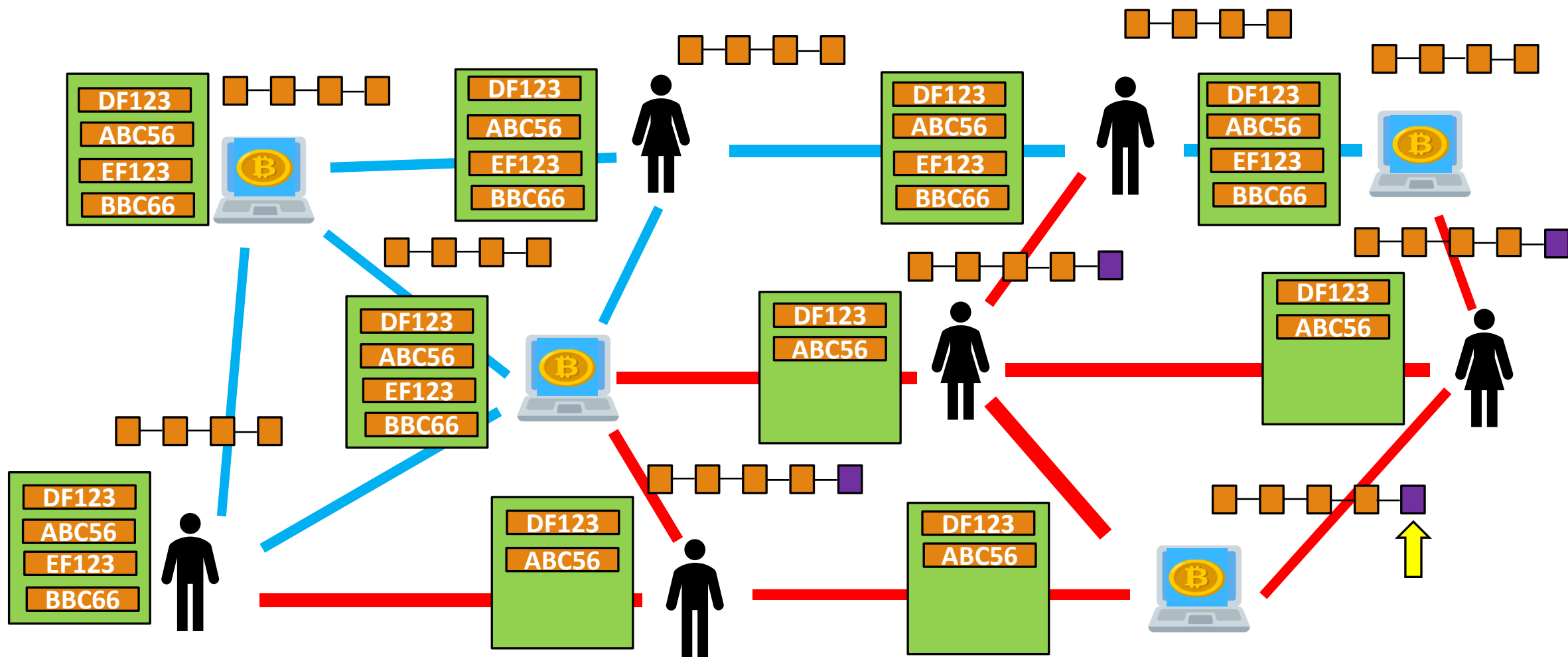


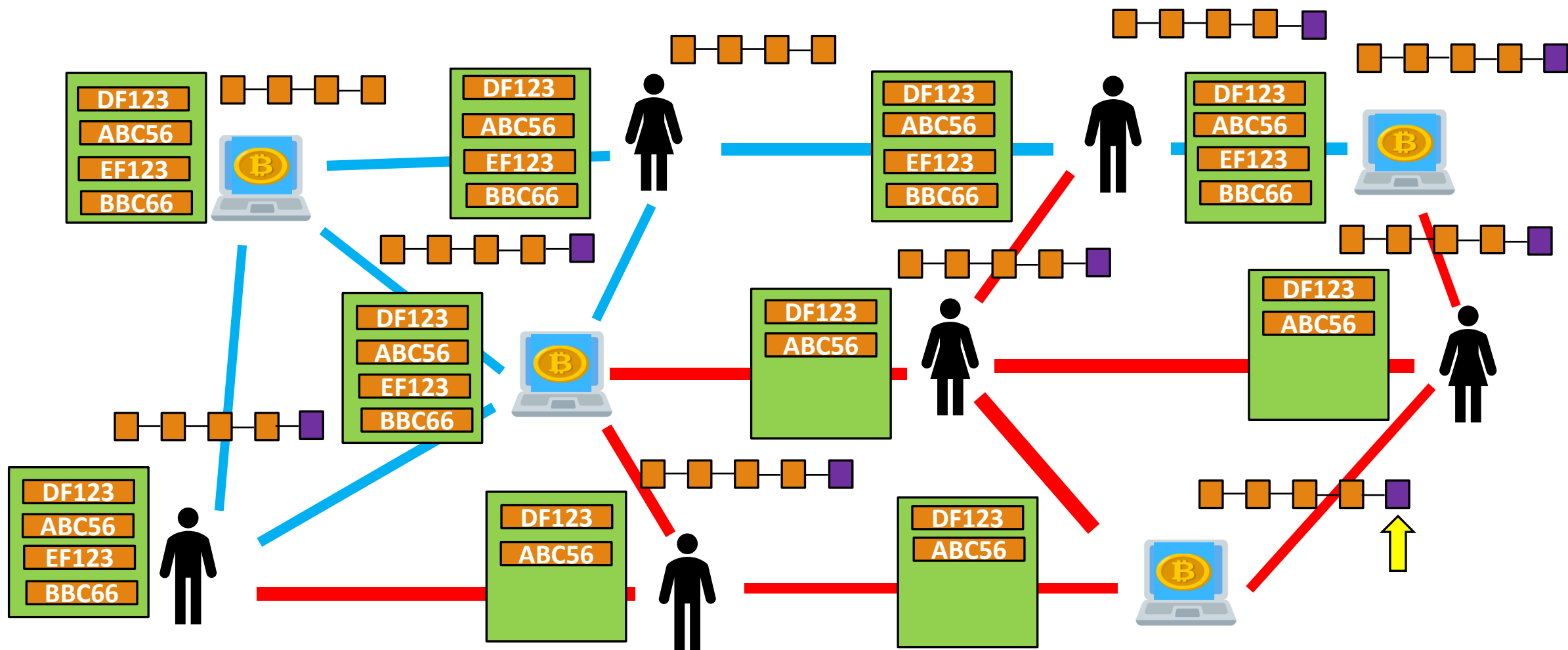


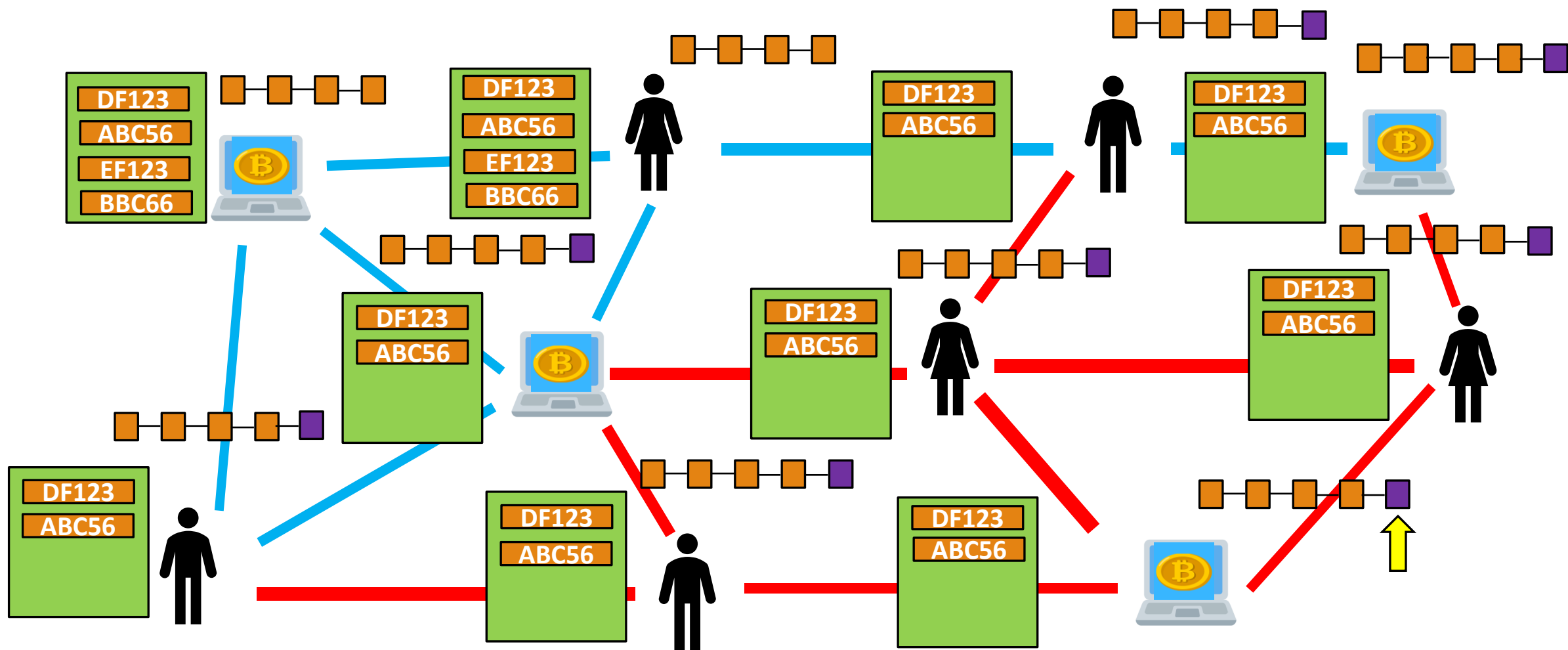


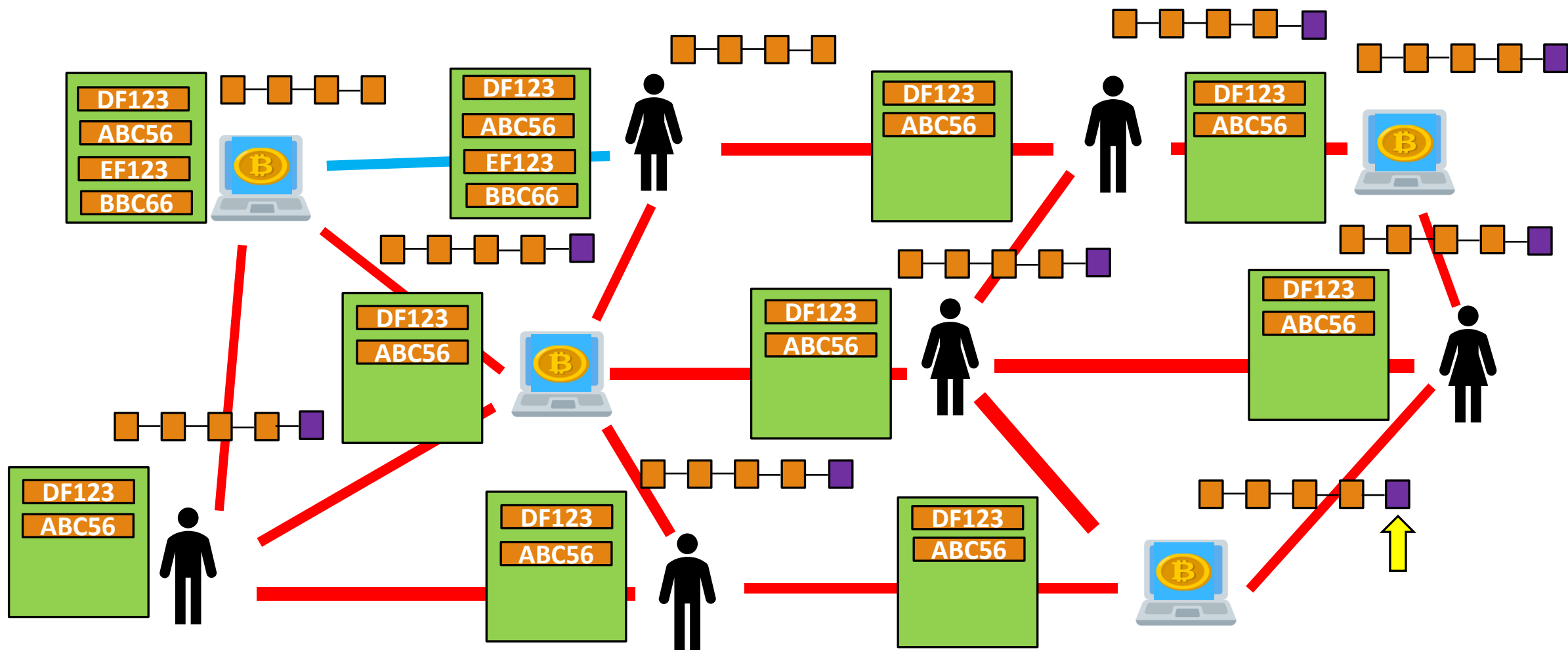


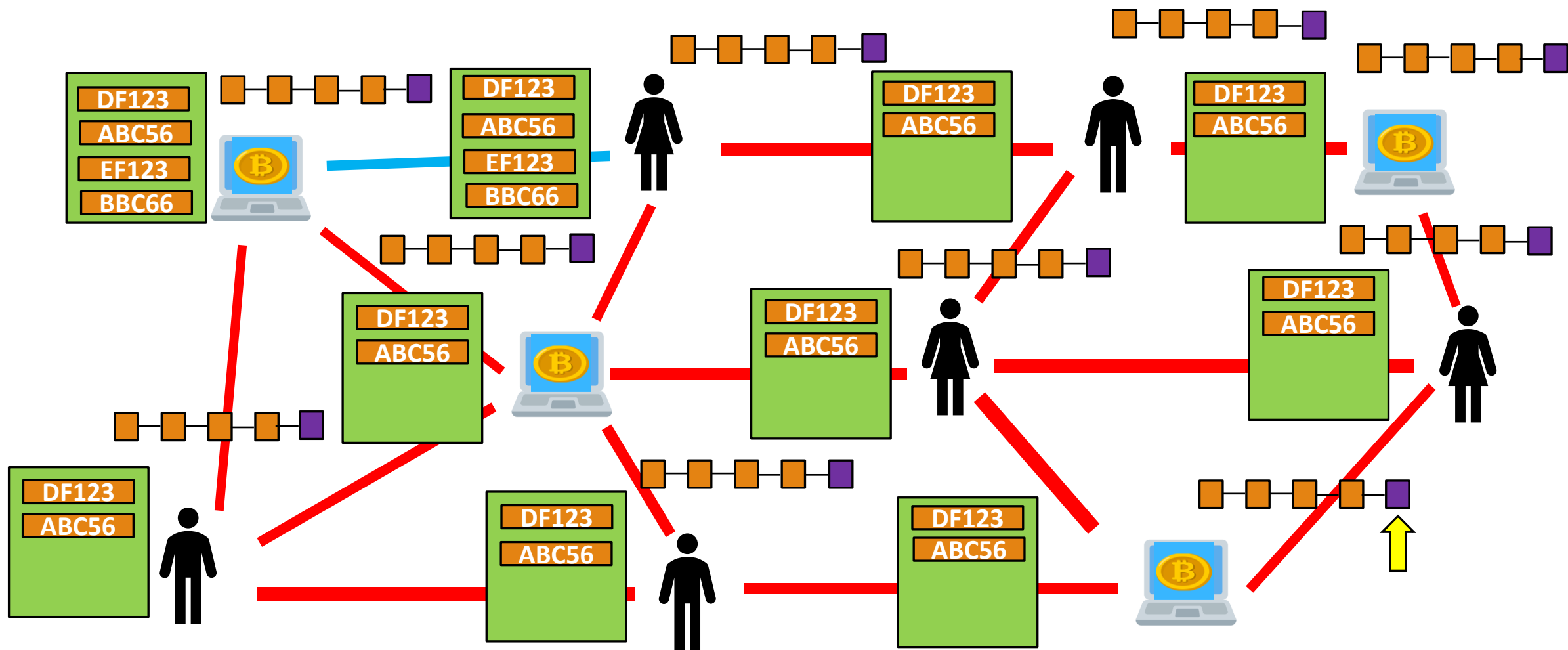


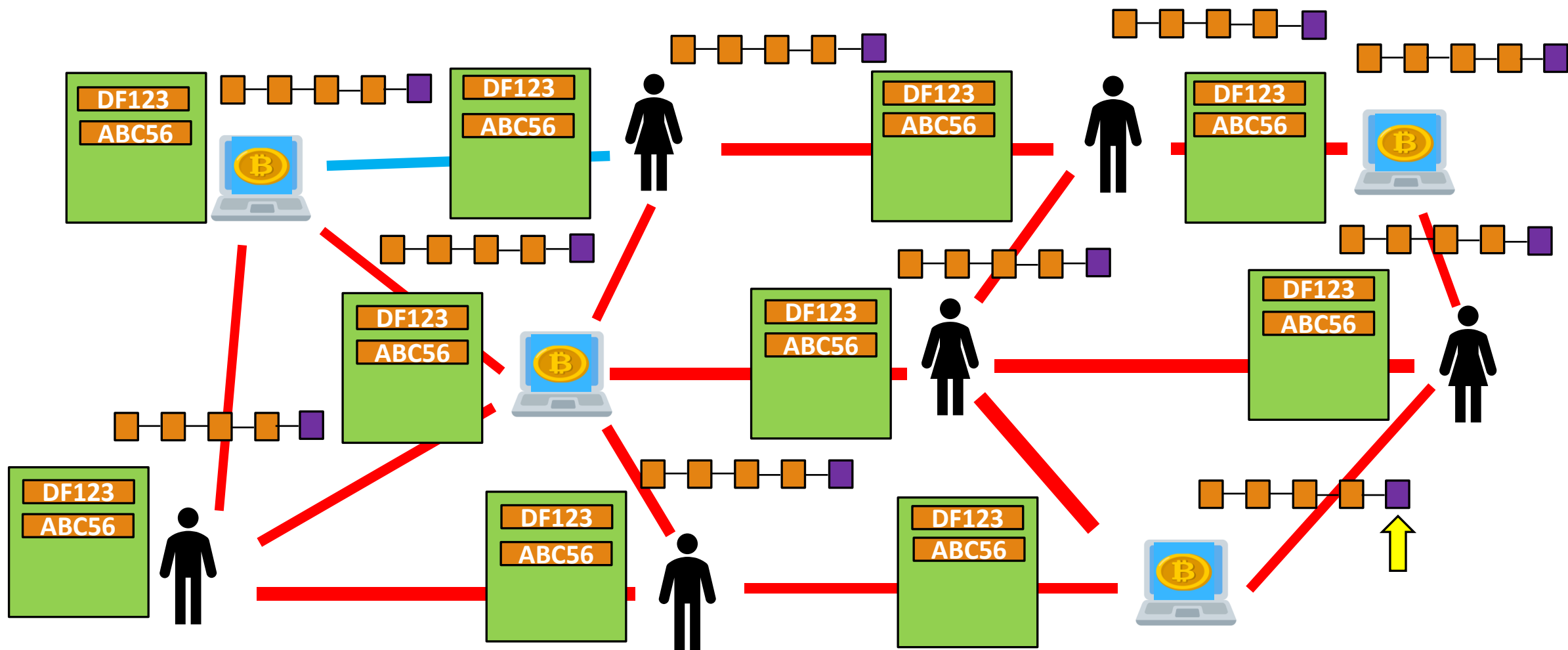


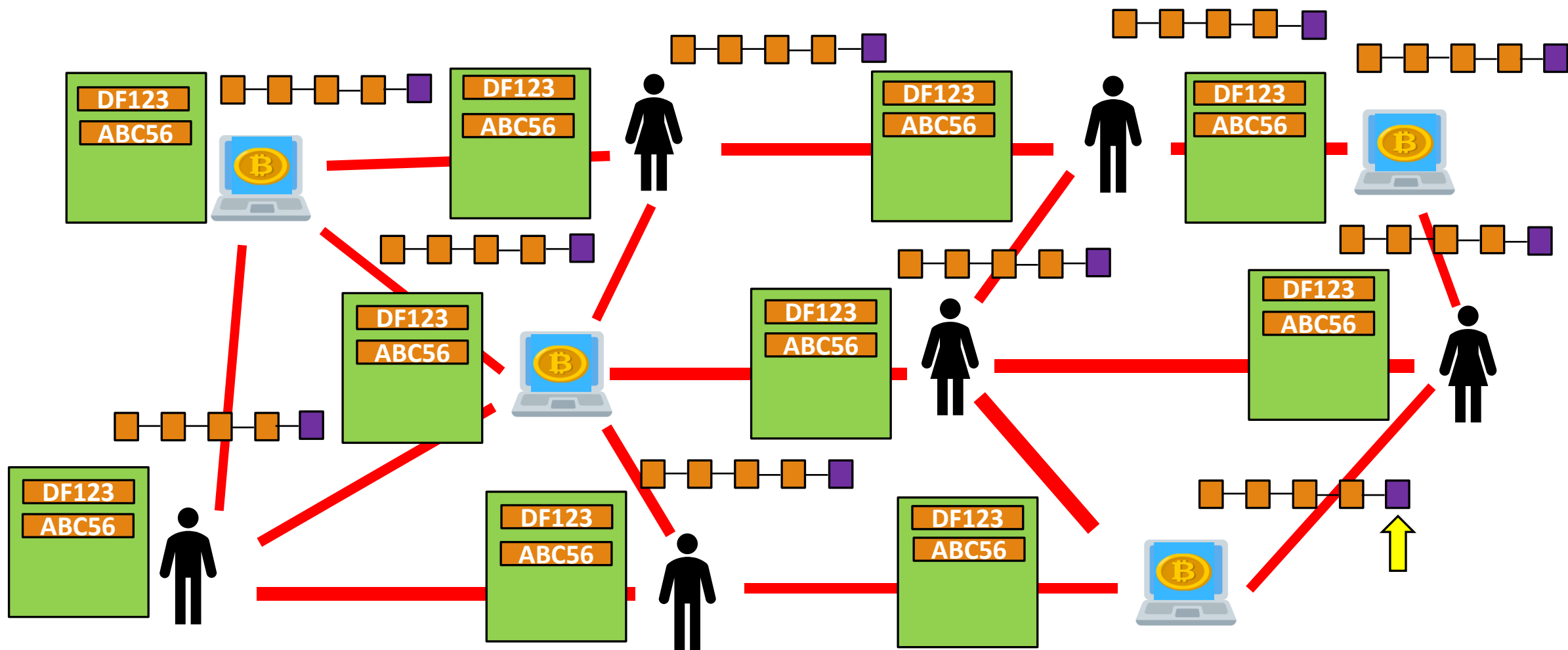


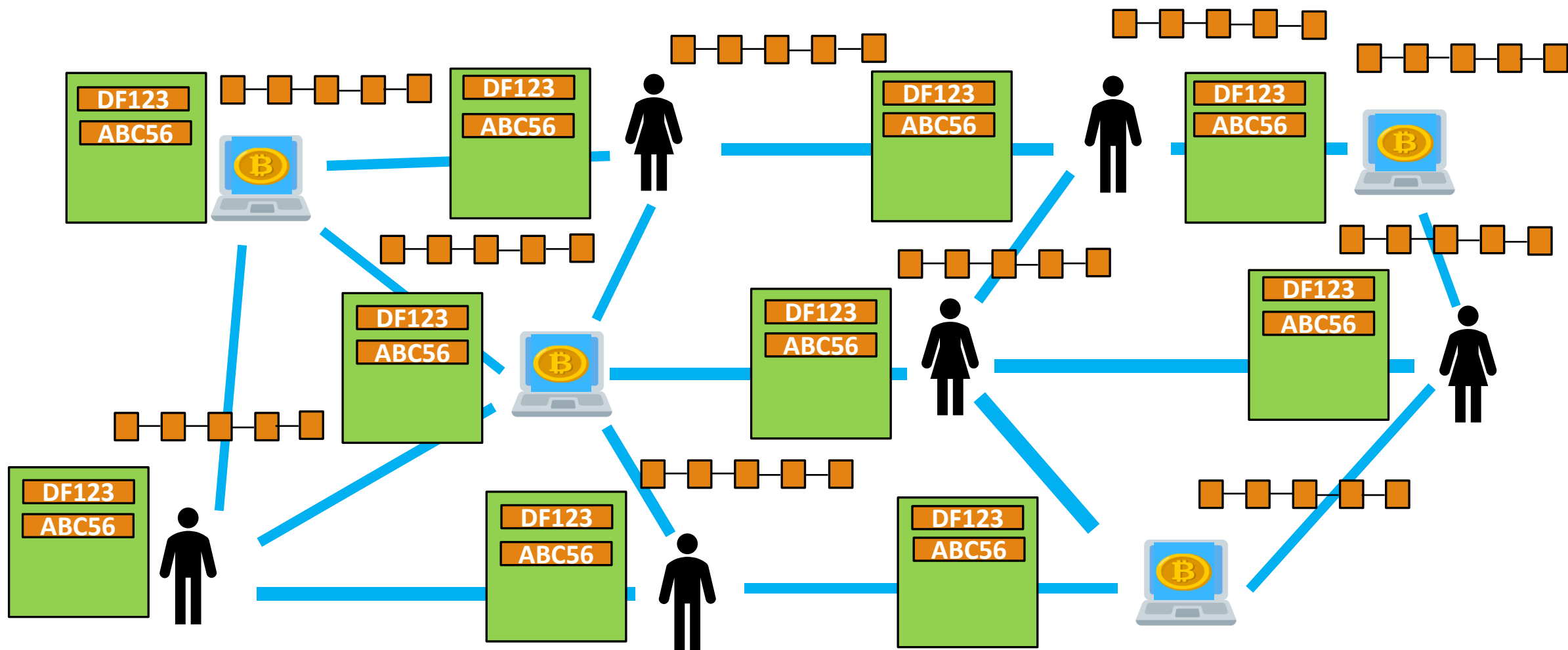


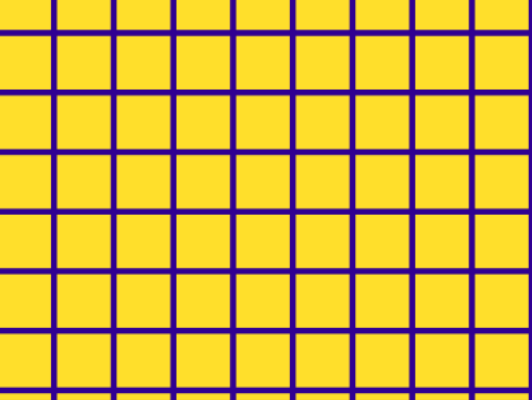












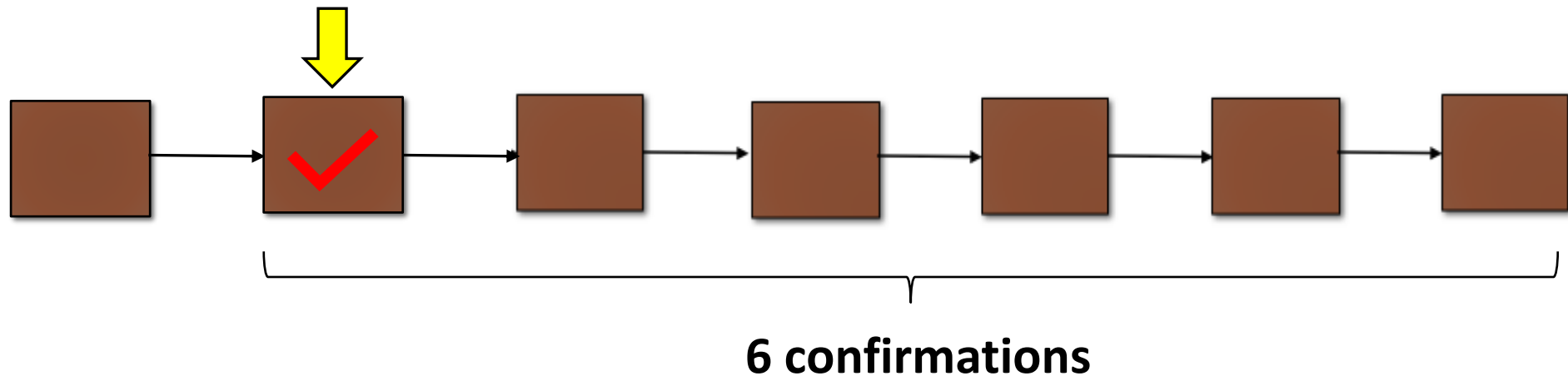
GIVE THIS VIDEO A THUMBS-UP !

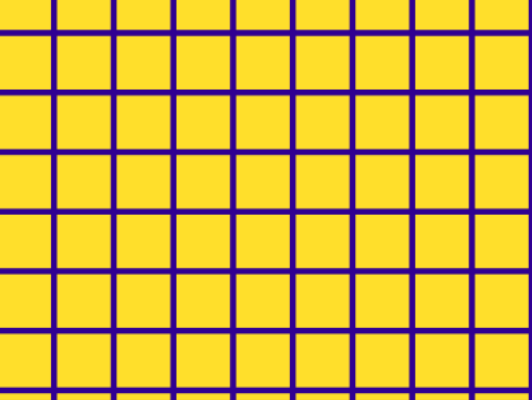
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Orphaned Blocks

- It does not matter who mined the block first. The only thing that matters is who has the longest chain.
- Ideally wait for 6 confirmations before considering the transaction to be successful.



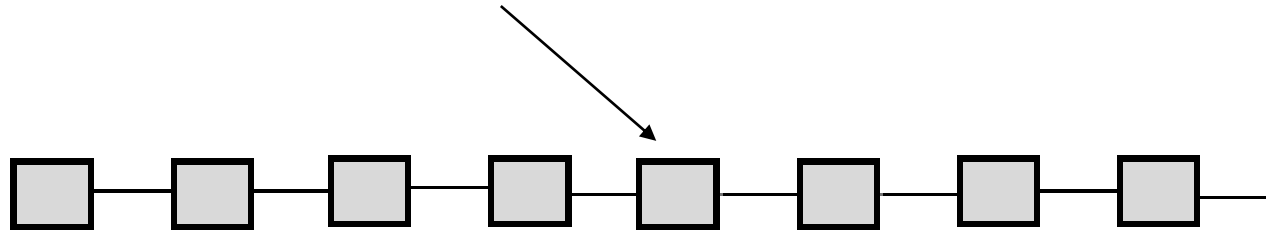


GIVE THIS VIDEO A THUMBS-UP !

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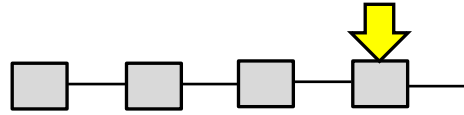


The 51% Attack

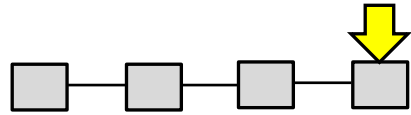


- The 51% attack is not about 51% of the network conspiracy against a single node as it is practically impossible.
- The 51% attack is about having the control over the hashing power.

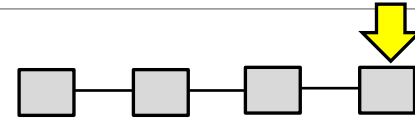
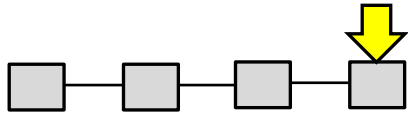
The 51% Attack



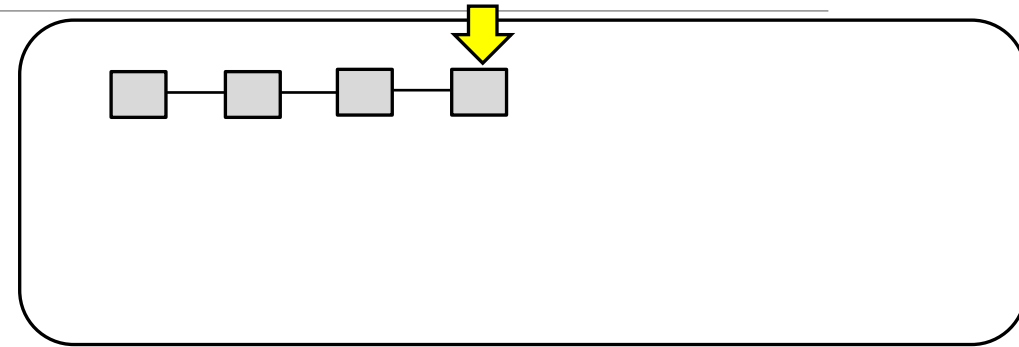
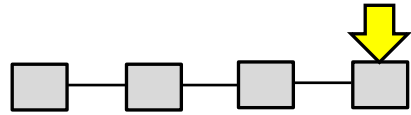
The 51% Attack



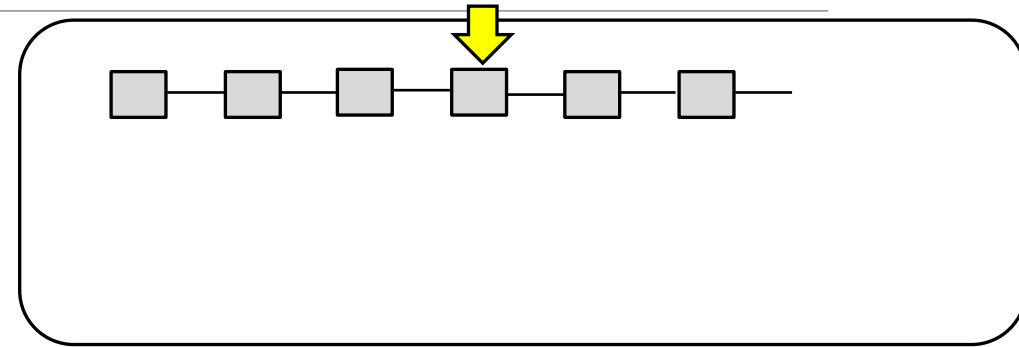
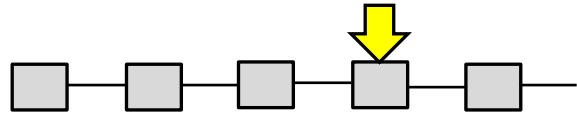
The 51% Attack



The 51% Attack



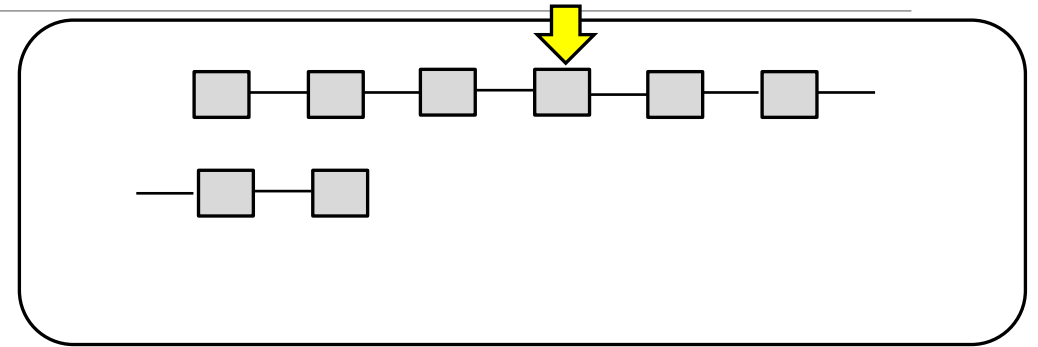
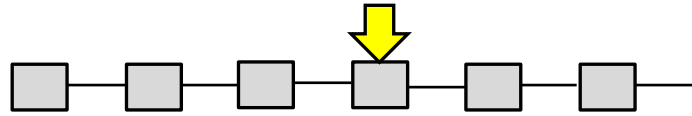
The 51% Attack



~~Broadcast~~



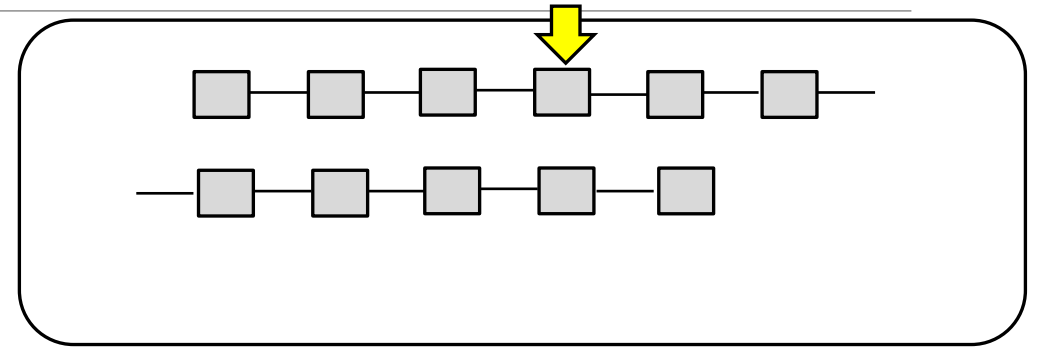
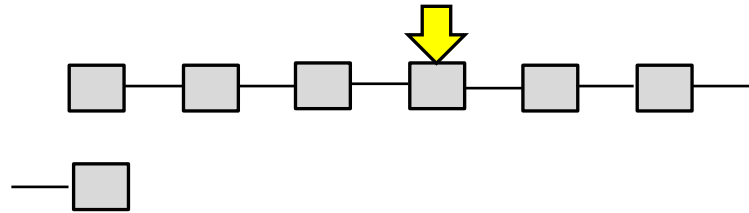
The 51% Attack



Broadcast



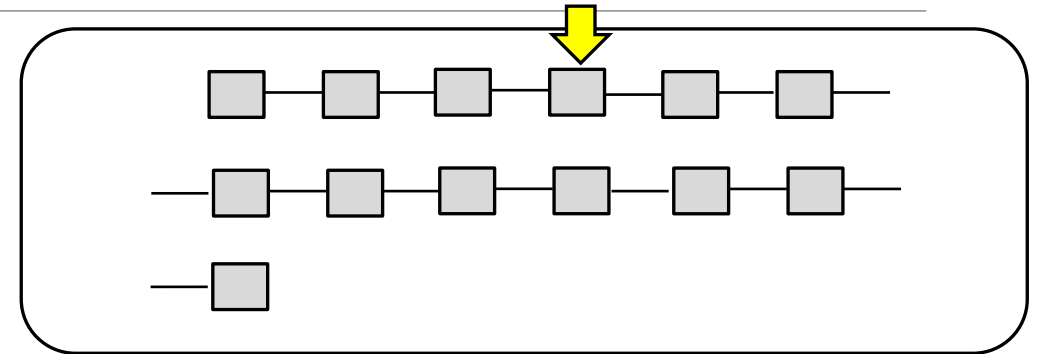
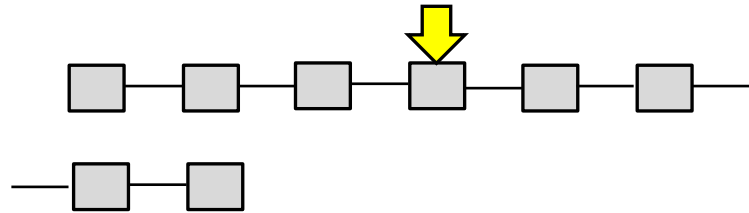
The 51% Attack



Broadcast



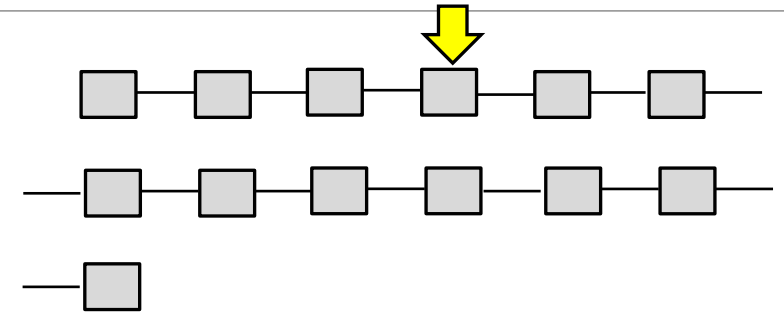
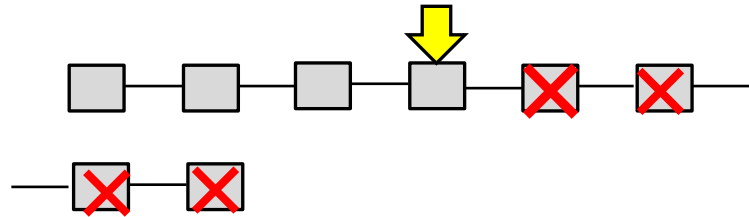
The 51% Attack



Broadcast

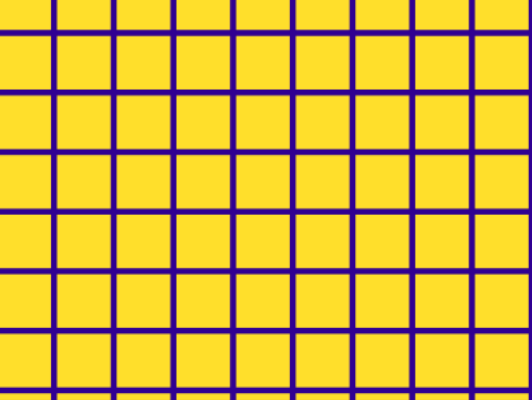


The 51% Attack



Broadcast





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Transaction and UTXOs

Transaction and UTXOs

Arjun-> Me 0.4 BTC

Raj -> Me 0.3 BTC

Alice -> Me 0.7 BTC

Bob -> Me 0.1 BTC

UTXOs

Let say I buy coffee for 0.5 BTC.



Transaction:

Input:

0.7 BTC from Alice

Output:

0.5 BTC to the coffee shop.

0.2 BTC back to me.

Transaction and UTXOs

Arjun-> Me 0.4 BTC

Raj -> Me 0.3 BTC

~~Alice -> Me 0.7 BTC~~

Bob -> Me 0.1 BTC

UTXOs

Let say I buy coffee for 0.5 BTC.



Transaction :

Input:

0.7 BTC from Alice

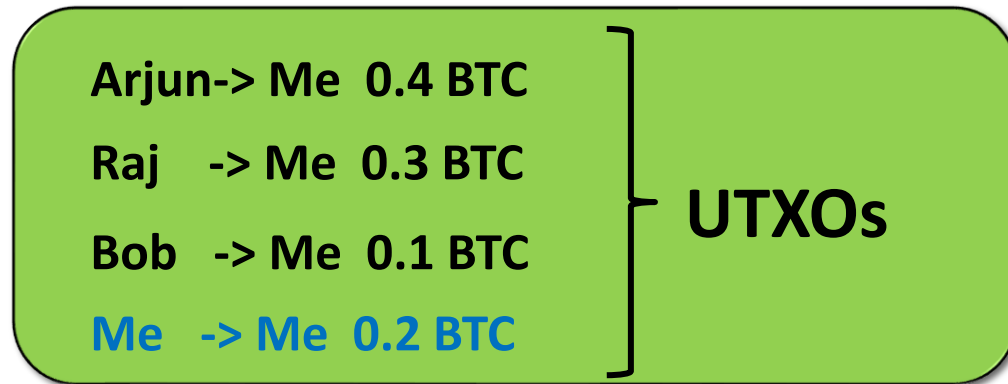
Output:

0.5 BTC to the coffee shop.

0.2 BTC back to me.

UTXO for the coffee shop.

Transaction and UTXOs



Transaction and UTXOs

Arjun-> Me 0.4 BTC

Raj -> Me 0.3 BTC

Alice -> Me 0.7 BTC

Bob -> Me 0.1 BTC

UTXOs

Let say I buy Noodles for 1.4 BTC.



Transaction :

Input:

Arjun-> Me 0.4 BTC

Raj -> Me 0.3 BTC

Alice -> Me 0.7 BTC

Output:

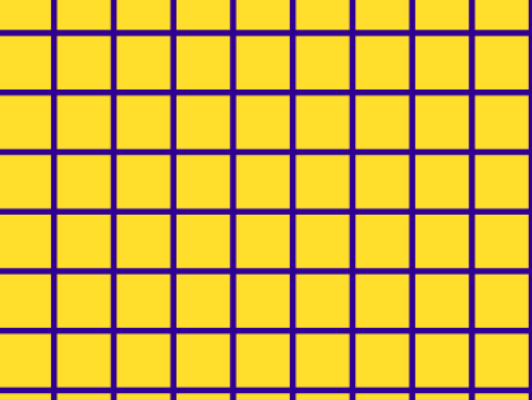
1.4 BTC to the noodles shop.

UTXO for the noodle shop.

Transaction and UTXOs

Bob -> Me 0.1 BTC } UTXOs





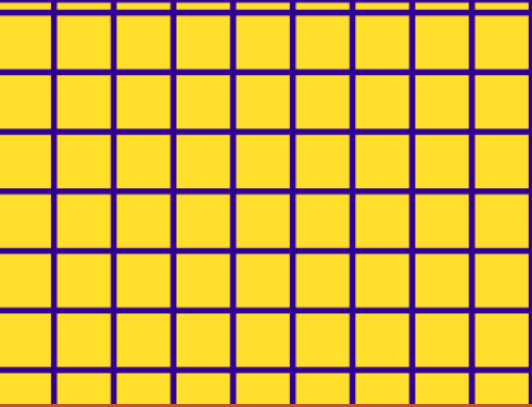
GIVE THIS VIDEO A THUMBS-UP !

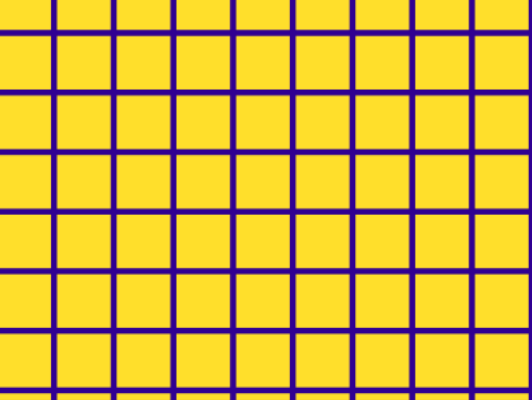
CODE EATER

Instagram - @codeeater21



Discord – Link in description





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Transaction Fee

Transaction Fee

Arjun-> Me 0.4 BTC

Raj -> Me 0.3 BTC

Alice -> Me 0.7 BTC

Bob -> Me 0.3 BTC

UTXOs

Let say I buy coffee for 0.5 BTC.



Transaction:

Input:

0.7 BTC from Alice

Output:

0.5 BTC to the coffee shop.

0.2 BTC back to me.

Transaction Fee

Arjun-> Me 0.4 BTC

Raj -> Me 0.3 BTC

Alice -> Me 0.7 BTC

Bob -> Me 0.3 BTC

UTXOs

Let say I buy coffee for 0.5 BTC.



Transaction:

Input:

0.7 BTC from Alice

Output:

0.5 BTC to the coffee shop.

0.1 BTC back to me.

Transaction Fee

Arjun-> Me 0.4 BTC

Raj -> Me 0.3 BTC

Alice -> Me 0.7 BTC

Bob -> Me 0.3 BTC

UTXOs

Let say I buy coffee for 0.5 BTC.



Transaction:

Input:

0.7 BTC from Alice

Output:

0.5 BTC to the coffee shop.

0.1 BTC back to me.

Fees: 0.1 BTC

Transaction Fee

Arjun-> Me 0.4 BTC

Raj -> Me 0.3 BTC

~~Alice-> Me 0.7 BTC~~

Bob -> Me 0.1 BTC

UTXOs

Let say I buy coffee for 0.5 BTC.



Transaction :

Input:

0.7 BTC from Alice

Output:

0.5 BTC to the coffee shop.

0.1 BTC back to me.

UTXO for the coffee shop.

UTXO for the miner.

Fees: 0.1 BTC

Transaction Fee

Arjun-> Me 0.4 BTC

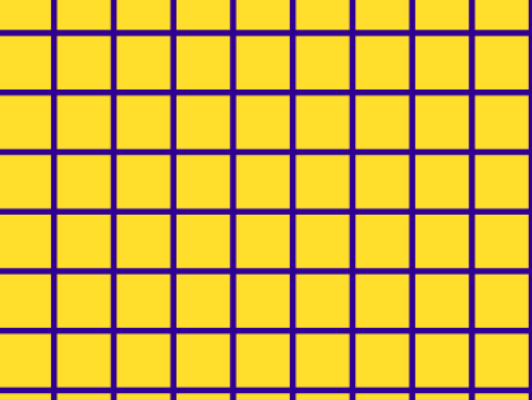
Raj -> Me 0.3 BTC

Bob -> Me 0.1 BTC

Me -> Me 0.1 BTC

UTXOs





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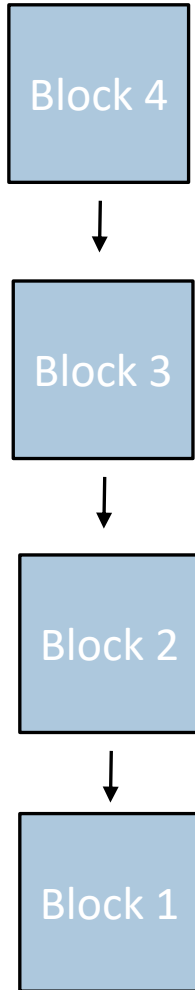
CODE EATER





Cryptocurrency Wallet

Cryptocurrency Wallets



Cryptocurrency Wallets

Block 4



Block 3



Block 2

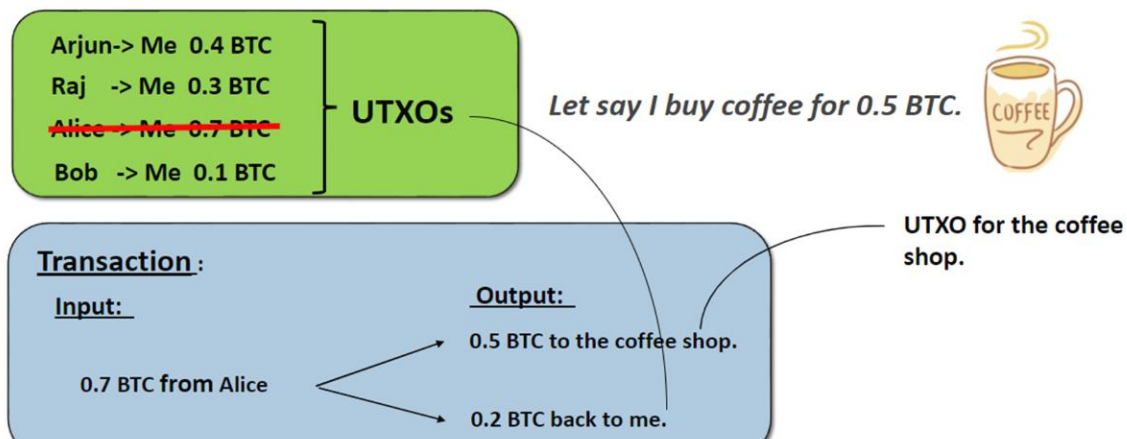


Block 1

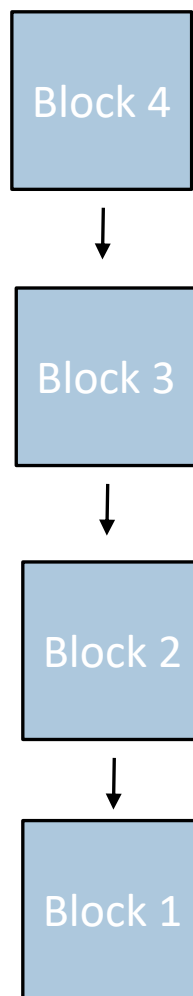
Me->coffee shop 0.5 BTC
Me->Me 0.2 BTC

Arjun-> Me 0.4 BTC
Raj -> Me 0.3 BTC
Alice -> Me 0.7 BTC
Bob -> Me 0.1 BTC

Transaction and UTXOs

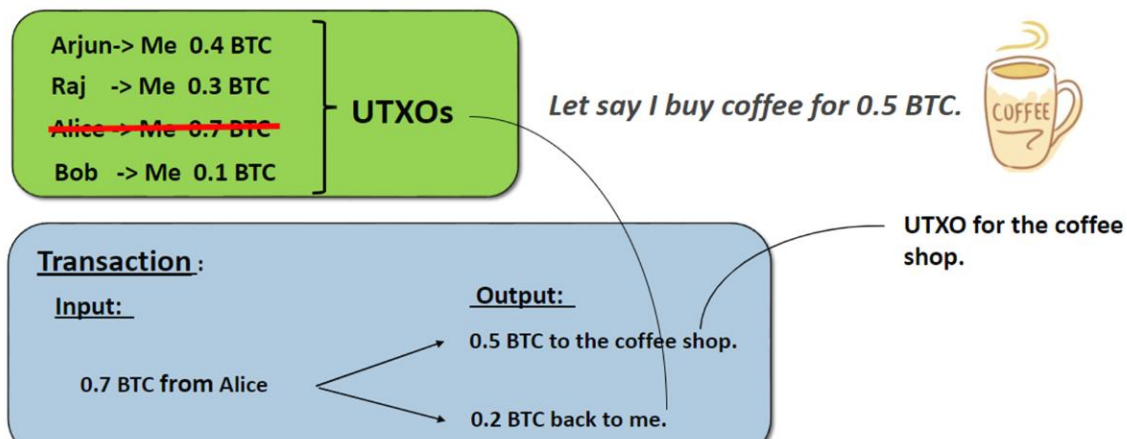


Cryptocurrency Wallets

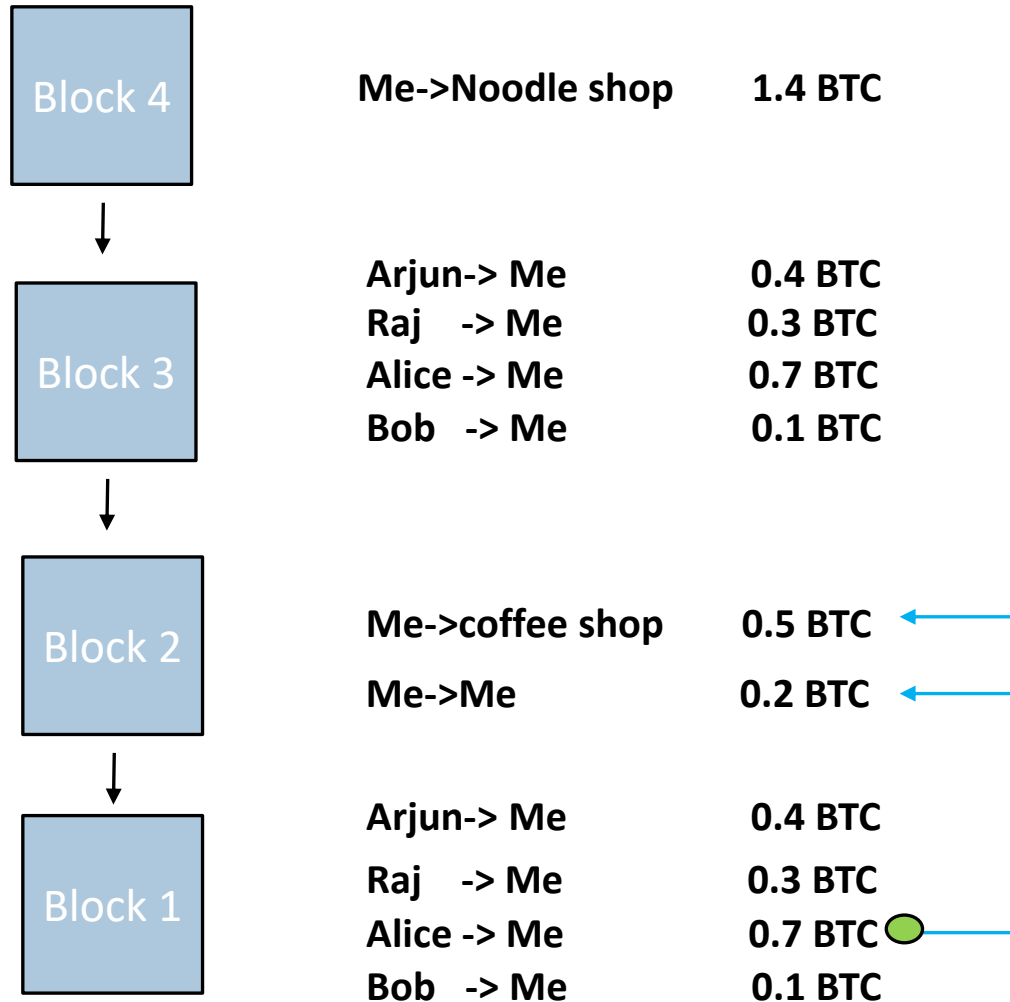


Me->coffee shop	0.5 BTC	←
Me->Me	0.2 BTC	←
Arjun-> Me	0.4 BTC	
Raj -> Me	0.3 BTC	
Alice -> Me	0.7 BTC	● ←
Bob -> Me	0.1 BTC	

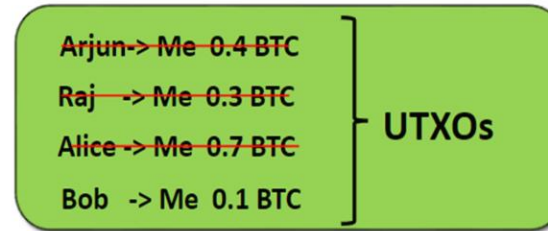
Transaction and UTXOs



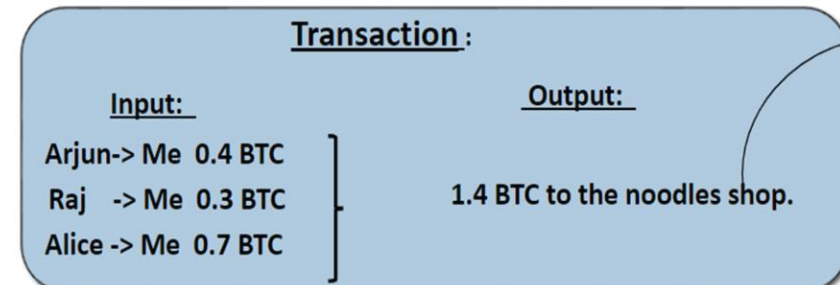
Cryptocurrency Wallets



Transaction and UTXOs

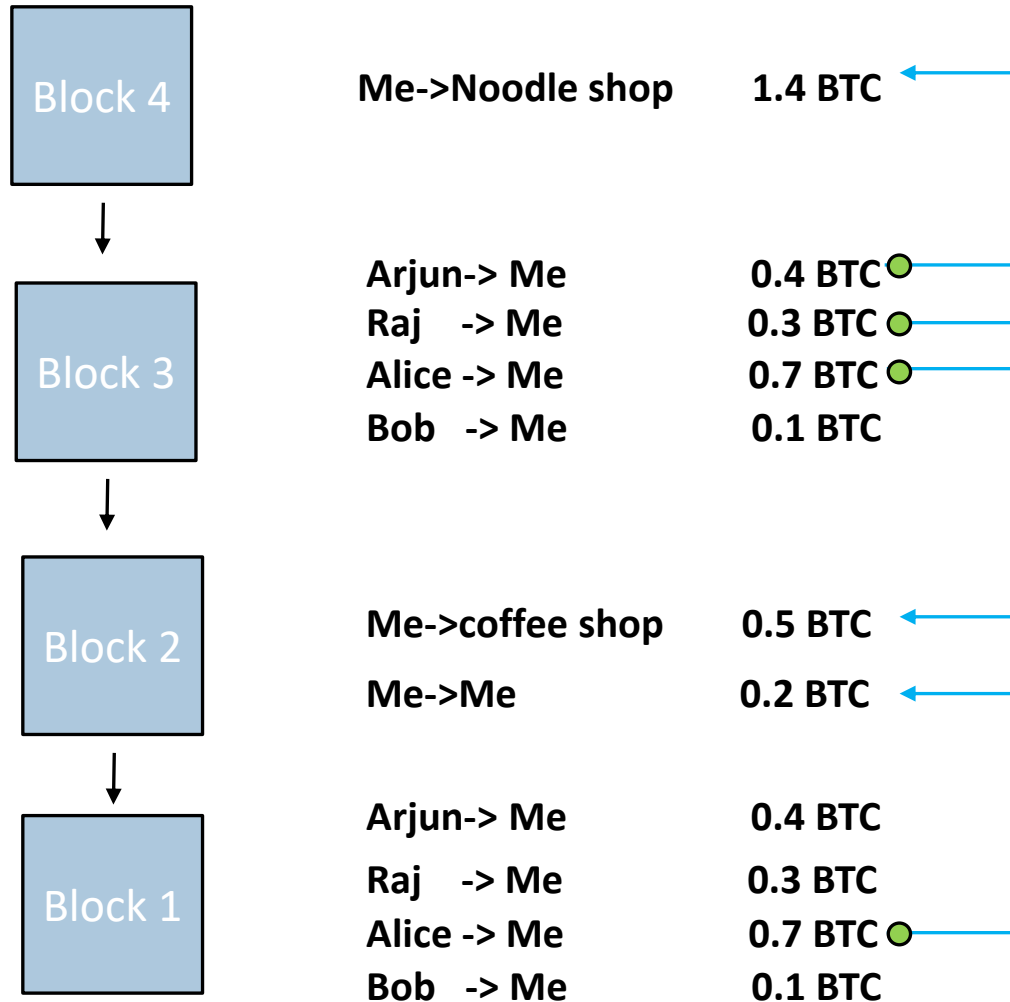


Let say I buy Noodles for 1.4 BTC.

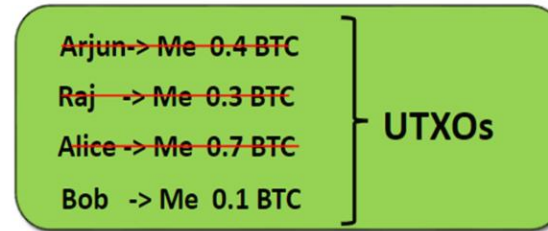


UTXO for the noodle shop.

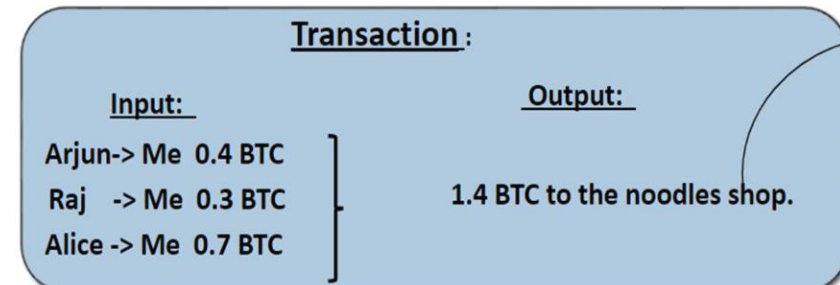
Cryptocurrency Wallets



Transaction and UTXOs

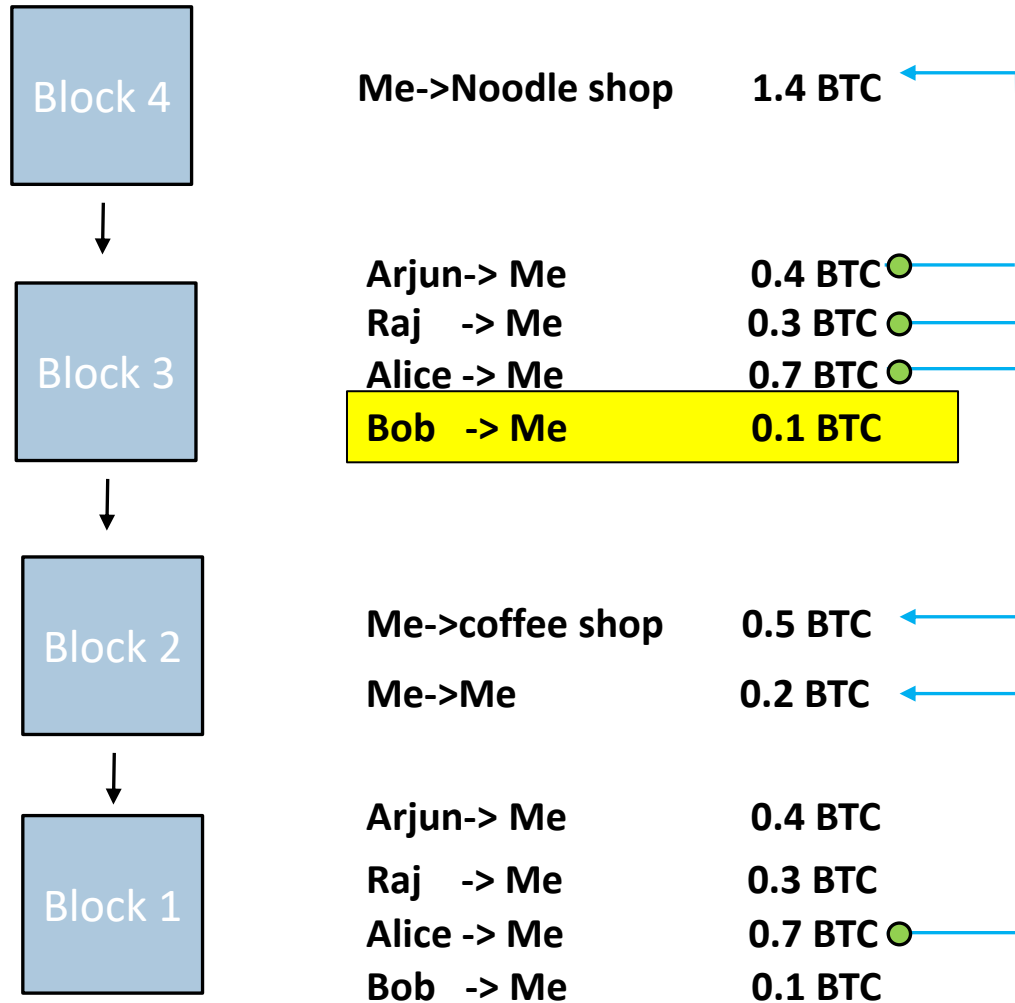


Let say I buy Noodles for 1.4 BTC.

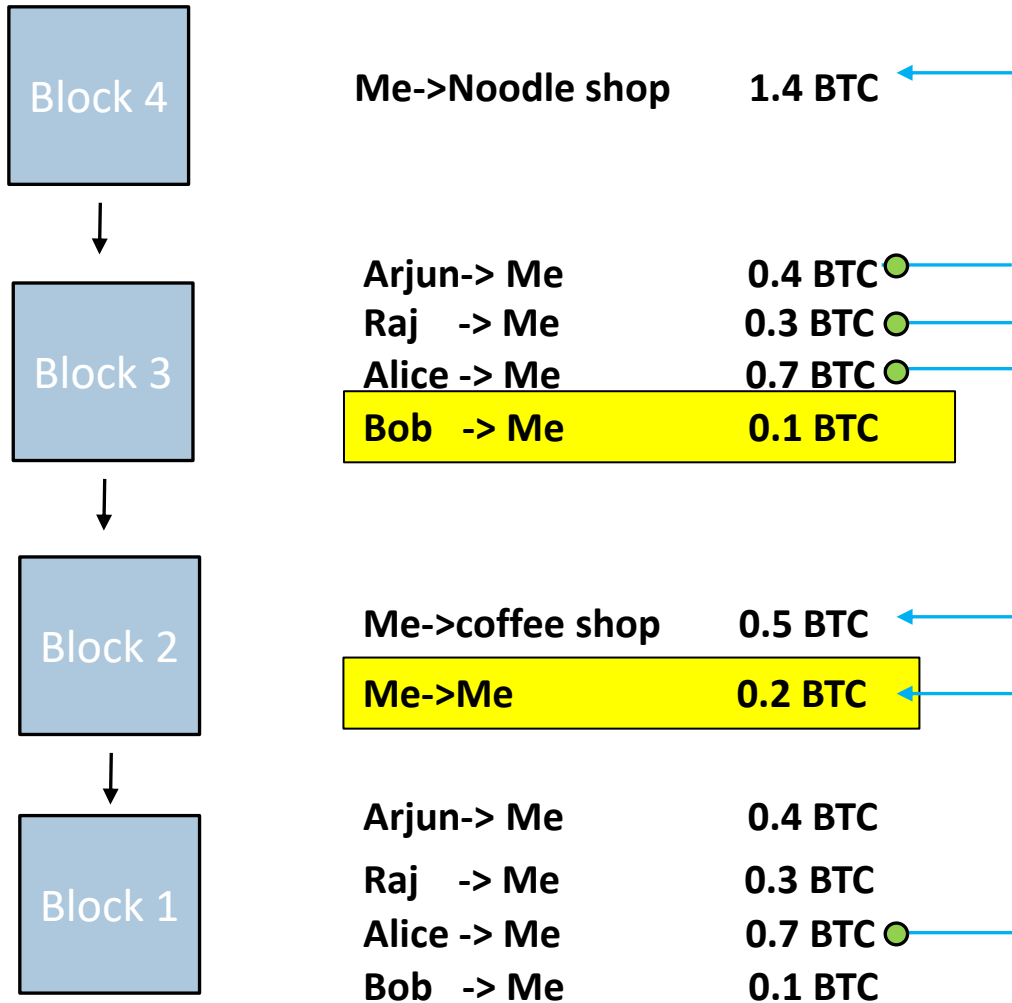


UTXO for the noodle shop.

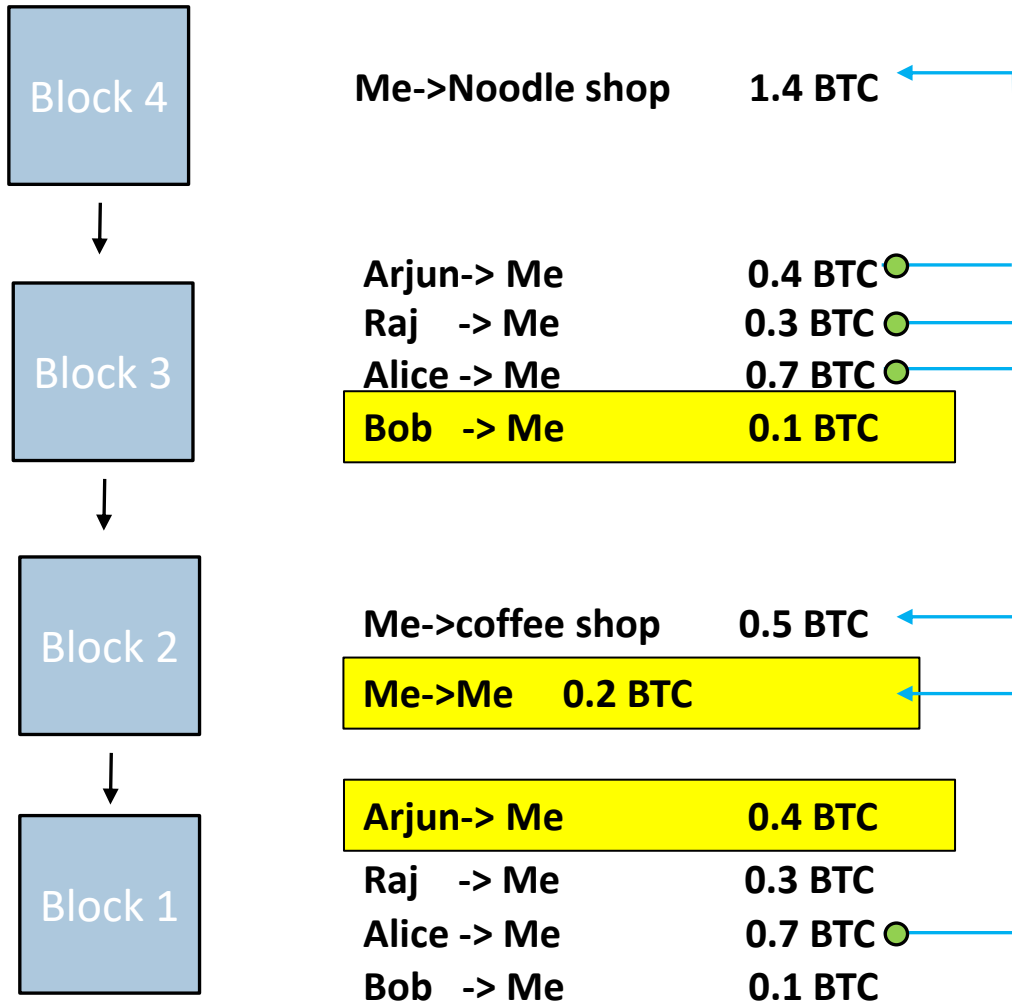
Cryptocurrency Wallets



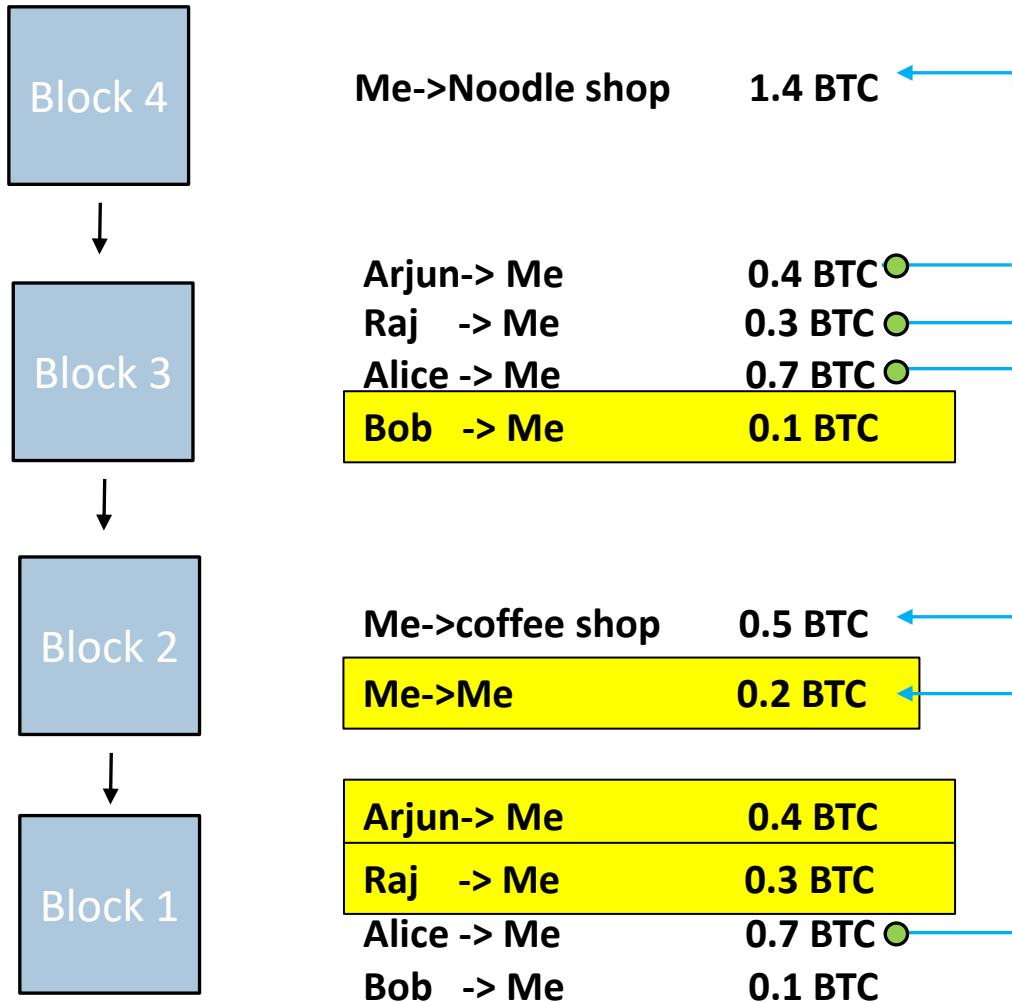
Cryptocurrency Wallets



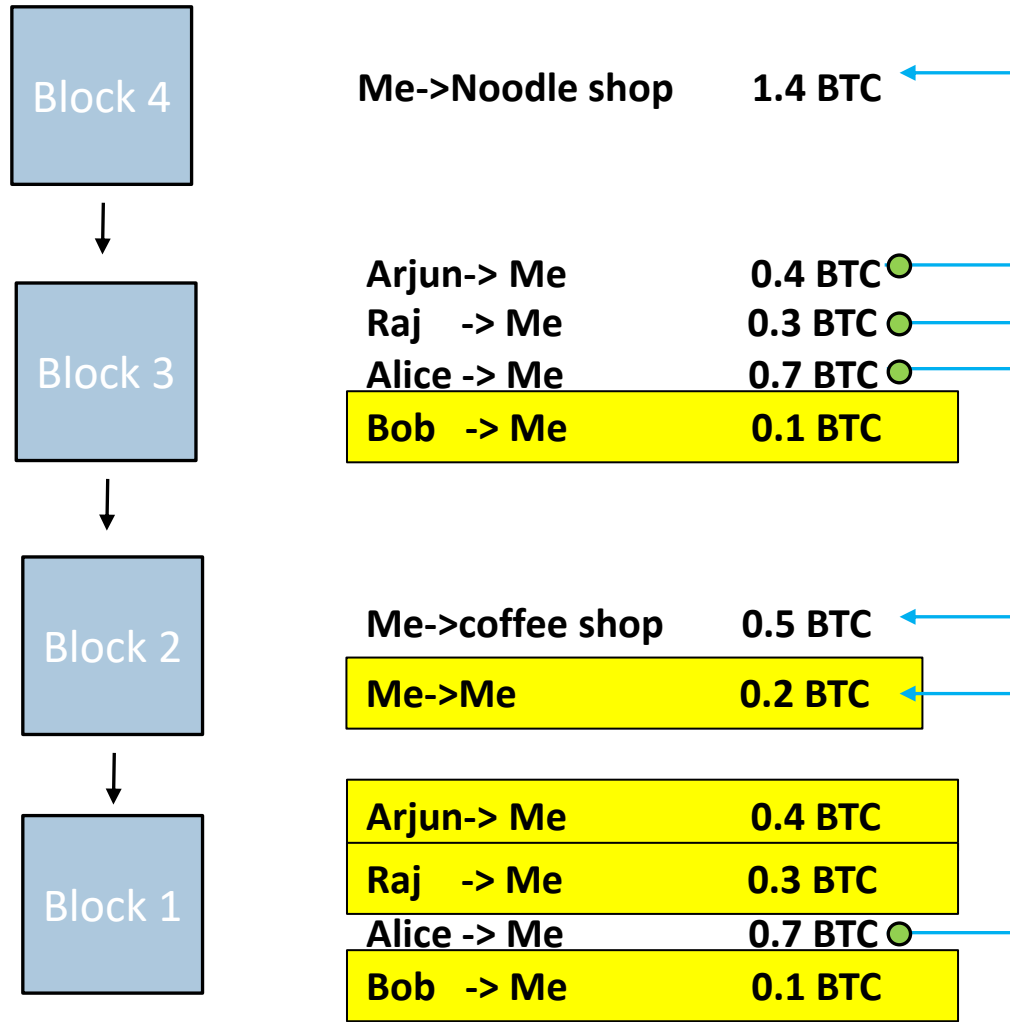
Cryptocurrency Wallets

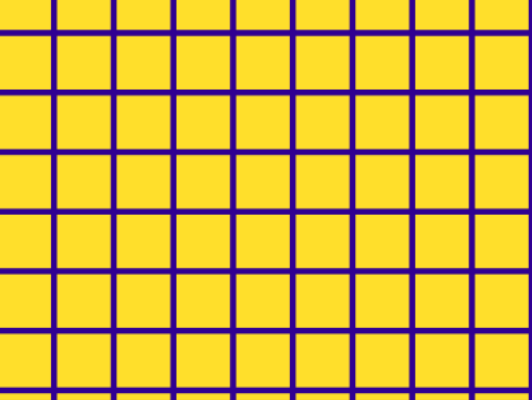


Cryptocurrency Wallets



Cryptocurrency Wallets



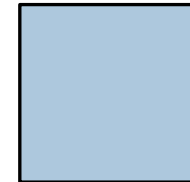
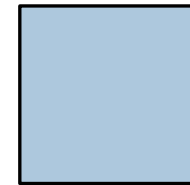
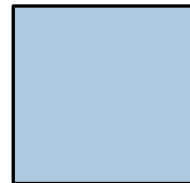
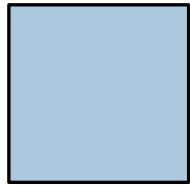


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Cryptocurrency Wallets



Me->Noodle shop 1.4 BTC

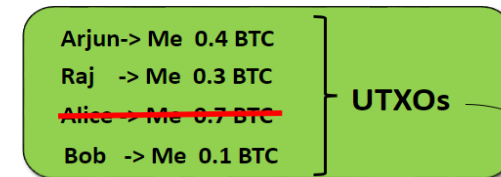
Arjun-> Me 0.4 BTC
Raj -> Me 0.3 BTC
Alice -> Me 0.7 BTC
Bob -> Me 0.1 BTC

Me->coffee shop 0.5 BTC

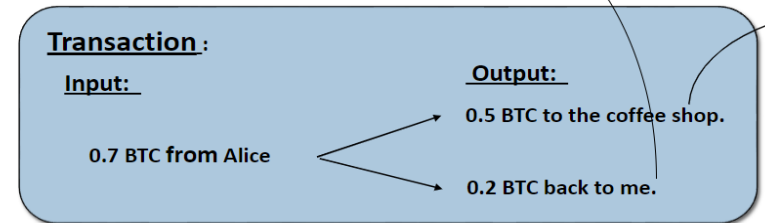
Me->Me 0.2 BTC

Arjun-> Me 0.4 BTC
Raj -> Me 0.3 BTC
Alice -> Me 0.7 BTC
Bob -> Me 0.1 BTC

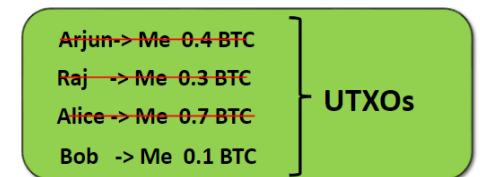
Transaction and UTXOs



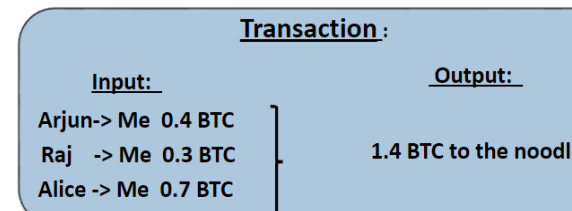
Let say I buy coffee for 0.5 BTC



Transaction and UTXOs



Let say

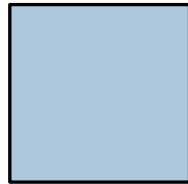
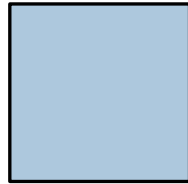
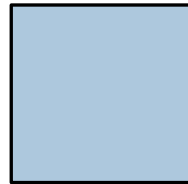
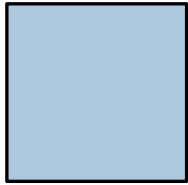


Cryptocurrency Wallets



Private and Public Key

Cryptocurrency Wallets



Me->Noodle shop 1.4 BTC

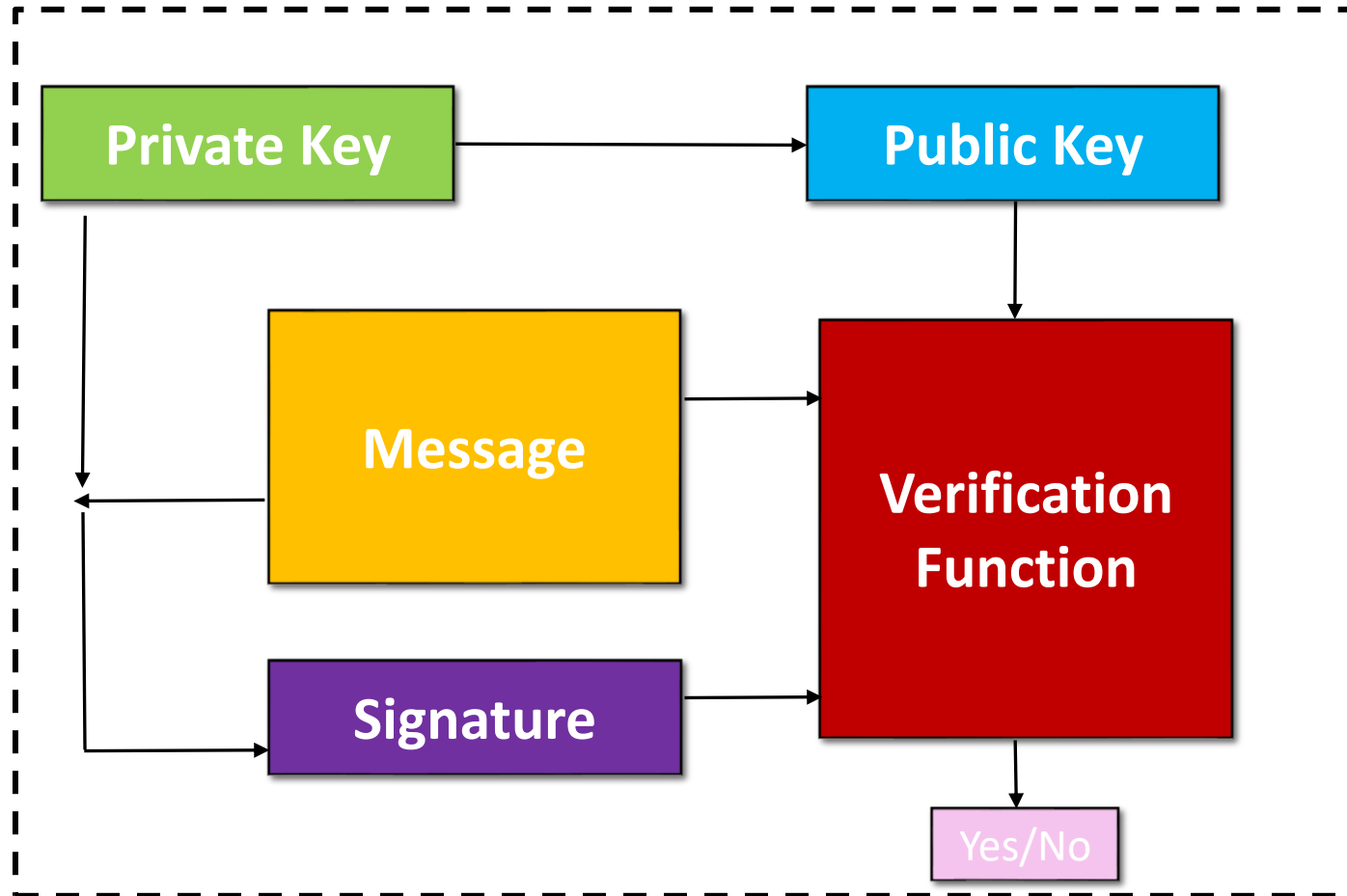
Arjun-> Me 0.4 BTC
Raj -> Me 0.3 BTC
Alice-> Me 0.7 BTC
Bob -> Me 0.1 BTC

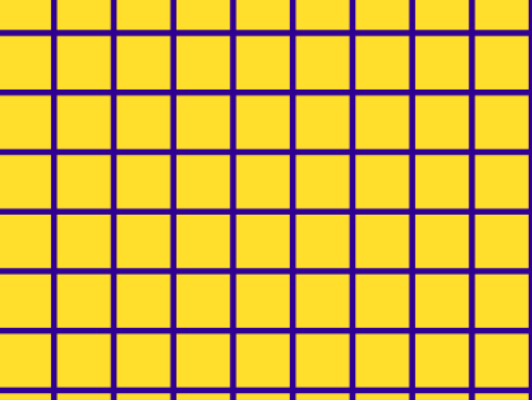
Me->coffee shop 0.5 BTC
Me->Me 0.2 BTC

Arjun-> Me 0.4 BTC
Raj -> Me 0.3 BTC
Alice-> Me 0.7 BTC
Bob -> Me 0.1 BTC

Private and Public Key

Private and Public Key





GIVE THIS VIDEO A THUMBS-UP !

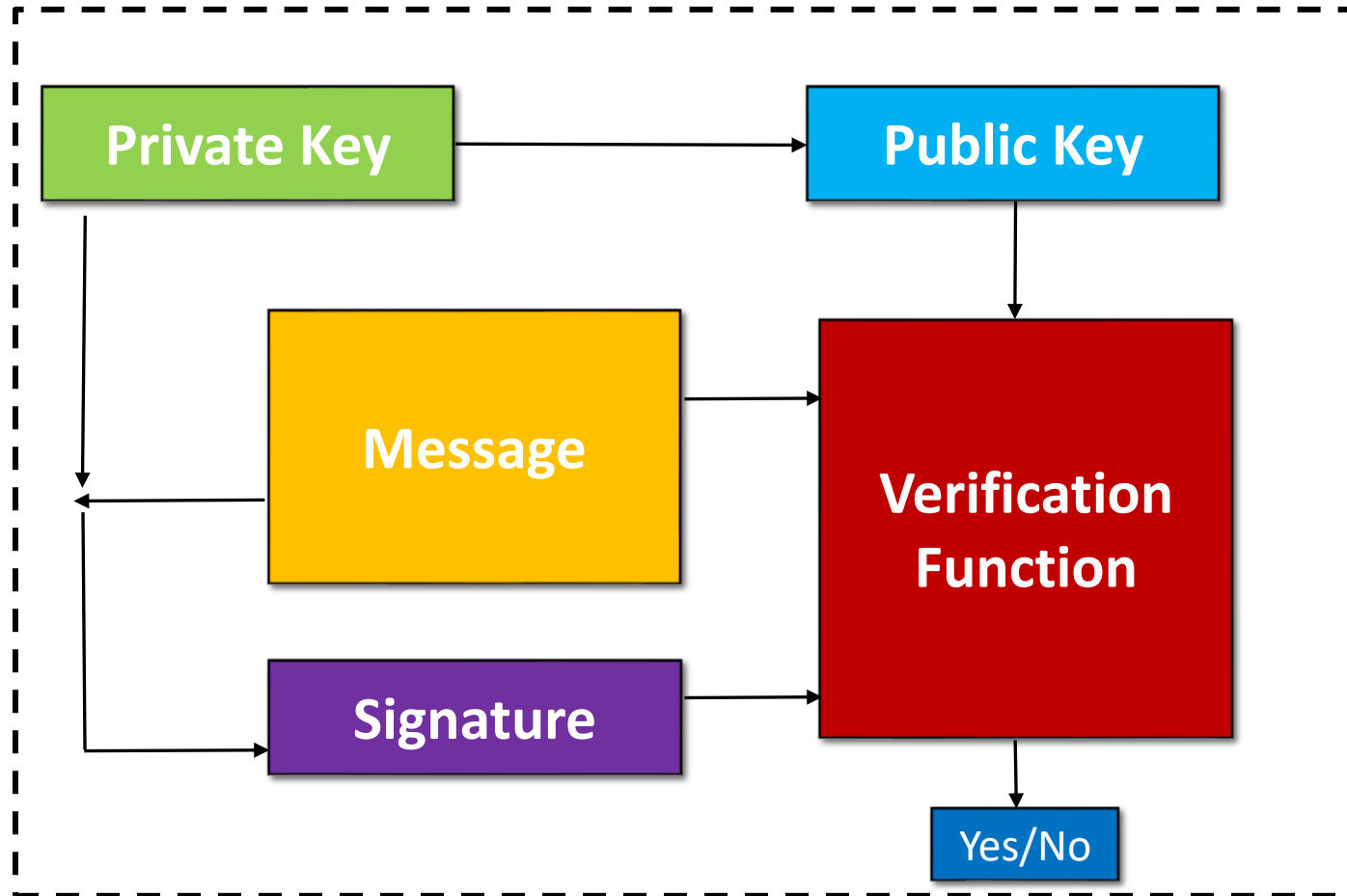
CODE EATER



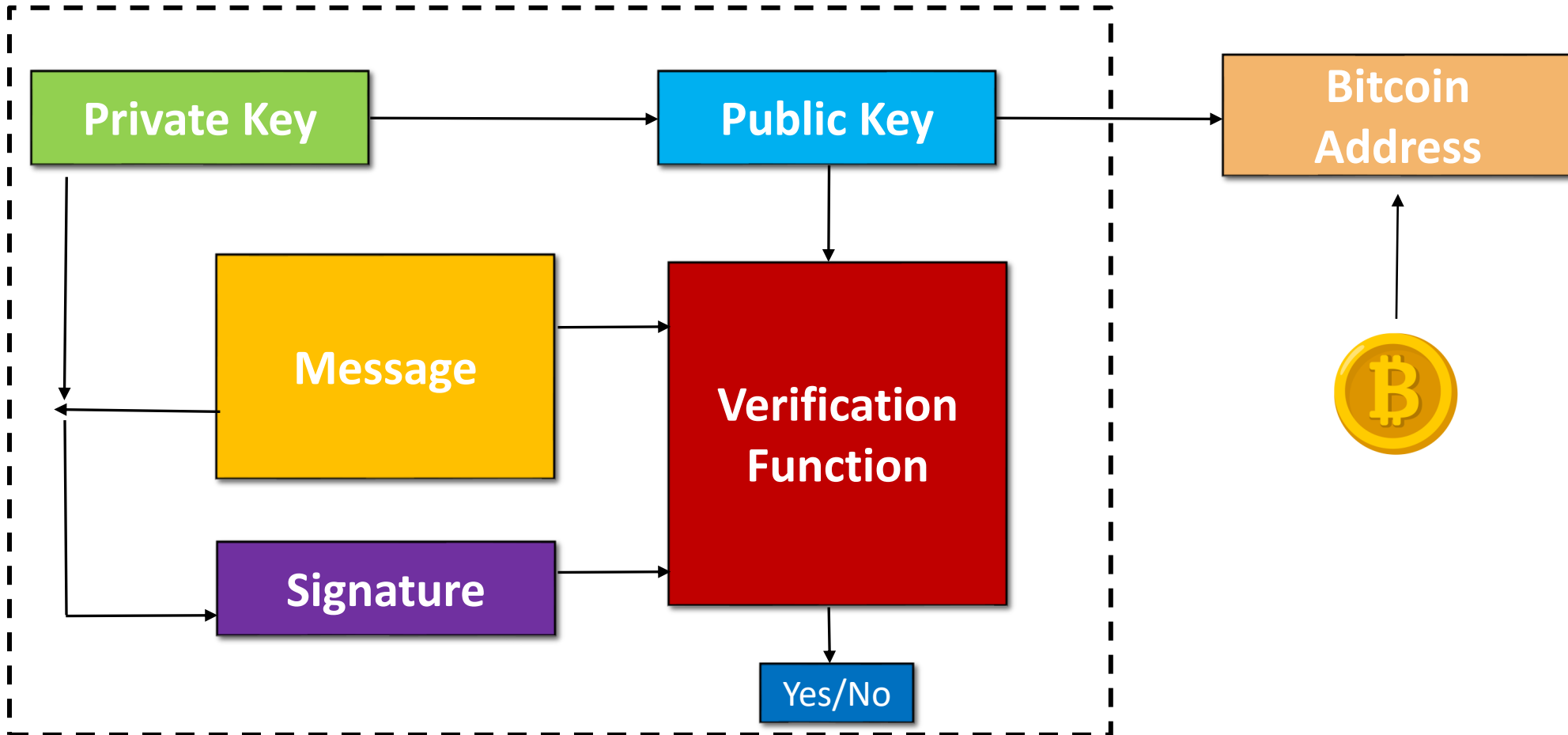
The background of the slide features a large, faint Bitcoin logo in the center. Overlaid on this logo are several concentric circles and lines of binary code (0s and 1s) that appear to be part of a digital or cryptographic theme. The overall color scheme is light gray with a white Bitcoin logo and binary text.

Public Key vs Bitcoin Address

Private and Public Key



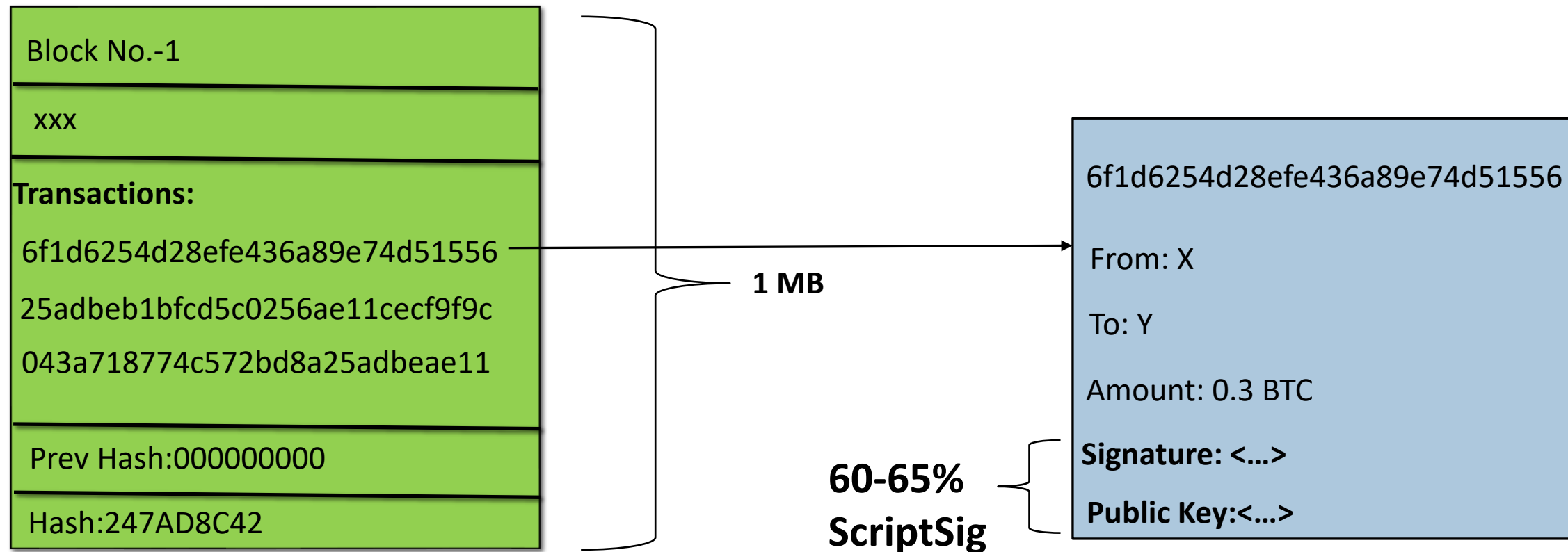
Private and Public Key

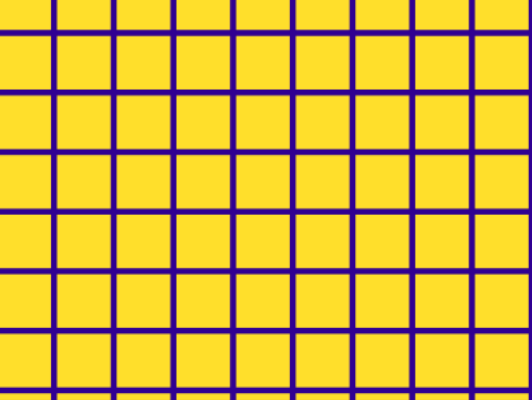




Segregated Witness

Segregated Witness





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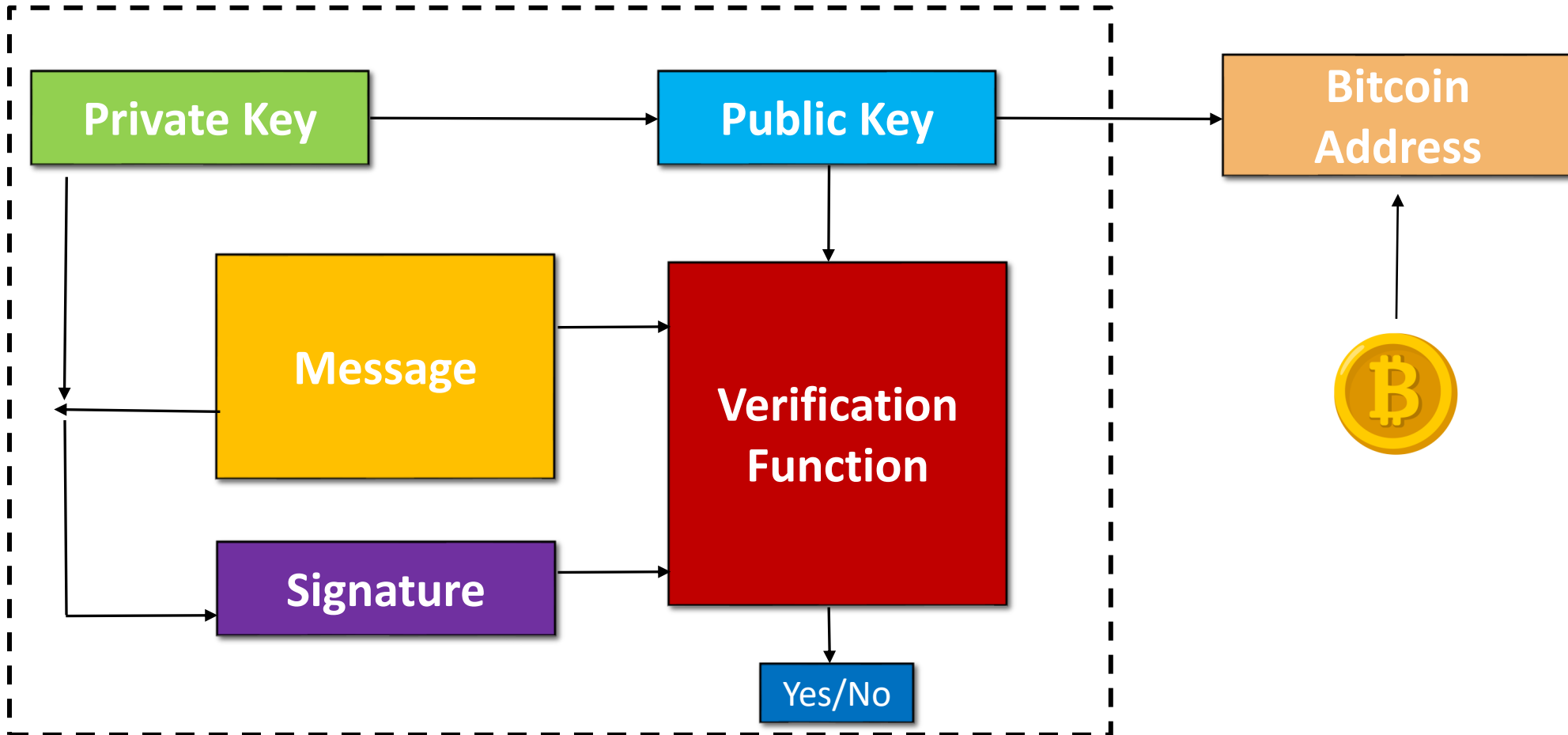
CODE EATER



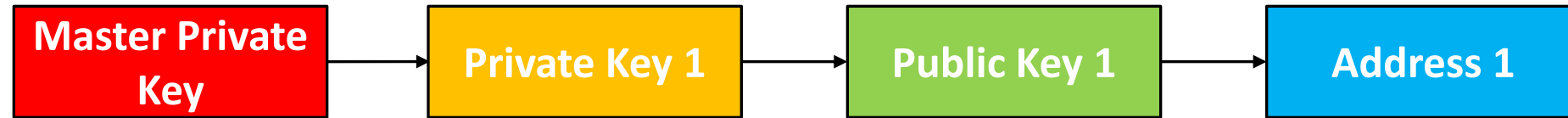


Hierarchically Deterministic Wallet

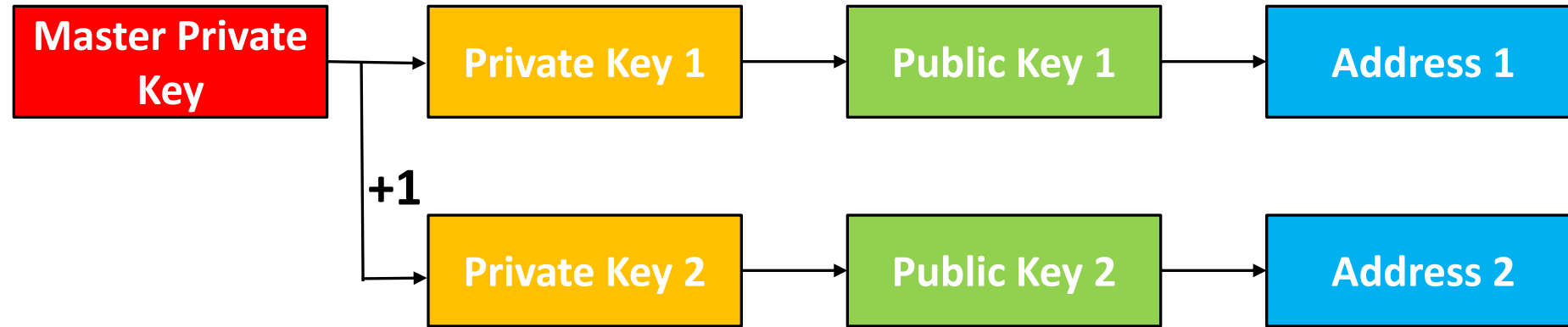
Private and Public Key



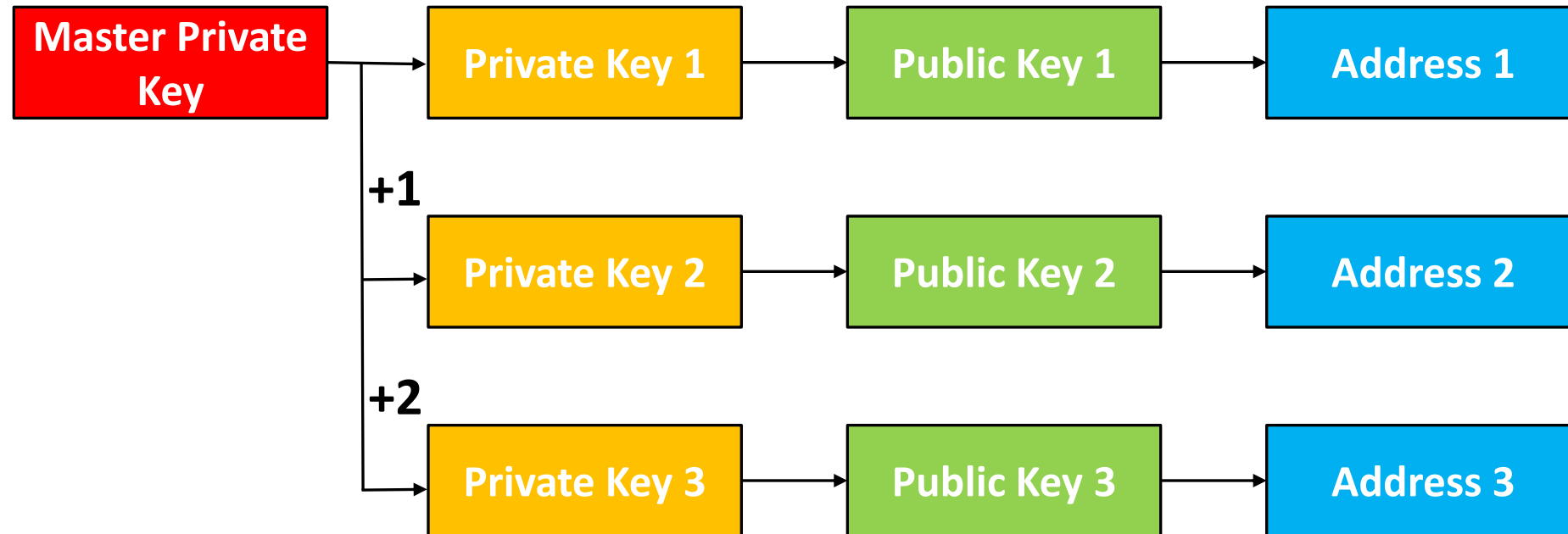
Hierarchically Deterministic (HD) Wallets



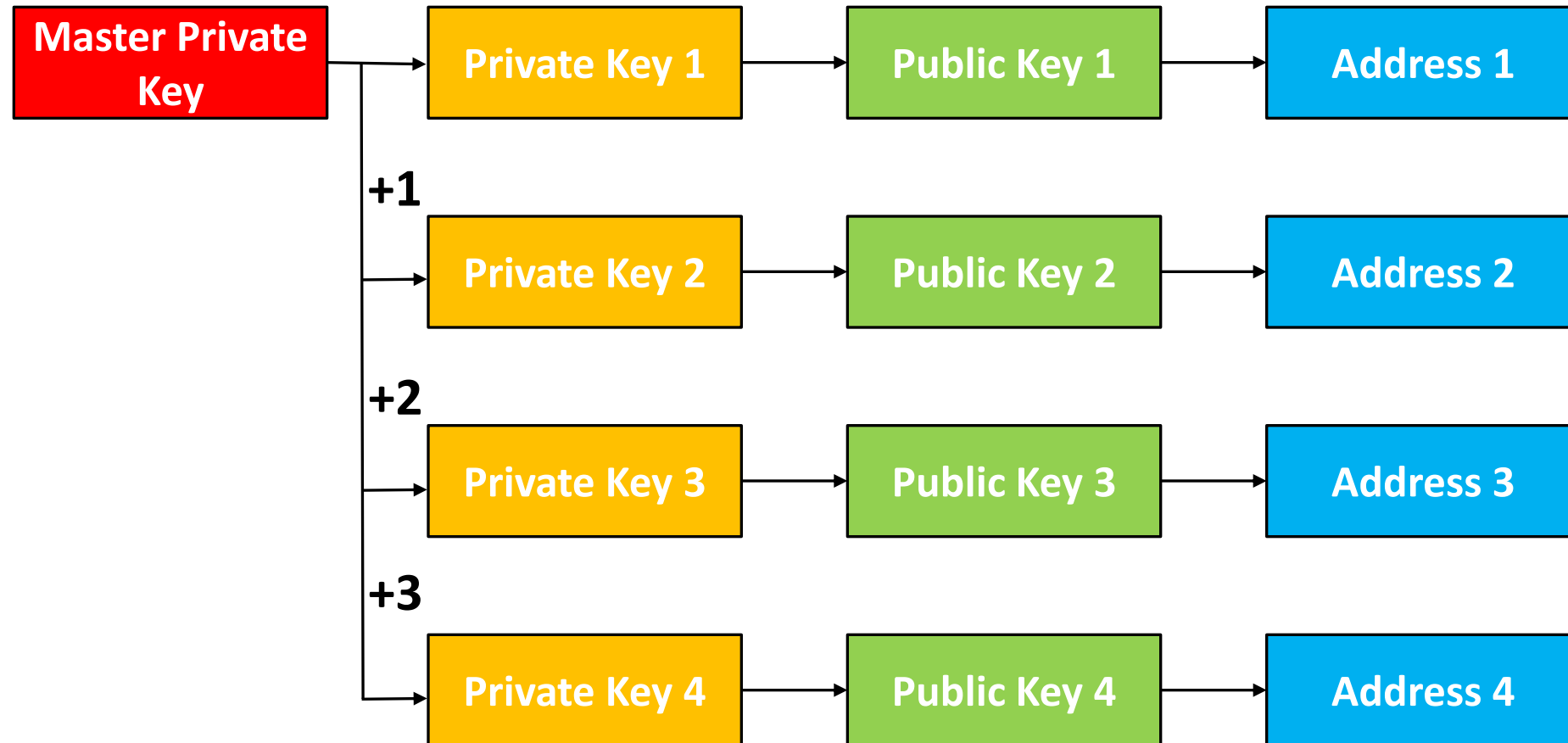
Hierarchically Deterministic (HD) Wallets



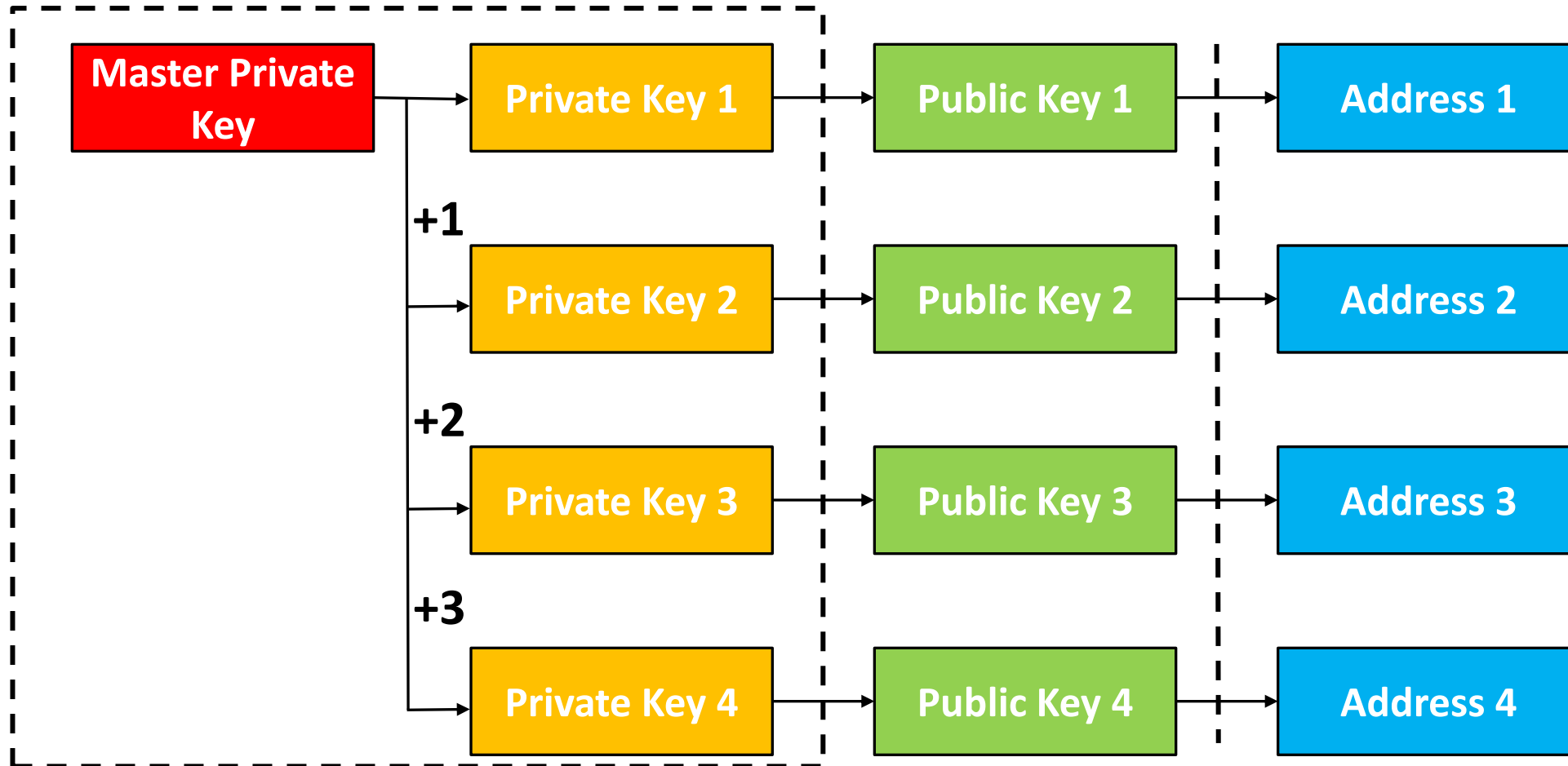
Hierarchically Deterministic (HD) Wallets

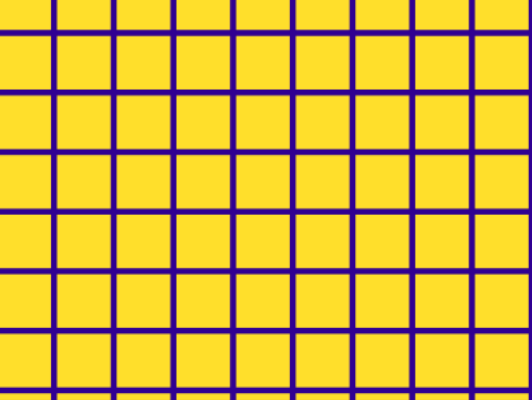


Hierarchically Deterministic (HD) Wallets



Hierarchically Deterministic (HD) Wallets

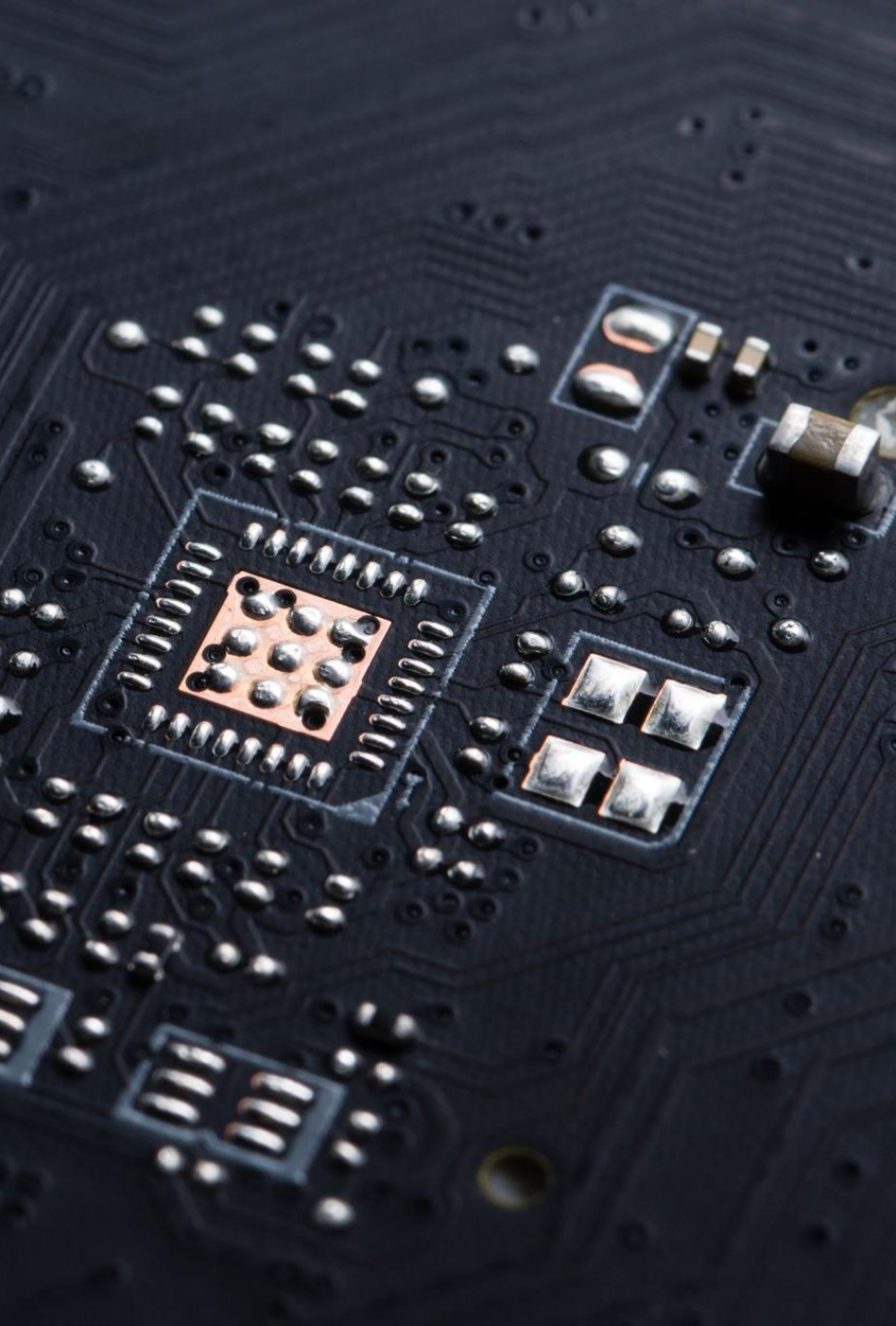




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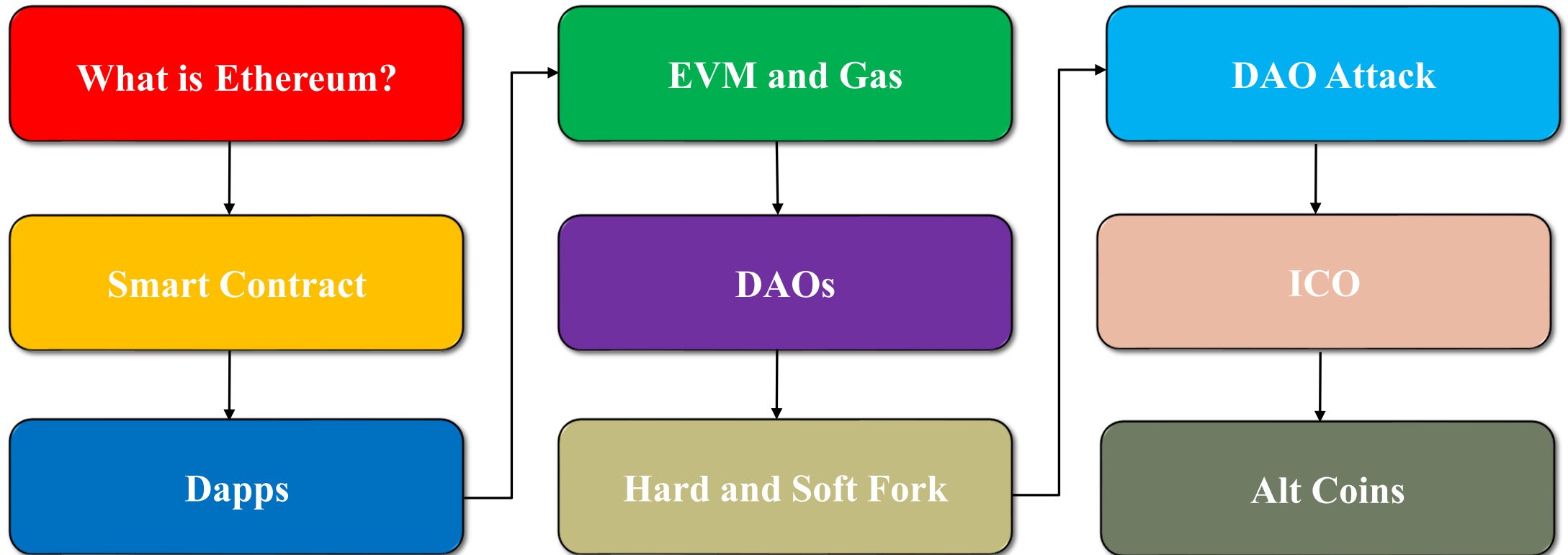
CODE EATER





Ethereum

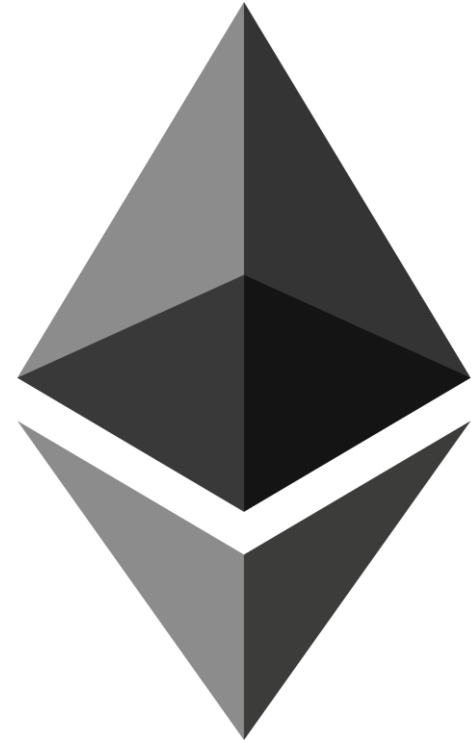
Contents – Module C



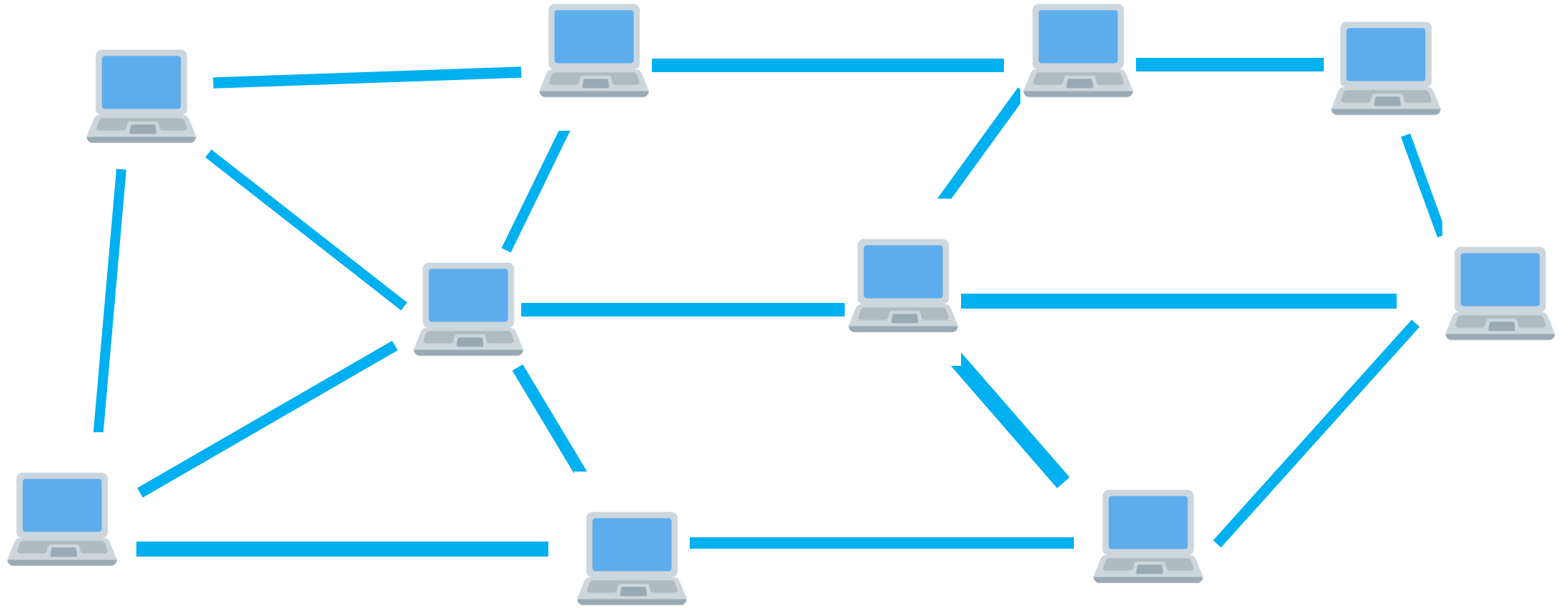
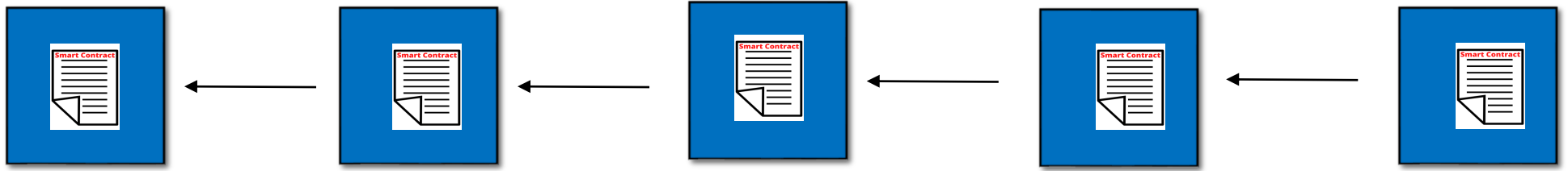
Ethereum



Vitalik Buterin



Ethereum



Bitcoin



Ethereum



What is Ethereum?

Technology

Blockchain

Protocol/Coin

Waves

Bitcoin

Ethereum

Token

WGB

BI

INTL

WGR

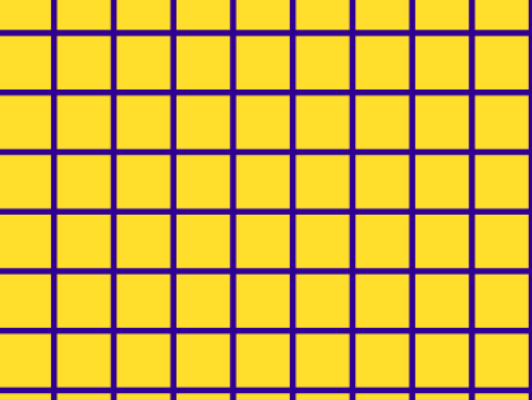


TRX

SNT

REP

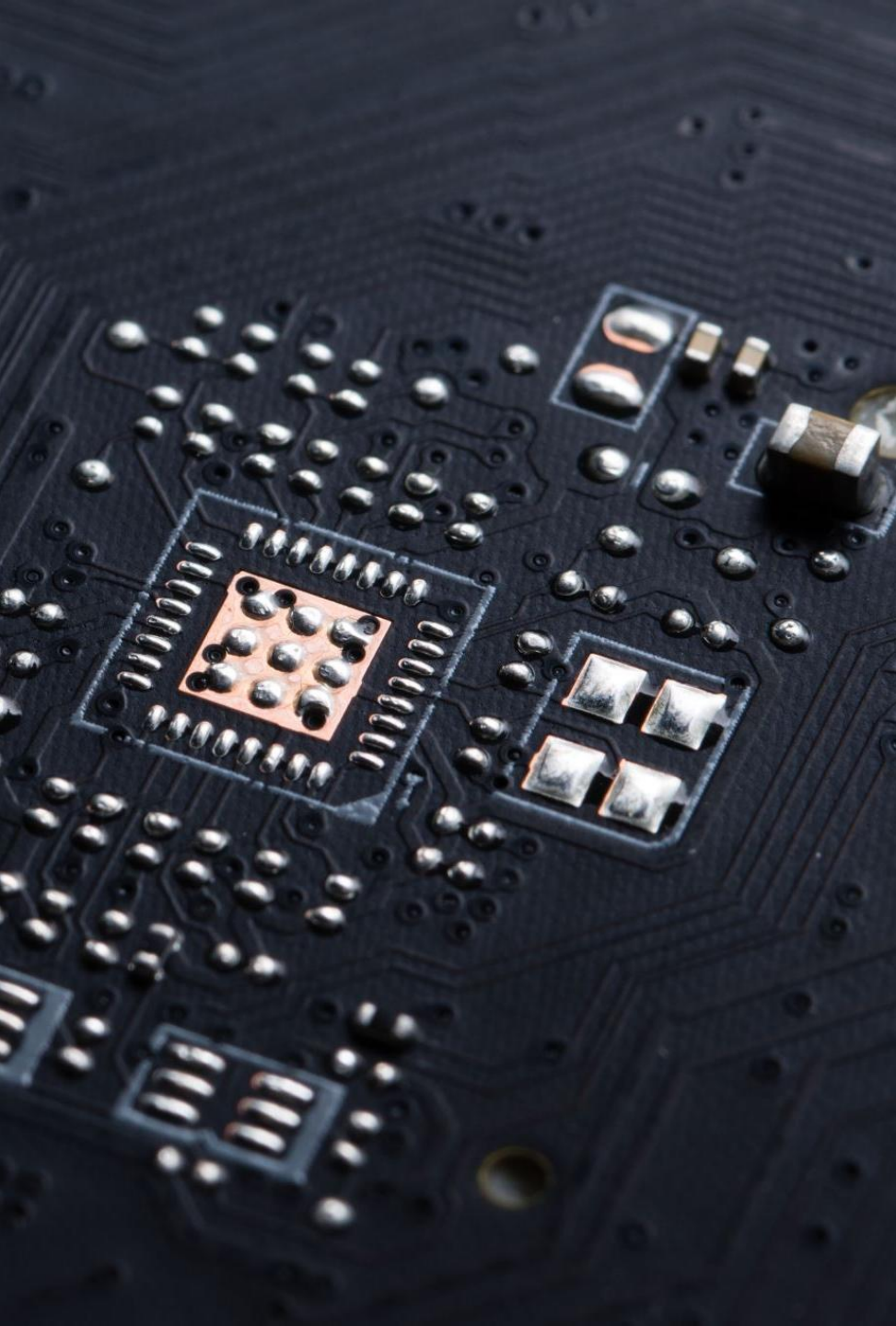
AE



GIVE THIS VIDEO A THUMBS-UP !

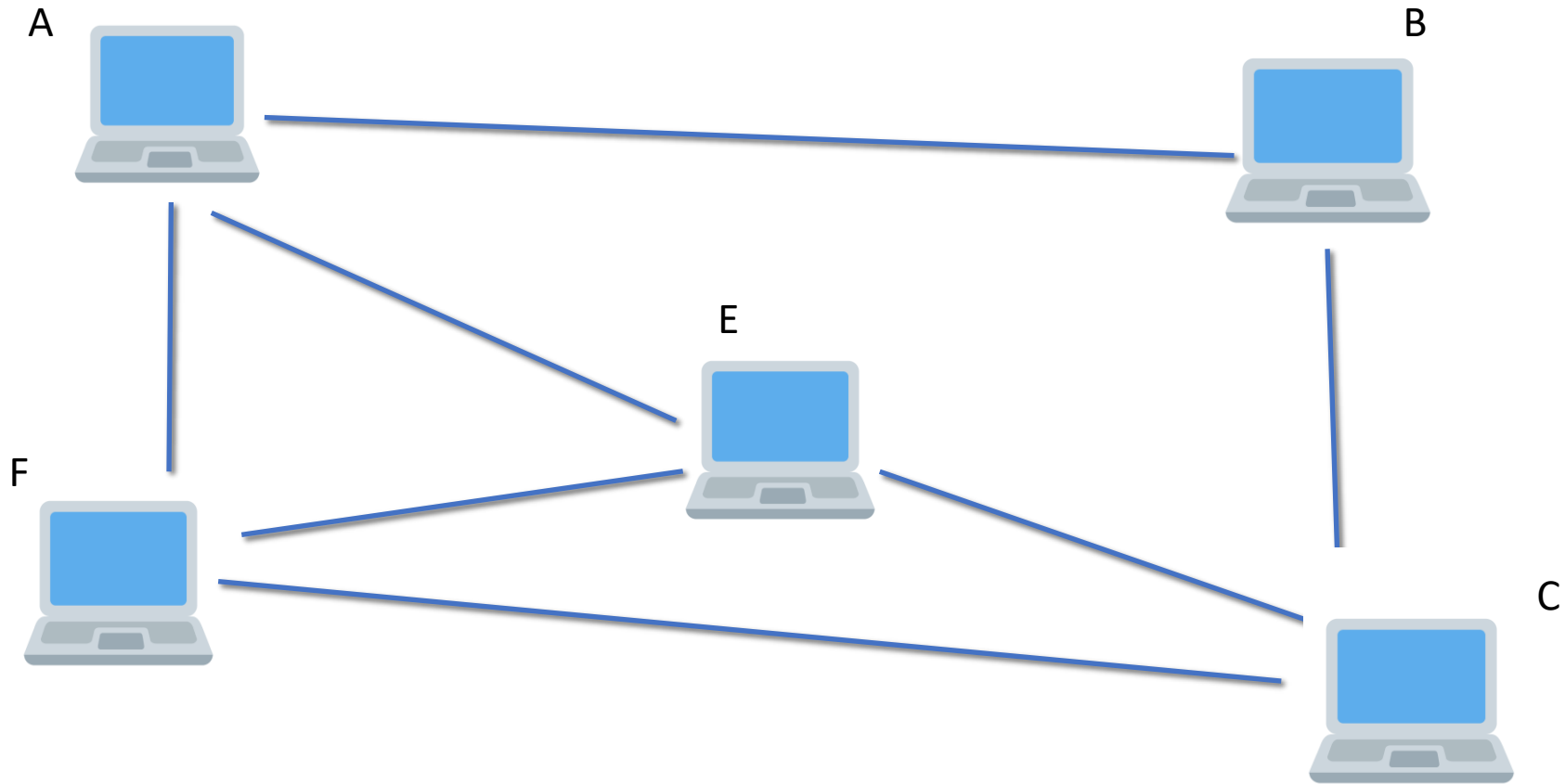
CODE EATER





Ethereum Nodes

Ethereum Nodes



Types of Nodes

Full Node

Light Node

Archive Node

Full Node

- Stores full blockchain data (although this is periodically pruned so a full node does not store all state data back to genesis)
- Participates in block validation, verifies all blocks and states.
- Serves the network and provides data on request.
- All states can be derived from a full node (although very old states are reconstructed from requests made to archive nodes)



Light Node

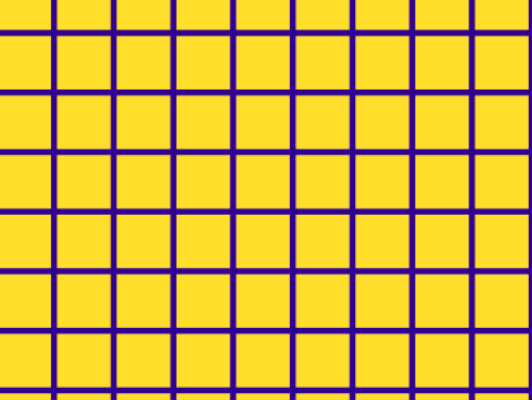
- Stores only the block header and depends on full node.
- For low capacity devices which cannot afford to store the gigabytes of data.
- The light nodes do not participate in consensus (i.e. they cannot be miners/validators), but they can access the Ethereum blockchain with the same functionality as a full node.



Archive Node

- Stores everything kept in the full node and built an archive of historical data.
- Requires terabytes of disk space.

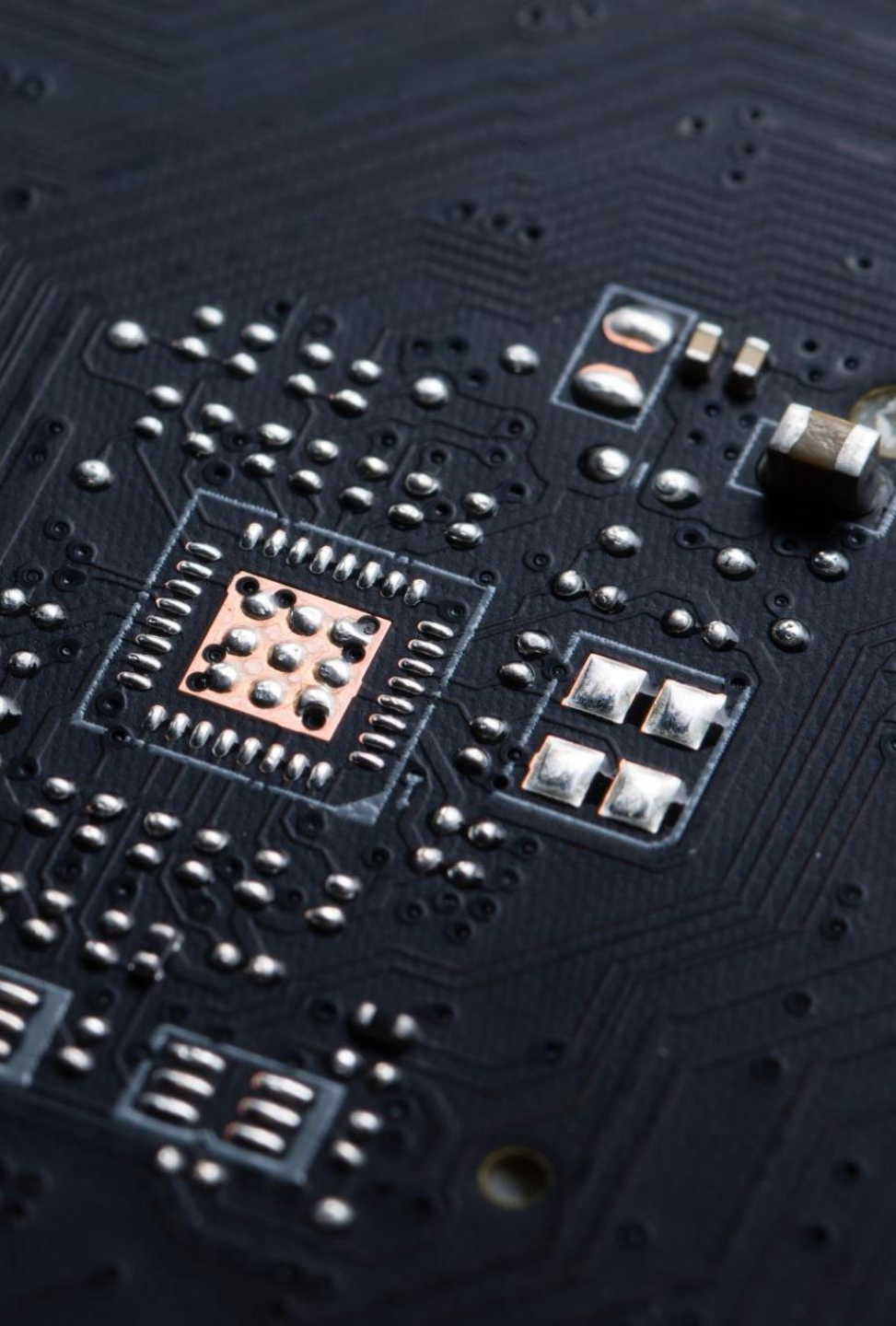




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CODE EATER





Ethereum Accounts

Ethereum Accounts

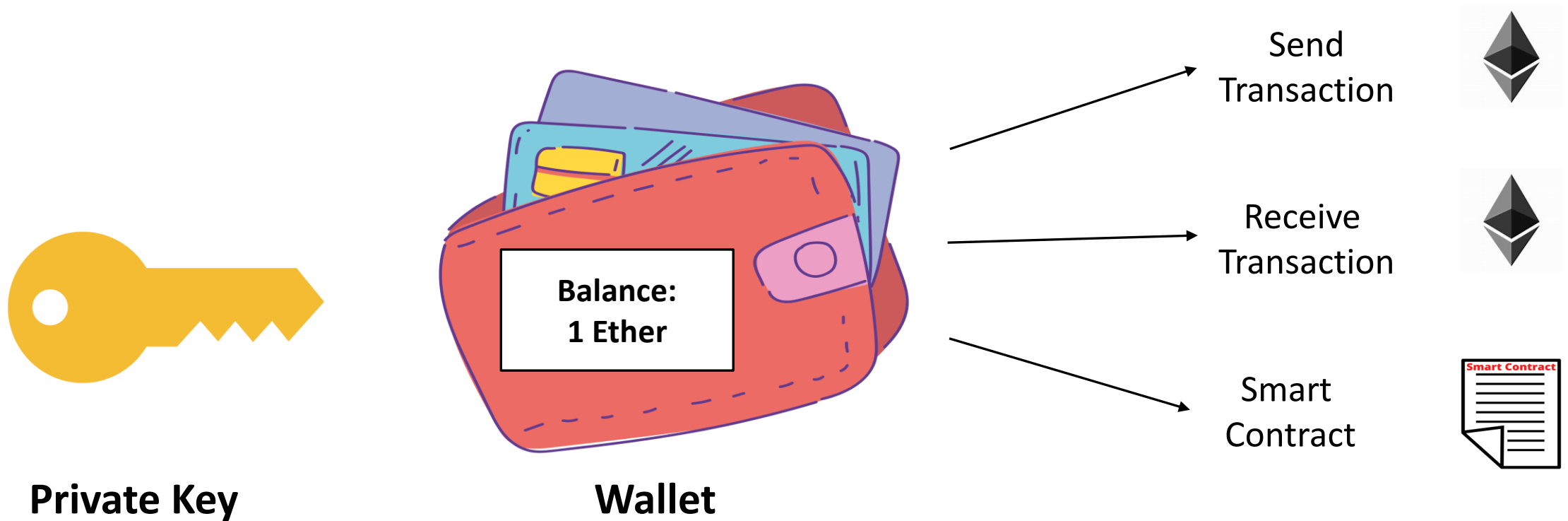
- An Ethereum account is an entity with an ether (ETH) balance that can send or receive transactions on Ethereum.

Types of Ethereum Accounts

**Externally Owned
Account(EOA)**

Contract Account(CA)

Externally Owned Account(EOA)

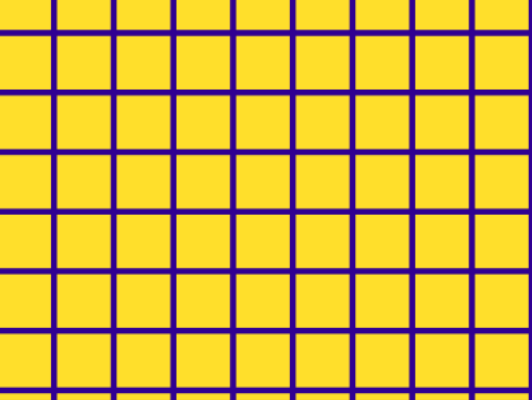


Contract Account (CA)

- Controlled by contract code.

EOA VS CA

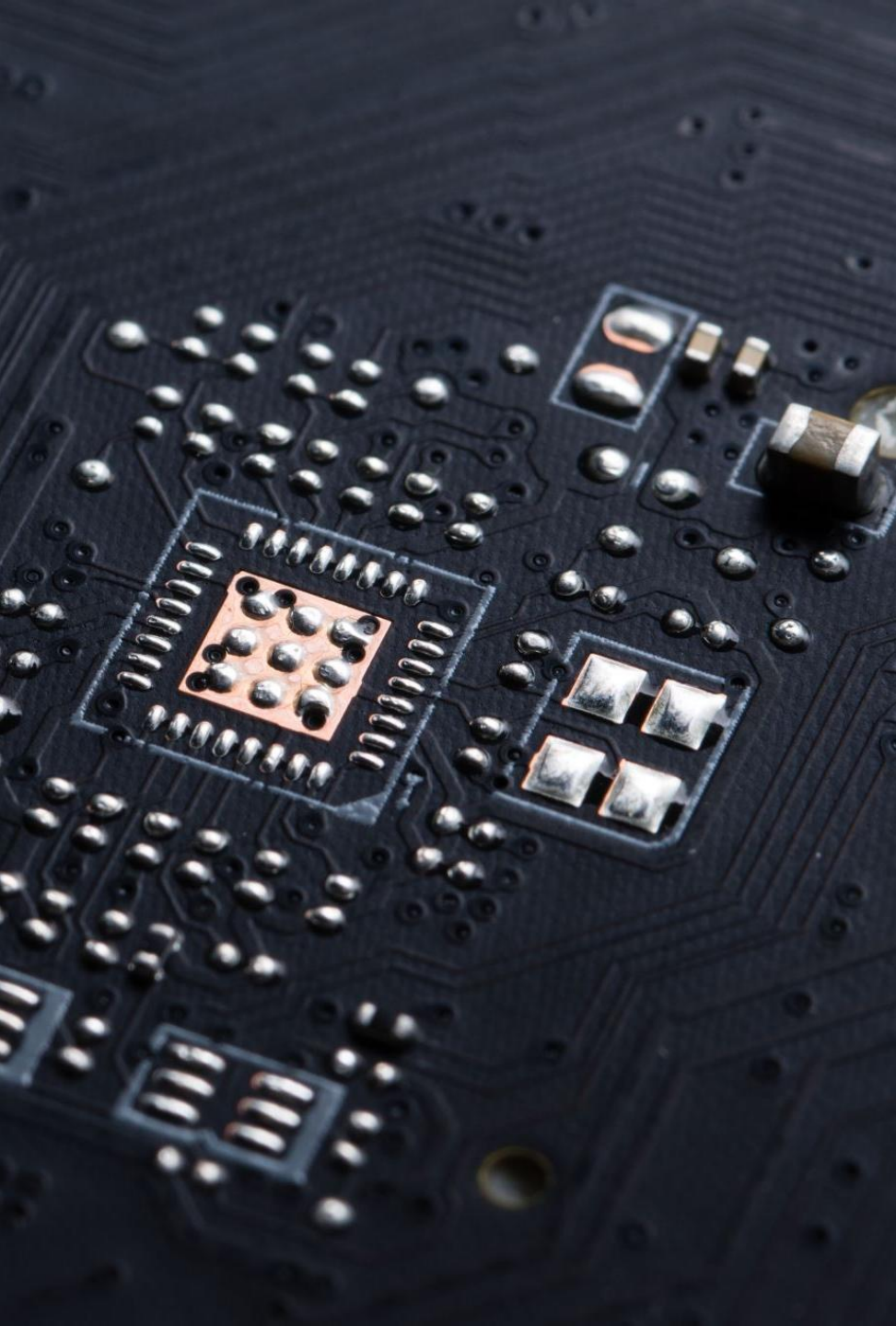
EOA	CA
Private Key is needed	No private or public key is needed.
Controlled by Human	Controlled by Contract code
Has a unique address	Has a unique address
Holds ETH balance	Holds ETH balance



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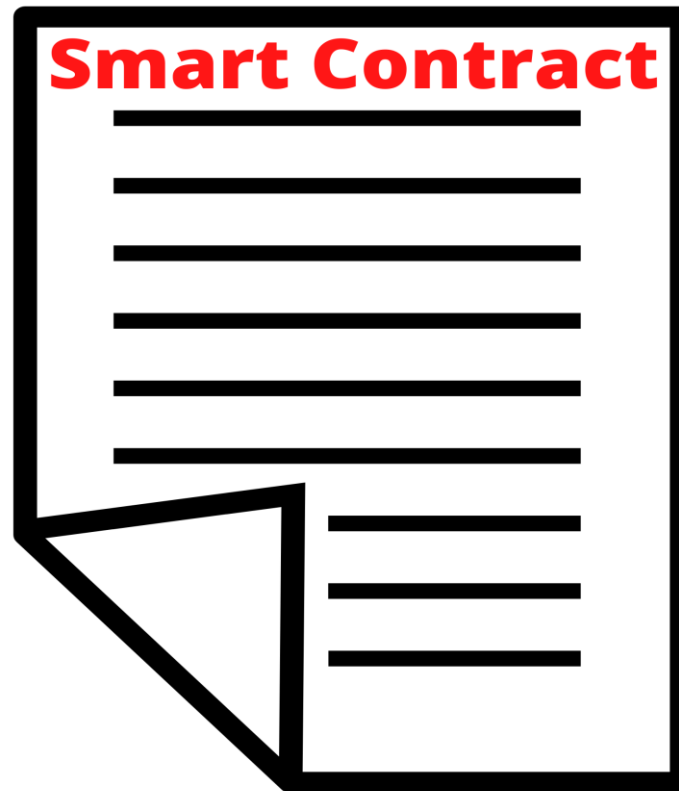
CODE EATER





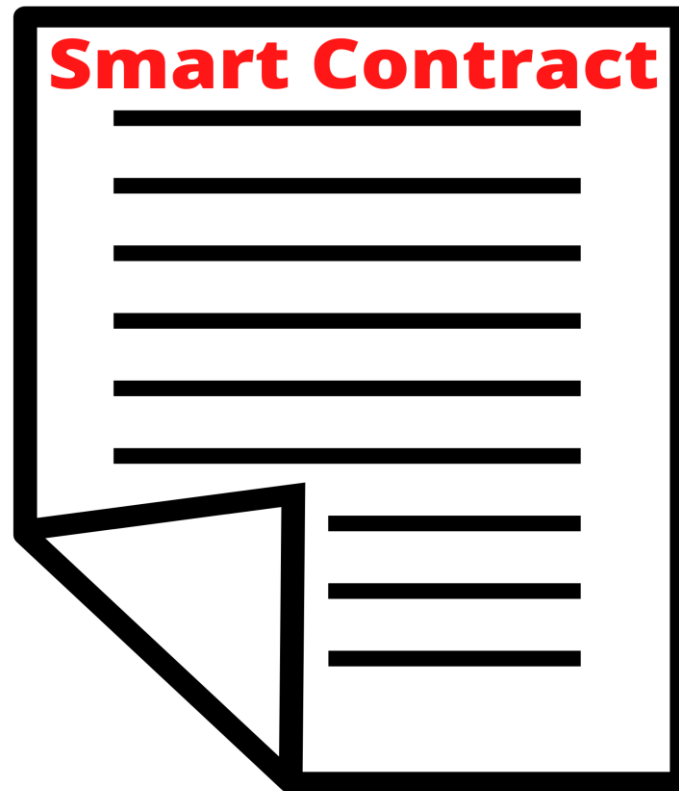
Smart Contract

Smart Contract





Smart Contract



Smart Contract

Bitcoin Script

Not Turing Complete

Solidity

Turing Complete

Smart Contract

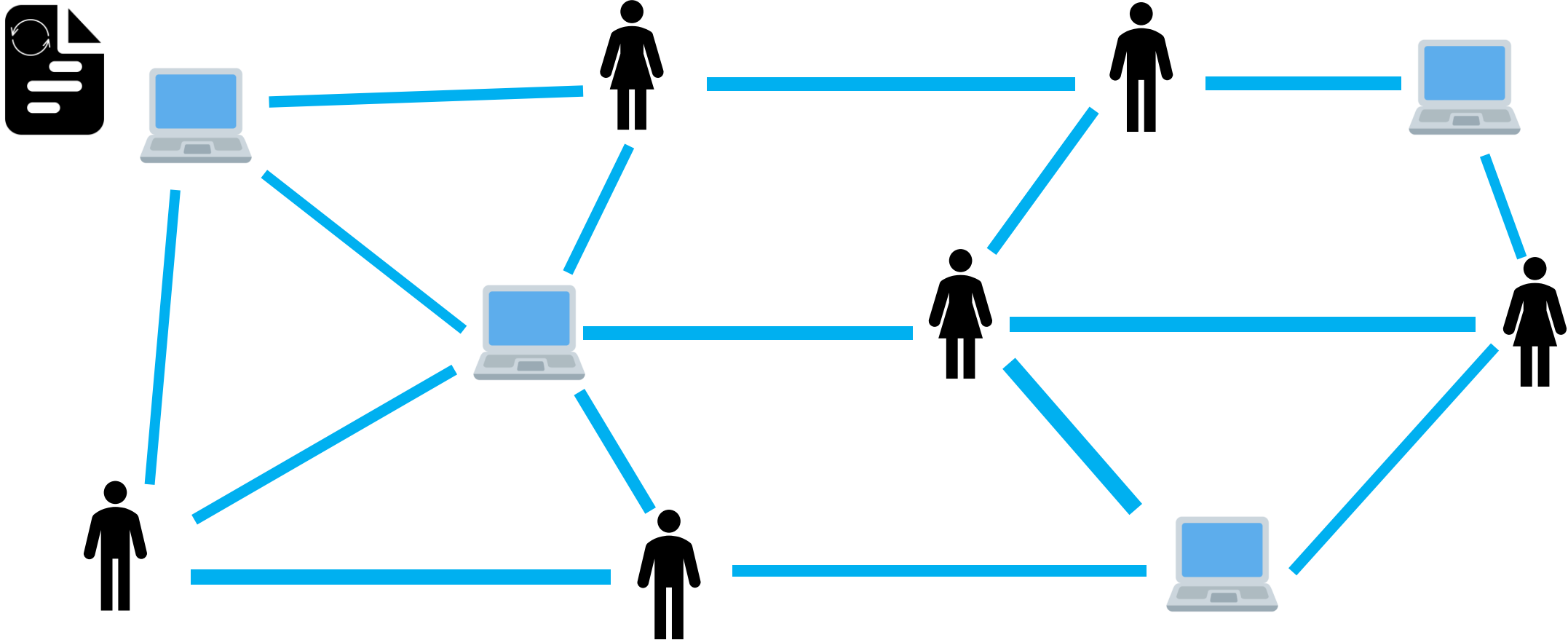
Bitcoin Script

Not Turing Complete

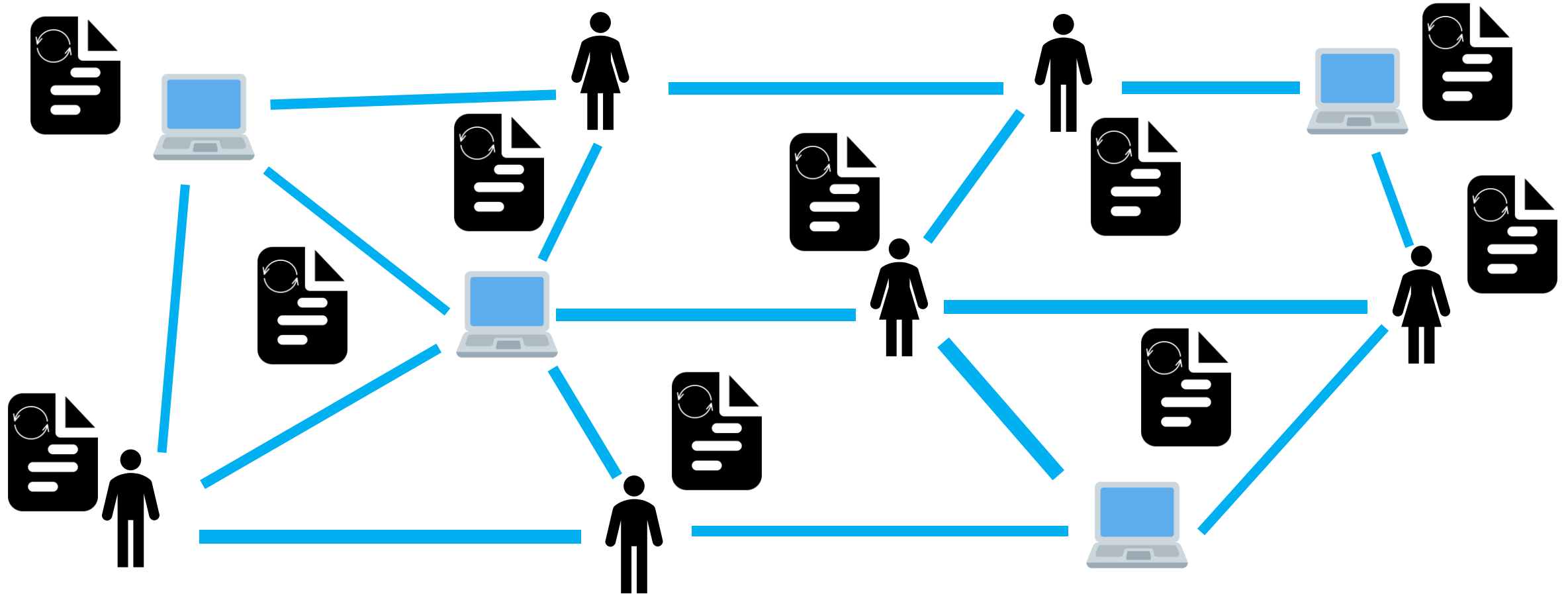
Solidity

Turing Complete

Smart Contract



Smart Contract

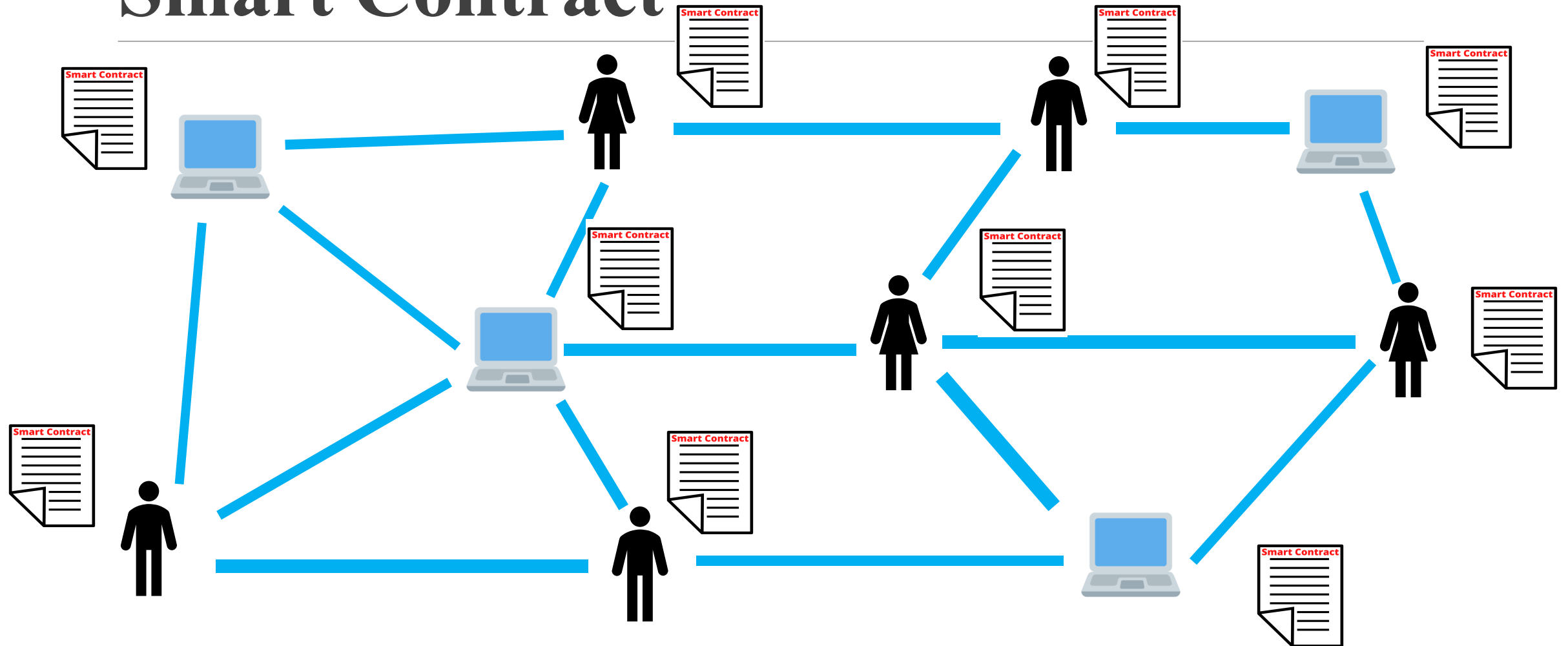


Smart Contract

Block No.-1
Nonce:
Timestamp:
Transactions: D67F232
Prev Hash:0000000000
Hash:



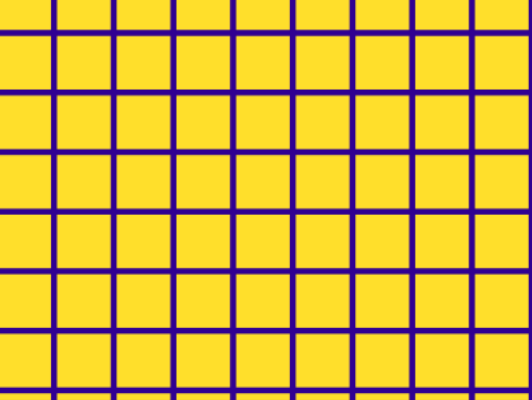
Smart Contract



Smart Contract

Each node has the following-:

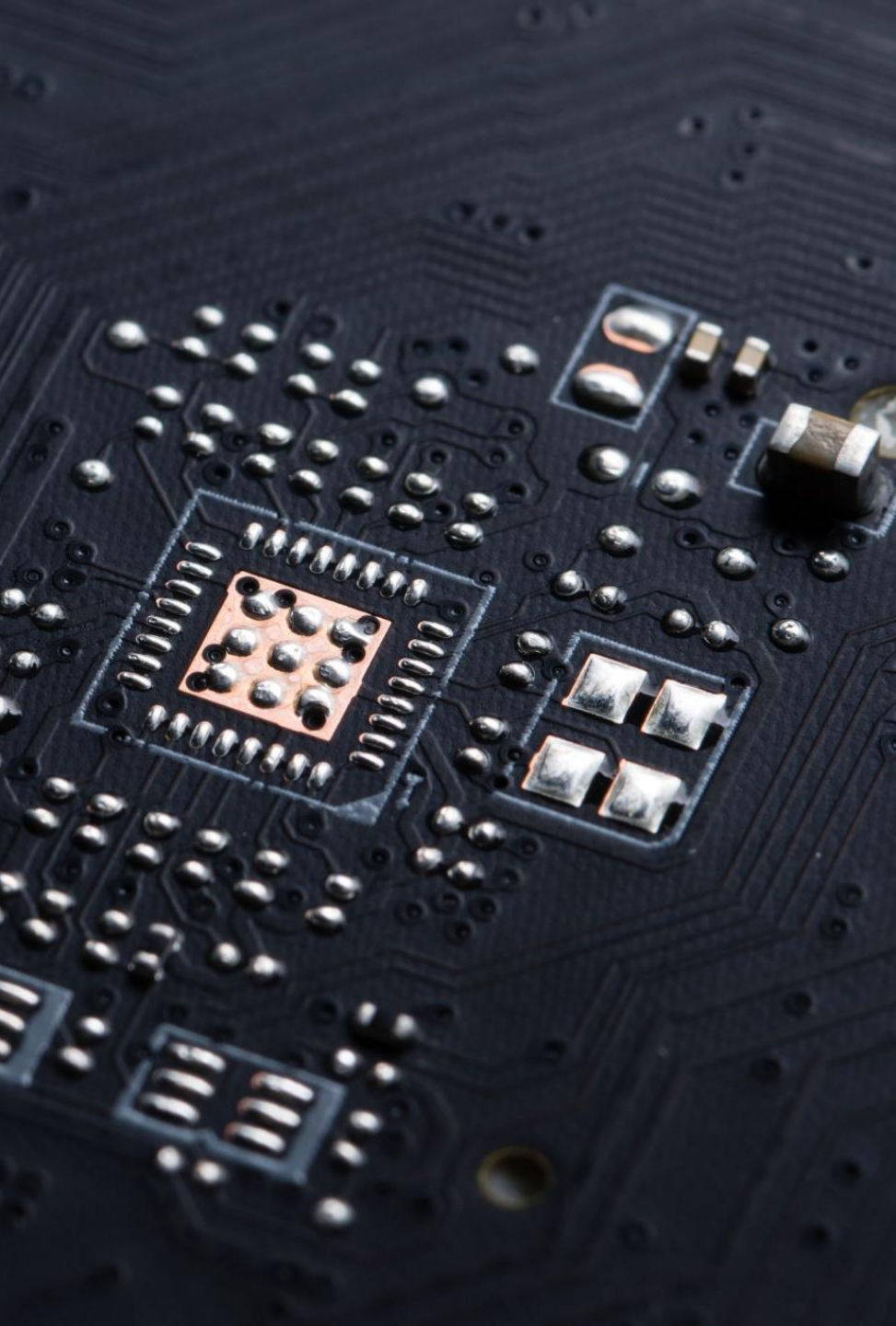
- Current state of all smart contracts.
- History of both transaction and smart contract.



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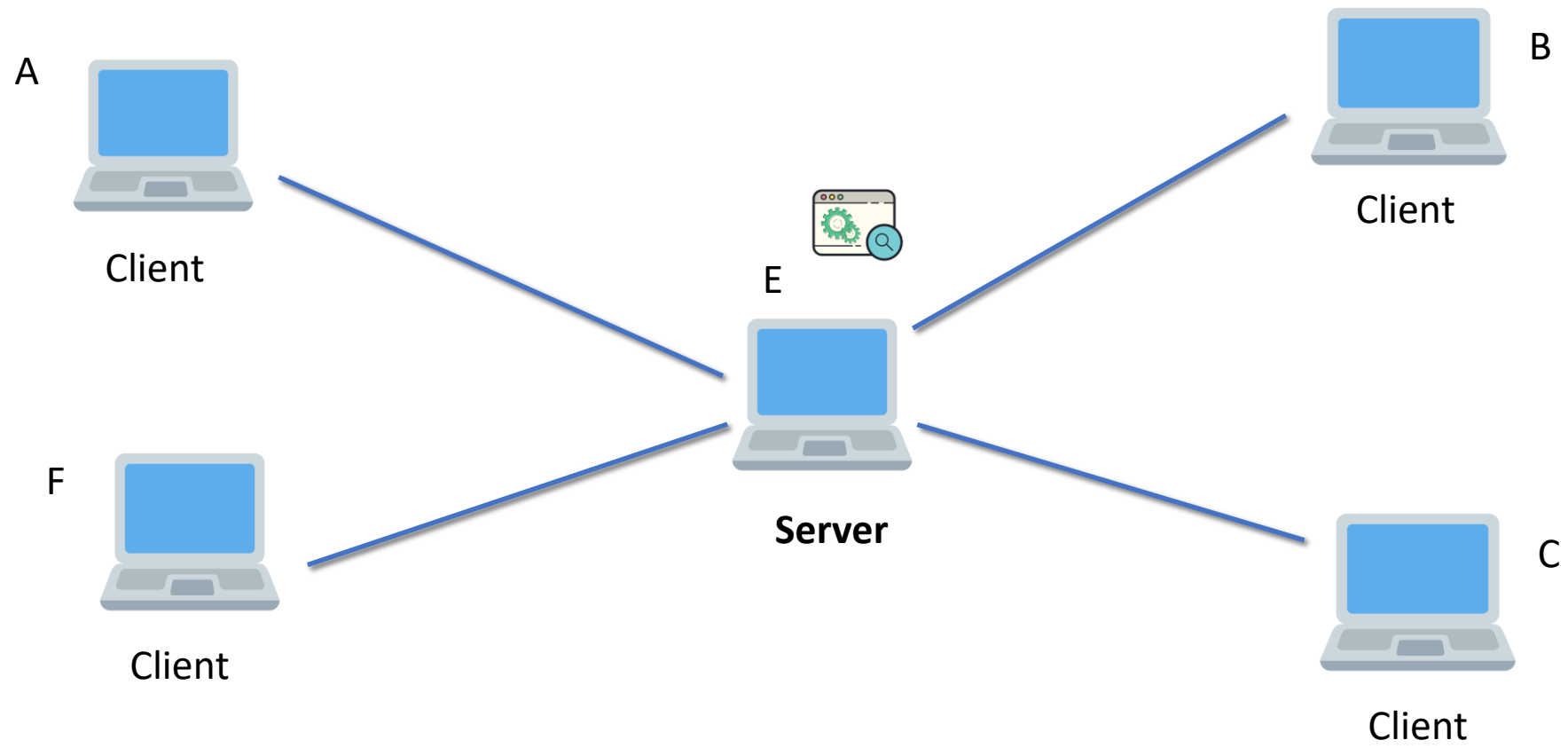
CODE EATER



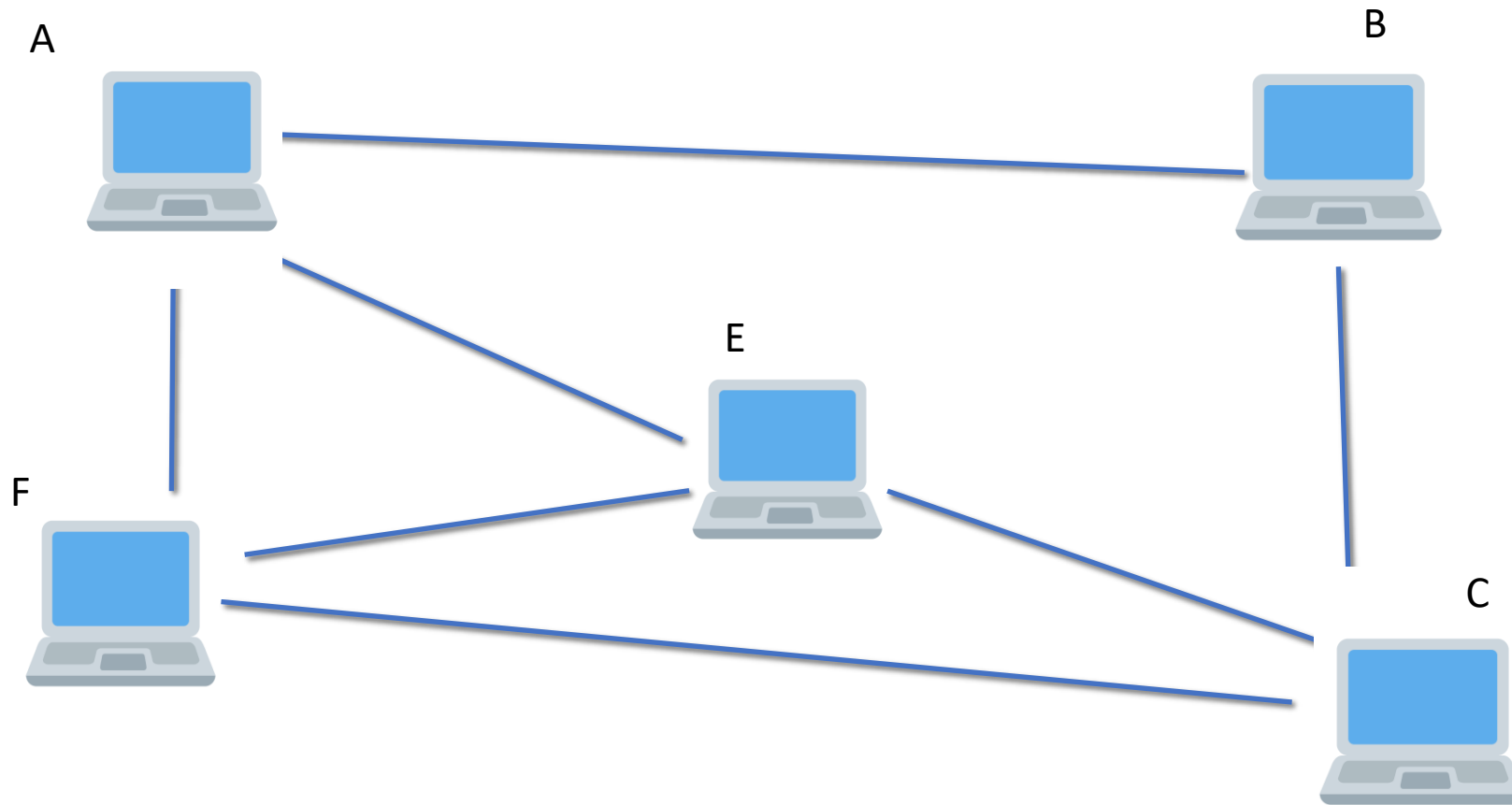


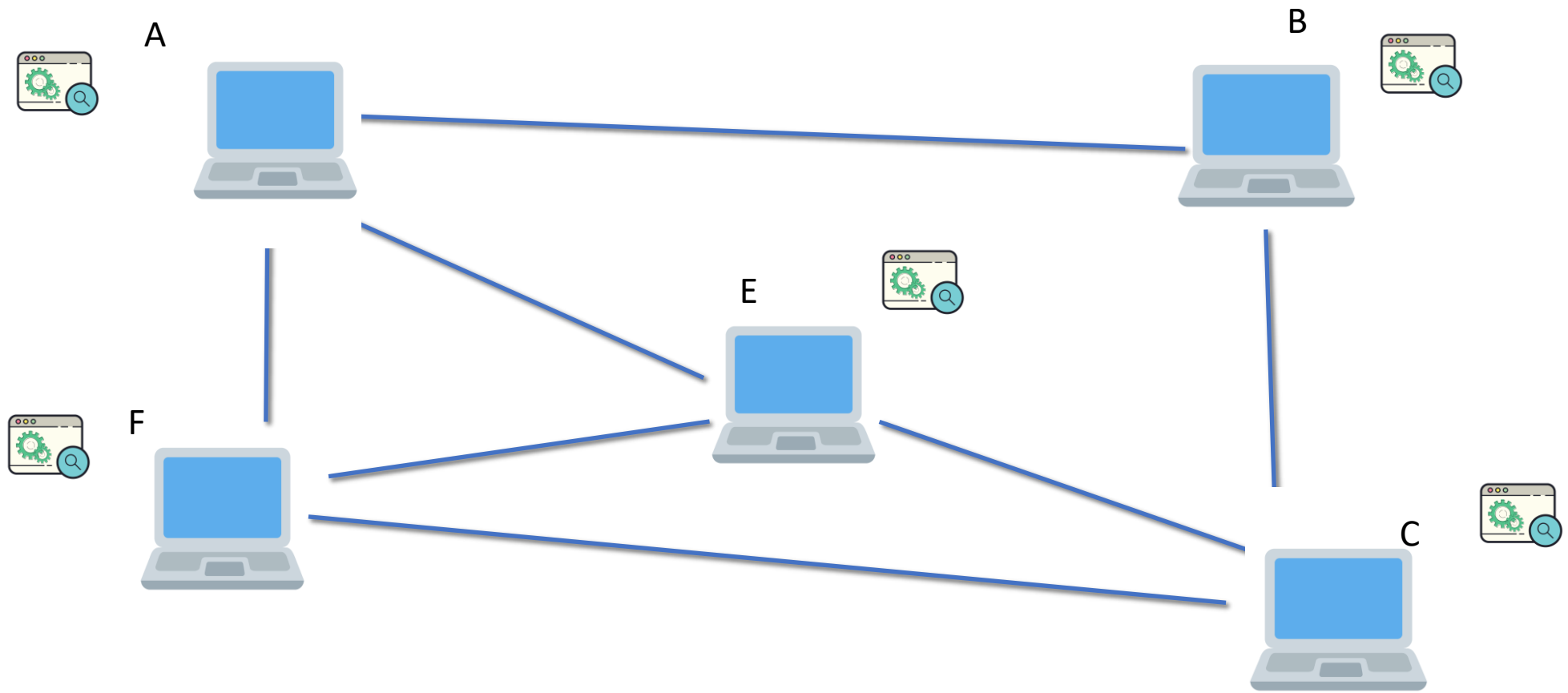
Decentralized Apps(Dapps)

Decentralized Apps(Dapps)



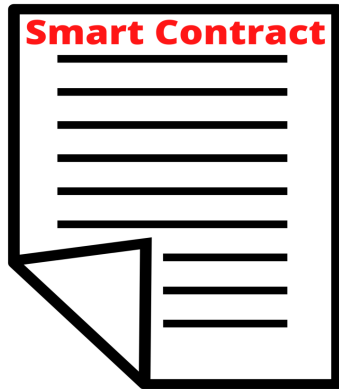
Decentralized Apps(Dapps)





Decentralized Apps(Dapps)

Decentralized Network



Smart Contract

+

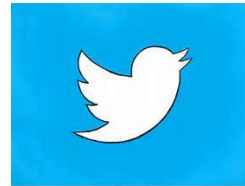


Front End

Search Engine



Social Media

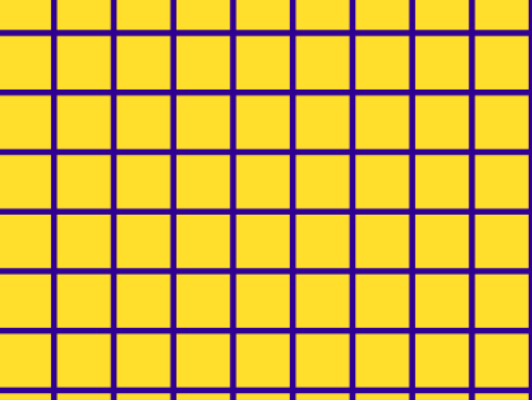


Video Platform



Decentralized Apps(Dapps)

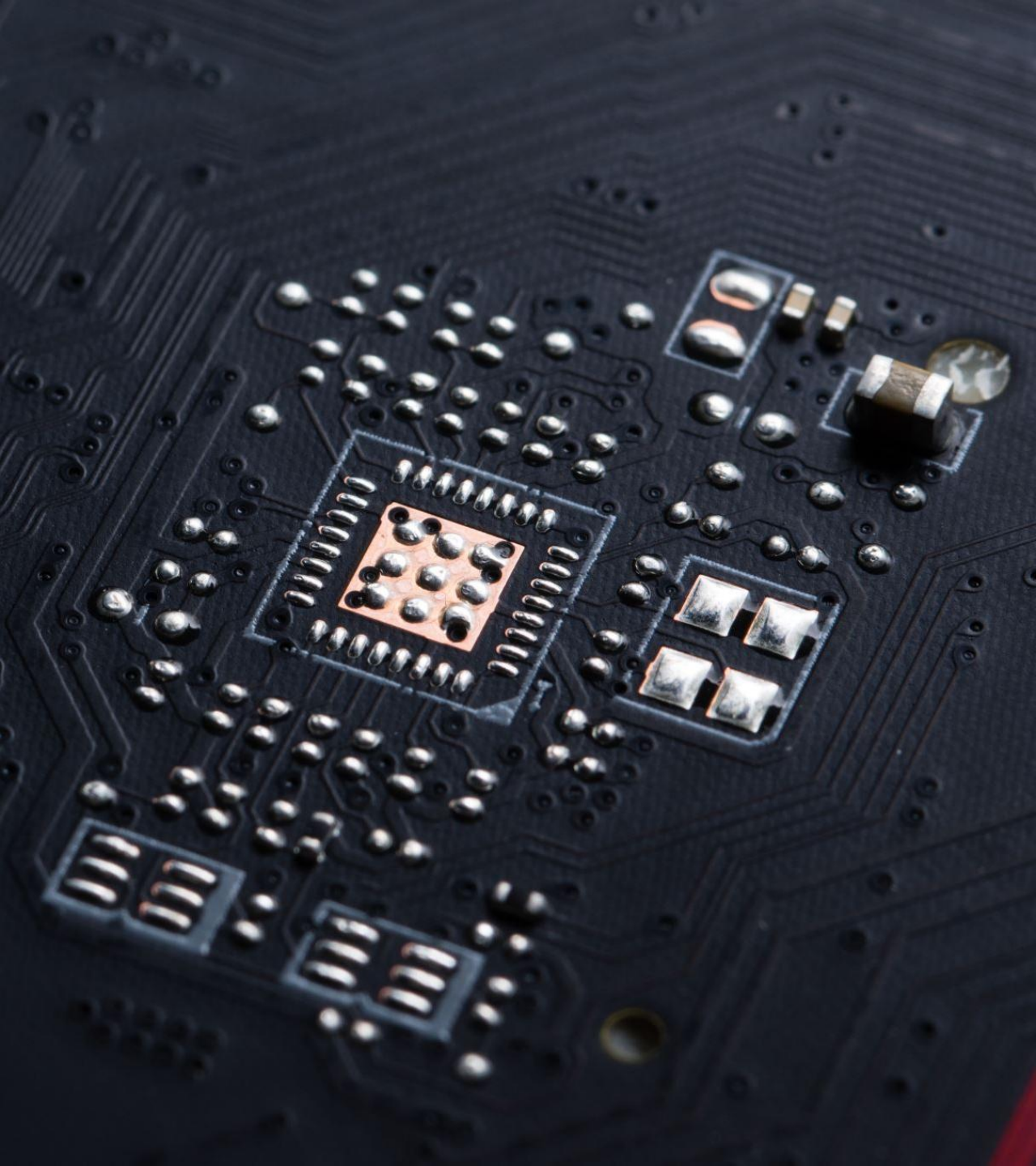
Centralized Apps	Decentralized Apps
Not Trustworthy	Trustworthy
Censorship	No censorship
You pay	They pay
Go down	Can never go down



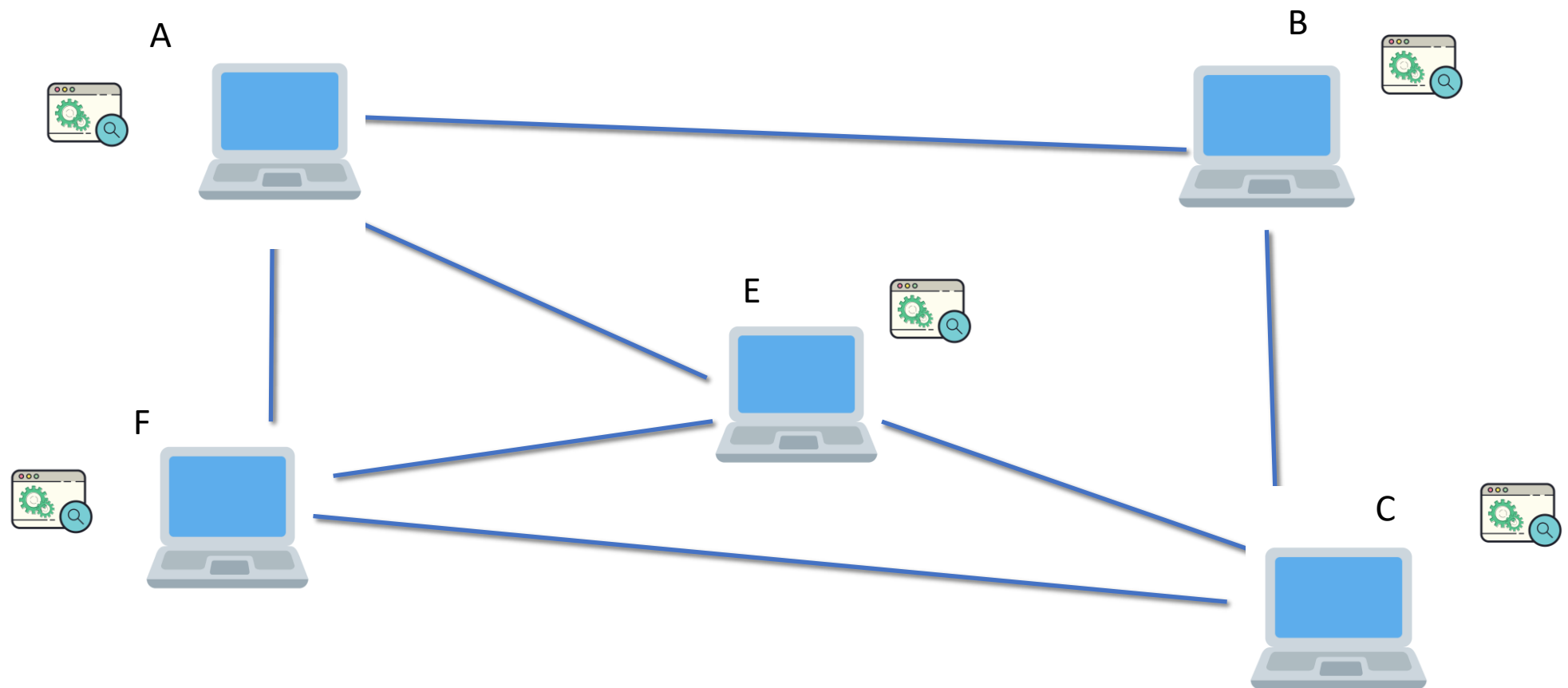
GIVE THIS VIDEO A THUMBS-UP !

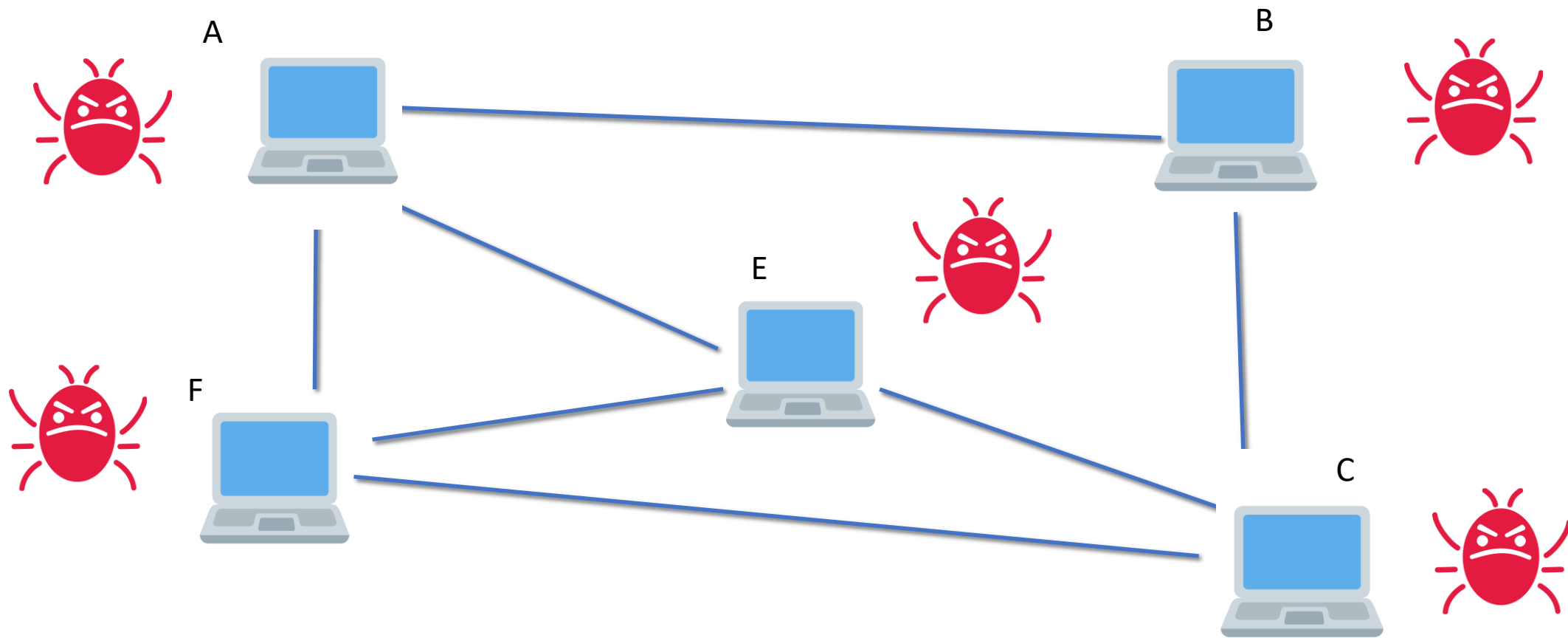
CODE EATER





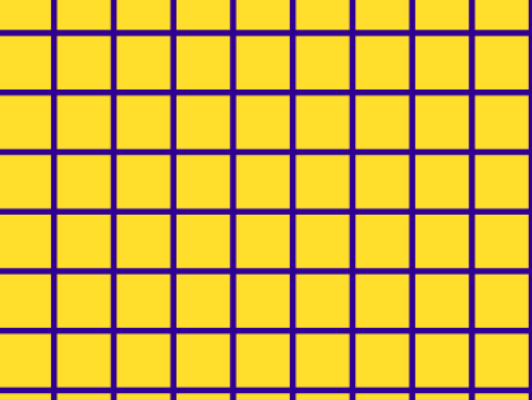
Ethereum Virtual Machine(EVM)





Ethereum Virtual Machine

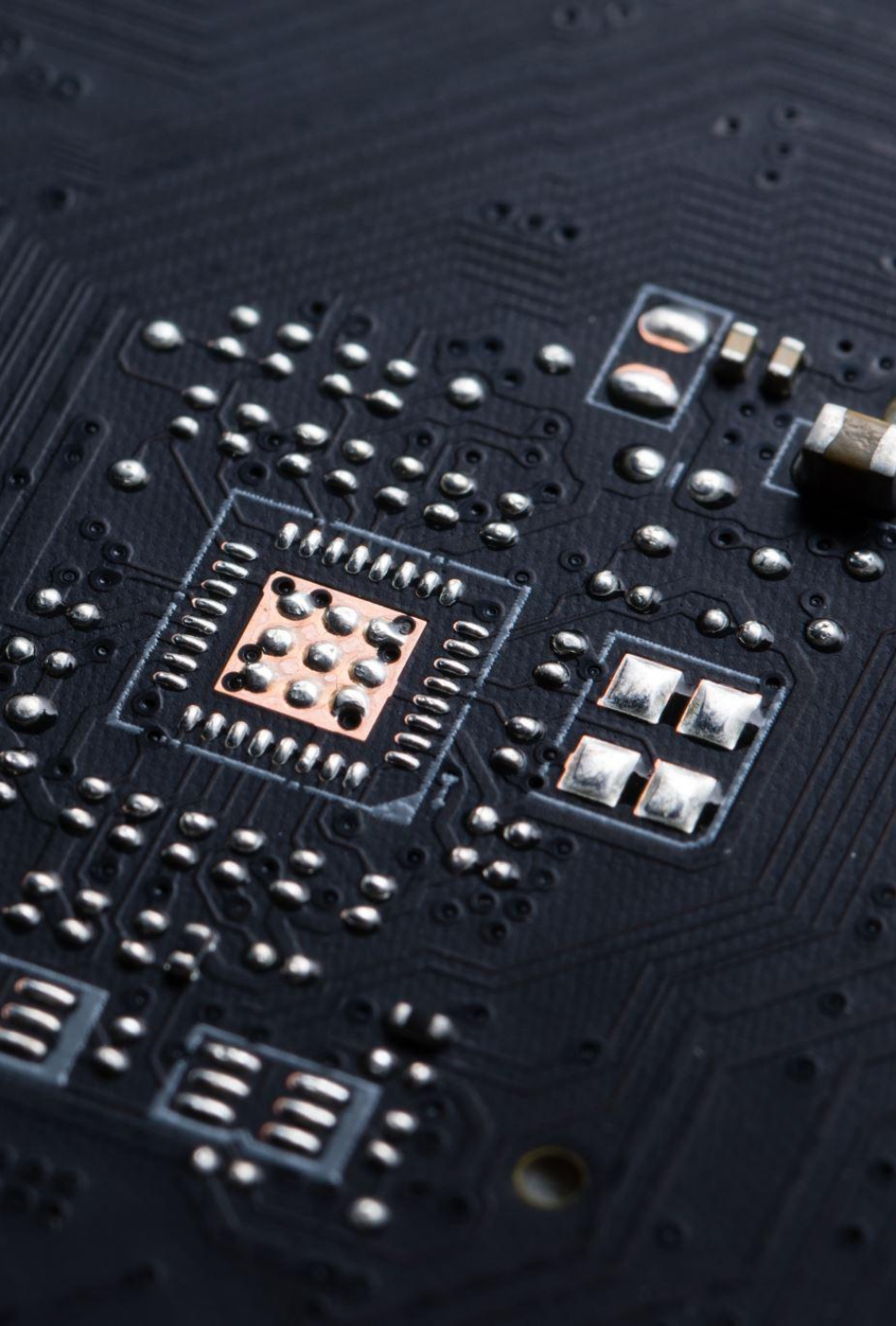




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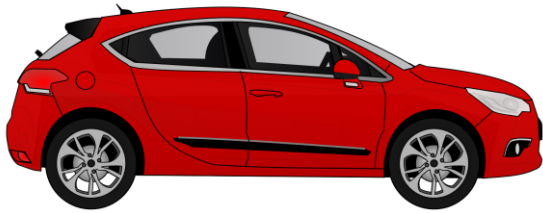
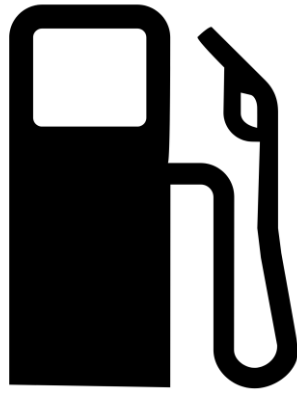
CODE EATER





Ethereum Gas

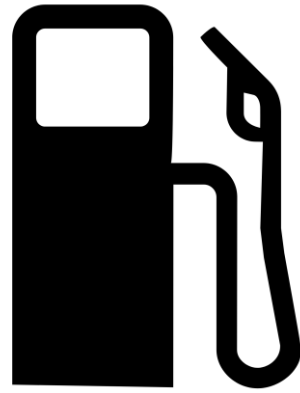
Ethereum Gas



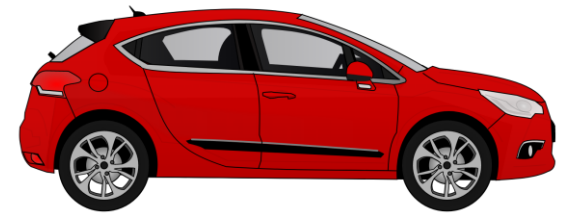
A

B

Ethereum Gas

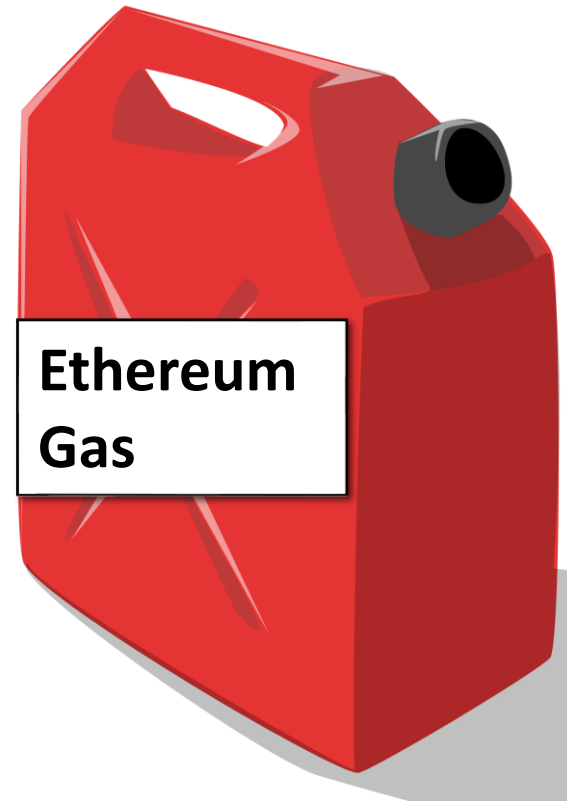
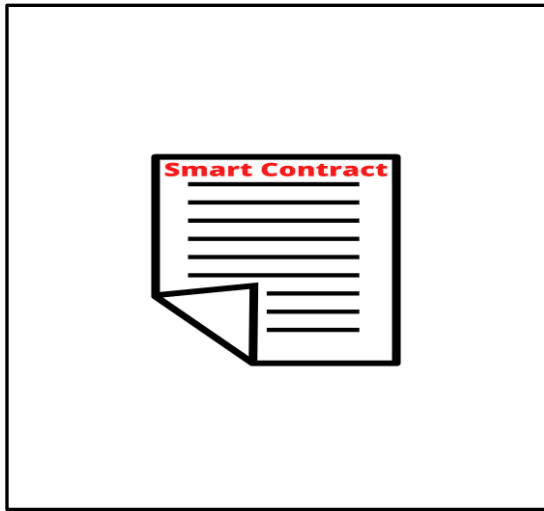


A



B

Ethereum Gas



Ethereum Gas

$$10 * 3 - 6 = ?$$

Multiplication – 5 gas

Subtraction – 3 gas

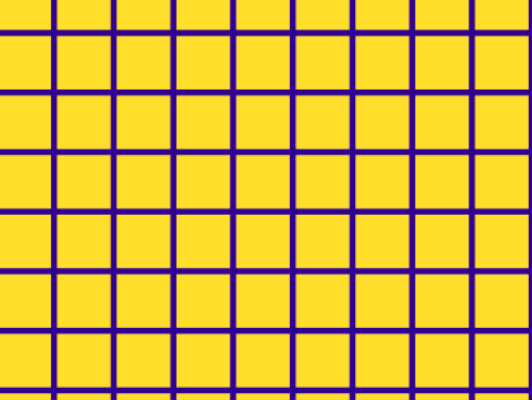
Equal to – 3 gas

Total gas → 5 + 3 + 3 = 11 Gas

Ethereum Gas

Some important points to note -

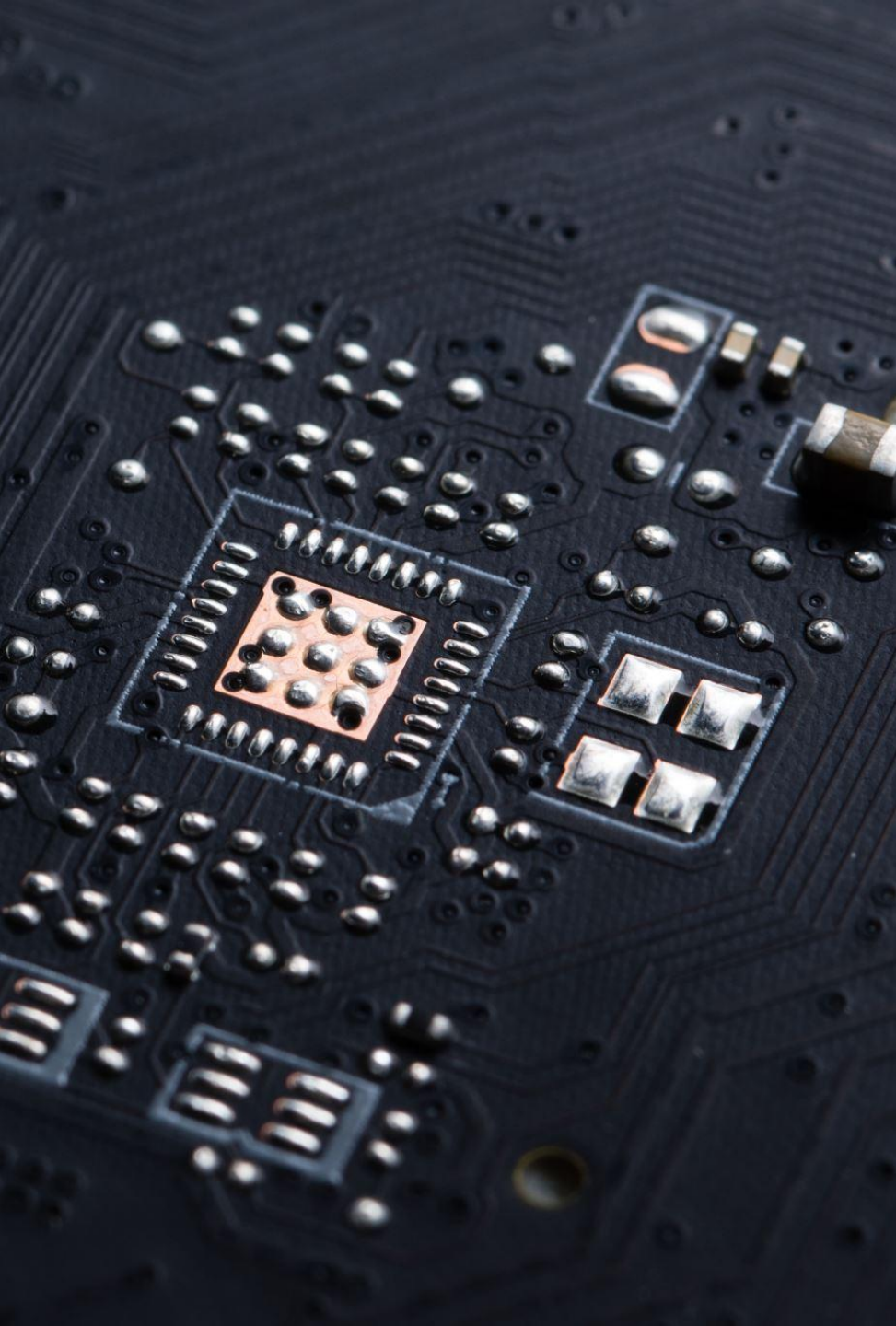
- Any transaction that modifies the blockchain costs gas.
- The user that generated the transaction pays for the gas.



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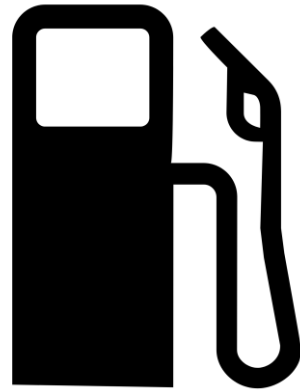
CODE EATER



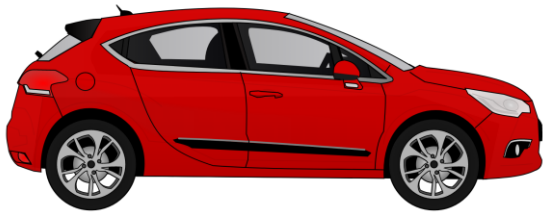


Ethereum Gas Price

Gas Price



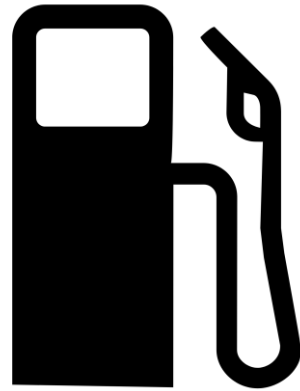
Petrol – 10 liters



A

B

Gas Price



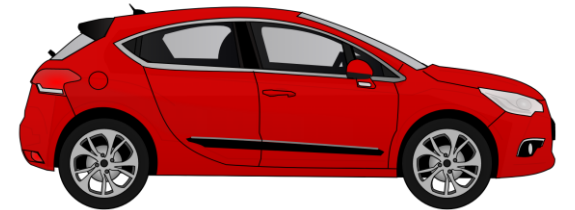
A

Petrol – 10 liters

Total price= ?

1 liter – Rs.5

**Total price= $10 \times 5 =$
Rs. 50**



B

Gas Price

$$10 * 3 - 6 = ?$$

Multiplication – 5 gas

Subtraction – 3 gas

Equal to – 3 gas

Total gas → 5 + 3 + 3 = 11 Gas

Gas Price

- It is the amount the sender wants to pay per unit of gas to get the transaction mined. gasPrice is set by the sender.
- Gas prices are denoted in gwei. (1 gwei = 10^{-9} ETH)

1 Gas price = 10 gwei

Gas Price

- It is the amount the sender wants to pay per unit of gas to get the transaction mined. gasPrice is set by the sender.
- Gas prices are denoted in gwei. (1 gwei = 10^{-9} ETH)

1 Gas price = 100 gwei

Gas Price

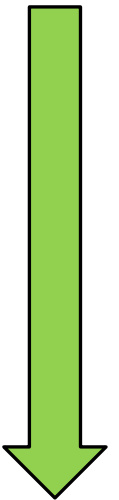
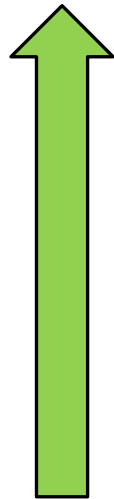
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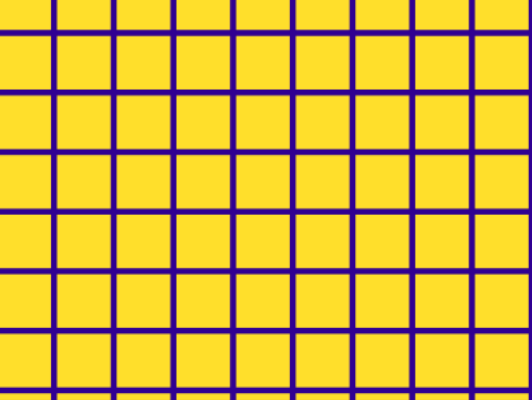
1 Gas price = 1000 gwei



Gas Price

- **The higher the gas price the faster the transaction will be mined.** It just like the transaction in Bitcoin.

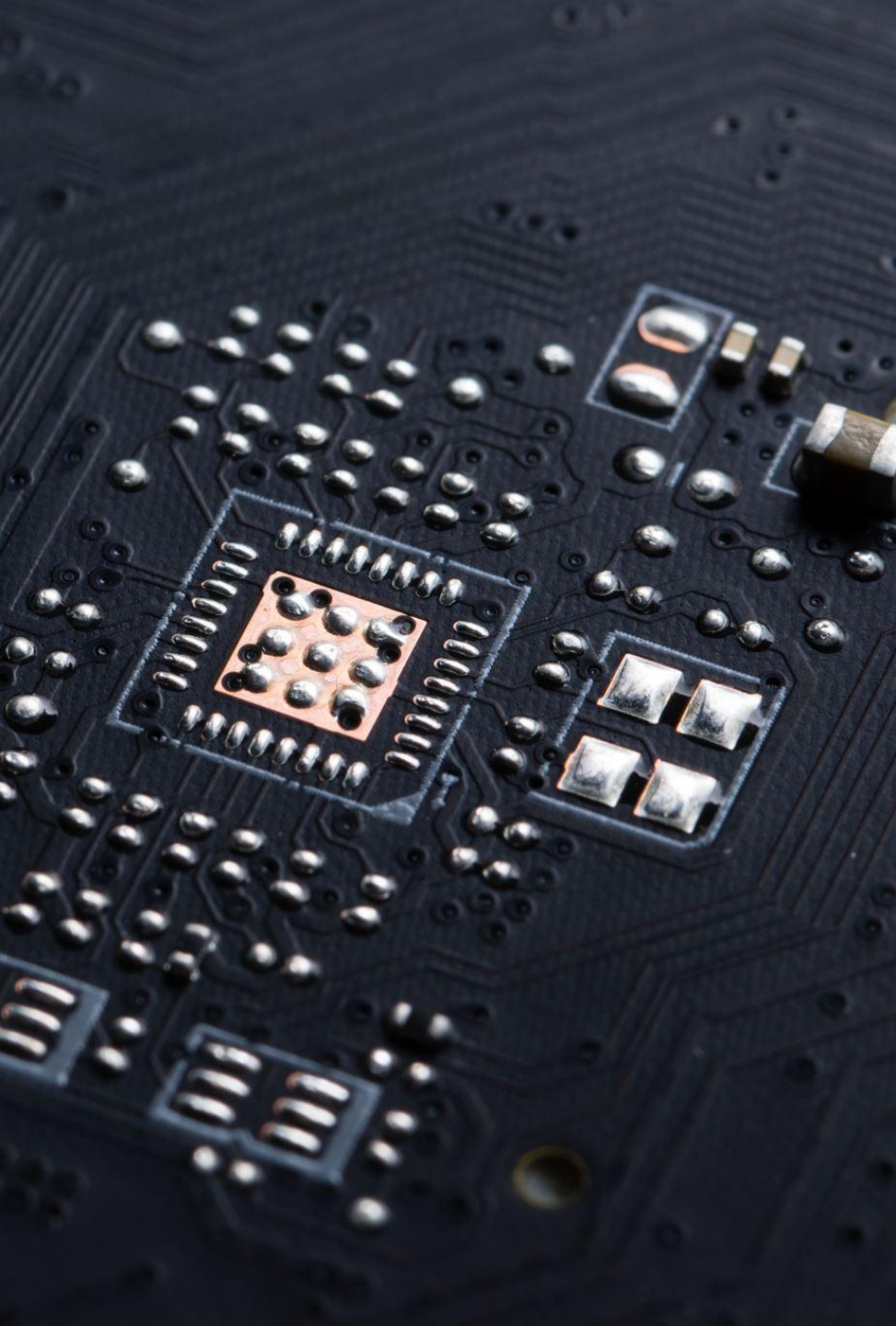




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CODE EATER

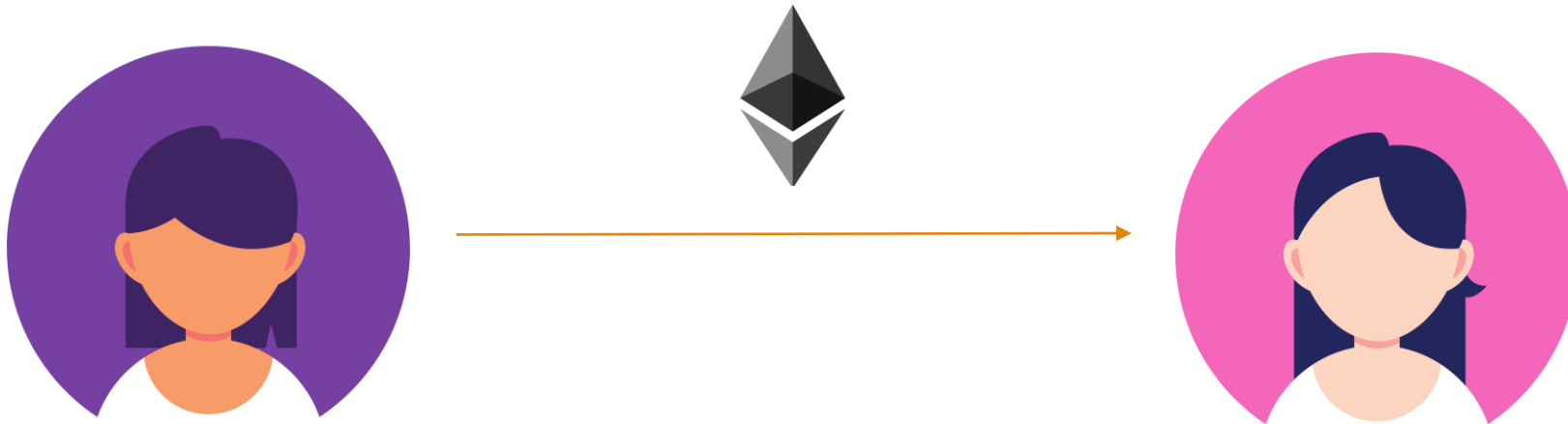




Ethereum Gas Limit

Gas Limit

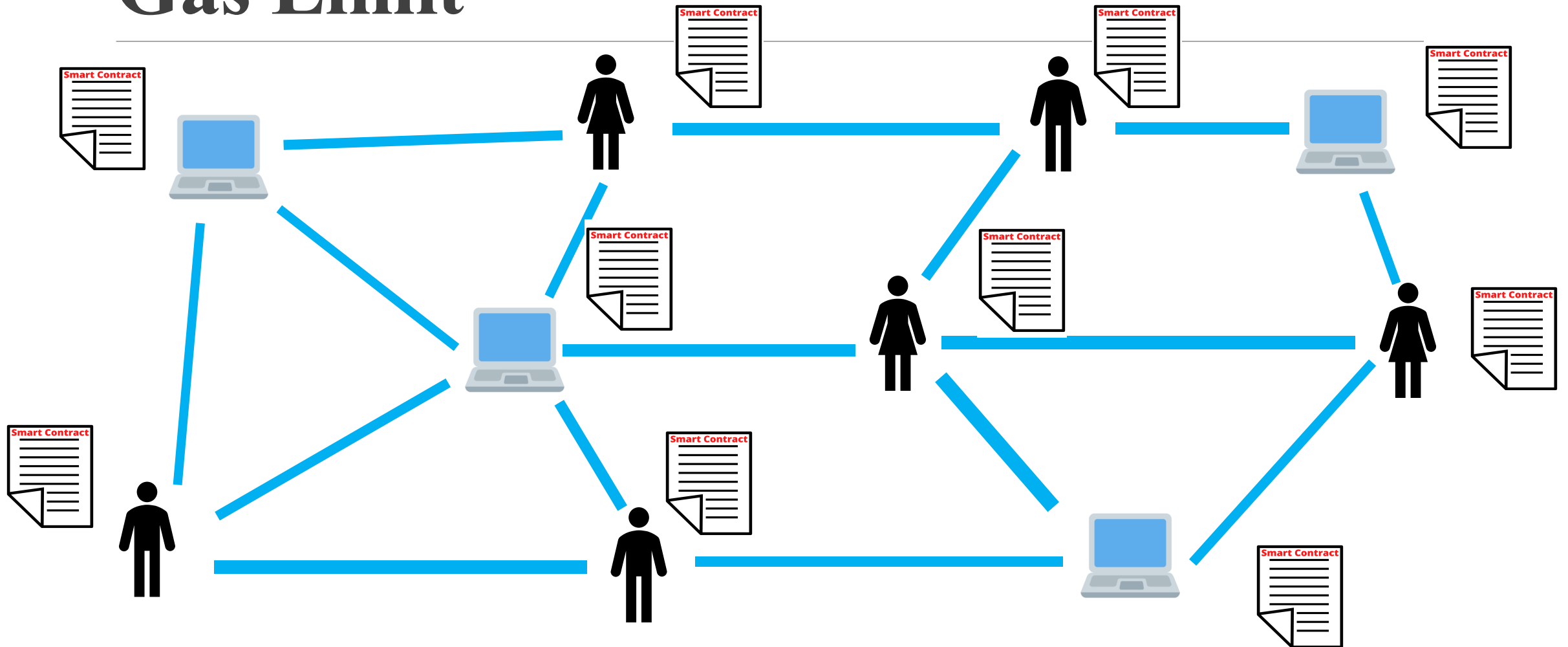
- It is the maximum gas the transaction can consume.
- Set by the sender.

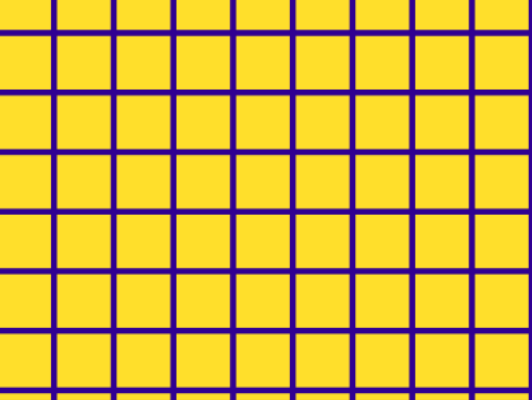


Gas Limit



Gas Limit





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CODE EATER



Gas Limit

Let say A wants to send B 2 ETH. So what will be the total fees A that has to pay ?

Case 1: When transaction gas limit is 21,000 units.

A sets the gas price per unit = 100 gwei.

Transaction gas limit = 21,000 units.

Total fee will be: Gas units(limit) * Gas price per unit

Total fee will be: 21,000 * 100 = 210,000 gwei or 0.0021 ETH

Gas Limit

Let say A wants to send B 2 ETH. So what will be the total fees A that has to pay ?

Case 2: When gas transaction limit < 21000 units.

Transaction gas limit = 20,000 units.

Transaction Fail

Gas Limit

Let say A wants to send B 2 ETH. So what will be the total fees A that has to pay ?

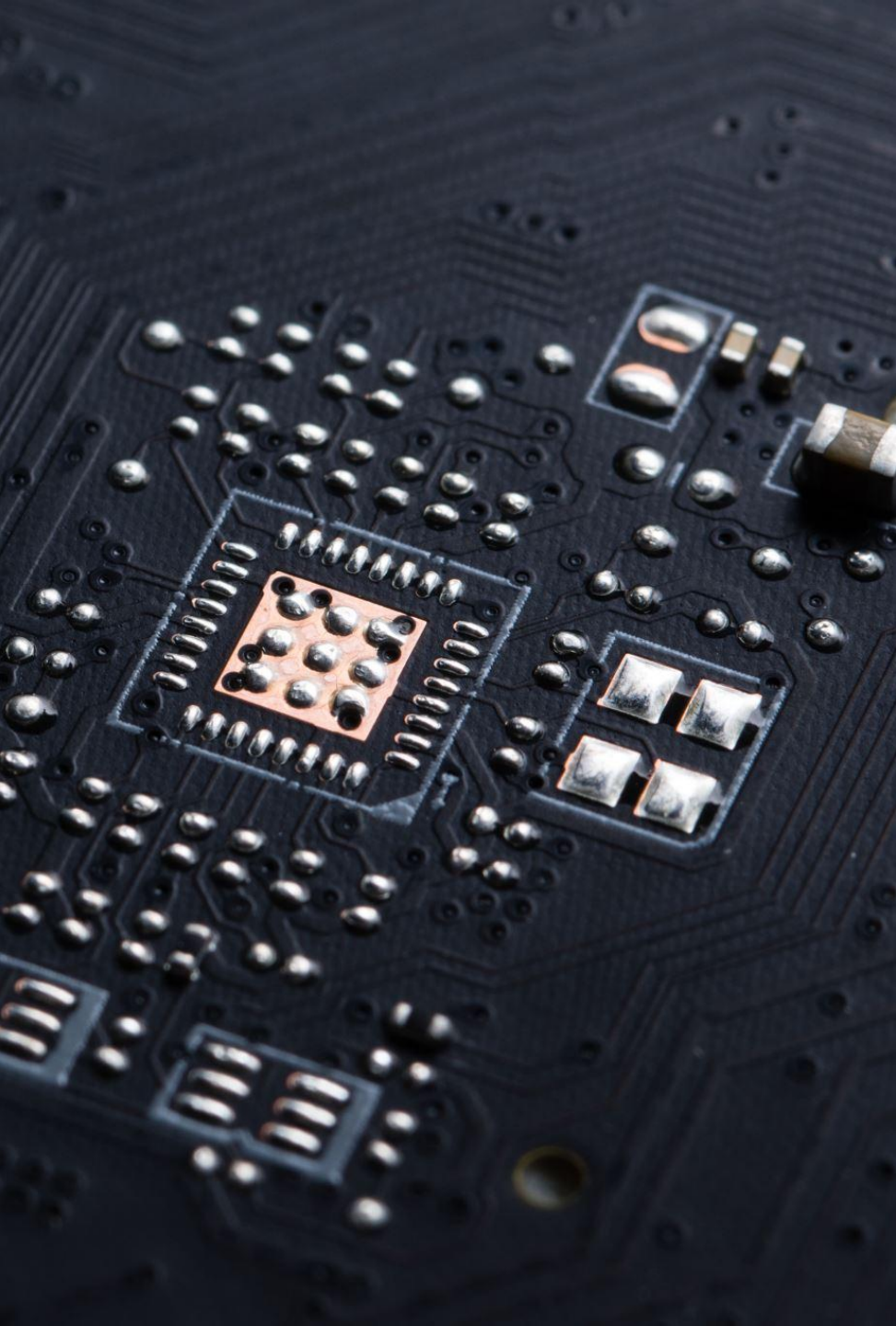
Case 3: When gas transaction limit > 21000 units.

Transaction gas limit = 22,000 units.

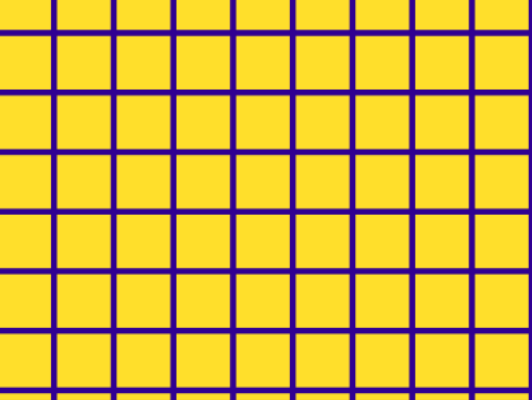
22,000 – 21000 = 1000 will be returned

Gas Limit

Q) What is the use of Gas Limit ?



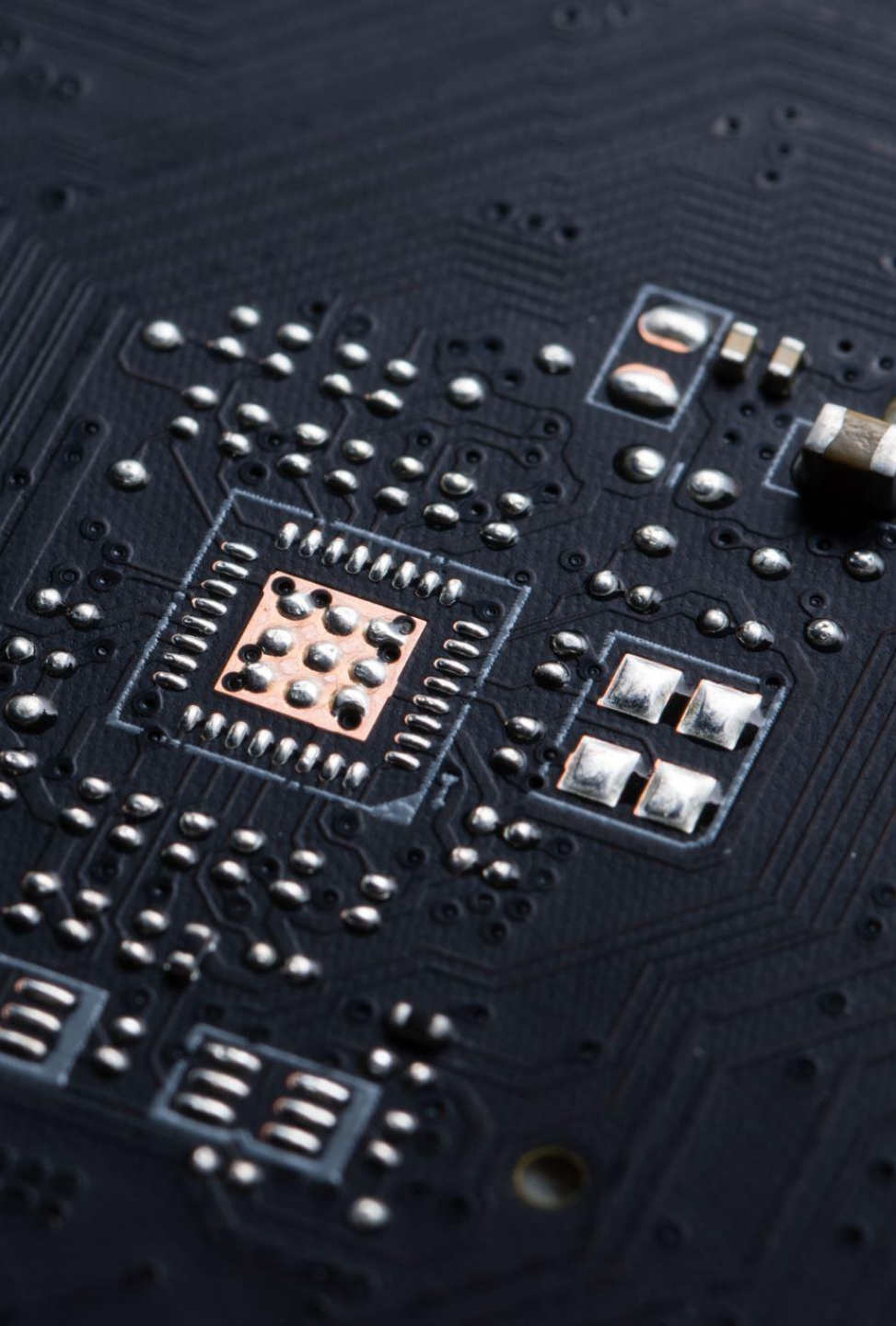
Ethereum Demo



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CODE EATER



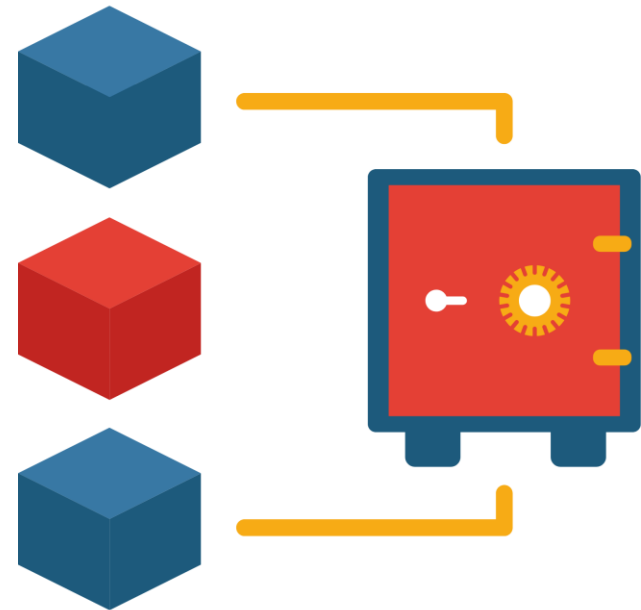


Ethereum 2.0

ETH 2.0



POW

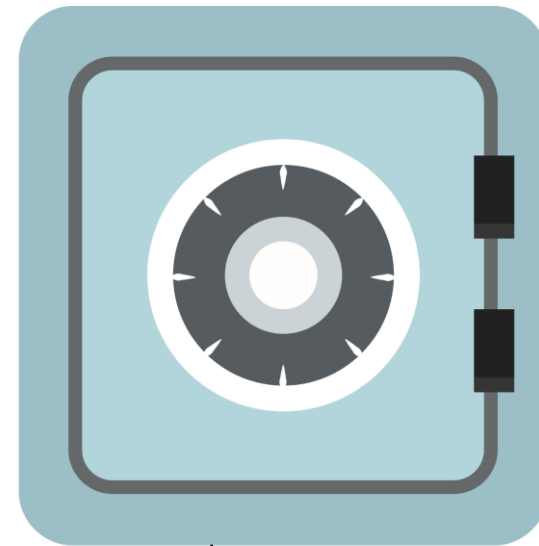


POS

Proof of Stake



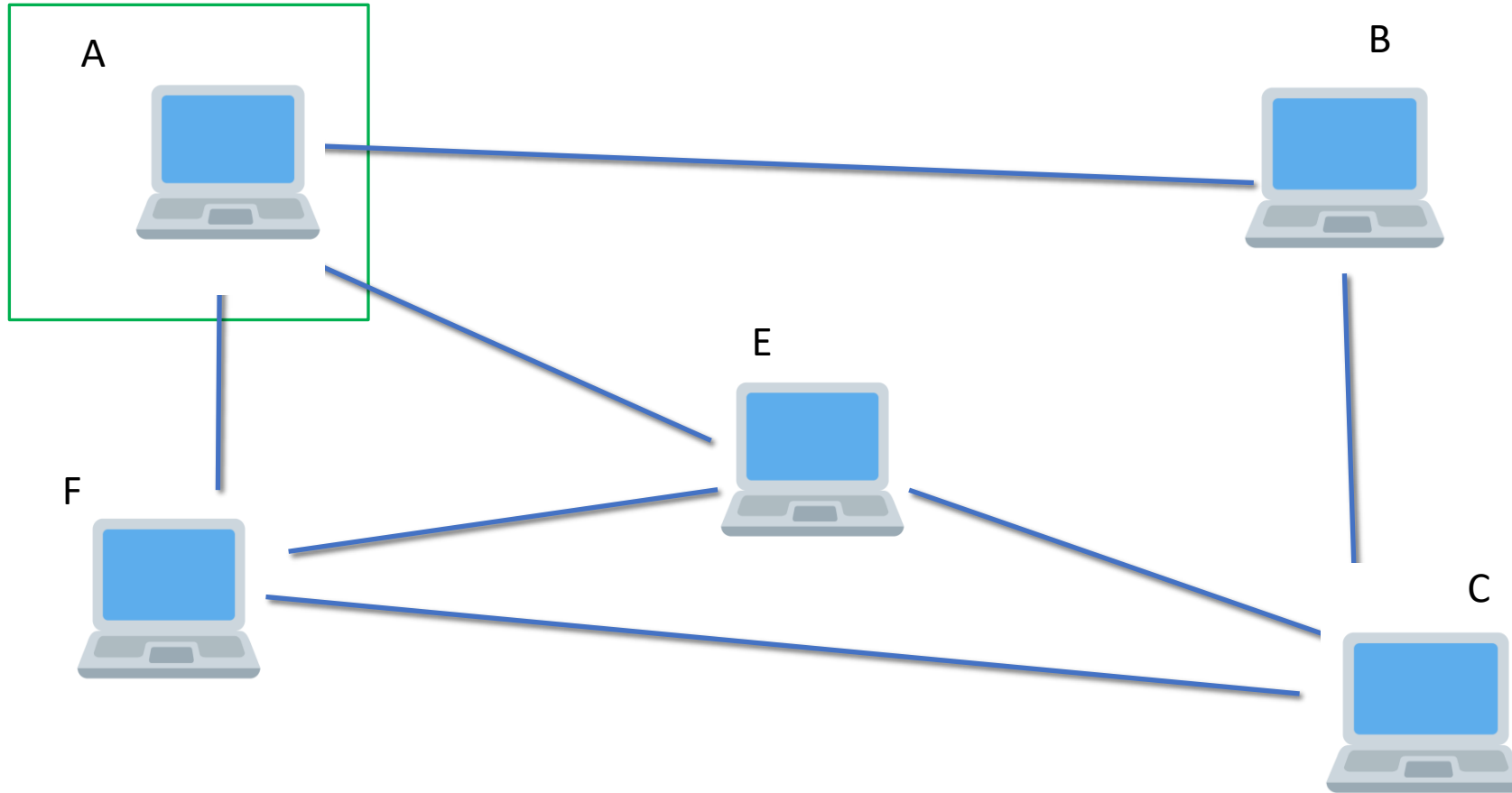
Validator



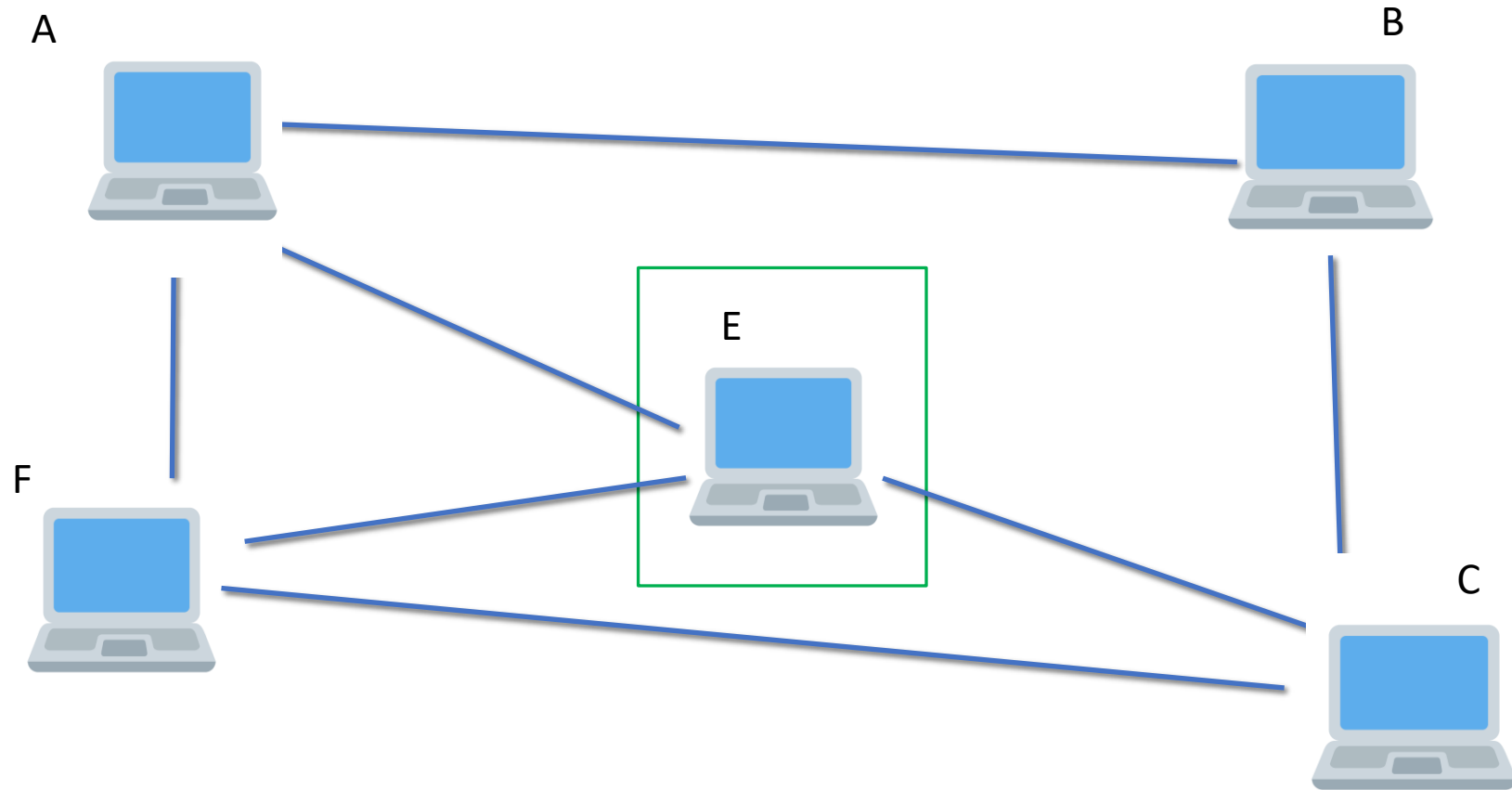
+ Transaction Fee



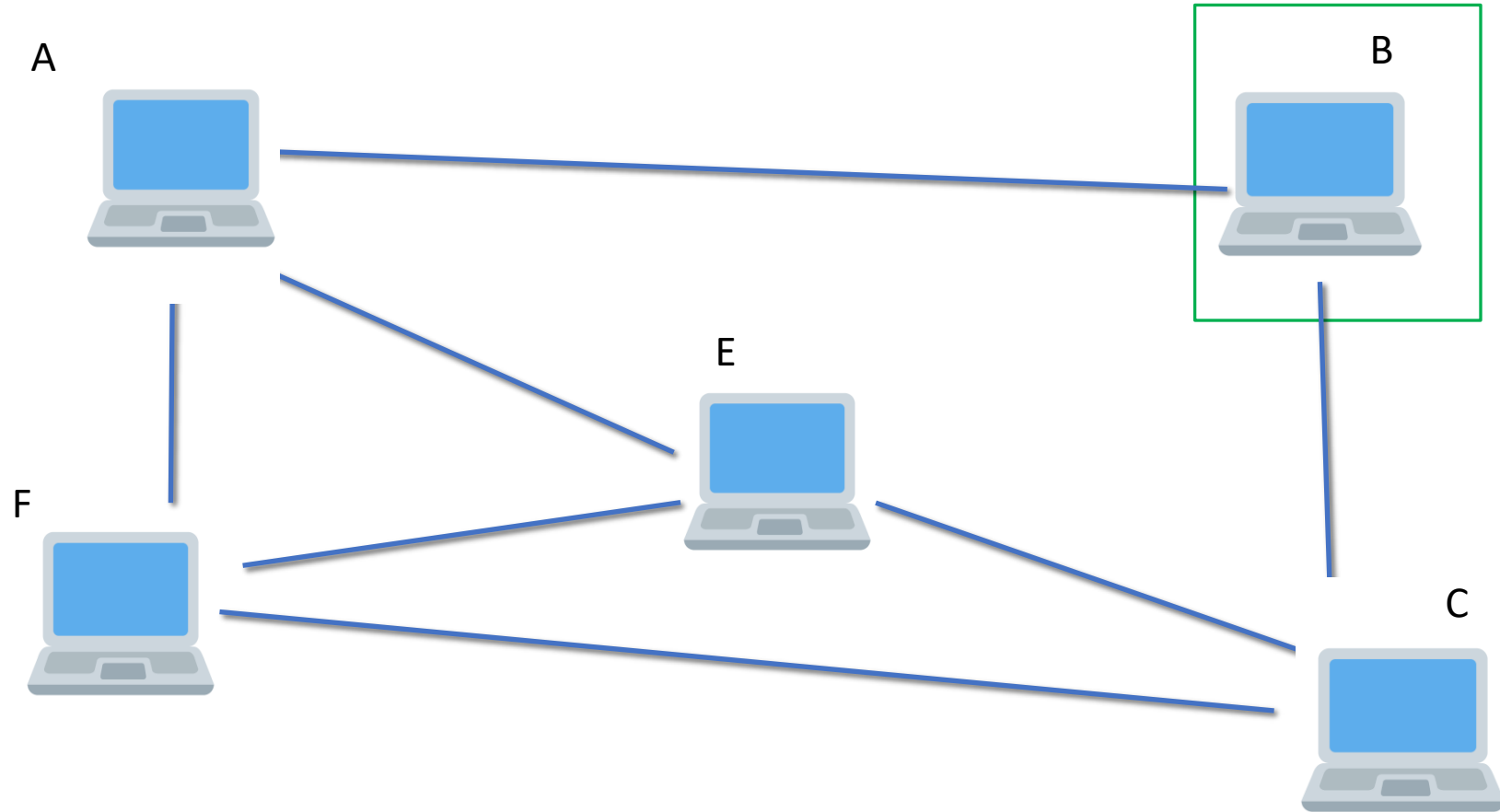
Proof of Stake



Proof of Stake



Proof of Stake

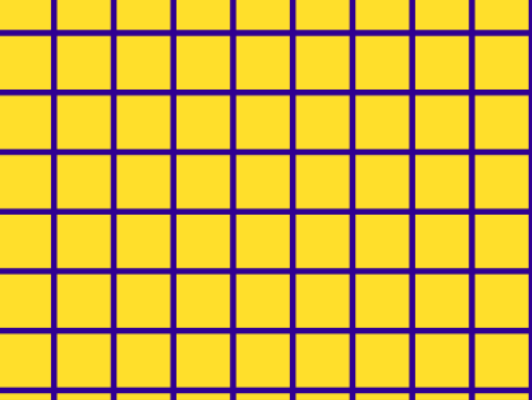


Proof of Stake

- The more ether you pay the more chances of getting randomly selected you have.

POW vs POS

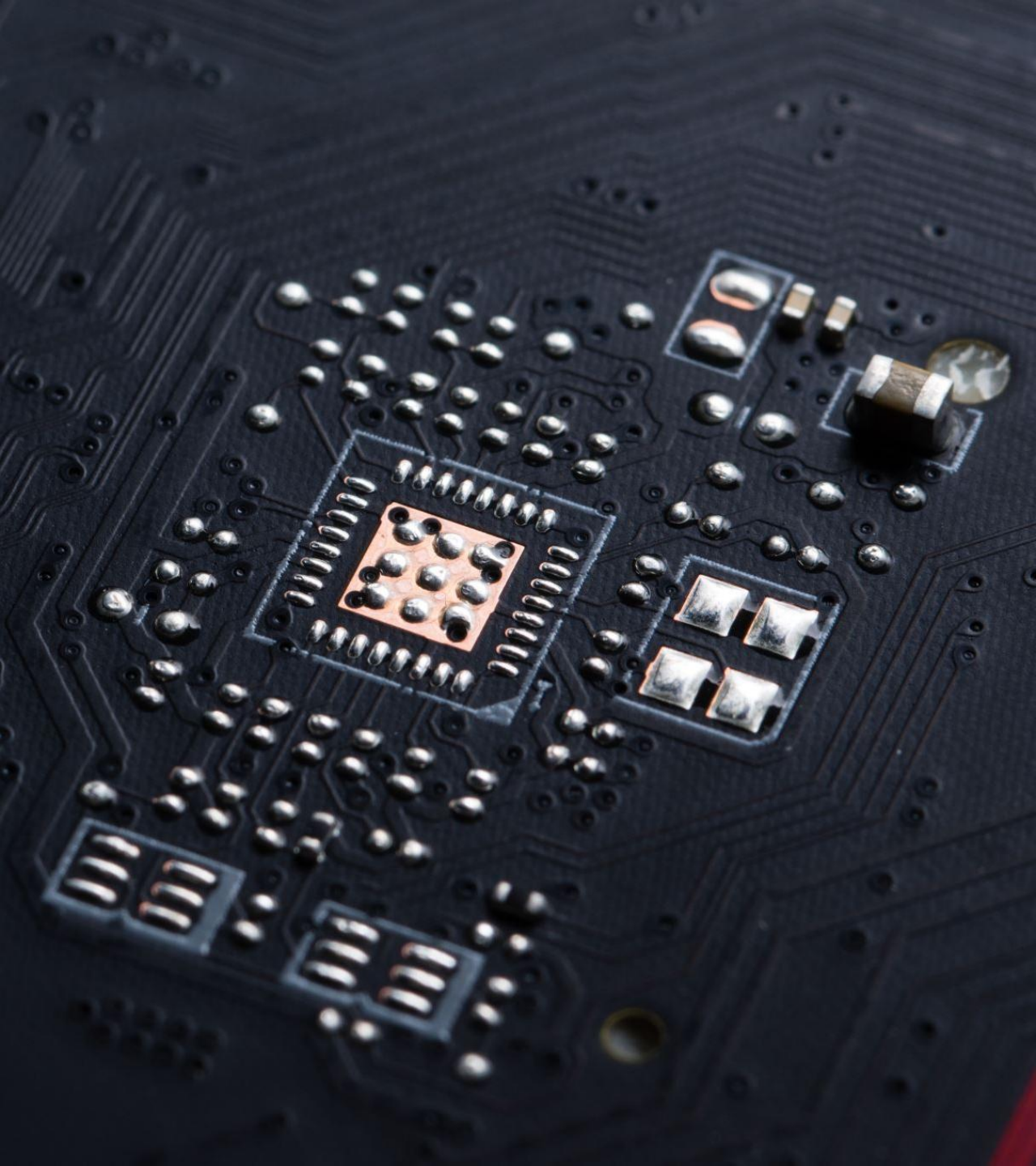
Proof Of Work(PoW)	Proof Of Stake(PoS)
Miners	Validators
High performance hardware required.	Mobile or Laptop are enough.
Lots of electricity required.	Not much electricity is required.
The more hashing power you have the more blocks you can validate.	The more ETH you stake the more blocks you can validate.
Attack to happen 51% hashing power is required.	Attack to happen 51% of stake is required.
Competition is there.	Random selection is there.



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CODE EATER

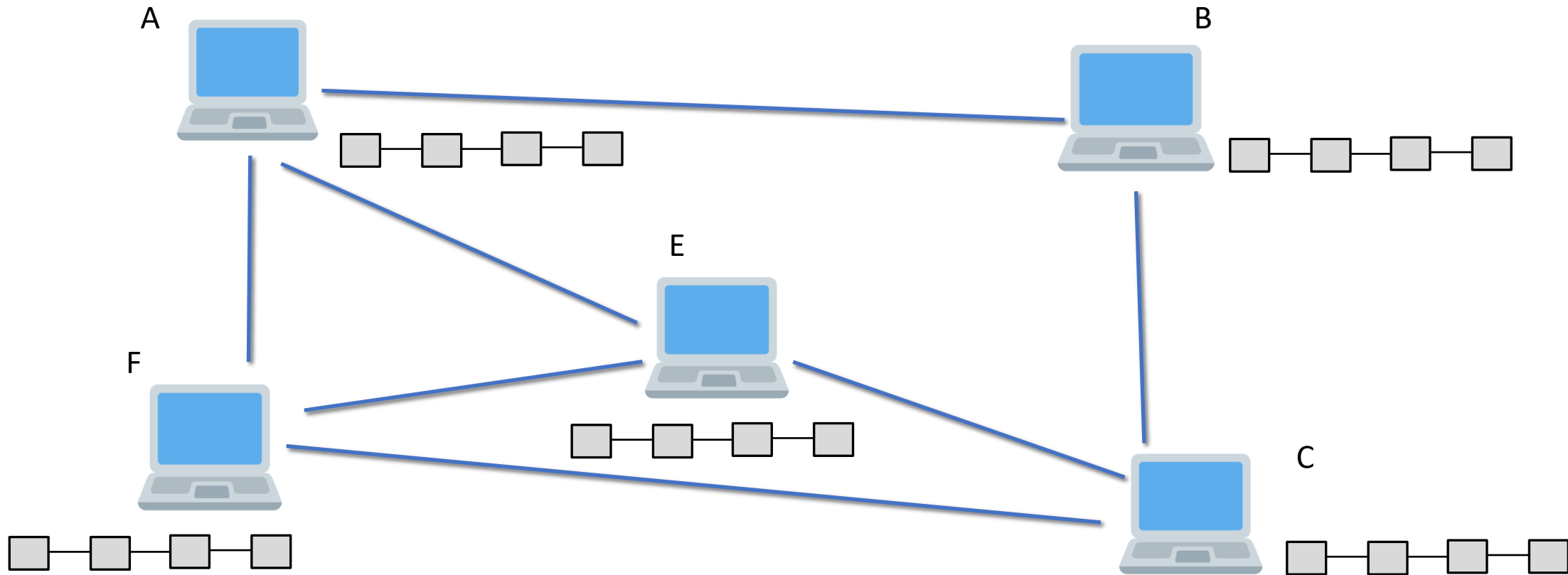




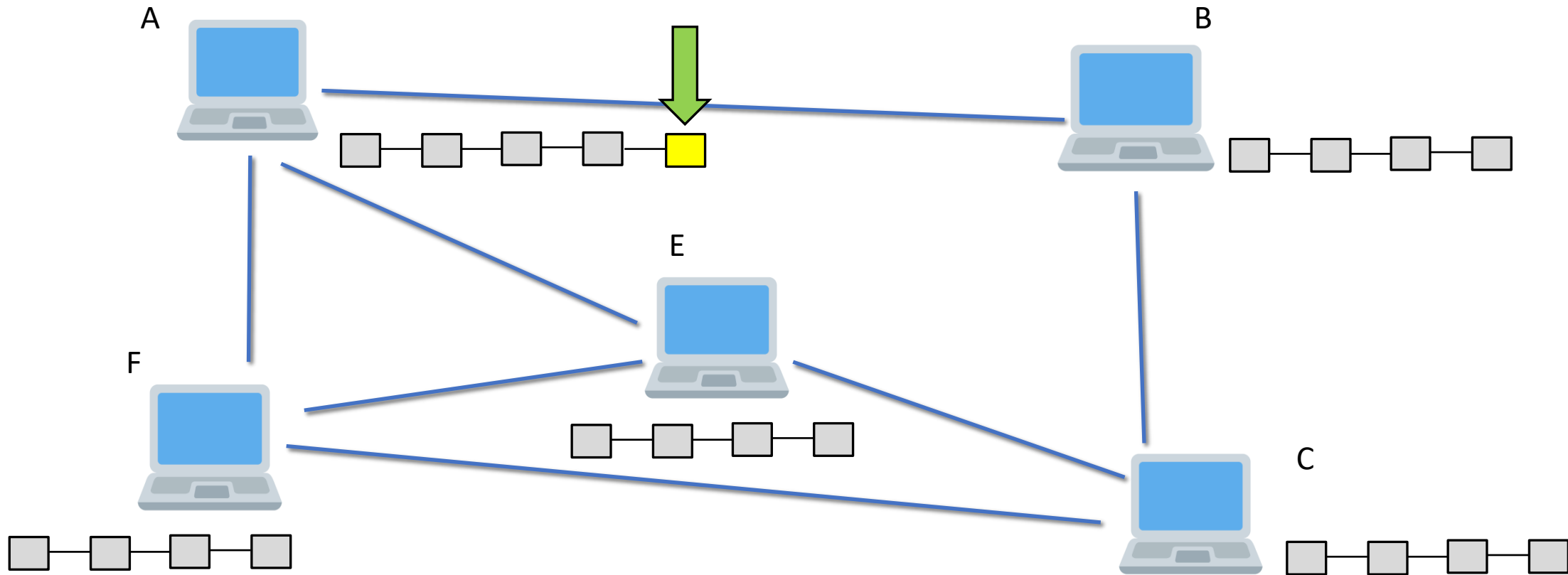
Sharding

Sharding

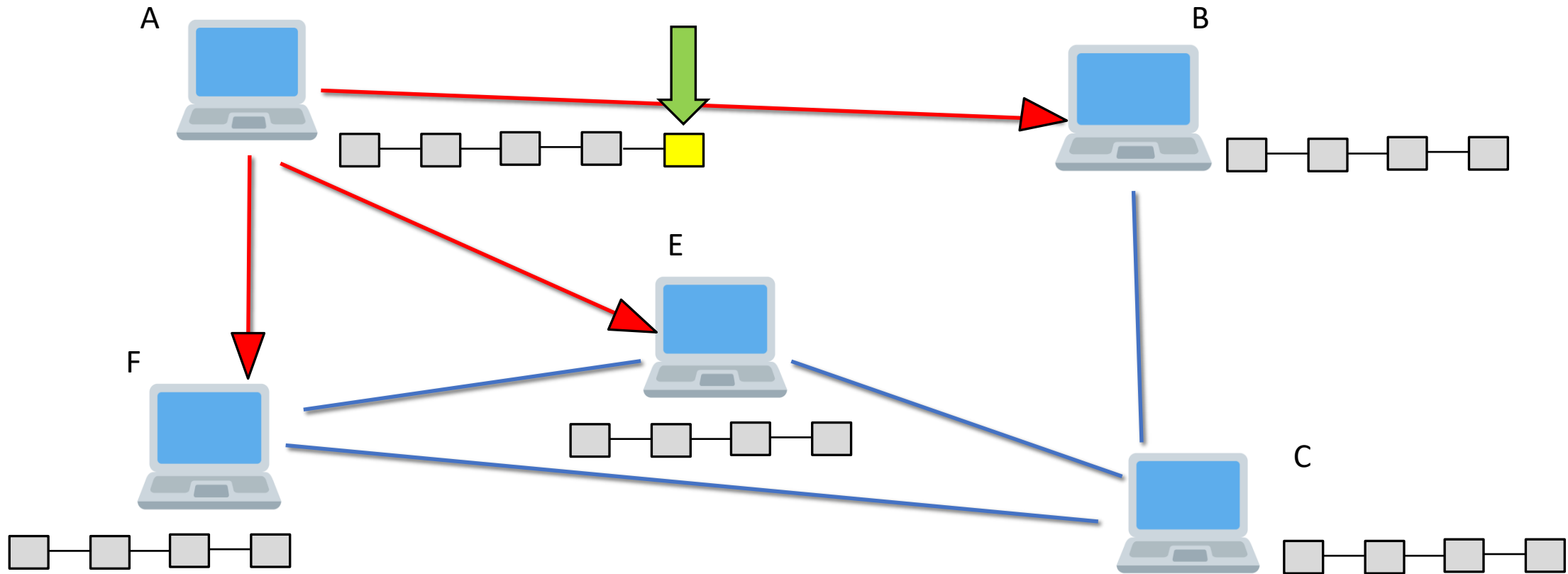
Consensus Protocol



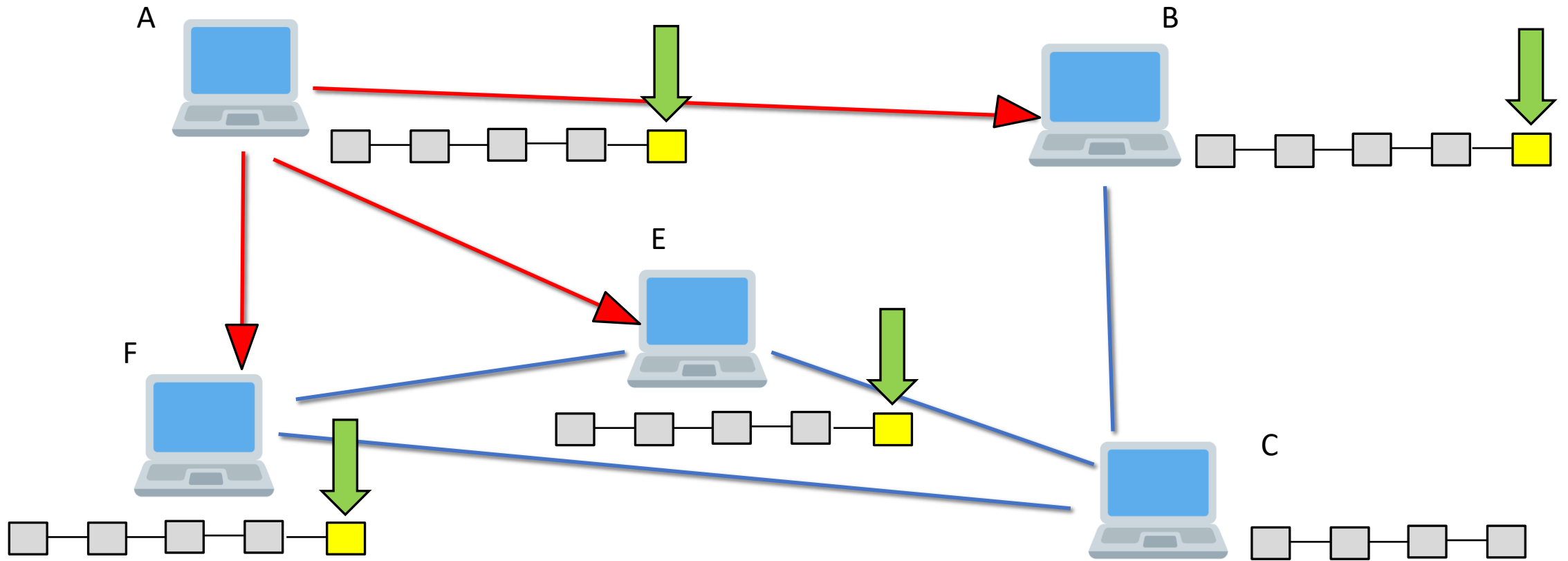
Consensus Protocol



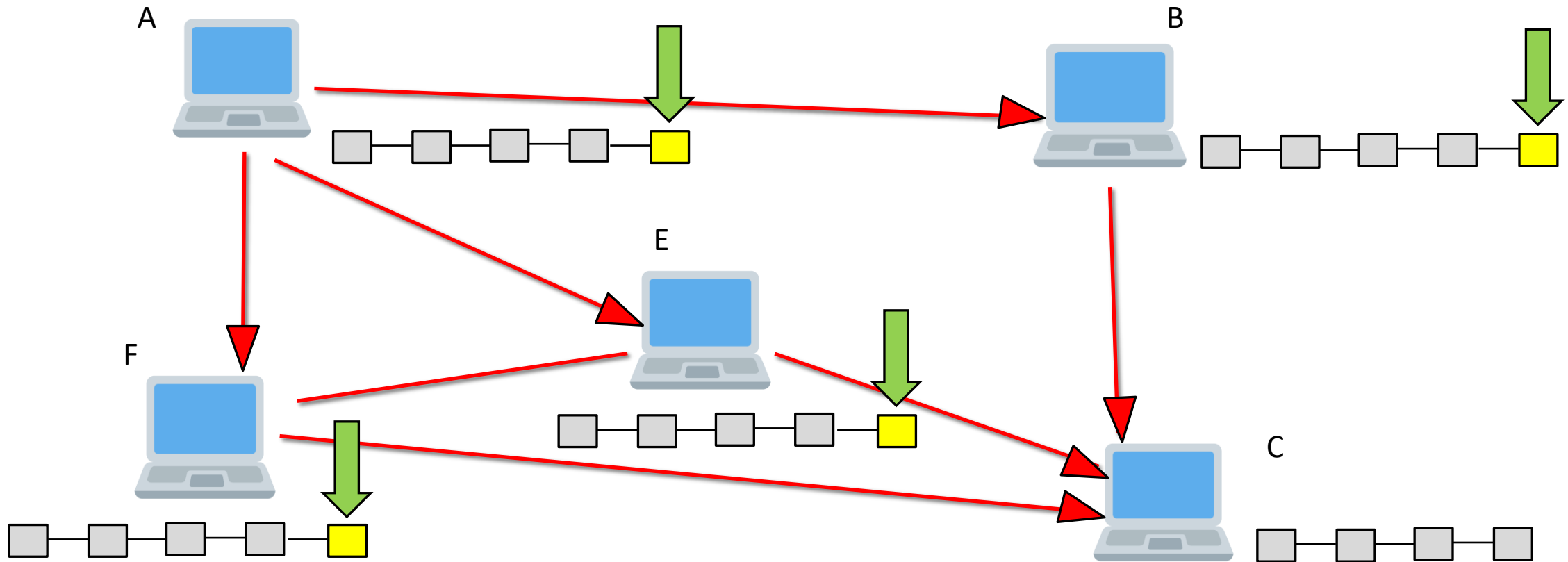
Consensus Protocol



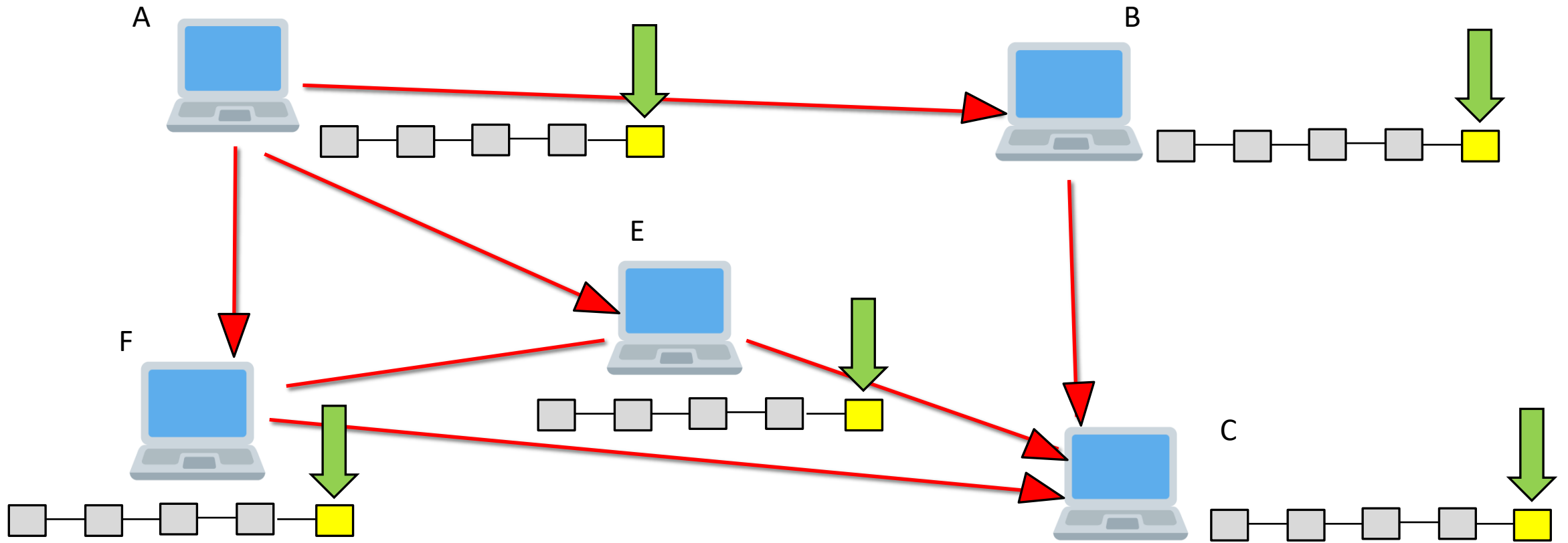
Consensus Protocol



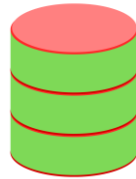
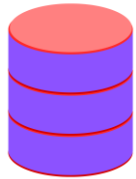
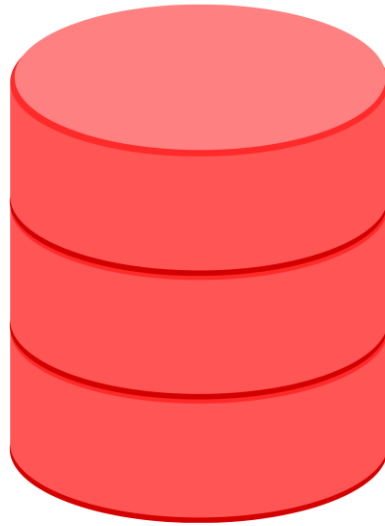
Consensus Protocol



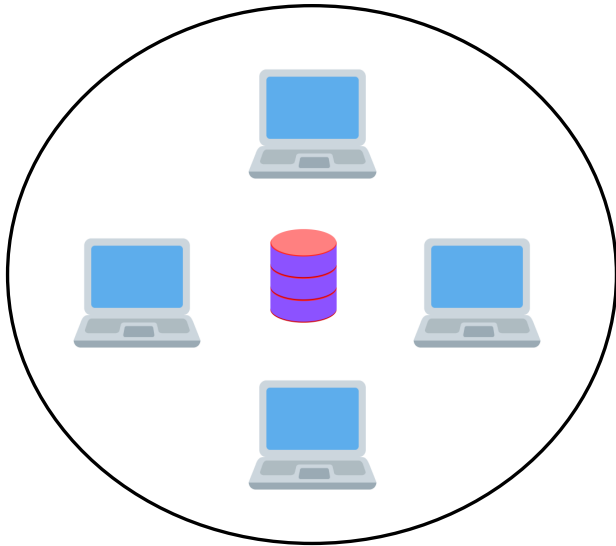
Consensus Protocol



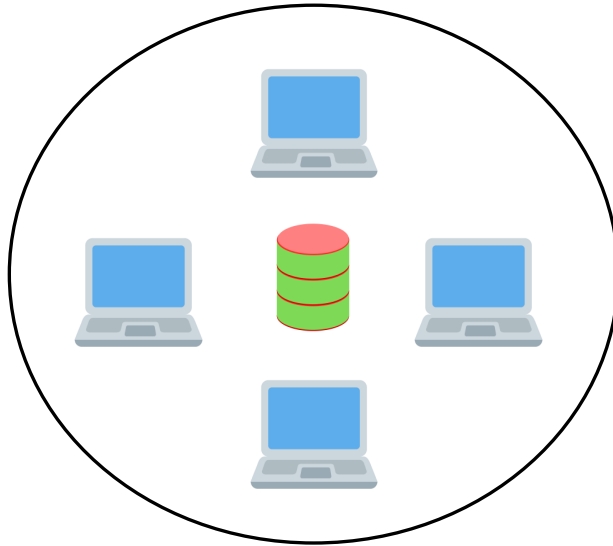
Sharding



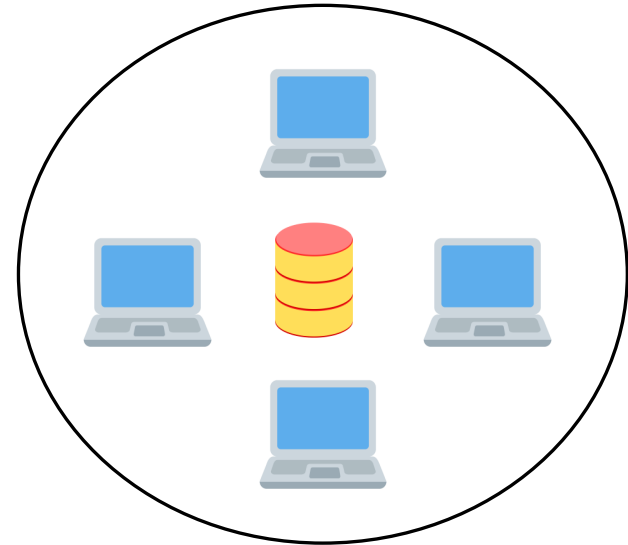
Sharding



Network A



Network B

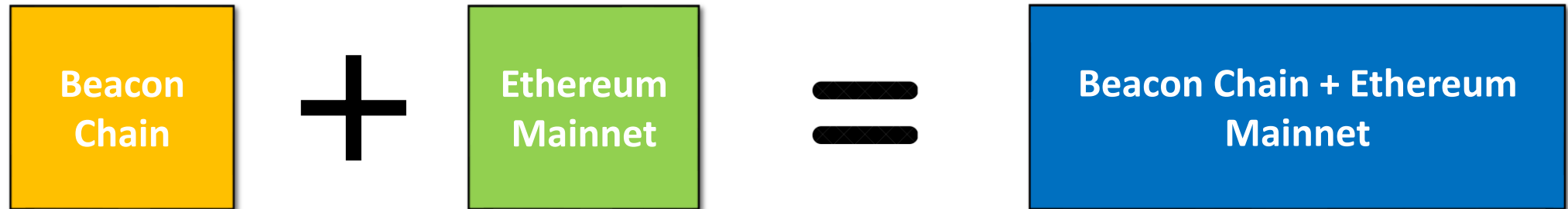


Network C

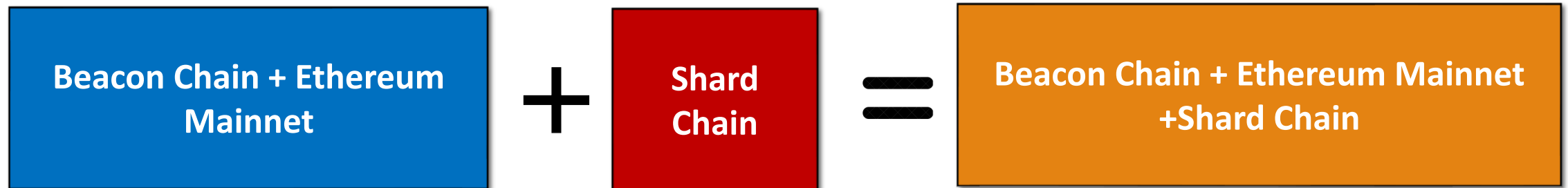
Major benefits

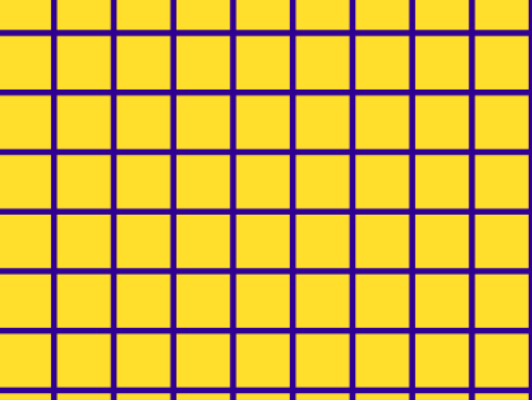
- Transactions per second increase.
- Powerful and expensive computers will not be needed.
- More validators will join.
- Energy consumption will reduce.

Sharding



Sharding

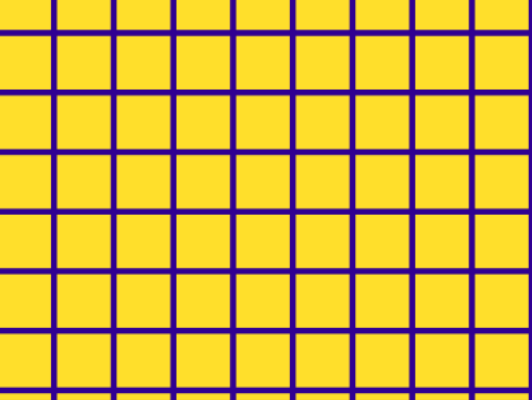




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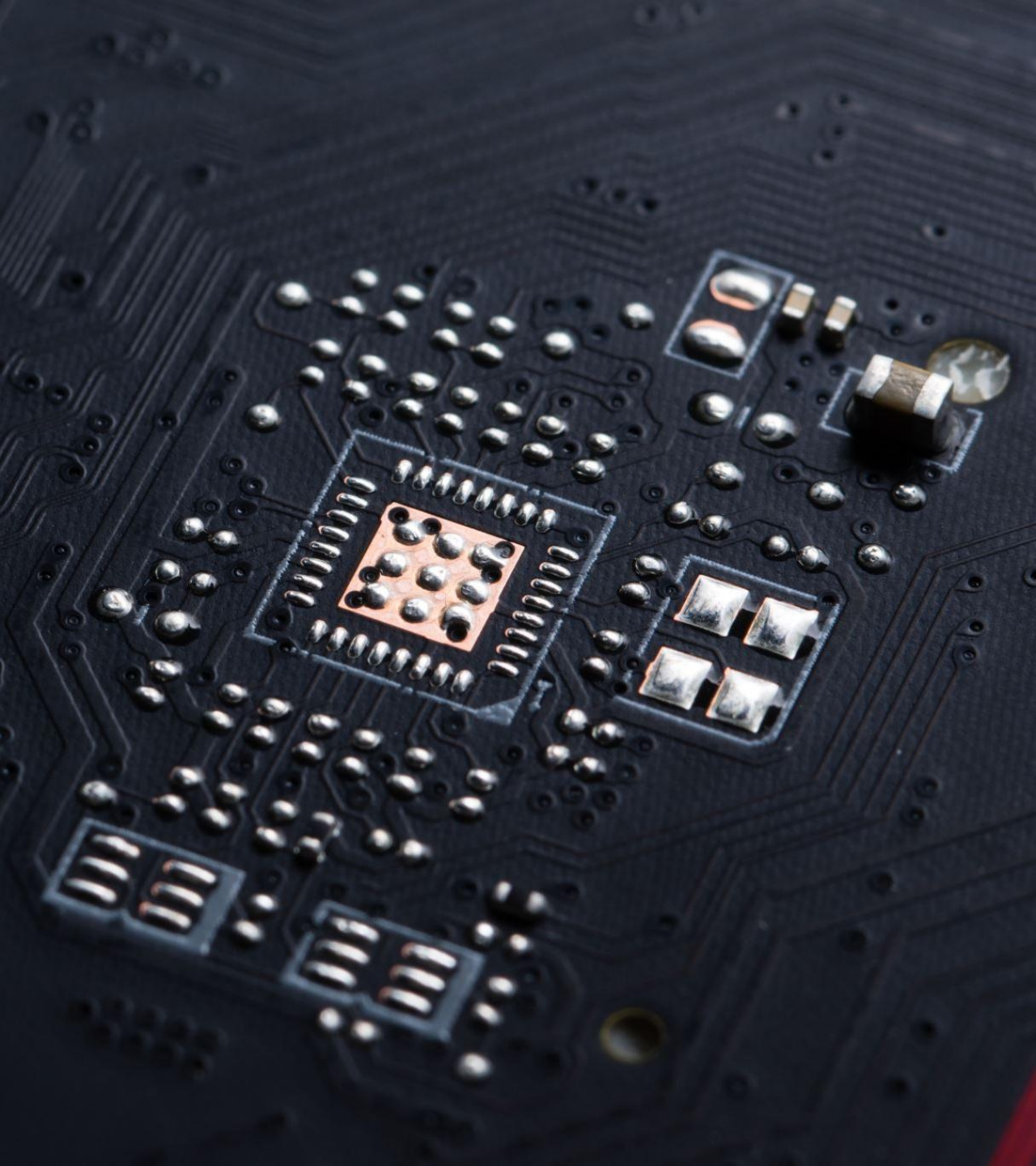




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Decentralized Autonomous Organization (DAOs)

Decentralized Autonomous Organization (DAOs)

Director



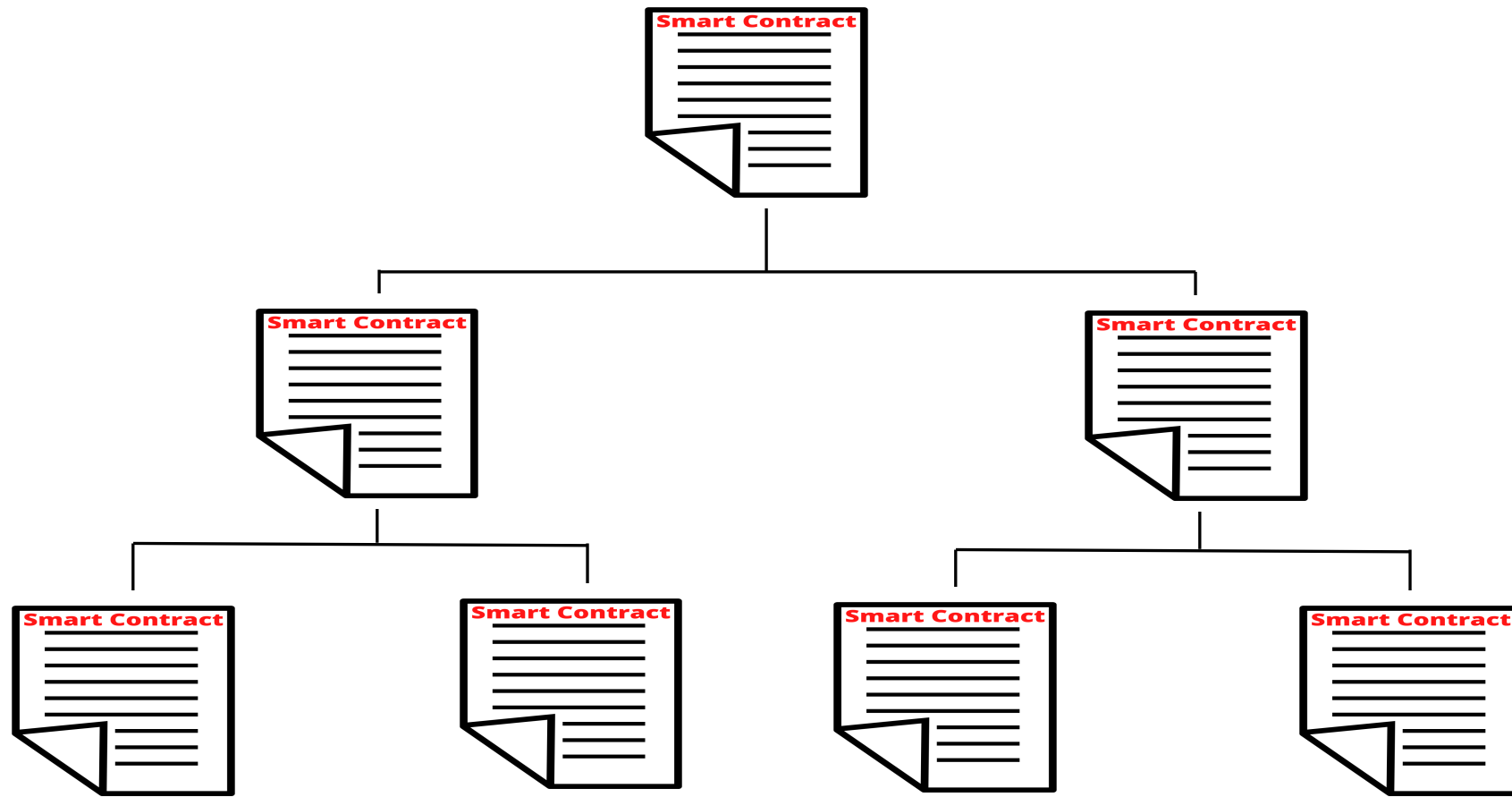
Manager



Employee



Decentralized Autonomous Organization (DAOs)



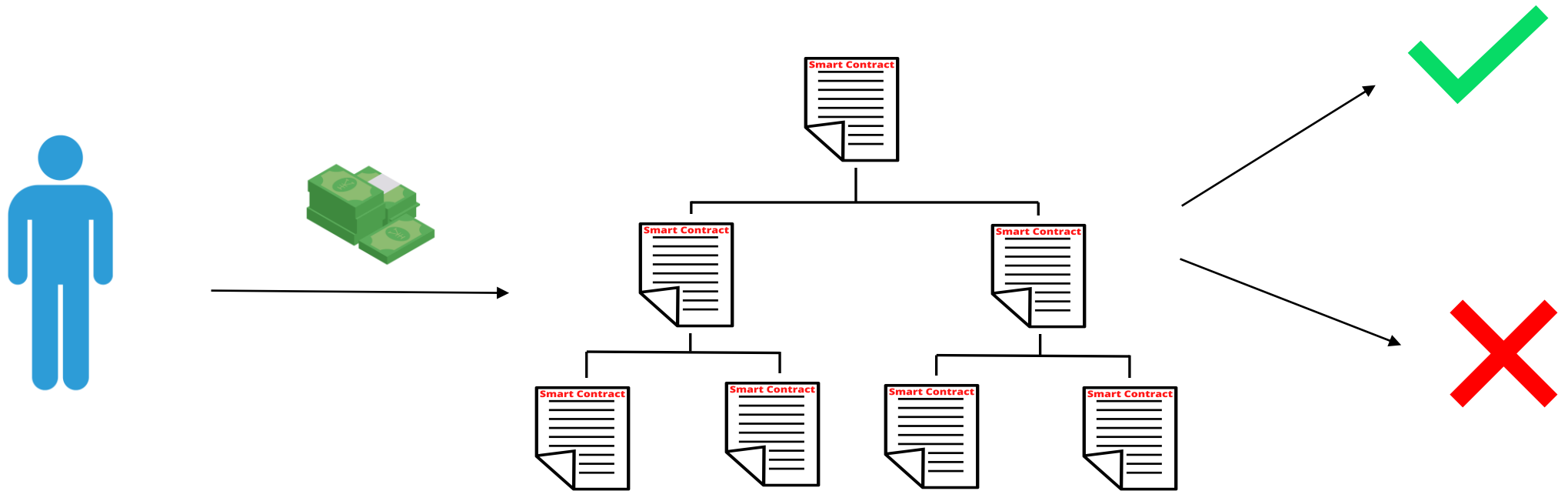
DAO vs Organization

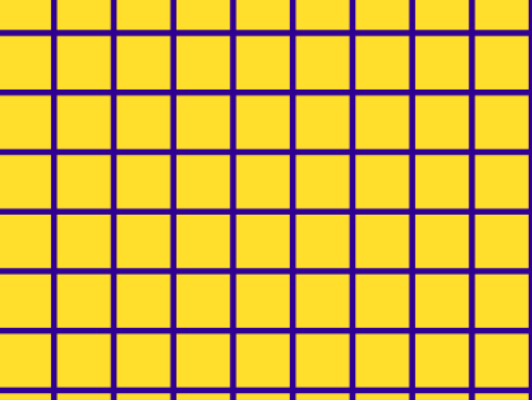
DAO	A traditional organization
Fully democratized.	Usually hierarchical.
Voting required .	Voting may or may not require.
No trusted intermediary to count vote.	Outcome of voting must be handled manually.
Services offered are handled automatically.	Requires human handling, or centrally controlled automation.
All activity is transparent and fully public.	Activity is typically private, and limited to the public.

Decentralized Autonomous Organization (DAOs)



Decentralized Autonomous Organization (DAOs)

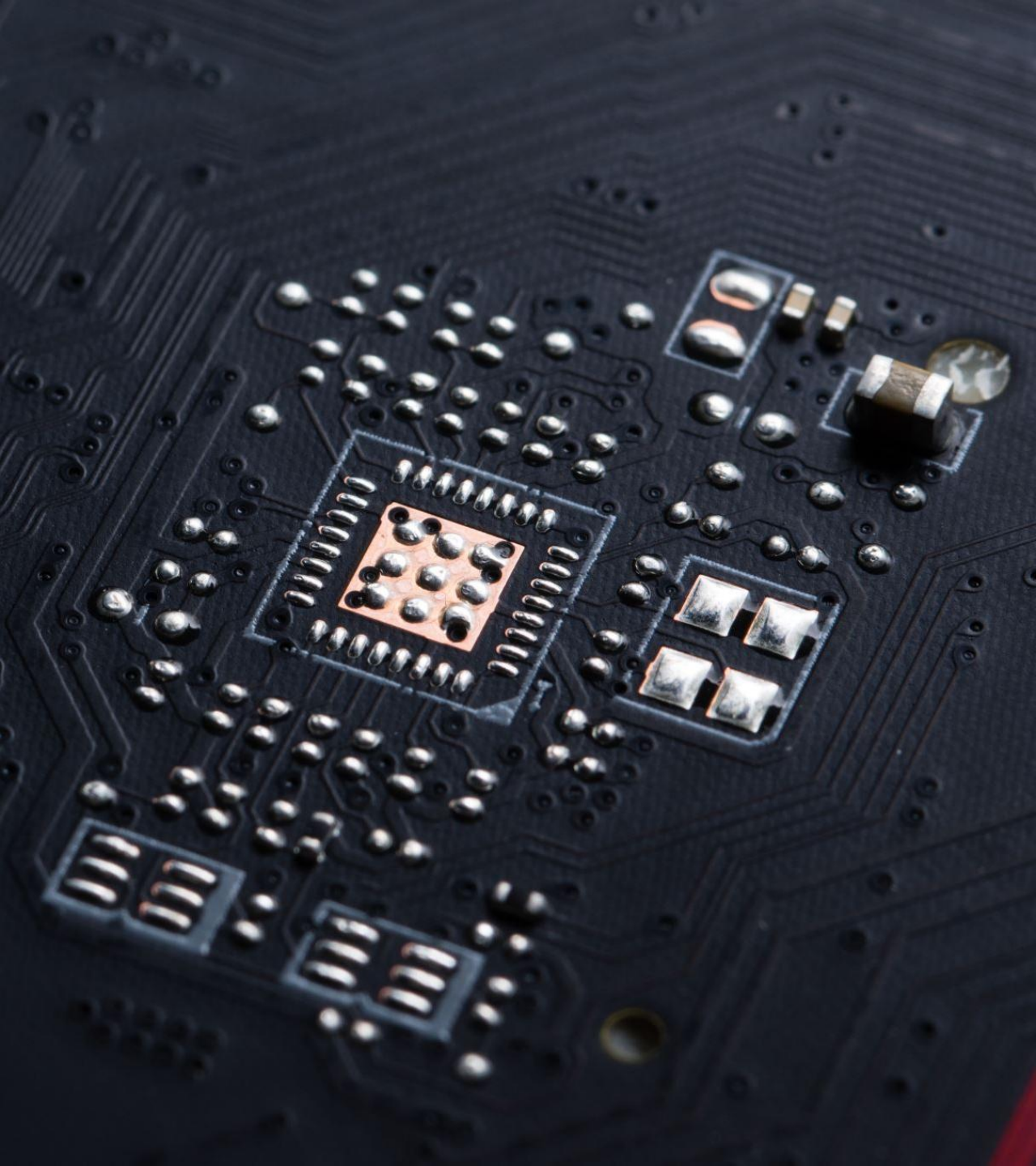




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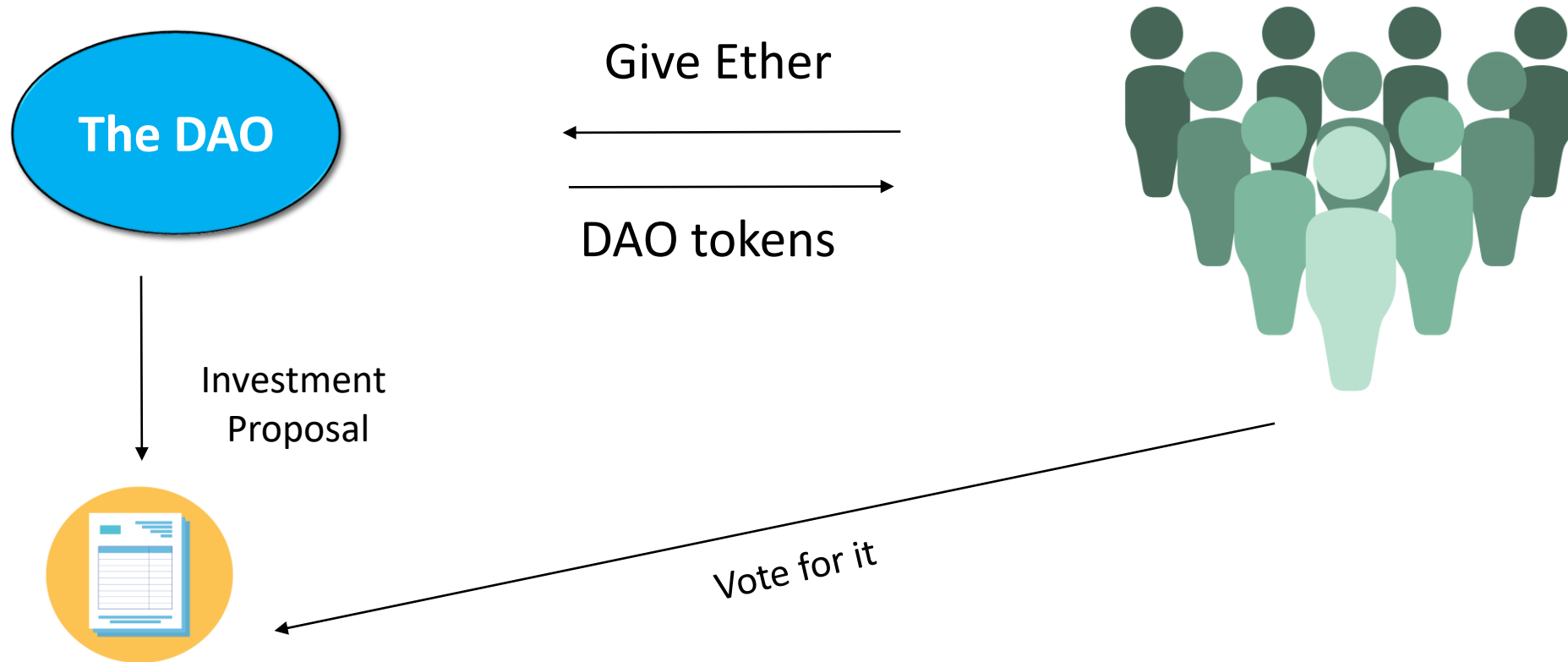
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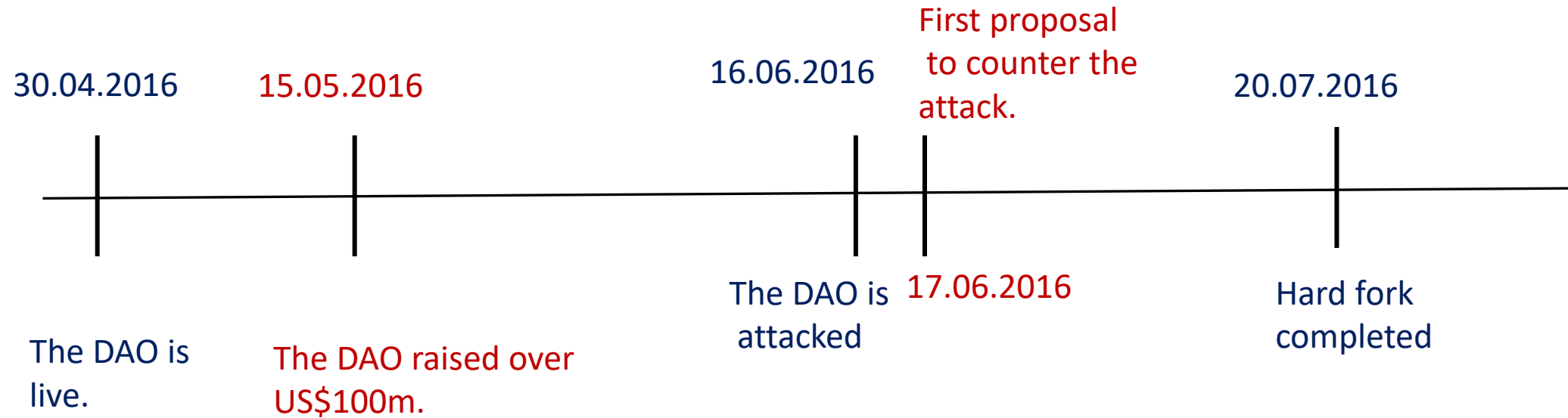


The DAO attack

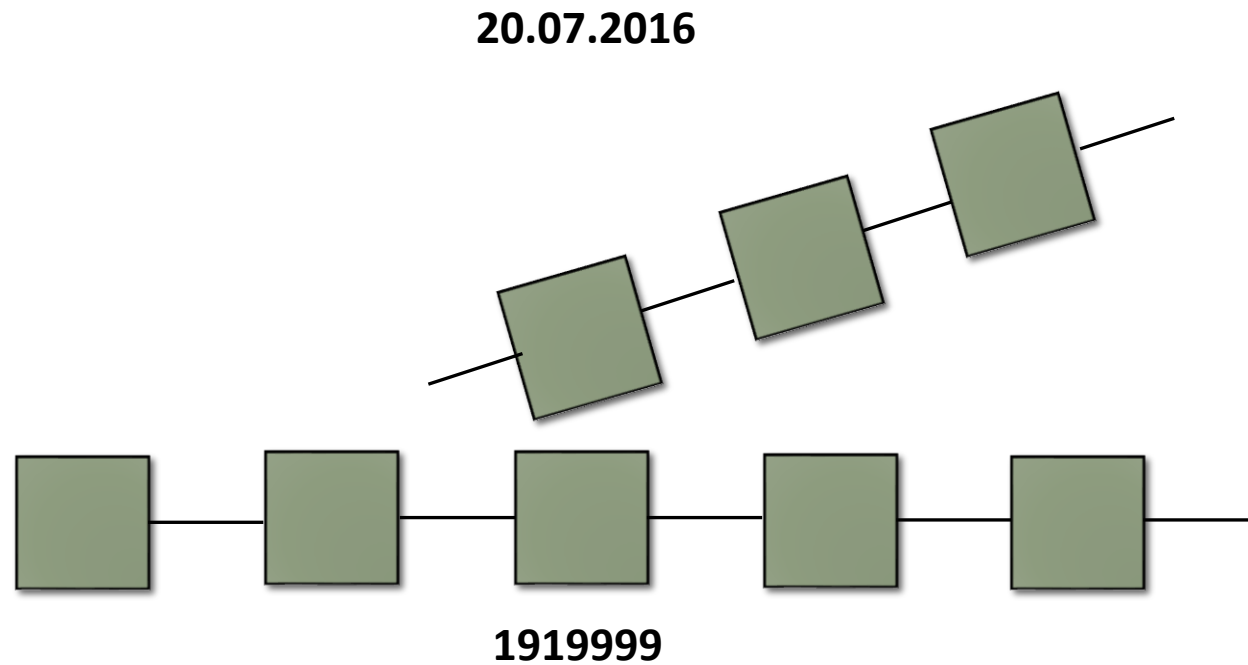
The DAO Attack



The DAO Attack

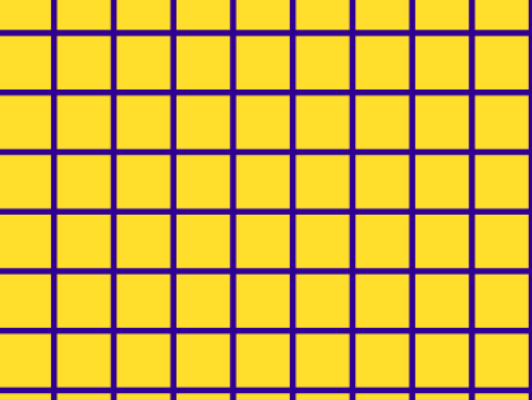


The DAO Attack



ethereum
classic

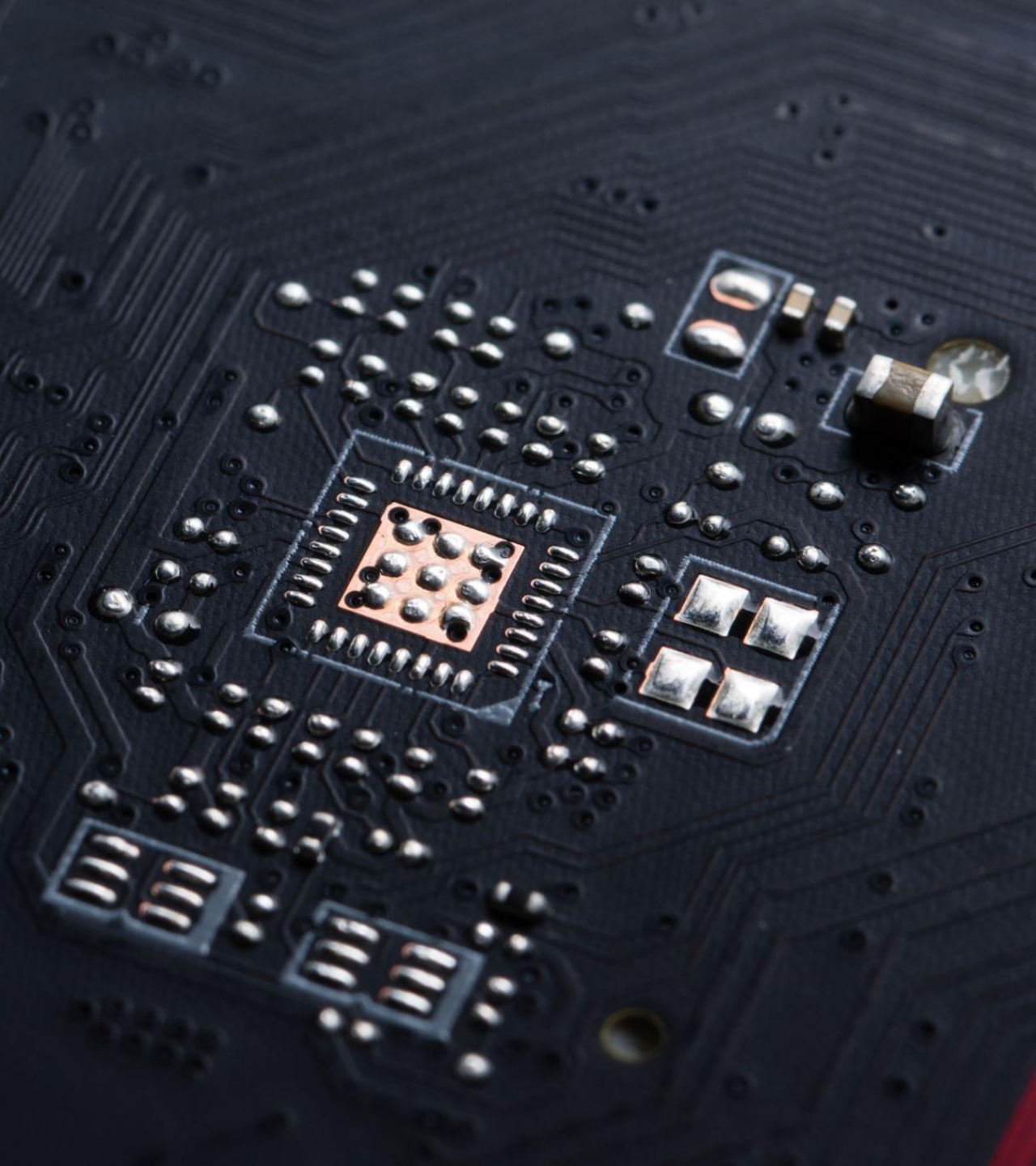




GIVE THIS VIDEO A THUMBS-UP !

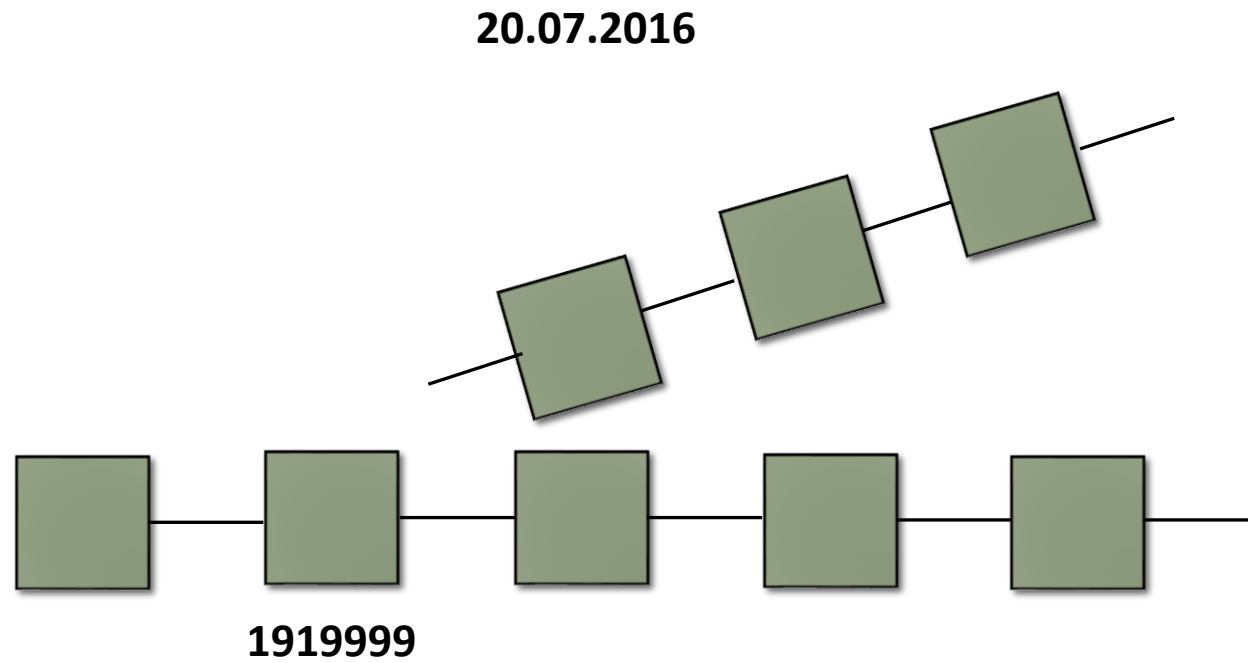
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Hard Fork

Hard Fork



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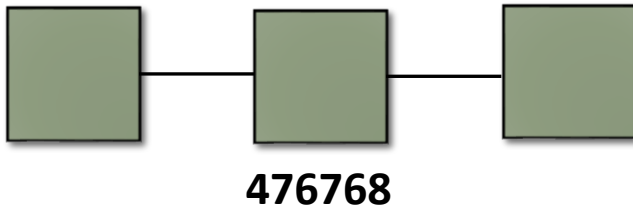


Hard Fork

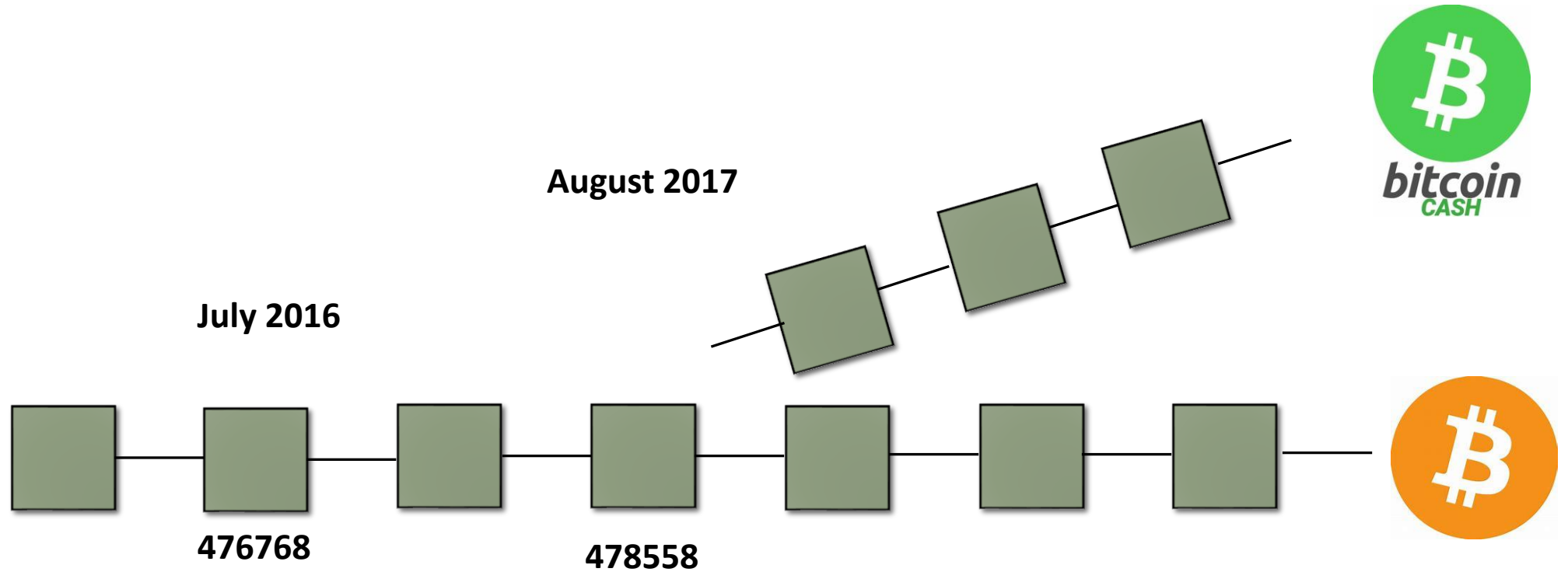
- During a hard fork, software implementing a protocol and its mining procedures is upgraded.
- Once a user upgrades their software, that version rejects all transactions from older software, effectively creating a new branch of the blockchain.
- However, those users who retain the old software continue to process transactions.

Hard Fork

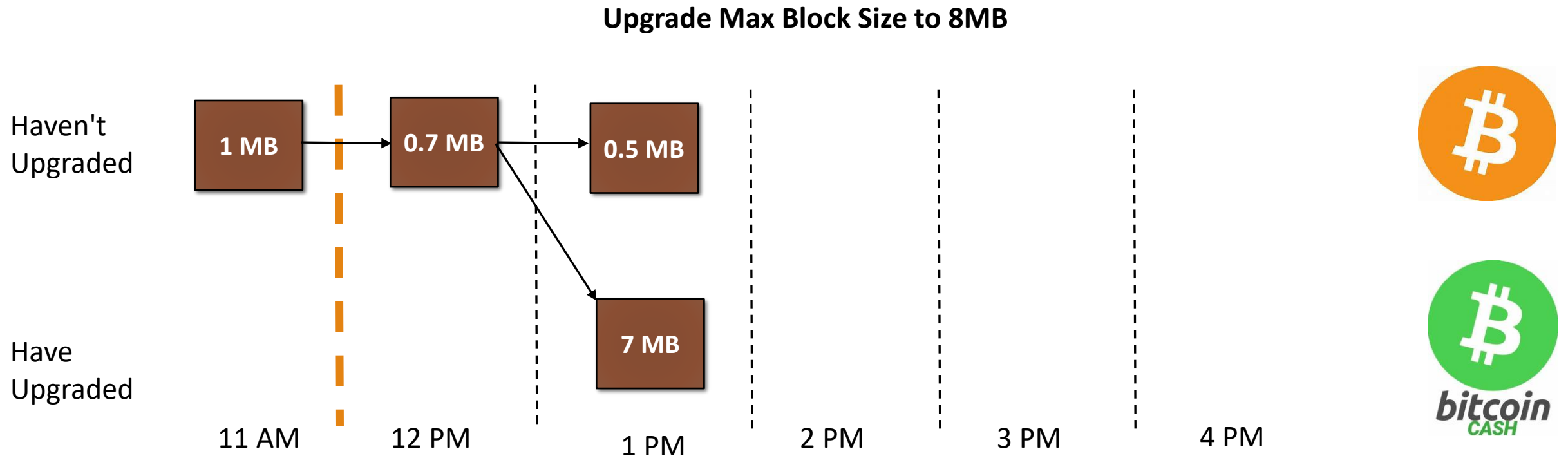
July 2016



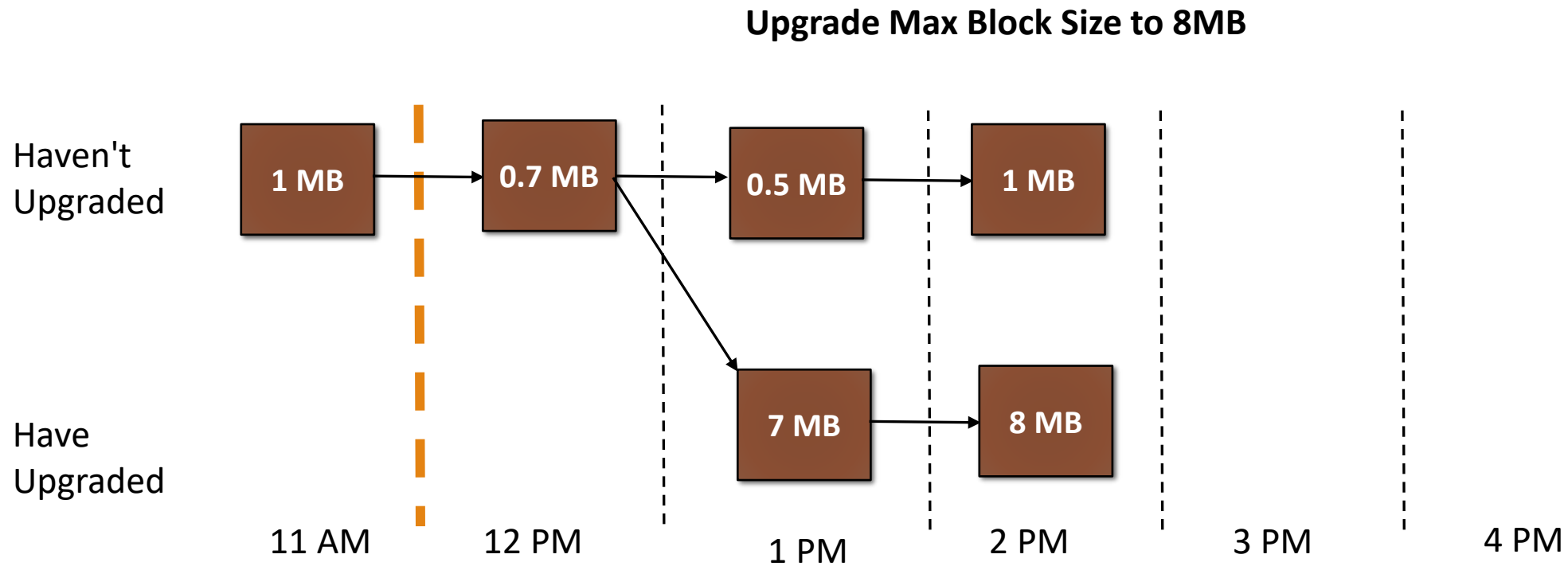
Hard Fork



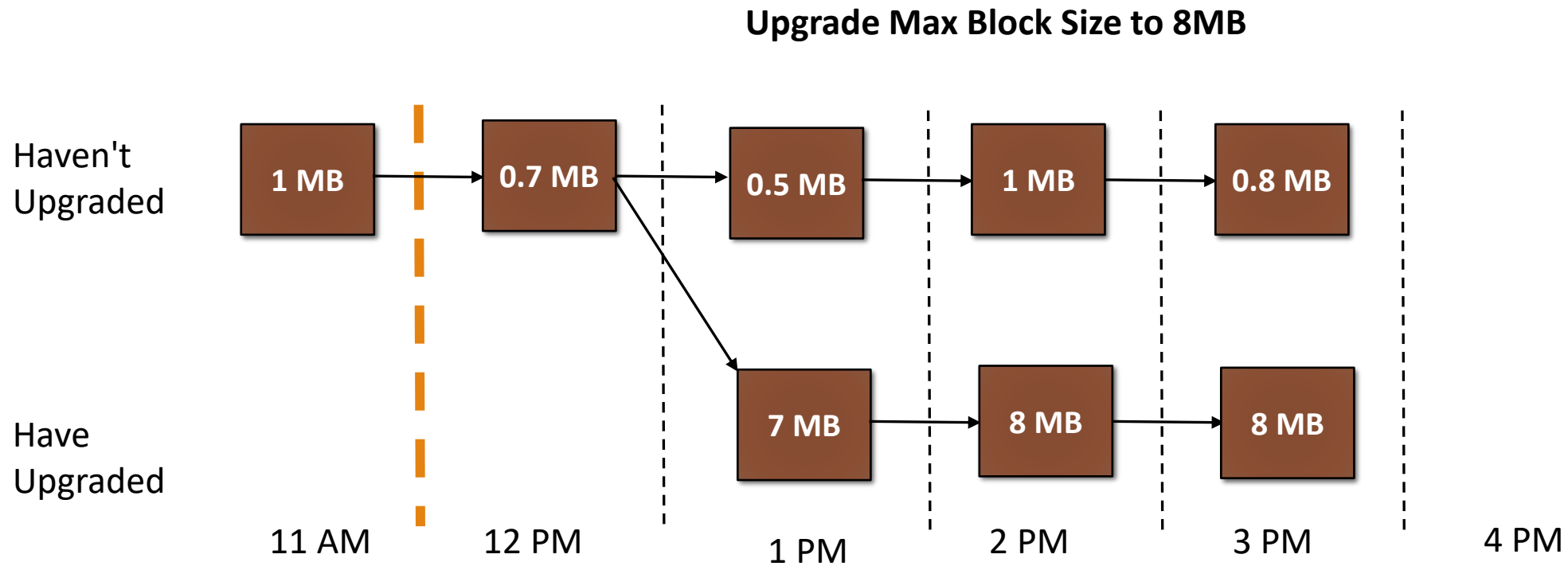
Hard Fork



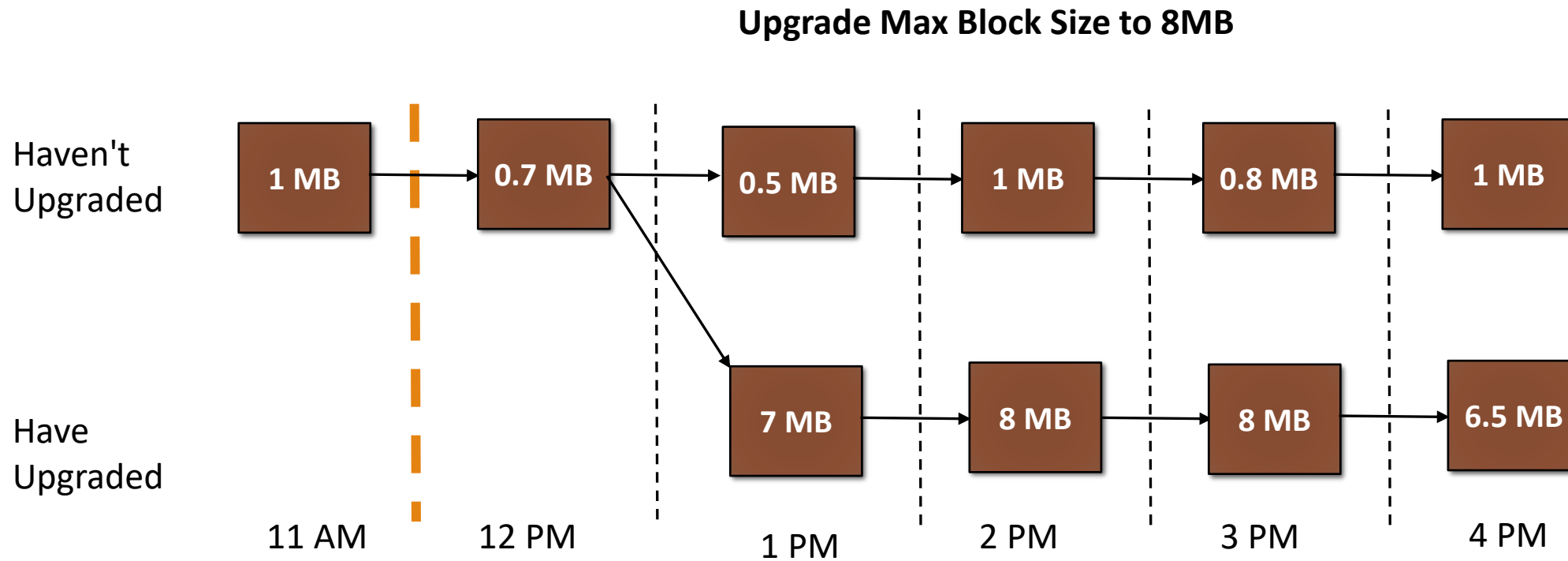
Hard Fork

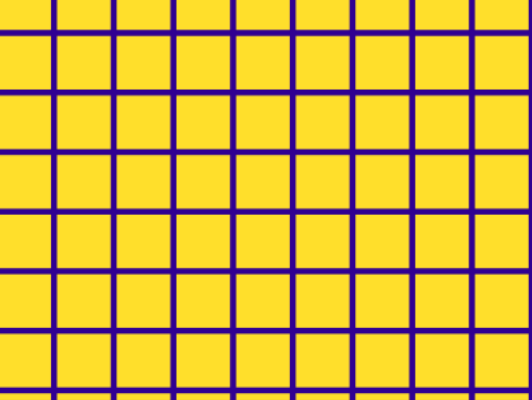


Hard Fork



Hard Fork

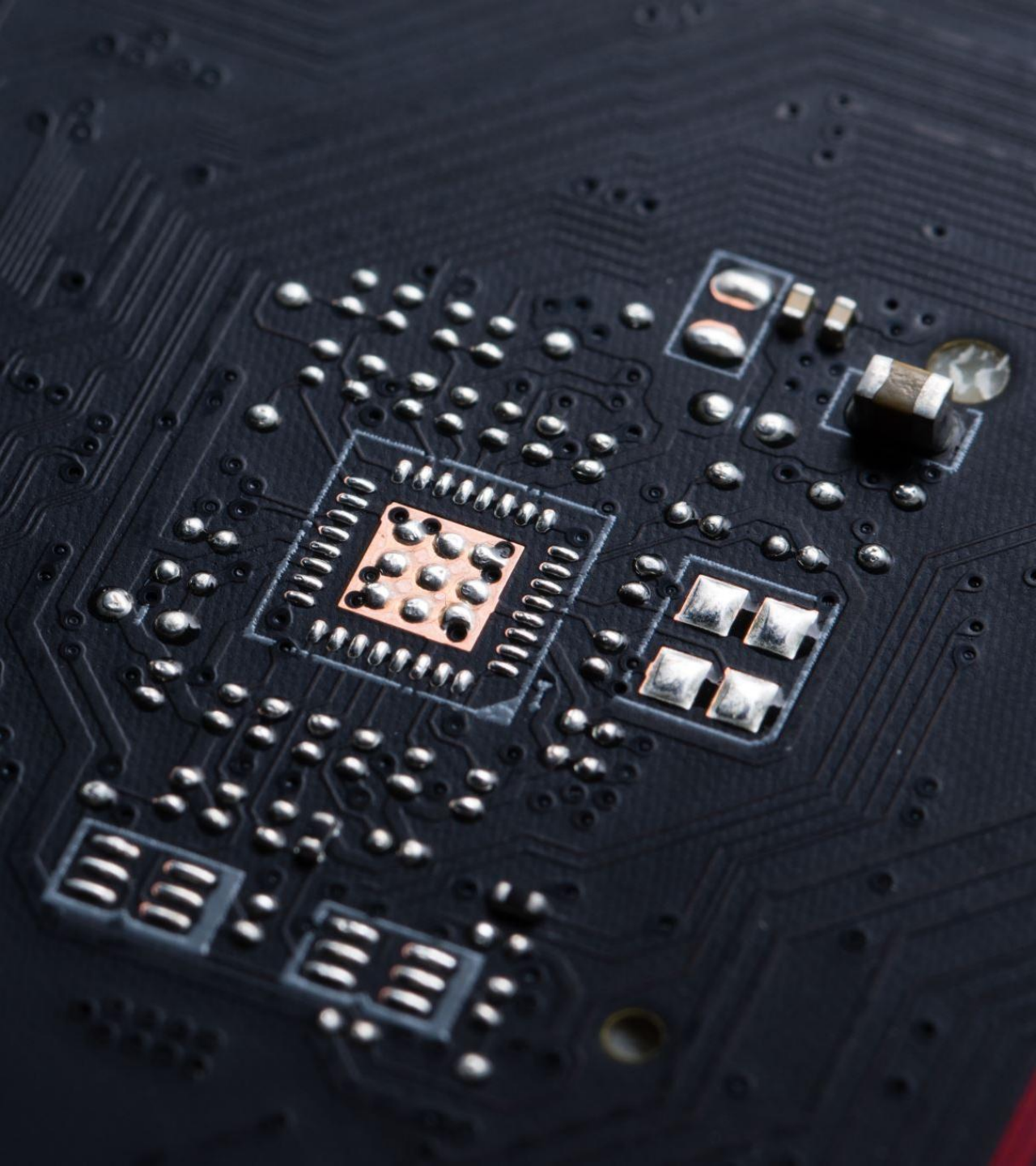




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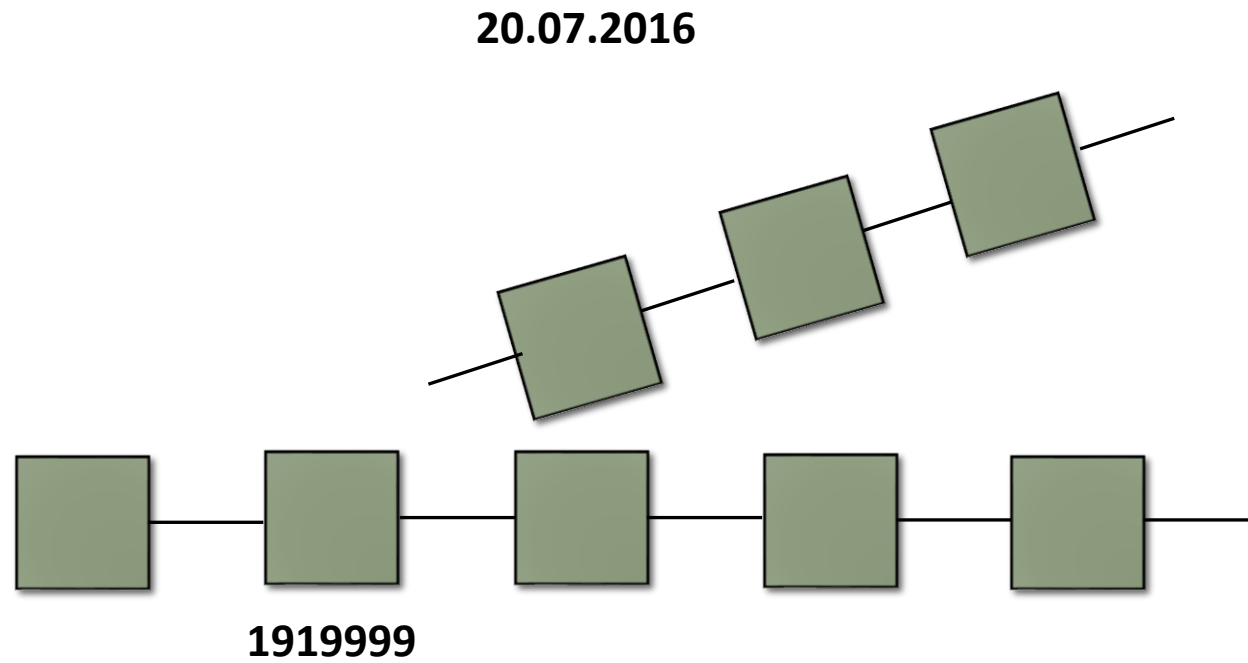


Soft Fork

Soft Fork

- Soft forks are a change to the protocol, but the end product remains unchanged.

Hard Fork



ethereum
classic



Soft Fork

- Soft forks are a change to the protocol, **but the end product remains unchanged.**
- A soft fork is a *backward-compatible* upgrade, meaning that the upgraded nodes can still communicate with the non-upgraded ones.
- Old nodes(not upgraded nodes) could still validate blocks and transactions (the formatting didn't break the rules), but they just wouldn't understand them.

Soft Fork



Soft Fork



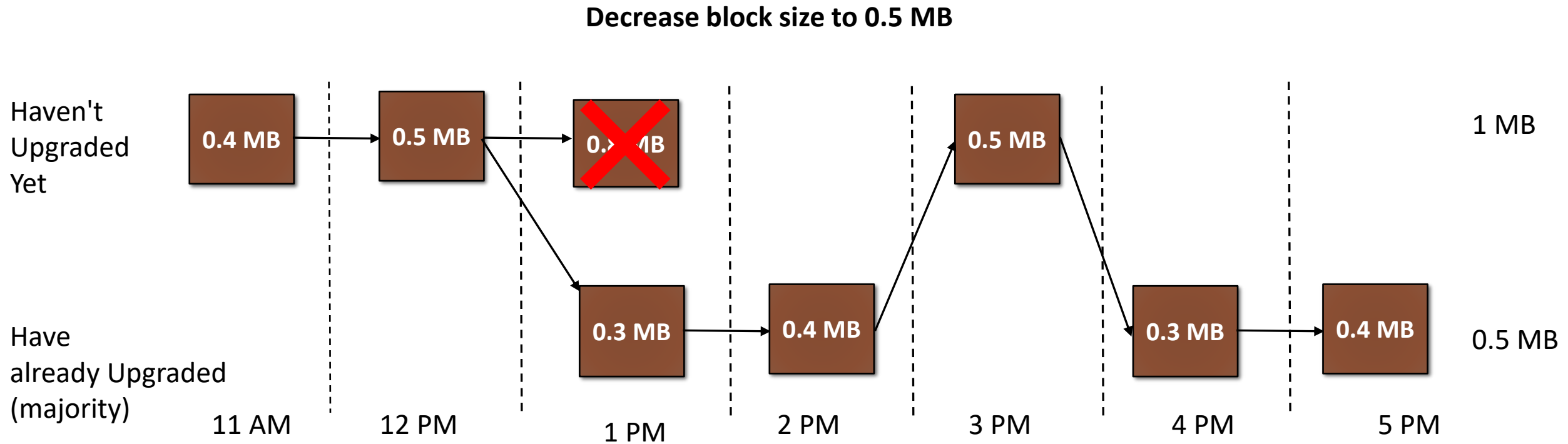
Soft Fork

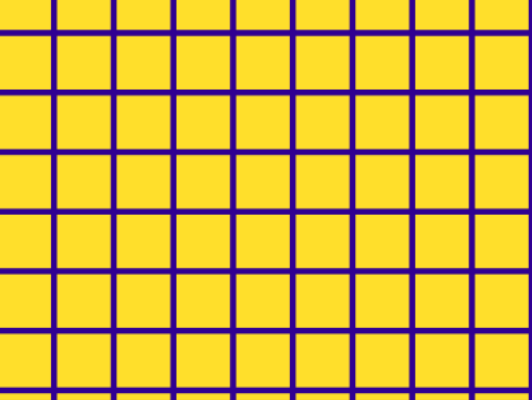


Soft Fork



Soft Fork



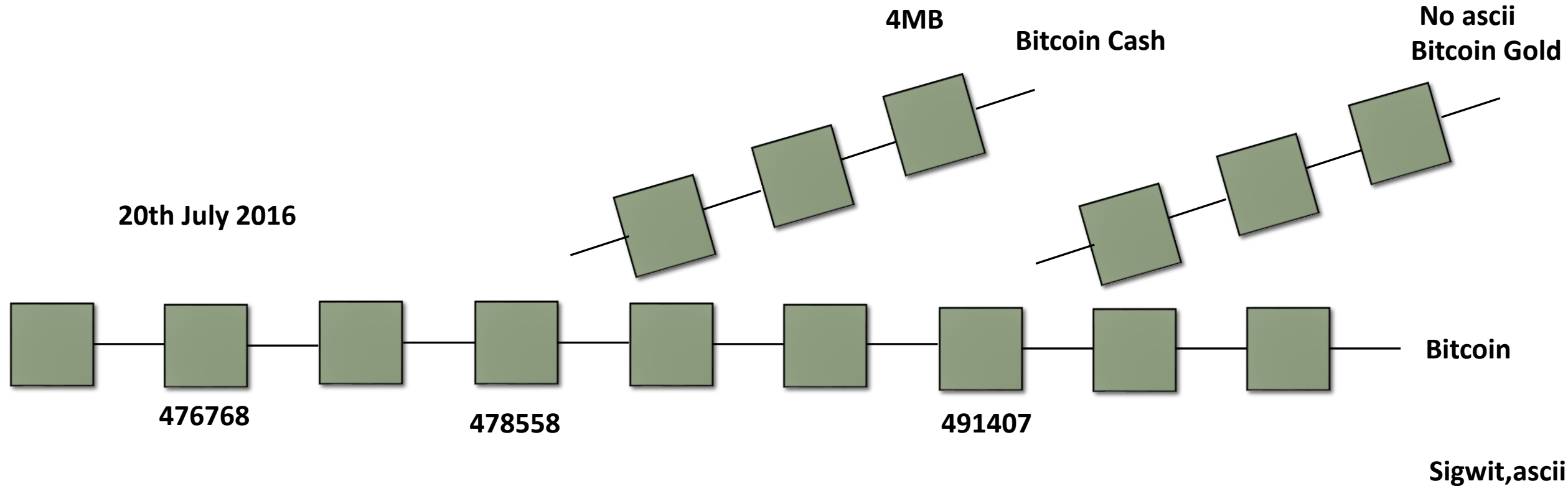


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The DAO Attack





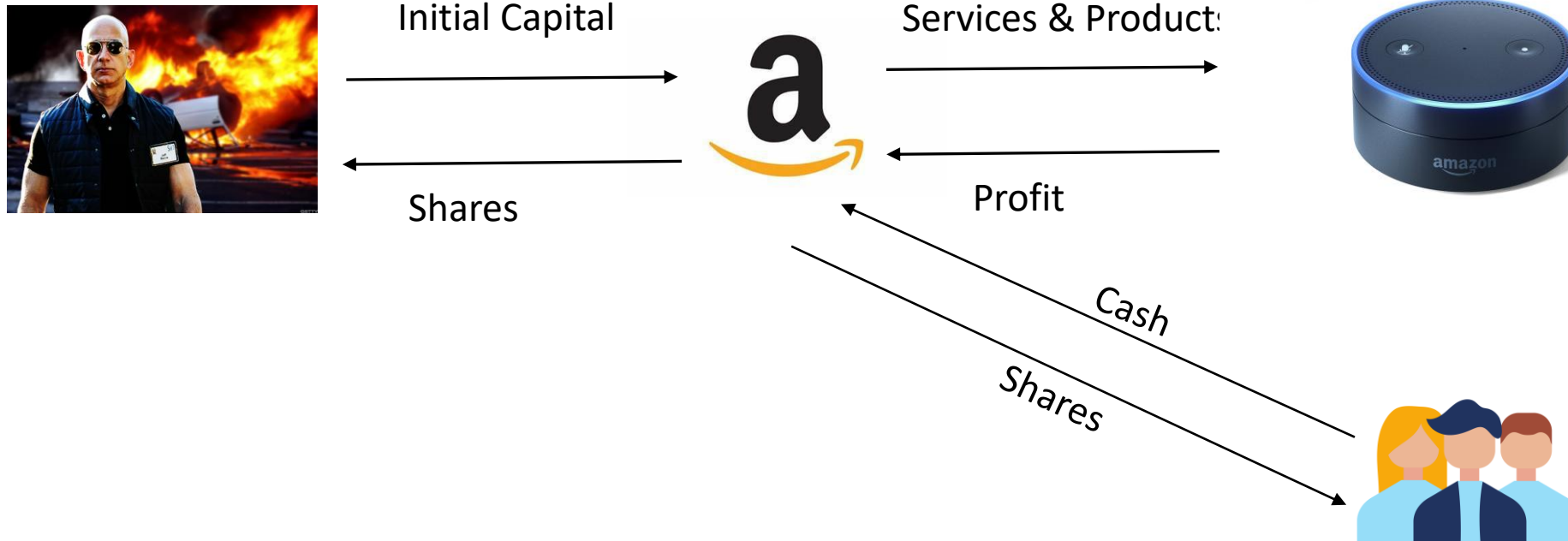
ICO's

Token

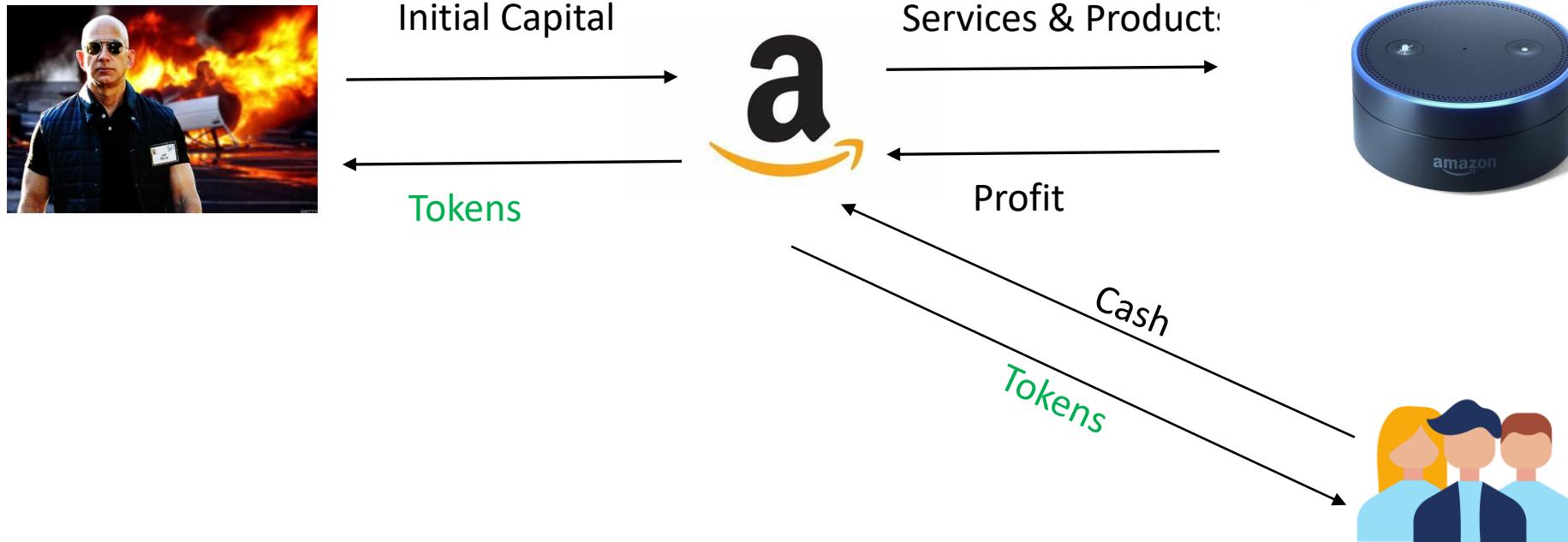
- **Tokens** are a type of **cryptocurrency** that represents an asset or specific use and resides on their **blockchain**

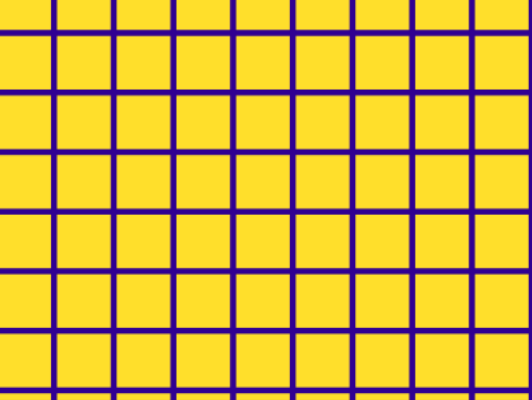


Initial Coin Offering (ICO)



Initial Coin Offering (ICO)

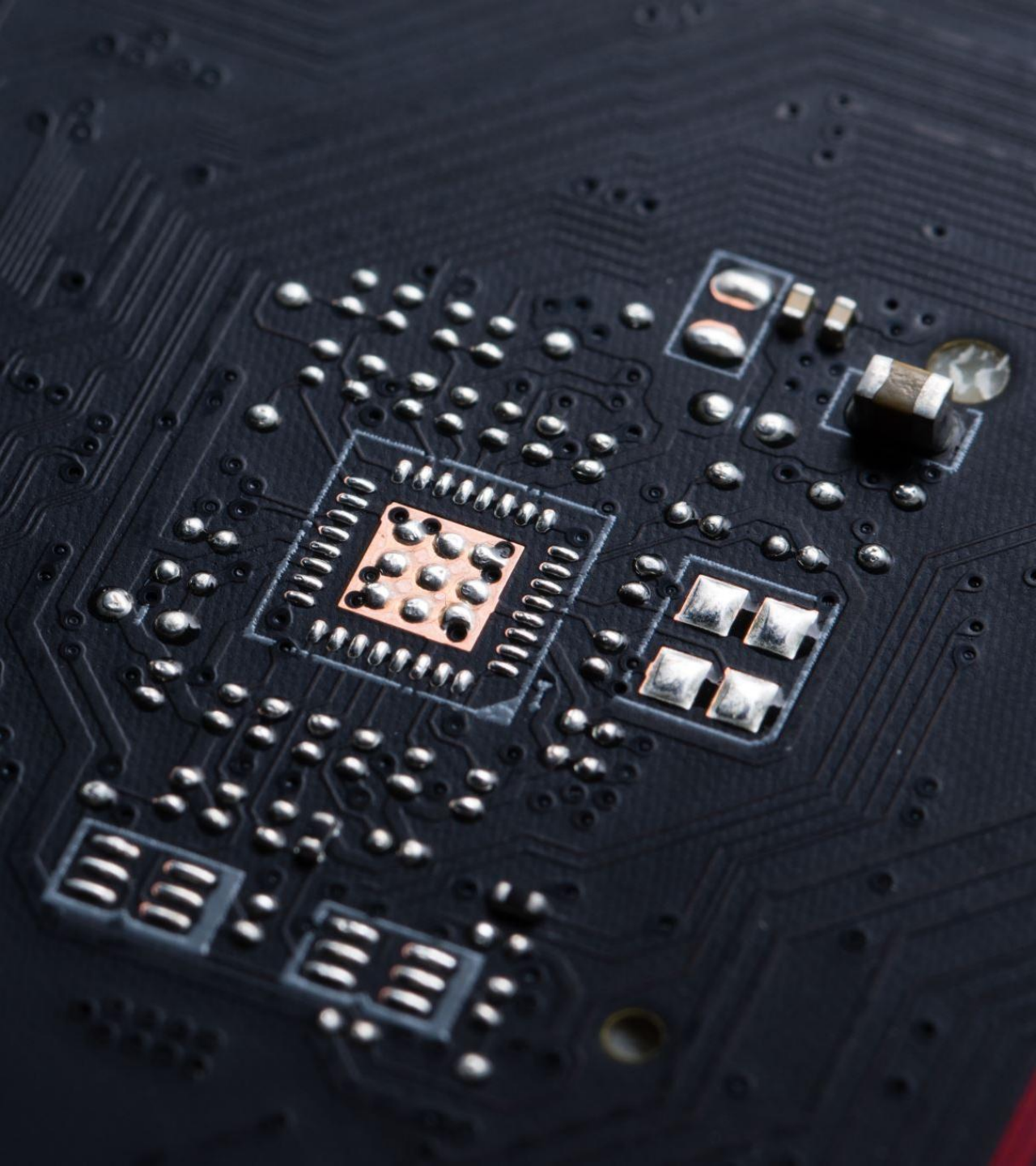




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Alt Coins

Alt Coins

- Altcoins are generally defined as **all cryptocurrencies other than Bitcoin (BTC)**.

Alt Coins



What is the need of Alt Coins ?

Application and Uses

Protocol

Technology

CONGRATULATIONS



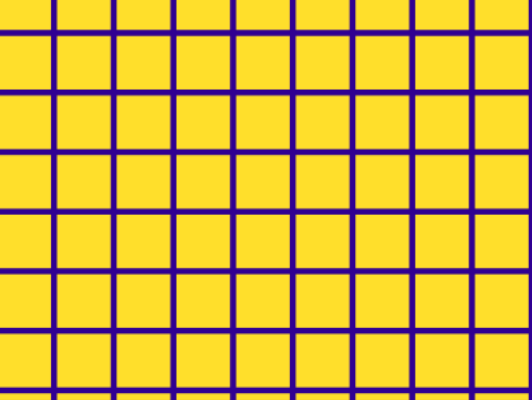
Mode of Contact



Instagram - @codeeater21



www.facebook.com/groups/codeeater21/

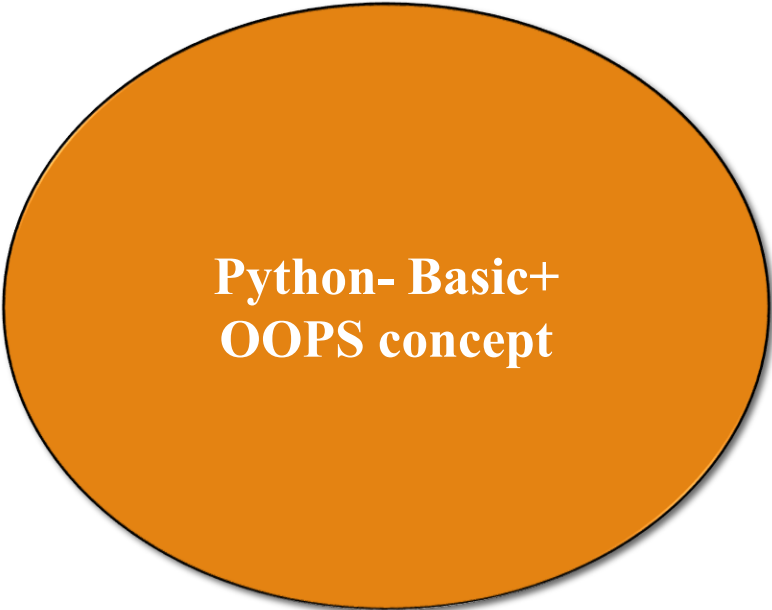


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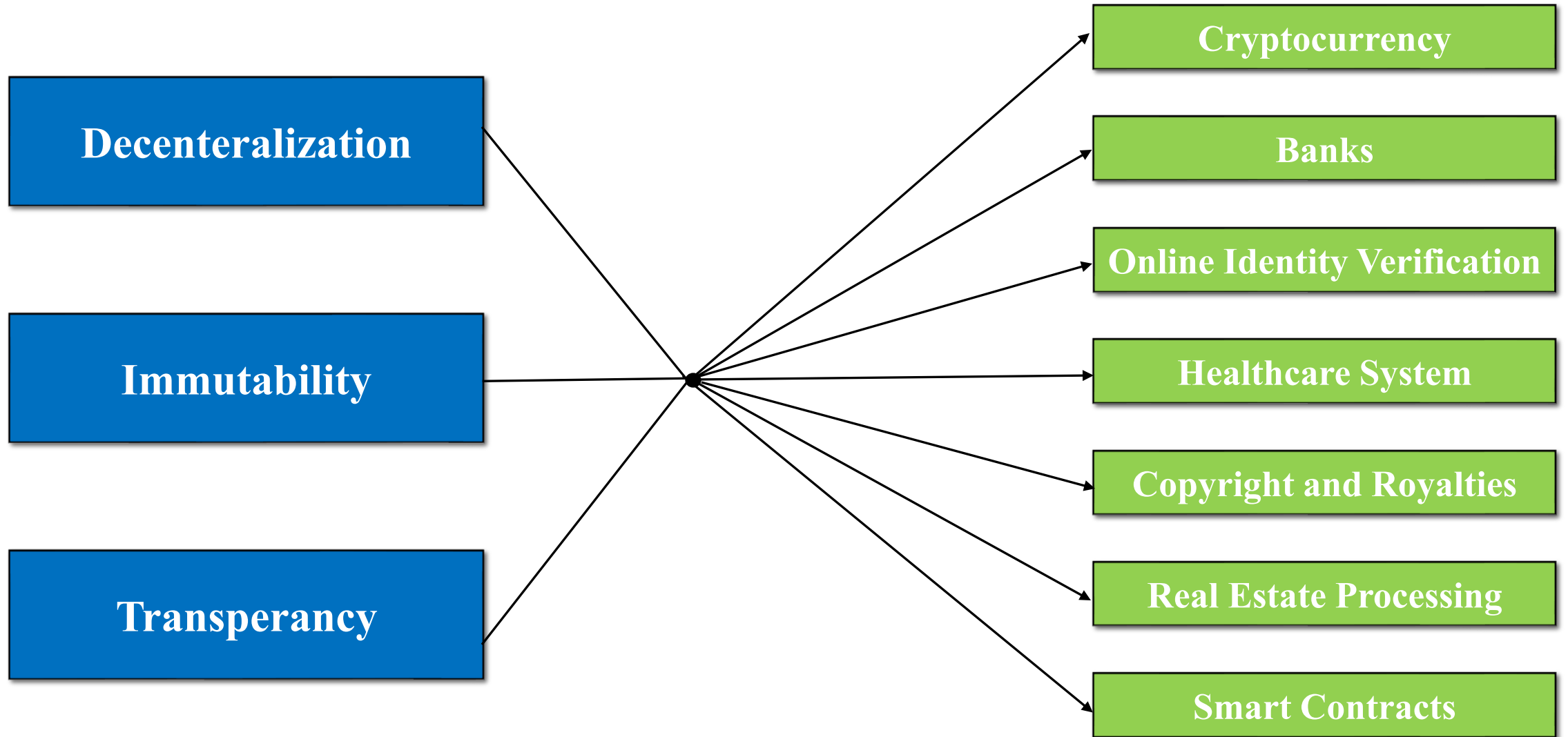
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Prerequisite for Implementation of Blockchain(Coding Part)



**Python- Basic+
OOPS concept**



Properties of Blockchain

Decentralization

Immutability

Transperancy